Can Place-Based Subsidies Improve Labor Market Efficiency?*

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Abstract

Place-based subsidies are often targeted to firms in distressed regions on the premise that local labor is underutilized. Yet whether such policies improve labor market efficiency remains less understood. In this paper, we explore a different theoretical rationale—*labor rationing*—and study its empirical and quantitative implications for the incidence and efficiency of a large investment subsidy program in Italy. To identify these effects, we exploit program allocation rules at two levels: a regression discontinuity across applicant firms and a recentered instrument across labor markets. Combining application records, longitudinal firm data, and matched employer–employee data, we show that funded firms expand employment with little change in average wages or workforce composition. Incumbent workers benefit primarily through additional weeks of employment, especially those with weaker labor market attachment, while wage gains are concentrated among workers earning above their contractual floors. We find no evidence of competitor firms increasing wages in the same labor market, and no detectable change in the mobility of workers in these markets. Together, these findings suggest that contractual minima bind for most workers, yielding employment rationing. By alleviating the underemployment generated by rationing, the subsidies improve labor market efficiency; yet workers capture a small share of the program's value, and whether its estimated social benefits exceed its costs hinges on the magnitude of investment distortions and the persistence of the created jobs.

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1 Introduction

> *"... a state of surplus labor cannot be unambiguously identified without reference to a specific wage rate."*
>
> Louis Lefeber, "Planning in a Surplus Labor Economy" (1968)

Persistent geographic disparities in employment pose a longstanding challenge for policymakers (Austin et al., 2018; Bartik, 2020). Across the world, place-based policies have become a prominent response, subsidizing firms to invest and create jobs in distressed regions at an annual cost of approximately $100 billion in the United States and $60–70 billion in the European Union (Hanson et al., 2024; Berkowitz et al., 2025). [1] While a large empirical literature has examined whether these policies deliver the employment growth they promise (Criscuolo et al., 2019; Siegloch et al., 2024; Hyman et al., 2022), the efficiency implications of the jobs they create remain sharply contested. In standard models without pre-existing distortions, subsidizing job creation distorts otherwise efficient private decisions (Glaeser and Gottlieb, 2008; Kline and Moretti, 2014b; Busso et al., 2013), leaving the justification for such interventions to rest primarily on redistributive grounds (Kline and Moretti, 2014b; Gaubert et al., 2025).

The theory of the second best, however, indicates that public interventions can generate efficiency gains in the presence of pre-existing distortions (Lipsey and Lancaster, 1956). While the literature has emphasized production-side externalities, the persistent joblessness of targeted markets points to the connected idea that local labor markets may fail to clear, providing an overlooked efficiency rationale centered on *labor rationing*. When prevailing wages exceed market-clearing levels, firms hire fewer workers than are willing to work at those wages, and employment is determined by labor demand rather than supply. This creates a first-order wedge between the market wage and workers' reservation value of time. Under such a wedge, subsidizing employment does not merely distort choices; by drawing rationed individuals into productive work, the policy directly attenuates pre-existing deadweight loss, creating scope for first-order efficiency gains.

In this paper, we study the *theoretical, empirical,* and *quantitative* implications of labor rationing for place-based policies. Whether labor rationing matters in practice hinges on an empirical assessment of how local labor markets accommodate subsidy-induced idiosyncratic

[1] The annual figure aggregates major federal, state and local programs spanning business recruitment, workforce development, community redevelopment, small business promotion and technological innovation; it excludes smaller programs and tax expenditures such as Opportunity Zones, and so understates the total (Hanson et al., 2024). On the narrower category of business incentives alone, Bartik (2020) estimates $47.1 billion of state and local and $5.4 billion of federal incentives in 2015, with their value tripling between 1990 and 2015; see also Slattery and Zidar (2020).

demand shocks. We examine these adjustments across three complementary levels: within recipient firms, among individual workers, and across non-applicant competitors in the same local labor market. Comparing these empirical responses to the predictions of alternative theoretical benchmarks allows us to determine whether the market accommodates expansion through prices or quantities. These comparisons in turn provide the empirical moments needed to value the efficiency gains of the policy and weigh them against the public resources required to create them.

We conduct our analysis in the context of the *Incentives for Disadvantaged Areas* (IDA), one of Europe's largest place-based investment subsidy programs. Operating within the European Union's state-aid framework, IDA committed €26 billion in grants across 21 calls issued between 1996 and 2007 to firms establishing or expanding production in economically distressed regions of Italy. Projects were ranked within funding pools according to predetermined criteria and financed in order until each pool's budget was exhausted. We combine the universe of applications with matched employer-employee records, workers' longitudinal income histories, and contractual wage floors.

Two features make this setting uniquely suited to evaluating the role of labor rationing. First, Italy combines stark regional disparities with severe downward wage rigidity, which can be directly traced back to its wage-setting rules (Boeri et al., 2021). As we directly observe in our data uniform contractual wage floors that confront firms facing vastly different levels of local productivity and demand, this institutional constraint is empirically measurable, allowing us to explicitly link observed downward wage rigidity to formal institutional rules. Second, IDA's allocation rules generate sharp quasi-experimental variation in grant receipt, providing exogenous, firm-level labor-demand expansions. This policy design allows us to conduct distinct empirical tests for the presence and role of labor rationing, tracing how idiosyncratic demand shocks propagate across recipient firms, individual workers, and non-applicant competitors in the surrounding local labor market.

A theoretical framework clarifies the economic mechanism and guides our empirical strategy. In an undistorted competitive labor market, wages equal workers' marginal opportunity costs; by the envelope theorem, expanding employment yields no first-order labor-allocation gain. When rigid wages bind above market-clearing levels, however, firms face queues of willing workers. An outward shift in labor demand can then expand employment along a flat labor supply schedule without increasing wage rates, directly attenuating the pre-existing distortion. A flat wage response alone, however, does not establish rationing, as substantial employment growth with little wage adjustment may also arise when labor supply is highly elastic and markets clear. Distinguishing among these alternatives requires micro-data on where workers

are recruited from, how contractual wage floors bind across workers, and whether competing firms experience wage pressure or worker poaching. The framework also makes clear that any efficiency gain from additional employment must ultimately be assessed against the resources required to induce it.

Guided by this framework, we structure our empirical analysis around two tests. The first test begins at the participating firm in the IDA subsidy bids. Identifying these effects is challenging because firms receiving investment subsidies may have different growth prospects from those that do not. We exploit discontinuities in IDA's funding rankings, comparing applicants that narrowly received or lost a grant. Funded firms increase employment by about 13 percent within three years, while average compensation per worker rises by only about 1.5 percent. Such a large employment response relative to compensation could reflect highly elastic labor supply, as commonly inferred in standard evaluations, rather than subsidized firms expanding employment at institutionally constrained wages. Firm-level responses alone cannot distinguish these explanations or reveal how individual workers benefit.

We therefore examine the subsidy incidence on individual workers using matched employer-employee data. Employment grows through both new hiring and increased retention, with a substantial share of new hires coming from nonemployment. Incumbent workers at subsidized firms capture a 7 percent increase in annual earnings (about €1,040 per year), with flat pre-trends. This gain is driven almost entirely by the quantity margin: weeks worked expand by 5.5 percent, while weekly wage rates remain statistically unchanged. Simultaneously, dependency on unemployment benefits and short-time work falls. Recently hired incumbents gain entirely through weeks, the gains are concentrated among shorter-tenured workers, and starting wages for new hires remain flat. A worker on her supply curve changes her weeks only if her wage changes; more weeks at an unchanged weekly wage, together with fewer weeks on benefits, therefore indicate that incumbents wanted more work at that wage and could not get it. Workers thus benefit substantially from the policy—not through higher pay rates, but through greater employment continuity and access to work.

Matching individual workers to digitized national collective agreements confirms that wage-setting institutions, present in many economies across Europe and Latin America, govern this margin of adjustment. National agreements establish fine occupational wage grids, which we decompose into statutory minima and discretionary firm-level top-ups. In targeted regions, these floors account for a substantial share of compensation, and firm surveys confirm the importance of nationally negotiated wages. We find that weekly wages rise only for workers paid further above their contractual minimums, and by about two log points, whereas workers paid close to their floors benefit entirely through additional weeks worked. Institutional

wage floors shift the incidence of labor demand shocks away from wage rates and toward employment quantities. Where the wage can move, it moves: the flat wage elsewhere is held up by the floor, not a property of labor supply.

A single firm hiring at the going wage is not, on its own, evidence of rationing: in a market that clears, a small firm is a wage taker and also hires at the going wage. Rationing is a property of the market, so our second test looks at the firms that never applied. If the market cleared, the new jobs would have to be filled by someone: competitors would either pay more to keep their workers or lose them. To estimate these market-level spillovers cleanly, we isolate the luck in the allocation: the funded jobs a market actually received relative to what it could expect given its near-cutoff applicants under the program's ranking and budget rules (Borusyak and Hull, 2023). We find that competitors do neither: their wages do not move ($\pi_w = 0.014$, s.e. 0.031), and their employment actually rises, by 15 to 28 jobs per 100 direct jobs. Under the null of a clearing market, the wage estimate implies a local supply elasticity of at least 13; the market test alone cannot separate rationing from a perfectly elastic supply of indifferent workers, which is what the floor split and reservation wages do. Together with the large share of new hires who were not in private-sector employment the year before, and with no detectable change in the mobility of workers attached to these markets, these findings suggest that labor markets accommodate additional labor demand largely through previously underutilized workers.

Finally, we ask what these jobs are worth, separating allocative efficiency *within* the labor market from the policy's *net social value*. Evaluations usually report the public outlay per job; benefit–cost analyses of local job creation value a job at its pay net of the worker's opportunity cost of time, which they must assume (Bartik, 2012). We measure it. In the Italian Labour Force Survey, 79 percent of job seekers in the targeted areas report a reservation wage below the average net pay of new hires (€974 a month). Worker surplus is therefore 23.5 percent of net pay under our most conservative, linear benchmark (€9,500 per direct job over six years), and at least 47 percent if the jobs go first to those who value them most.

This gain is first order, but it does not pay for the program. The grants cost €176,000 per direct job (comparable to Slattery and Zidar, 2020), mostly a transfer to firm owners; raising them at a 25 percent excess burden costs €44,000 per job. Adding the jobs at non-applicant firms and the fiscal gain, labor-market gains over six years cover 56 to 62 percent of that cost under the linear benchmark and at least 81 to 90 percent under efficient rationing, where they break even if the jobs last through year twelve. The marginal value of public funds (MVPF; Hendren and Sprung-Keyser, 2020) is 1.13 to 1.15 without a distortion of investment and about 0.90 with it. Workers receive 6 to 9 percent of the measured value under the linear

and uniform benchmarks, and firm owners the rest.

Our findings reveal that place-based policies and national labor institutions are closely linked. When uniform wage floors prevent compensation from adjusting to local conditions, place-based subsidies act as a second best, expanding employment where prices cannot clear the market. In such rationed environments, mobilizing idle labor generates a first-order social surplus that standard market-clearing evaluations miss. Yet capital subsidies are a costly and indirect instrument: the cost of raising public funds absorbs a large part of these labor gains, and the subsidy may also distort firms' investment. Ultimately, the social return to place-based policy depends on how directly its tools target the wage-setting frictions that keep labor idle.

Related literature. This paper connects to three strands of literature.

First, we contribute to the empirical literature evaluating place-based subsidies (Greenstone et al., 2010; Busso et al., 2013; Criscuolo et al., 2019; Slattery and Zidar, 2020; Atalay et al., 2023; Freedman et al., 2023; Siegloch et al., 2024; Grunau et al., 2025)[2] and the normative literature on their second-best efficiency rationale (Kline and Moretti, 2014b). Existing models typically ground efficiency gains in agglomeration economies (Gaubert, 2018), search frictions (Kline and Moretti, 2013; Bilal, 2023), or employer wage-setting power (Lindenlaub et al., 2025), but none feature rationed employment. We show that binding uniform wage floors provide an efficiency rationale on its own, generating first-order worker surplus even absent agglomeration or search frictions. In doing so, we quantify a benefit missed by conventional cost-per-job metrics-the surplus of workers hired above their opportunity cost of time-moving beyond firm-level employment additionality, the primary focus of the majority of the established literature on place-based subsidies, including earlier evaluations of the IDA (Bronzini and De Blasio, 2006; Cerqua and Pellegrini, 2014; Cingano et al., 2025).[3]

To assess the distortion and the value of alleviating it, we depart from existing work in two directions. First, we link employer–employee data to contractual wage floors bringing wage-setting institutions into the empirical evaluation and examining how they shape policy incidence. Second, we exploit funding thresholds to construct a market-level labor-demand instrument and test whether employment gains generate wage pressure. Together with reservation-wage evidence, these exercises distinguish labor rationing from market clearing

[2]See Neumark and Simpson (2015) for a survey, and Autor et al. (2025) on local adjustment mechanisms.

[3]Bronzini and De Blasio (2006) examine investment additionality, finding that subsidies primarily brought forward projects that firms would otherwise have undertaken later. Both Cerqua and Pellegrini (2014) and Cingano et al. (2025) instead find positive effects on investment and employment. In particular, Cingano et al. (2025) exploit the same funding thresholds used in the first part of our analysis to estimate direct effects on recipient firms and compare the cost per job under political discretion versus objective criteria.

with elastic labor supply, which provides an efficiency rationale for place-based policies and allows us to quantify the worker surplus generated by additional employment.

Second, our analysis contributes to research on the transmission of labor-demand shocks within and across firms. Within firms, we build on the literature on rent sharing and shock pass-through (Van Reenen, 1996; Bertrand and Mullainathan, 1999; Guiso et al., 2005; Card et al., 2014, 2018; Kline et al., 2019; Howell and Brown, 2023; Garin and Silvério, 2024; Friedrich et al., 2024), including recent work emphasizing adjustment through employment margins (Lombardi et al., 2026; Maibom and Vejlin, 2026a). Closely related work shows how institutional constraints shape wage and employment responses and complicate their interpretation as evidence of employer market power (Berger et al., 2025; Bhuller et al., 2025). We bring this perspective to job-creation policies, asking who benefits from an expansion in labor demand when wage floors are fixed and what surplus additional employment generates. Unlike studies of changes in contractual wages or bargaining arrangements (Fanfani, 2023; Dustmann et al., 2025; Schoefer, 2025), we study a demand expansion under a fixed institution.

Beyond recipient firms, our analysis relates to research on local demand propagation and employment multipliers (Blanchard and Katz, 1992; Moretti, 2010; Autor et al., 2013; Feyrer et al., 2017), particularly the responsiveness of employment to demand in distressed regions (Nakamura and Steinsson, 2014; Austin et al., 2018). We examine wage-setting constraints as a mechanism through which additional labor demand translates into employment growth with limited wage adjustment. Our design complements approaches based on procurement and fiscal shocks (Suárez Serrato and Wingender, 2016; Wilson, 2012; Shoag, 2013; Pennings, 2021). We combine employment pledges with funding-pool thresholds to construct a recentered instrument, based on the quasi-random ranking of projects near cutoffs, to estimate employment and wage spillovers onto non-applicant firms (Borusyak and Hull, 2023; Borusyak and Kolerman-Shemer, 2024).

Third, by tracing a labor-demand expansion where the wage-setting constraint is observable, we give empirical content to the classical surplus-labor argument (Lewis, 1954; Sen, 1960; Lefeber, 1968; Ahluwalia, 1973): when the social opportunity cost of labor lies below the prevailing wage, creating employment has a second-best rationale (Lipsey and Lancaster, 1956; Bhagwati and Ramaswami, 1963). Our evidence is also an empirical counterpart to optimal minimum-wage theory, in which a binding minimum wage complements, rather than substitutes for, demand-side subsidies (Lee and Saez, 2012). And because limited wage adjustment can reflect institutional constraints rather than highly elastic supply, our results bear on reading wage and employment responses as evidence on labor supply elasticities and employer power (Manning, 2021; Lamadon et al., 2022; Berger et al., 2025; Loertscher and

Muir, 2025).

The rest of the paper proceeds as follows. Section 2 presents the framework and the two tests, and Section 3 details the institutional setting and data. Section 4 carries out the first test, at funded firms and their workers, including the role of contractual floors (Section 4.3). Section 5 carries out the second test, in the labor market. Section 6 quantifies the worker surplus from the new jobs and compares it with their cost, and Section 7 concludes.

2 Theoretical Framework

We present a framework to analyze how a binding wage floor changes the employment and welfare effects of a labor-demand expansion. Section 2.1 sets out the model, Section 2.2 states its predictions for employment and wages and its implications for welfare, and Section 2.3 draws them. Appendix C states the assumptions in full, proves the results and develops extensions.

2.1 Model

Firms. A firm produces $F\big(K,\, n\, g(h)\big)$ with capital K, headcount n and weeks per worker h, where additional weeks of a given worker add less than proportionately to effective labor. It pays the weekly wage $\bar{w}$ set by the national contract, faces an increasing and convex cost $\kappa(n - n_0)$ of expanding headcount above its initial level n_0 ($\kappa' > 0$, $\kappa'' \geq 0$), and receives a subsidy equal to a share σ of the user cost of capital ρ. When it has more willing applicants than it employs, it solves

$$\max_{K,n,h} \; F\big(K,\, n\, g(h)\big) \;-\; (1-\sigma)\rho K \;-\; \bar{w}\, nh \;-\; \kappa(n - n_0). \tag{1}$$

It invests until the marginal product of capital equals its subsidized cost, hires until an additional worker's output covers her pay and the marginal hiring cost, and adds weeks until the marginal product of a week equals $\bar{w}$. The model charges every worker the floor. We expect, but do not model, that workers paid above it, whose top-ups the firm sets itself, can see their pay rise; Section 4.3 examines this.

Workers. Workers are equally productive but differ in the value of their time outside formal employment. Working h weeks a year costs worker i an amount $\psi_i(h)$-forgone home

production, leisure and informal earnings-and taking a job may mean giving up a benefit b_i. She takes a job offering h weeks at $\bar{w}$ if

$$\bar{w} \;\geq\; \zeta_i^R(h) \;\equiv\; \frac{\psi_i(h) + b_i}{h}, \tag{2}$$

where $\zeta_i^R(h)$ is her private reservation weekly wage, and once employed she wants an additional week if

$$\psi_i'(h) \;<\; \bar{w}. \tag{3}$$

The first condition decides who is willing to work at the floor; the second, whether those employed would welcome more weeks. The second does not follow from the first: a worker can prefer a job of 30 weeks to non-employment and still prefer 30 weeks to 40.

From firms to the market. Fixing the employment package, let $L^D(w; A)$ be market labor demand for a standardized job, increasing in a demand shifter A, and $L^S(w)$ the number of workers who would accept that job at wage w. Without a floor, the wage $w^*(A)$ equates the two, and if supply slopes upward a demand expansion creates wage pressure. With a floor $\bar{w}$ above w^*, firms employ fewer workers than are willing to work at it,

$$Q \;=\; L^S(\bar{w}) - L^D(\bar{w}; A) \;>\; 0, \tag{4}$$

so employment is set by demand: employment is *rationed*, and the Q workers who would accept the floor but are not employed form a queue. Surplus labor is thus defined relative to the floor (Lefeber, 1968). A demand expansion that leaves excess supply then raises employment without wage pressure.

Social versus private incentives. Order workers by when they obtain employment, so that the employed are $i \in [0, n]$, and let $C(n, h) = \int_0^n \psi_i(h)\, di$ be their aggregate opportunity cost. With quasi-linear utility, wages and the grant are transfers, and social surplus is

$$W \;=\; F\big(K,\, n\, g(h)\big) \;-\; \rho K \;-\; \kappa(n - n_0) \;-\; C(n, h) \;-\; \lambda G, \tag{5}$$

with capital at its full resource cost, $G = \sigma\rho K$ the gross outlay and λ the excess burden per euro of public funds. Private and social incentives differ on both labor margins. The firm hires until the marginal worker's output covers $\bar{w}$, but the resource cost of employing her is $\bar{\zeta}_{\text{entrant}} = \psi_{i=n}(h)/h$, the weekly value of the time given up by the n-th worker, the marginal hire. This is her reservation wage (2) without the benefit: the benefit is a transfer,

which the government saves when she is hired, not a resource cost. The firm adds weeks until their marginal product equals $\bar{w}$, but a week costs those employed $\bar{\zeta}_{\text{week}} = n^{-1} \int_0^n \psi_i'(h)\, di$ on average.

2.2 Employment, Wages, and Welfare

Proposition 1 describes how the funded firm adjusts, Proposition 2 values that adjustment, and Proposition 3 says what the response of other employers in the same market reveals about how wages are set.

Proposition 1. Suppose the firm pays the floor $\bar{w}$ and has more willing applicants than it employs (Assumptions 1–4). An increase in the subsidy rate σ

(i) leaves the wage unchanged, $\delta_w = 0$;
(ii) raises headcount, $dn/d\sigma > 0$;
(iii) raises weeks per worker, $dh/d\sigma \geq 0$, strictly if the marginal cost of hiring rises with the pace of hiring ($\kappa'' > 0$); with linear hiring costs, the adjustment falls entirely on headcount.

Proof. See Appendix C.2. ■

The subsidy lowers the cost of capital and, because capital and labor are complements, raises the marginal product of labor at the firm's current employment. How the firm responds depends on what it must pay to hire. If it faces an upward-sloping supply of labor, it can attract additional workers only by offering a higher wage, so part of the higher marginal product is paid out in wages and employment rises by less. Under a binding floor, it already has more willing applicants than it employs at $\bar{w}$, so it can hire more without raising pay, and the whole adjustment falls on quantities: the firm adds workers and, when hiring quickly is costly, adds weeks for the workers it already employs. Appendix Corollary 1 compares the two responses under a supply curve of constant elasticity. A flat wage at the funded firm is, however, also what a market with very elastic labor supply would produce, so Proposition 1 alone does not reveal a queue; Proposition 3 provides the test. Section 4 takes these predictions to the funded firms and their workers.

Proposition 2. Under the conditions of Proposition 1, the effect of the subsidy on social

surplus (5) is

$$\frac{dW}{d\sigma} = \underbrace{h\left(\bar{w} - \bar{\zeta}_{\text{entrant}}\right)\frac{dn}{d\sigma}}_{\text{additional jobs}} + \underbrace{n\left(\bar{w} - \bar{\zeta}_{\text{week}}\right)\frac{dh}{d\sigma}}_{\text{additional weeks}} - \underbrace{\sigma\rho\frac{dK}{d\sigma}}_{\text{distortion of capital}} - \underbrace{\lambda\frac{dG}{d\sigma}}_{\text{cost of funds}} . \tag{6}$$

Proof. See Appendix C.3. ■

Each labor term is additional work multiplied by the gap between what it is paid and what it costs. The jobs term is positive when the marginal hire's resource opportunity cost lies below the pay attached to the job; the weeks term is positive when those employed would welcome more weeks at the contractual wage (Assumption 5), which the floor and excess applicants do not by themselves establish. Without a floor, and with no benefits or taxes separating private reservation pay from the resource cost of work, both gaps are zero at the margin and only the two costs remain. With rationing at the prevailing wage, both can be positive even though the wage does not rise: the wage paid is then above what the worker's time is worth in its next best use.[4] We call the sum of the two labor terms the labor-market efficiency gain of the subsidy: the part of its welfare effect that comes from reducing rationing, and the part a clearing-market evaluation sets to zero. It can be positive even when the subsidy lowers welfare once the cost of funds and the capital distortion are subtracted. Section 6 measures it.

Proposition 3. Consider a firm in the same labor market that did not apply, and let the funded firm expand.

(i) If the floor binds at that firm and its own queue is longer than the number of applicants the expansion draws away, its wage and employment do not respond through the labor market.

(ii) If instead it sets its wage above the floor and faces a labor supply of constant elasticity, the expansion raises its wage and, absent other channels, lowers its employment.

Proof. See Appendix C.5. ■

A queue insulates other employers: the funded firm hires from it rather than from them, so they face no pressure to raise pay. In a market that clears, the funded firm can grow only by attracting workers who would otherwise work elsewhere or not at all, and other employers must bid to keep theirs. Because other employers' employment can also move

[4] The development planning literature calls this difference the shadow-wage gap (Lewis, 1954; Lefeber, 1968; Ahluwalia, 1973).

through demand linkages under either arrangement (Appendix C.5), it does not by itself separate the two; their wage response does. An unchanged wage together with a non-negative employment response is therefore informative against a clearing market with a finite supply elasticity, in which competitors could expand only by bidding for workers.[5]

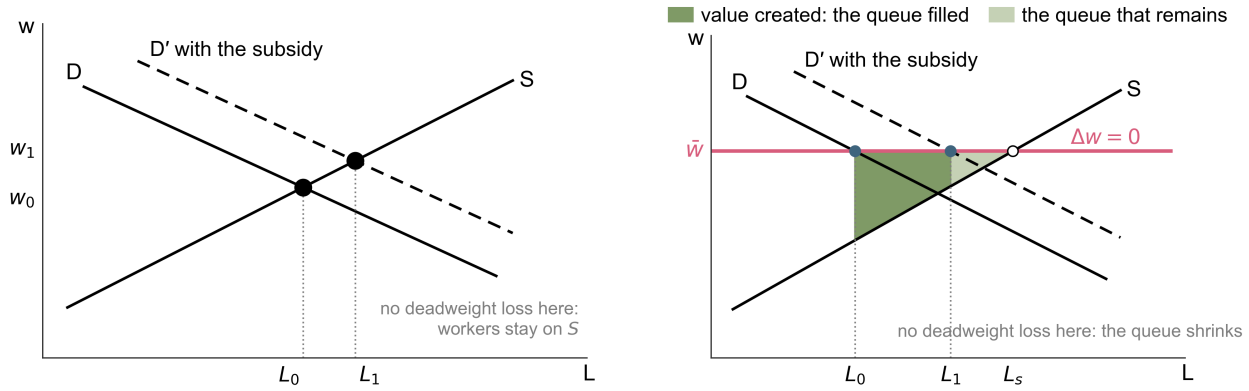
2.3 Graphical Analysis

Figure 1 draws the same demand shift for a standardized job under the two arrangements. Without a floor (panel i), the wage rises along the supply curve and the marginal entrant's opportunity cost equals the wage. With a binding floor (panel ii), employment rises at a fixed wage, and the area between the floor and the supply curve over the new jobs is the workers' surplus; how much of it is realized depends on which members of the queue get the jobs (Lee and Saez, 2012). The figure is for illustration: whether the wage moves, how other employers respond, where the new workers come from and their reservation wages are to be estimated. A flat wage at the funded firm does not by itself reveal a queue (Figure 2; Lemma 1); what separates the cases is whether other employers face wage pressure and where the workers come from.

Two tests. The framework turns the contrast between the two arrangements into four questions for the evidence, and two tests. The first test is at the funded firm (Section 4): where new hires come from, whether incumbents work more weeks at an unchanged weekly wage, and whether the wage moves where the contractual floor does not bind (Section 4.3). A flat wage at the funded firm is also what a small firm in a clearing market would show (Figure 2), so the second test looks at the market (Section 5): whether firms that never applied pay more or lose workers, and whether workers arrive from elsewhere. Section 6 then asks what the opportunity costs of the workers who obtain the new jobs are.

[5]It is not a universal test: with frictionless mobility across regions, supply to the local market is perfectly elastic and wages do not move either, which is why Section 5.4 also traces where workers come from. Appendix Table C.1 collects the predictions for the four coefficients we estimate, next to the estimates.

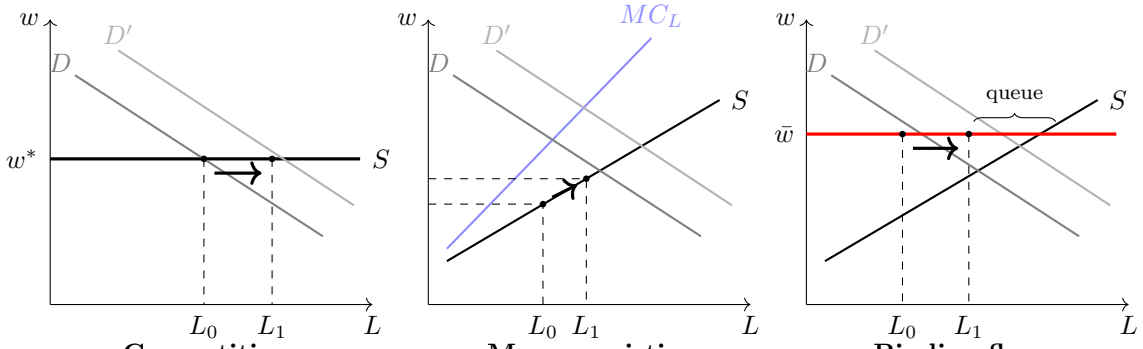
Figure 1: Graphical Analysis: A Labor Demand Shock in the Labor Market



(i) Market clearing (ii) Wage floor

Notes: This figure shows the same shift in labor demand, from D to D', in one labor market under two wage-setting arrangements. In panel (i), without a floor, the market clears: the wage rises from w_0 to w_1 and employment from L_0 to L_1 along the supply curve S; the raise is a transfer to incumbents, and the labor market adds no deadweight loss. In panel (ii), with a binding floor, $\bar{w}$ sits above the clearing wage: employment moves from L_0 to L_1 along the floor while the wage does not move. The dark area, between the floor and the supply curve from L_0 to L_1, is the workers' surplus as the queue is filled; the light area, up to supply at the floor L_s, is the queue that remains. Appendix C derives both panels from one model.

Figure 2: Graphical Analysis: A Labor Demand Shock at the Firm



Notes: This figure shows the response of a single firm to the same outward shift in its labor demand, from D to D', under three views of how the wage is set. In the left panel the firm is a price taker in a competitive market and faces a supply of labor that is flat at the market wage w^*: employment rises from L_0 to L_1 and the wage does not move. In the middle panel the firm faces an upward-sloping supply curve S and sets employment where the marginal cost of labor MC_L meets demand: employment and the wage both rise, and the wage stays below the marginal revenue product. In the right panel a contractual floor $\bar{w}$ sits above the wage the firm would otherwise pay, and a queue of workers is willing to work at it: employment rises out of the queue at an unchanged wage. A flat wage is therefore predicted by both the competitive and the binding-floor views; what separates them is what happens elsewhere in the market (Figure 1).

3 Setting and Data

3.1 The IDA Program and Institutional Setting

Our empirical setting is the *Incentivi per lo Sviluppo delle Aree Depresse* (IDA; Law 488/1992), Italy's primary place-based industrial policy active between 1996 and 2007. With €26 billion in grants committed across its 21 calls (about €2.2 billion a year; nominal euros, of which about €22 billion in the South; Appendix Figure D.2), triggering €87 billion in private investment, IDA provided investment grants to firms in European Commission-designated eligible areas, predominantly European Union Objective 1 regions in Italy's lagging South (*Mezzogiorno*).[6] In contrast to prior discretionary transfers, IDA allocated grants through a competitive ranking on predetermined criteria, one of which reflected regional policy priorities. Grants were disbursed in installments conditional on certified investment progress, and approximately one-quarter of committed funds was ultimately withheld due to non-completion.

The allocation mechanism operated through annual regional calls for tenders. Eligible firms submitted project proposals detailing the project type (new plant versus capacity expansion), location, total investment, own-capital contribution, and promised job creation. Projects were evaluated along five standardized criteria: the ratio of own capital to total investment, the ratio of promised jobs to total investment, the requested grant relative to the maximum allowable subsidy, undisclosed regional policy priorities, and an environmental impact assessment. For firm j applying in competition $c(j)$ (a region-year pair), these components were standardized within the competition and summed into a composite score:

$$\text{ProjectScore}_j = \sum_{d=1}^{D} \frac{i_{d,j} - \mu_{d,c(j)}}{\sigma_{d,c(j)}} \tag{7}$$

Within each regional competition, projects did not compete in a single ranking. Instead, they were partitioned into distinct *funding pools* f defined by firm size (small versus medium and large), sector (manufacturing versus services), EU co-financing eligibility, and EU objective area. Each pool was assigned an earmarked budget b_f and ranked from highest to lowest score. Grants were allocated sequentially until the budget was exhausted, generating a pool-specific threshold score c_f. Firm j in pool $f(j)$ received funding if and only if its normalized score satisfied $S_j \equiv \text{ProjectScore}_j - c_{f(j)} \geq 0$.

[6]The South encompasses eight regions (Abruzzo, Molise, Campania, Apulia, Basilicata, Calabria, Sicily, and Sardinia) characterized by longstanding economic distress. In the early 1990s, Southern GDP per capita was roughly 60% of Northern levels, with unemployment exceeding 20% in several regions.

This institutional design provides the foundation for our empirical identification. Our regression discontinuity design compares marginal winners and losers within the same administrative funding pool. These funding pools, however, span across different labor markets, which we define following the literature as a province crossed with a 2-digit NACE industry.[7] The average funding pool includes applicants from 27 distinct labor markets, while each market typically hosts only a small number of winning firms. This disconnect between administrative competition boundaries and geographic labor markets allows us to exploit quasi-random variation in subsidy receipt within funding pools while absorbing labor market shocks with fixed effects.

Understanding the incidence of IDA grants requires accounting for Italy's wage-setting institutions. Wages are governed by a two-tier collective bargaining system. At the national level, industry federations and trade unions negotiate sector-wide collective agreements (*contratti collettivi nazionali di lavoro*, CCNL) that establish legally binding nominal wage floors $\underline{w}_{\ell s}$ for each occupational grade ℓ within sector s.[8] These agreements specify monthly salaries and standard weekly hours, leaving annual earnings free to vary only with weeks worked. Contractual wage floors carry no territorial differentiation[9]. Second-tier firm bargaining can only augment compensation above the national floor due to the statutory *favorability principle*, though in practice only about 10% of firms maintain formal company-level agreements (Boeri et al., 2021). Total compensation for worker i at grade $\ell(i)$ in firm $f(i)$ and sector $s(i)$ therefore decomposes as:

$$\text{Wage}_i = \underbrace{\underline{w}_{\ell(i)s(i)}}_{\substack{\text{CBA floor} \\ \text{(national, binding)}}} + \underbrace{\text{top-up}_{if(i)}}_{\substack{\text{firm-worker specific} \\ \text{(discretionary)}}} \tag{8}$$

Because national contracts are negotiated predominantly in northern industrial centers, nominal floors might be anchored close to market-clearing wages in the high-productivity regions, but bind tightly above market-clearing levels in more depressed areas (Boeri et al., 2021; Caponi, 2008). This spatial rigidity may prevent wages from adjusting downward to clear the market, shifting the incidence of local shocks to employment and unemployment (Daruich et al., 2024). If this friction binds, it has direct consequences for place-based subsidies. An investment grant lowers the cost of capital, increasing labor demand via scale effects. If contractual floors exceed market-clearing levels, expanding firms can recruit new

[7] A province is broadly analogous to a US county. Appendix D.4 discusses labor market definitions and checks using inter-firm worker mobility.

[8] Collective bargaining agreements cover approximately 98% of employees in Italian firms with over ten workers (Boeri et al., 2021; Bovini et al., 2026).

[9] Territorial wage zones (*gabbie salariali*) were eliminated in the early 1970s

labor at the floor $\underline{w}_{\ell(i)s(i)}$ plus an entry top-up that does not respond to the subsidy, without bidding up entry wages. The pay rate of a worker at the floor can then rise only through reclassification or through the discretionary top-ups of workers already above it, although annual earnings can still rise through weeks worked. Whether the floor binds in the treated markets is what Sections 4 and 5 test.

3.2 Data

We construct a panel combining administrative application registries, social security employer-employee records, balance sheets, and digitized collective bargaining agreements. Two surveys complement them: the Bank of Italy's survey of industrial and service firms (Invind), which reports how much of firms' wage bill national contracts determine (Section 4.3), and the Italian Labour Force Survey, which reports the net monthly reservation wages of job seekers in the South in 1996, with the 2004 wave as a check, and on which the welfare calibration of Section 6 rests.

Data on Bids and Grants. Our first source is the administrative universe of IDA applications across all 21 regional calls issued between 1996 and 2007. This records 62,631 bids submitted by 47,200 firms, of which 21,271 projects (submitted by 18,545 firms) cleared their pool threshold and received funding. The data report standardized project scores, pool-specific cutoffs, awarded and requested grant amounts, proposed investment and employment pledges, project type (new establishment versus expansion), target municipalities, and firm tax identifiers.

Matched Employer-Employee and Firm Records. We link application records to two administrative registers. First, the Italian firm registry provides longitudinal census data for all non-agricultural firms with at least one employee from 1983 onward, recording establishment-level employment, average earnings, 6-digit NACE sector, and province. Second, we access matched employer-employee records from the National Social Security Institute (INPS) covering 1990–2013 through the VisitINPS program. These records encompass approximately 93% of private-sector dependent employees, detailing gross earnings, full-time-equivalent weeks worked, contract type, occupational classification, and basic demographics. We also access a 13% representative longitudinal sample recording all social security income, including unemployment benefits, short-time compensation (*Cassa Integrazione Guadagni*), and public-sector earnings. Firm balance sheet records from Cerved provide capital stock,

value added, EBITDA, and gross sales.

Data on Collective Bargaining Agreements. To isolate contractual floors from discretionary firm top-ups, we employ nominal wage floors from all major national collective agreements from 1990 onward by sector, occupational grade, and effective calendar date. Matching these contractual tables to individual job spells allows us to observe both actual compensation and statutory floors, defining each worker's wage cushion ($\text{Wage}_{it} - \underline{w}_{\ell(i)s(i),t}$) as in equation (8).

Summary Statistics. Appendix Table A.2 describes the application sample, applicant firms, and workforce composition at baseline (the year before the application). Three patterns emerge. First, while creating jobs was not a necessary condition, firms still promised to create jobs in almost all applications: 98% of funded projects involved either new plant construction or capacity expansions with explicit job creation commitments. Second, the policy promoted local firm growth rather than outside entry. New plants accounted for over half of total grant expenditure (averaging €617,000 per project), but 97.5% were undertaken by local incumbent firms or newly formed local startups, with established outside entrants accounting for only 2.5% of projects. Third, grant receipt was concentrated in high-unemployment southern manufacturing districts. Because exposure to IDA varies systematically across labor markets, our market-level spillover analysis relies on a recentered instrument following Borusyak and Hull (2023) to isolate quasi-random local policy exposure.

4 Effects on Applicant Firms

We begin by estimating the direct effects of investment subsidies on firm employment and compensation, which is the basis of our first test. These responses provide a first indication of how labor demand shocks are absorbed: through changes in employment, wages, or both. We focus on incumbent firms applying to expand an existing establishment or establish a new plant in a market where they already operate.

4.1 Empirical Strategy

Regression discontinuity design. We exploit the discontinuity in grant eligibility at the cutoff of each funding pool. As described in Section 3.1, grants were awarded in score order within each pool until its budget ran out, so every pool has its own cutoff.

Let $Y_j(1)$ and $Y_j(0)$ be the potential outcomes of applicant firm j, competing in funding pool $f(j)$, when it is awarded the subsidy and when it is not. Let $S_j = \text{ProjectScore}_j - c_{f(j)}$ be firm j's score relative to the cutoff $c_{f(j)}$ of its pool, so that $S_j = 0$ at the cutoff, and let $\text{Win}_j = \mathbf{1}\{S_j > 0\}$ indicate an application above it. If applicants cannot precisely manipulate their position relative to the cutoff, potential outcomes vary smoothly through it, and the difference in average outcomes between applicants just above and just below the cutoff identifies the effect of scoring above it for firms at the cutoff (Hahn et al., 2001).

Let t_j^* be the year in which firm j applies, and k the number of years since then, so that $t_j^* + k$ is a calendar year. Firms apply, are ranked and start receiving the grant in the same year, and we use $k = -1$, the last year before the subsidy, as the baseline year. We write t^* when the firm is clear from the context. We pool observations across funding pools and estimate the following local-linear specification separately for each post-application horizon k:

$$Y_{jk} = \beta_0 + \beta_1 \text{Win}_j + \beta_2 S_j + \beta_3 \text{Win}_j S_j + \gamma_{f(j)} + \alpha_{m(j)} + \varepsilon_{jk}, \tag{9}$$

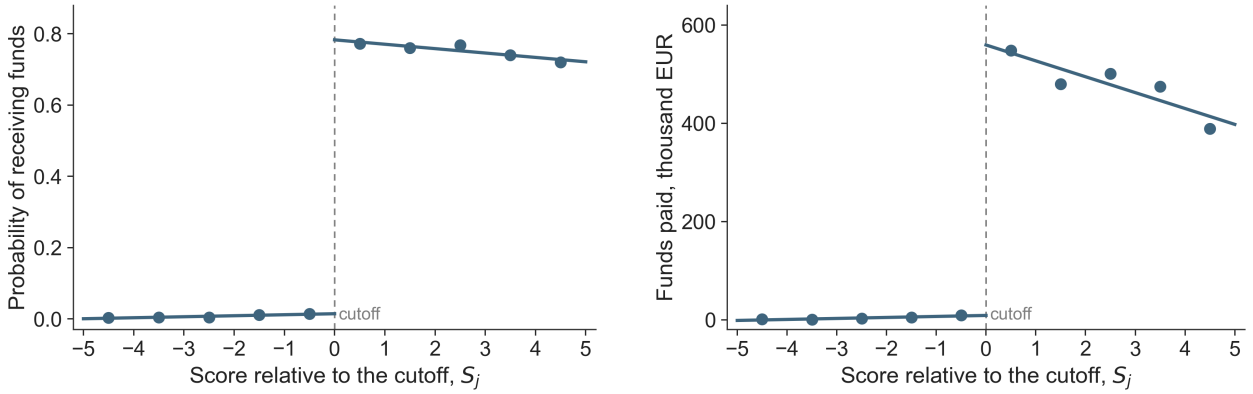
where Y_{jk} is the change in an outcome of firm j between the year before the application and k years after it. For employment, $Y_{jk} = \log E_{j,t_j^*+k} - \log E_{j,t_j^*-1}$, the change in log employment; for pay, it is the change in log average compensation per worker. The coefficient β_1 estimates this jump at the cutoff. Funding-pool fixed effects $\gamma_{f(j)}$ account for differences across pools, and are needed because each cutoff is not set in advance but determined by the pool's budget and the scores of competing projects (Fort et al., 2022). Market fixed effects $\alpha_{m(j)}$, for each province $\times$ industry, absorb common market-level changes.

Validity. Identification requires that firms cannot precisely sort around the funding threshold. While applicants know the scoring criteria, the cutoff depends on the scores of competing projects and the budget available within each funding pool, limiting their ability to predict its exact realization. We assess this assumption by examining the distribution of application scores and the continuity of predetermined firm characteristics around the threshold. Appendix Figure A.1 presents the density of the running variable normalized by each pool's cutoff. We find no evidence of bunching immediately above the threshold, consistent with applicants being unable to precisely manipulate their funding status.

First stage and interpretation. Winning a grant does not guarantee that funds are ultimately received. Disbursements are made in tranches following verification of eligible investments and compliance with program requirements, and some awarded projects are subsequently withdrawn or never completed. Figure 3 reports the discontinuity in the

probability of receiving funds and in the amount paid. The probability of receiving funding increases by approximately 80 percentage points at the cutoff, indicating substantial but incomplete take-up among winners. Non-compliance is close to one-sided: some applicants above the cutoff never receive the subsidy, and almost no applicant below it is funded.

Figure 3: First Stage: Funding at the Cutoff



(i) Probability of receiving funds

(ii) Funds paid (thousand euros)

Notes: This figure plots the first stage of the regression discontinuity design: the probability of receiving funds (panel i) and the funds paid, in euros (panel ii), against the score relative to the pool's funding cutoff, S_j. Dots are binned means; lines are linear fits estimated separately on each side; the dashed line is the cutoff. The discontinuity at the cutoff is 0.762 (s.e. 0.008) in the probability of receiving funds and €541,000 (s.e. €13,000) in funds paid.

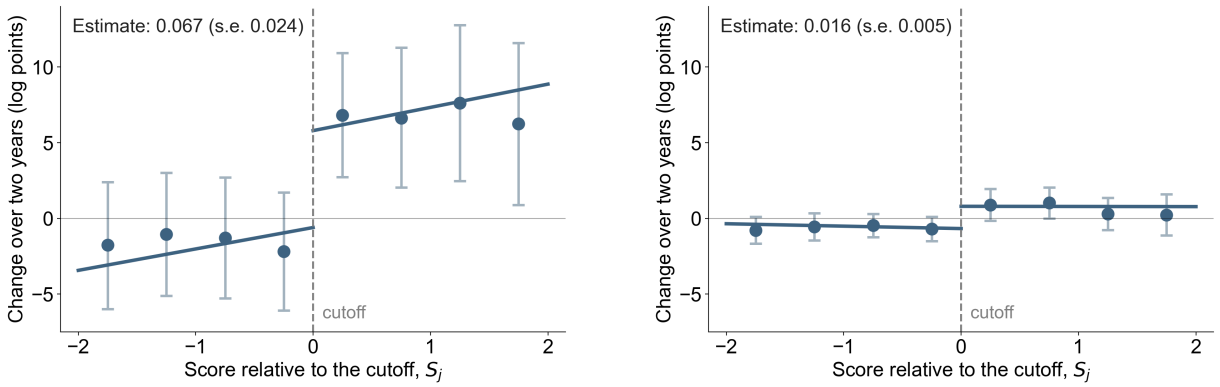
We therefore interpret β_1 as the intention-to-treat effect of crossing the funding threshold: the effect of being awarded the subsidy under the program's disbursement rules, not of receiving it or of an additional euro of funding. Because non-compliance is close to one-sided, it is a lower bound on the effect of receiving the subsidy at the cutoff; if scoring above the cutoff affects outcomes only through receipt, that effect equals the intention-to-treat effect divided by the share of compliers, about 0.8.

Results. Figure 4 and Table 1 report the effects of grant eligibility on employment and pay. Three years after the application, employment at funded firms is 12.6 log points higher (s.e. 0.030), having risen by 2.9 and 6.7 log points after one and two years (Table A.4); pay per worker rises by only 1.5 log points (s.e. 0.005) over the same three years.[10] Both estimates are

[10]The expansion is accompanied by an increase in firm survival. Because employment growth is measured among firms remaining active at each horizon, the estimation sample declines from 22,473 firms at t^*+1 to 21,611 at t^*+3. We therefore report survival effects separately to distinguish growth among continuing firms from changes in firm exit. Results are also robust to restricting the worker analysis to application cohorts observed at every event-time horizon (Appendix B.2). Complementary estimates using Cerved balance-sheet

close to zero before the application and similar across linear, quadratic and local-polynomial specifications (0.143 and 0.152 for employment in the last two). We estimate the specification within a fixed bandwidth around each cutoff, allowing the slope of the running variable to differ on either side, and report quadratic estimates and nonparametric estimates with the MSE-optimal bandwidth of Calonico et al. (2014) as robustness (Table 1).[11]

Figure 4: Regression Discontinuity Estimates: Employment and Pay per Worker



(i) Employment

(ii) Wage bill per worker

Notes: This figure plots the change in log employment (panel i) and in log wage bill per worker (panel ii) from the year before the application to two years after, in log points, against the score relative to the funding cutoff, S_j, for Southern applicants. The two panels share the same axes. Outcomes are net of funding-pool and market (province × two-digit industry) fixed effects. Dots are means within bins of 0.5, with 95% confidence intervals; lines are linear fits on each side of the cutoff within the fixed bandwidth, and the dashed line is the cutoff. Both panels show applicants within two points of the cutoff; for the wage bill per worker, the fit uses the full fixed bandwidth. The estimates reported in each panel are the linear fixed-bandwidth estimates at two years of Table 1, Panel (a). Estimates at every horizon, with pre-application placebos, are in Appendix Figure A.30.

In a standard monopsony model, such a pattern might suggest highly elastic labor supply and limited wage markdowns (Appendix B.1 computes the elasticity this reading implies). Yet this interpretation assumes that firms adjust along their labor supply curves. The following sections examine whether the muted wage response instead reflects institutional wage-setting constraints and how workers benefit from the additional labor demand.

Dynamics, heterogeneity, and survival. The employment effects build gradually, as the grants fund multi-year investment projects, and they persist. At longer horizons the

data, reported in Appendix B.1, document changes in capital, revenue, value added, margins, and profitability associated with this expansion.

[11]We keep the polynomial in the score of low order, as Gelman and Imbens (2019) recommend.

Table 1: Firm-Level Regression Discontinuity Estimates: Employment and Pay per Worker

	Δ Log Employment		Δ Log Wage	
Outcome variable:	(1) 2y	(2) 3y	(3) 2y	(4) 3y
Panel (a): Parametric (linear) estimates - fixed bandwidth				
RD estimate	0.067***	0.126***	0.016***	0.015***
	(0.024)	(0.030)	(0.005)	(0.005)
Panel (b): Parametric (quadratic) estimates - fixed bandwidth				
RD estimate	0.067**	0.143***	0.019**	0.021**
	(0.029)	(0.037)	(0.009)	(0.009)
Panel (c): Non-parametric estimates - full sample, MSE-optimal bandwidth				
RD estimate	0.077**	0.152***	0.017*	0.029***
	(0.035)	(0.037)	(0.009)	(0.010)
Bandwidth (h) – non-parametric	1.85	1.78	2.09	1.69
Observations within h	13,562	12,887	14,854	12,401
Mean (within h)	0.346	0.393	0.067	0.093
Standard deviation	0.997	1.121	0.218	0.247

Notes: This table reports RD estimates of β_1 in equation (9) for log employment and log wage bill per worker at horizons t^*+2 and t^*+3. Panel (a): parametric linear RD, fixed bandwidth. Panel (b): parametric quadratic RD. Panel (c): local polynomial with MSE-optimal bandwidth. Funding-pool and market fixed effects; standard errors clustered at the funding-pool level. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

employment effect stays at about 9 to 11 log points from four to seven years after the application, while pay per worker rises slowly, to about 5 log points at seven years (Appendix Figure A.30). Employment expands for both blue- and white-collar workers (Appendix Table A.4 and Figure A.6), while the employment response is stronger in new-plant projects compared to expansion of current facilities (Appendix Figure A.7).[12]

4.2 Decomposing Firm-Level Effects

While the regression discontinuity estimates in Section 4.1 establish that subsidized firms expand, they do not reveal how that growth occurs: whether firms expand through hiring or retention, whether workforce composition changes, how earnings divide between weeks and wage rates, or how workers' outside income responds.

Investigating these margins requires linking the IDA applications to the INPS matched employer-employee records (Appendix D.4). Because administrative confidentiality precludes conditioning continuously on evaluation scores within the worker-level data, we implement an event-study version of a local-constant regression discontinuity. This design restricts the sample to near-cutoff applicants and tracks marginal winners and losers over event time $k = t - t_j^*$, where t_j^* is the application year and $k = -1$ serves as the omitted reference period.[13] All specifications in this subsection are estimated at the firm level; the corresponding worker-level regressions are reported in Appendix B.2.1.

4.2.1 Hiring versus Retention

We decompose the cumulative percentage change in employment since the pre-application year $(t_j^* - 1)$ into accumulated new hires and retained baseline incumbents:

$$\frac{\text{Emp}_{j,t} - \text{Emp}_j^0}{\text{Emp}_j^0} = \underbrace{\frac{\text{Hires}_{j,t}}{\text{Emp}_j^0}}_{\text{hiring}} + \underbrace{\frac{\text{Retained}_{j,t}}{\text{Emp}_j^0}}_{\text{retention}} - 1, \tag{10}$$

where Emp_j^0 is employment at firm j in $t_j^* - 1$, $\text{Hires}_{j,t}$ denotes post-baseline entrants, and $\text{Retained}_{j,t}$ denotes surviving baseline incumbents.[14] We estimate each margin using the

[12] Survival at different horizons is reported in Appendix, Figure A.28).

[13] Appendix D.4 details the linkage protocol and variable definitions. Appendix B.1 demonstrates that this event-study specification exhibits parallel pre-trends and closely reproduces the cross-sectional RD estimates of Section 4.1.

[14] By construction, the components sum to total net growth after application. Prior to $t_j^* - 1$, $\text{Hires}_{j,t}$ captures historical turnover and $\text{Retained}_{j,t}$ reflects baseline worker tenure; pre-trends are therefore evaluated

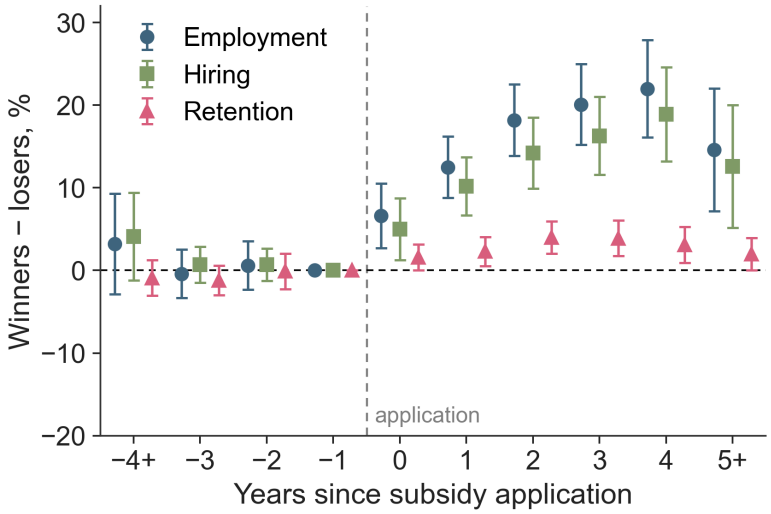
firm-level event study:

$$Y_{jt} \;=\; \sum_{k\neq -1} \beta_k \, \mathbf{1}\{t - t_j^* = k\} \cdot \text{Win}_j \;+\; \mu_{t-t_j^*} \;+\; \tau_t \;+\; \delta_{c(j),t} \;+\; \varepsilon_{jt}, \tag{11}$$

where Win_j indicates a near-cutoff winner, $\mu_{t-t_j^*}$ are event-time fixed effects, τ_t are calendar-year fixed effects, and $\delta_{c(j),t}$ are funding pool $\times$ year fixed effects, so that β_k compares winners and losers in the same funding pool and year. Standard errors are clustered by funding pool.[15]

Figure 5 reports the results. Both margins are active: subsidized firms expand through external recruitment *and* by retaining workers who would otherwise have separated. This contrast with other firm shocks: for example, Portuguese exporters absorb positive demand shocks exclusively through hiring, with zero retention effect (Garin and Silvério, 2024). While hiring dominates initially, retention compounds over time: by years four and five, one in three subsidized jobs represents a job maintained rather than newly created (Appendix Table A.5).

Figure 5: Employment Growth at Funded Firms: Hiring and Retention



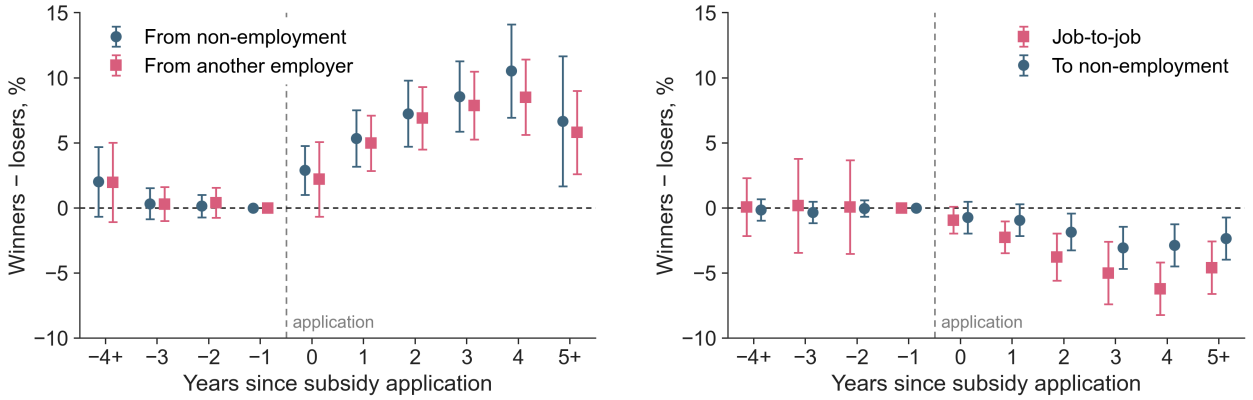
Notes: This figure plots event-study estimates of equation (11) for employment and its two components in equation (10): hires still present at each horizon, and baseline workers retained, each as a share of baseline employment, in percent. Winners minus near-cutoff losers, with 95% confidence intervals; t^*-1, the year before application, is the reference year. Both components rise, with no pre-trend: funded firms grow both by recruiting and by keeping workers who would otherwise have left. Separations by destination are in Figure 6ii; the split by project type is in Appendix Figure A.32.

directly on total employment.

[15] Because the decomposition outcomes are expressed as changes relative to $t_j^* - 1$, equation (11) omits firm fixed effects. Specifications for outcomes in levels (earnings, weeks, wages, and workforce composition) include firm fixed effects to absorb baseline differences.

Tracking worker origins and separation destinations confirms that new hires are drawn primarily from non-employment—workers with no formal private-sector record in the preceding year—rather than from local poaching (Figure 6). Symmetrically, avoided separations reduce both immediate job-to-job exits and, with a lag, exits into non-employment (Figure 6ii).

Figure 6: Hires by Origin and Separations by Destination



(i) Hires, by origin

(ii) Separations prevented, by destination

Notes: This figure plots event-study estimates of equation (11) for where funded firms' hires come from and where their separations go. As in Figure 5, each series is a number of workers divided by the firm's employment in the year before application, in percent. Panel (i) splits hires by the worker's state in the year before hiring: not formally employed, or moving from another employer; the two series add up to the hiring component of Figure 5. Panel (ii) splits separations between one year and the next by destination: another employer (job-to-job), or non-employment. Winners minus near-cutoff losers, with 95% confidence intervals; t^*-1, the year before application, is the reference year.

Workforce Composition. To examine whether expanding firms alter workforce quality, we estimate equation (11) on workforce characteristics in levels, adding firm fixed effects. Appendix Figure A.14 shows no change in gender or occupational mix. The overall workforce becomes slightly younger (average age declines by one-third of a year from t^*+1), and average worker quality (predicted earnings, as in Kline et al. (2019)) dips by less than one log point.[16]

Appendix Figure A.15 isolates the composition of new hires: post-application entrants are 0.25 to 0.5 years younger, exhibit a 1 to 2 percentage point higher blue-collar share, and have 1 to 2 log points lower predicted earnings. This modest compositional tilt toward entry-level workers explains why unadjusted starting wages decline slightly more than residualized wages (Section 4.2.2).

[16]Kline et al. (2019) document a similar pattern following patent awards, where firms expand by hiring younger workers with no change in observable worker quality.

4.2.2 Earnings Effects: Incumbents and New Hires

Following Kline et al. (2019), we hold workforce composition fixed by defining an *application cohort* comprising all workers employed at firm j in the pre-application year $t_j^* - 1$. Cohort members are tracked longitudinally regardless of whether they remain at the firm, with earnings set to zero in non-employment years. For firm j, year t, and cohort subgroup g, we estimate:

$$Y_{jt}^g = \alpha_j^g + \sum_{k \neq -1} \beta_k^g \mathbf{1}\{t - t_j^* = k\} \cdot \text{Win}_j + \mu_{t-t_j^*}^g + \tau_t^g + \delta_{c(j),t}^g + \varepsilon_{jt}^g, \tag{12}$$

where firm fixed effects α_j^g absorb baseline group averages, $\mu_{t-t_j^*}^g$, τ_t^g and $\delta_{c(j),t}^g$ are event-time, calendar-year and funding pool $\times$ year fixed effects as in equation (11), and $k = -1$ is the reference period.[17] Individual-level regressions tracking workers directly yield identical conclusions (Appendix B.2.1).

Annual earnings of the application cohort rise steadily after the application and are €1,037 per year higher at $t^* + 3$ (Table A.17), with flat pre-trends.[18] In logs, cohort earnings increase by 7% (Figure 7 and Table 2). The gain comes almost entirely from the quantity margin. Weekly wages remain statistically unchanged (+0.006 at $t^* + 3$), whereas weeks worked increase by 5.5% of the control mean. Increased weeks account for 78% to 91% of the earnings gain, and higher weekly wages for less than a tenth.[19]

This pattern is informative about labor supply. A worker on her supply curve chooses her weeks given her weekly wage, and in a competitive market she could already sell more weeks at that wage to another employer. If her weekly wage does not change, neither should her weeks, and her own employer's subsidy should not matter. Weeks that rise when her employer's demand rises, at an unchanged weekly wage, therefore indicate that she was working fewer weeks than she wanted at that wage: condition (i) of Section 2.1.

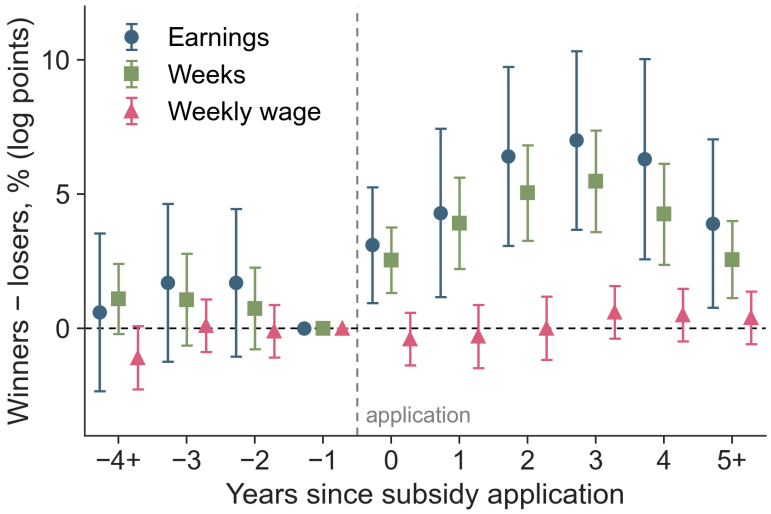
Disaggregating the cohort confirms this pattern (Table 3 and Appendix Figure A.22). Stayers gain 7.3% in annual earnings, driven predominantly by weeks. Workers who separate gain nothing (−1.4%, statistically insignificant), and recent hires gain 4.7%, entirely through weeks.

[17] Earnings are CPI-deflated to 2010 euros, summed across all private employers, and winsorized at the 1st and 99th percentiles.

[18] Interaction-weighted estimates (Sun and Abraham, 2021) yield €1,019 at $t^* + 3$, virtually identical to the baseline estimate of €1,037.

[19] Annual earnings and weeks include non-employment zeros, whereas weekly wages are conditional on employment. Estimating all three margins on a balanced sample of worker-years with positive work confirms that time worked accounts for the vast majority of the earnings response.

Figure 7: Incumbent Workers: Earnings, Weeks, and the Weekly Wage



Notes: This figure plots event-study estimates for the application cohort-all workers employed at the applicant firm at t^*-1, followed wherever they go-for log annual earnings, log weeks worked and the log weekly wage, in log points. Winners minus near-cutoff losers, with 95% confidence intervals; t^*-1, the year before application, is the reference year. Annual earnings and weeks keep years without work; the weekly wage is defined in years with work only, so the three series are not additive. Earnings in euros are in Table A.17; the monthly wage is in Appendix Figure A.21; the interaction-weighted version is in Appendix Figure A.35.

Table 2: Incumbent Workers: Earnings, Wage Rates, and Weeks

	Δ log earnings	Δ log weekly wage	Δ log monthly wage	Weeks (FTE), % of mean
t^*-4	0.006	−0.011*	−0.007	+1.1%
t^*-3	0.017	0.001	0.002	+1.1%
t^*-2	0.017	−0.001	0.001	+0.7%
t^*	0.031***	−0.004	0.000	+2.5%***
t^*+1	0.043***	−0.003	0.010	+3.9%***
t^*+2	0.064***	0.000	0.015*	+5.0%***
t^*+3	0.070***	0.006	0.020***	+5.5%***
t^*+4	0.063***	0.005	0.020**	+4.3%***
t^*+5	0.039**	0.004	0.017**	+2.6%***

Notes: This table reports event-study coefficients for the application cohort. Earnings, the weekly wage and the monthly wage in logs; weeks worked (full-time equivalent) as a percentage of the control-group mean. $^*p<0.10$, $^{**}p<0.05$, $^{***}p<0.01$.

New Hires and Within-Firm Inequality. Entry wages do not increase as well. New hires at winning firms earn the same as new hires at losing firms and work the same number of weeks (Appendix Figure A.11). Their raw entry weekly wage is 1.4% to 2.7% lower (Table A.18), a

Table 3: Who gains, and on which margin, at t^*+3

	Δ log earnings	Δ log weekly wage	Δ log monthly wage	Weeks, % of mean	Weeks share, approx.
Application cohort	0.070***	0.006	0.020***	+5.5%***	∼78%
Stayers	0.073***	0.001	0.013*	+5.9%***	∼81%
Recent hires	0.047***	0.002	0.014*	+4.8%***	∼100%
Leavers	−0.014	−0.008	−0.010	≈0	-

Notes: Each row is a group within the application cohort. Earnings and weeks keep years without work; the weekly and monthly wage are defined in years with work only, so the columns are not additive. "Weeks share, approx." is the weeks coefficient divided by the log-earnings coefficient, an approximate attribution. Stayers remain with the applicant firm; recent hires are cohort members hired shortly before the application; leavers are incumbents not observed at the applicant firm in year t. These groups are defined by outcomes the treatment affects and describe the pattern; the application cohort is the causal comparison.

gap that halves and becomes insignificant once residualized on worker observables (Appendix Figure A.13), because entrants at winning firms are slightly younger, more often blue-collar, and more often hired from within the province (Appendix Figures A.15–A.33). Because entry pay remains pinned to the contractual floor while incumbents gain weeks and discretionary top-ups, the within-firm 90-10 log wage ratio rises after the subsidy (Appendix Figure A.12).

4.2.3 Safety Net Take-Up and Broader Income

To determine whether private earnings gains crowd out public transfers, we evaluate individual workers using a 13% representative administrative panel recording comprehensive social security income (private earnings, unemployment benefits, short-time work, and public-sector wages). We estimate regressions in long differences from the baseline year $t_j^* - 1$, weighting worker i at firm j inversely by baseline firm size:

$$y_{ijt} - y_{ij,t_j^*-1} = \sum_{k \neq -1} \tilde{\beta}_k \, \mathbf{1}\{t - t_j^* = k\} \cdot \text{Win}_j + \tilde{\mu}_{t-t_j^*} + \tilde{\tau}_t + \tilde{\delta}_{c(j),t} + \eta_{ijt}, \qquad (13)$$

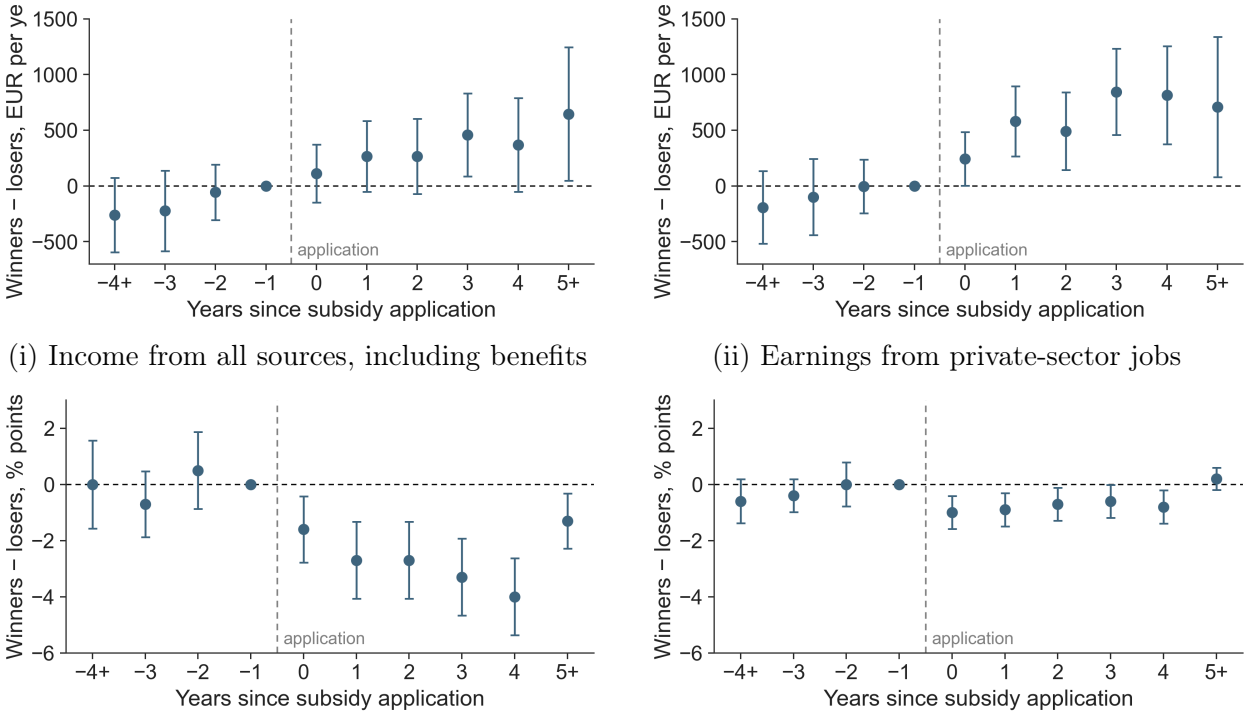
where $\tilde{\mu}_{t-t_j^*}$, $\tilde{\tau}_t$ and $\tilde{\delta}_{c(j),t}$ are event-time, calendar-year and funding pool × year fixed effects, as in equation (11), and differencing from baseline removes time-invariant worker and firm components. Standard errors are clustered by firm (or, alternatively, funding pool).[20]

Figure 8 and Table A.19 show that treated incumbents' private earnings expand by 3.7% to 6.1%, while dependence on the safety net falls sharply. Unemployment benefit receipt drops by 1.6 to 4.0 percentage points against a 9.5% control mean—a relative reduction of up to two-fifths. Short-time work compensation (*Cassa Integrazione*) falls by up to 1.0 percentage

[20]Differencing from baseline avoids the bias of two-way fixed effects in unbalanced panels with staggered treatment (Jäger et al., 2024).

point on a 3.1% mean, while public-sector earnings remain flat.

Figure 8: Incumbent Workers: Total Income and Safety-Net Receipt



(i) Income from all sources, including benefits

(ii) Earnings from private-sector jobs

(iii) Receiving unemployment benefits

(iv) Receiving short-time work benefits

Notes: This figure plots worker-level event-study estimates for incumbents on the 13% subsample of the social-security archives in which every source of income is observed. Panel (i) is income from all sources, including benefits, and panel (ii) earnings from private-sector jobs, both in euros per year. Panels (iii) and (iv) are the probability of receiving unemployment benefits and short-time work benefits, in percentage points, against control means of 9.5 and 3.1 percentage points. Winners minus near-cutoff losers, with 95% confidence intervals; t^*-1, the year before application, is the reference year.

These findings provide two key insights: first, earnings gains partly substitute for safety-net transfers, so net income gains to workers are smaller than gross earnings increases; second, saved unemployment benefits and expanded social security contributions directly reduce the net marginal cost of public funds in our welfare accounting (Section 6).

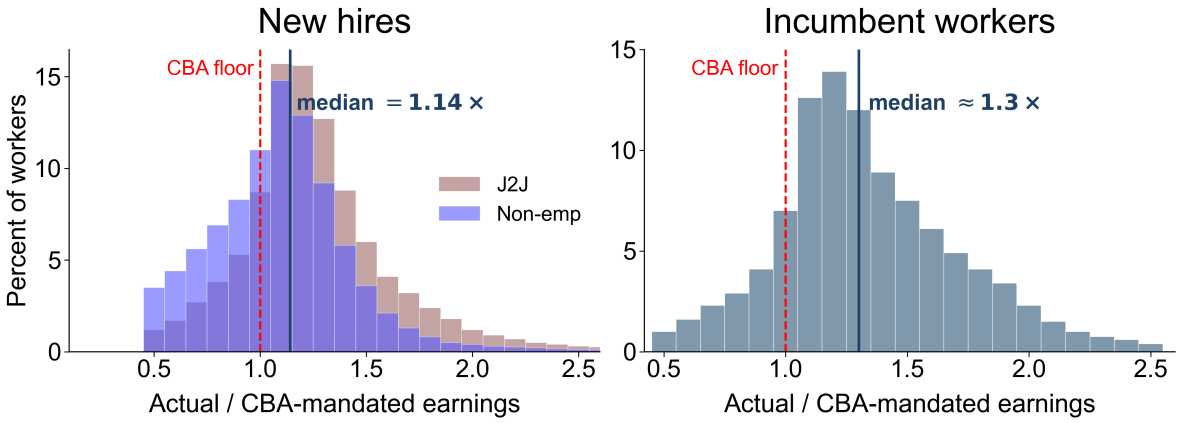
4.3 Employment Growth Without Wage Growth: The Role of Centralized Collective Bargaining

The finding that subsidized firms expand employment and weeks without raising hourly or weekly pay contrasts with standard evidence on firm-level shocks, where demand expansions typically pass through into worker earnings.

Contractual Bindingness and Wage Cushions. Because collective agreements constrain wages from below rather than above, a firm competing for scarce labor would respond to an expansion by bidding up starting pay. In the presence of labor market slack, however, firms can recruit at the contractual floor. The framework therefore predicts that entry wages remain pinned to the national schedule, with wage gains likely to concentrate among workers whose compensation already exceeds contractual minima.

Decomposing observed earnings into grade-specific national floors $\underline{w}_{\ell s}$ and discretionary firm top-ups confirms this asymmetry (Appendix Table A.12 and Figure 9). For the median new hire, the contractual floor accounts for 89% of earnings; for the median incumbent, 75%. We report medians because the mean share is pulled up by spells paid below the full-year floor: averaged over new hires, the floor accounts for 100% of earnings, and over incumbents for 78%. Defining a worker's *cushion* as the ratio of observed pay to the floor, the median cushion increases with establishment attachment (Appendix Table A.13): it is 1.10 for new hires from non-employment, 1.22 for job-to-job hires, 1.17 across all new entrants, and 1.32 for incumbents. Because workers recruited from non-employment are paid barely above the floor, most of the room for discretionary wage adjustment lies with incumbents.[21]

Figure 9: Pay Relative to the Contractual Minimum: New Hires and Incumbents

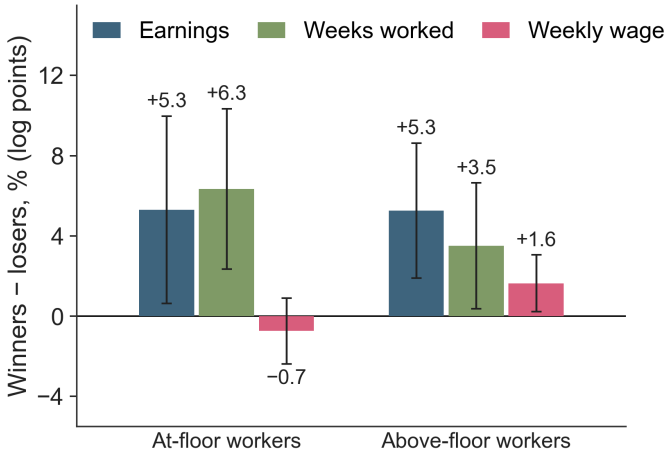


Notes: This figure plots the distribution of actual earnings divided by the earnings mandated by the worker's own collective agreement and grade, for new hires (left) and incumbents (right), trimmed at 0.5. A value of one means that the contract sets the whole of pay. New hires sit close to the floor-a median of 1.17 times the mandated amount, 1.10 for those arriving from non-employment and 1.22 for job-to-job movers-while incumbents are paid on the bargained top-up tier, with a median of 1.32 (Appendix Table A.13).

Survey and labor-market data confirm that national floors bind tightly in eligible regions.

[21] Contractual floors bind most tightly for blue-collar workers, whereas white-collar pay displays greater dispersion and larger top-ups. Because both earnings and floors are measured in nominal terms, the cushion does not require a deflator.

Figure 10: Responses by Distance to the Contractual Floor



Notes: This figure reports responses separately for incumbents paid close to and above their contractual minimum. Incumbents are split at the median distance between their baseline pay and their own contractual minimum into near-floor and cushioned workers. Bars are the average of the year 0–4 worker-level event-study coefficients (winners minus near-cutoff losers, in log points) for earnings, weeks and the weekly wage; t^*-1, the year before application, is the reference year. Intervals use the average of the yearly standard errors, which is conservative because the yearly estimates are positively correlated. Appendix Figure A.17 plots the yearly estimates, and Appendix Figure A.18 the difference between the two groups.

In the Bank of Italy's Invind survey, 63% of firms in eligible southern areas report that national contracts determine at least 99.5% of their total wage bill (median 100%), compared to 21% in non-eligible northern areas (Figure A.29), and only 15.8% report any firm-level agreement (versus 41.8% elsewhere). This contractual rigidity operates alongside severe structural slack: long-term unemployment affects 11.8% of the southern labor force (versus 2.4% in the Center-North; Table D.1) and voluntary quit rates are depressed (22.8% versus 29.9%), consistently with workers queuing for jobs at contractual floors rather than bidding down wages.

Empirical Test Across Wage Cushions. If national wage floors govern this wage rigidity, price adjustments should be confined strictly to workers whose pay exceeds them. We test this prediction by defining each worker's cushion as the ratio of their annual earnings to their grade-specific contractual floor in the year before the firm's application, $\mathrm{w}_{i,t^*-1}/\underline{w}_{\ell(i)s(i),t^*-1}$, and splitting incumbents at the median into *near-floor* and *cushioned* workers (Figure 10). The split is relative and assigns half of incumbents to each group by construction: since the median incumbent earns about 1.3 times the floor, near-floor workers are those closer to their contractual minimum, not a measure of how many workers the floor binds for. Because the cushion uses annual earnings, a worker with few weeks at baseline could appear close to

the floor; we therefore also rank firms by the average cushion of their full-time incumbents, which does not depend on individual weeks worked. The empirical response separates along this institutional margin. Appendix Figure A.18 plots the difference between the two groups directly; each point is the interaction coefficient of a pooled regression, so the comparison is a test of the differential response rather than a contrast between a significant and an insignificant estimate (Table A.14, bottom panel). Two to three years after the application, weekly wages of cushioned workers rise by 2.4 to 3.8 log points relative to near-floor workers, whose wages remain flat; both differences are significant at the 5 percent level. In levels, the weekly wages of cushioned workers rise by 1.5 to 2.6 log points in the three years after the application, while those of near-floor workers change by between -1.2 and -0.6 log points, never significantly (Table A.14). The two groups are halves of the incumbent workforce, so the rise among cushioned workers is not at odds with the flat wage of incumbents as a whole: averaging the two responses at $t^* + 3$ gives 0.5 log points, against 0.6 for the full application cohort (Section 4.2.2). Conversely, quantity adjustments dominate where the contract binds: in the first two years, weeks worked expand by 5.1 to 5.8 log points more for near-floor workers, again significant at the 5 percent level. Consequently, at $t^* + 3$ increased weeks account for 128% of the earnings gain of near-floor workers, compared to 73% for cushioned workers. The pattern is the same when firms, rather than workers, are ranked by the average cushion of their incumbents (Appendix Table A.15), and it is concentrated among shorter-tenured incumbents (Table A.16). Thus, place-based investment subsidies raise wages only where pay sits above the floor; where the floor binds, adjustment occurs through weeks and employment.

Alternative explanations. The asymmetry across cushions also helps rule out other mechanisms through which a subsidy could reach workers' pay.

- *Rent sharing.* When workers are costly to replace, firms share part of their surplus with them, so pay rises when surplus per worker rises (Card et al., 2018; Kline et al., 2019). A grant that raised the firm's surplus should then raise the pay of the workers it already employs. We find no increase in value added per worker (Appendix B.1); entrants' weekly wages are flat, and incumbents' weekly wages rise only where pay is above the contractual floor, as shown above.
- *Efficiency wages.* Firms may pay above the market-clearing wage to deter shirking, and the wage that does so rises when jobs become easier to find, because losing one costs the worker less (Shapiro and Stiglitz, 1984). An expansion that tightens the local market should then raise pay for entrants and incumbents alike. Entrants' weekly wages

do not move (Section 4.2.2), and neither do wages at other firms in the same market (Section 5).

- *Offer matching.* Pay may rise only when a worker receives an outside offer, which her employer either matches or loses her to (Postel-Vinay and Robin, 2002). An expanding firm would then hire by outbidding other employers, and pay would grow fastest for recently hired cohorts. Starting wages are flat, and the share of hires that funded firms draw from other local employers is unchanged (Figure 13).
- *Financing constraints.* A firm short of credit can pay its workers less today against the promise of more later, in effect borrowing from them, and a cash grant lets it raise pay (Michelacci and Quadrini, 2009; Howell and Brown, 2023). Responses should then be larger at financially constrained firms, yet employment expansions are uniform across credit-risk ratings. Under tenure-dependent bargaining the gains should be concentrated among senior workers, whereas earnings gains accrue to shorter-tenured incumbents (Appendix Table A.16). Nor are the gains disguised payments to owners: managers and the self-employed are excluded from the sample, and the gains go to recent hires.
- *Dynamic learning and long-term contracts.* Three further models work through time. If a job at a funded firm reveals a worker's ability to other employers (Gibbons and Murphy, 1992), workers who leave should earn more at their next employer; leavers' earnings do not rise. If worker and firm learn about the quality of their match, and matches that survive are paid more (Jovanovic, 1979), wages should grow for new hires; starting wages remain flat. If firms promise pay that rises with tenure to keep workers from quitting (Burdett and Coles, 2003), higher pay and lower turnover should come together; retention rises with no pay increase. Finally, one-off bonuses would produce transitory spikes in earnings, whereas the gains we observe persist (Figure 7).

None of these accounts explains why quantities adjust while prices remain rigid, nor why wage gains emerge only among workers paid above statutory minimums. This asymmetry is the signature of a binding institutional wage floor.

5 Effects on Labor Markets

A flat wage at the funded firm does not by itself reveal rationing: a small firm in a market that clears is a wage taker, and it too hires at the going wage. Rationing is a property of the market. We next ask whether the expansion of subsidized firms translates into net employment gains for the local market as a whole, or merely displaces workers from non-subsidized competitors. If subsidized firms compete for scarce labor, neighboring employers face upward wage pressure

and may be forced to downsize via poaching. Conversely, if underutilized labor is available at statutory wage floors, subsidized firms can expand without generating cost pressures elsewhere. The employment and wage responses of non-applicant firms therefore reveal how the broader labor market accommodates an outward shift in labor demand.

5.1 Empirical Specification and Theoretical Benchmarks

We define a local labor market m as a province $\times$ 2-digit NACE sector cell.[22] Within each market, we focus on *firms that never applied* to the program. Non-applicants represent the natural population for evaluating spillovers: they did not receive direct subsidies, even though they would have been eligible for them, and account for the vast majority of local employment.

For each non-applicant firm j, we relate the change in its outcome $Y_j \in \{W_j, L_j\}$ to total employment growth in its local labor market m:

$$\Delta_h \log Y_j \;=\; \pi\, \Delta \log E_{m(j)}^{\bar{t}\text{-}1995} \;+\; \alpha_{\text{prov}(j)} \;+\; \alpha_{\text{sec}(j)} \;+\; \varepsilon_{jh}, \qquad \bar{t} \in \{2001,\, 2007\}, \tag{14}$$

where $\Delta_h \log Y_j \equiv \log Y_{j,1996+h} - \log Y_{j,1995}$ denotes the change in firm j's wage rate (W_j) or employment (L_j) from the baseline year (1995) to year $1996 + h$. The term $\Delta \log E_m^{\bar{t}\text{-}1995}$ is total market employment growth (combining applicants and non-applicants) between 1995 and endpoint $\bar{t}$, while α_{prov} and α_{sec} are province and sector fixed effects. We report two cumulative endpoints: $\bar{t} = 2001$ (the short run, marking the end of initial disbursements) and $\bar{t} = 2007$ (the long run, marking program termination). These calendar endpoints include, in the shock, disbursements made late in the window whose employment response has not yet arrived. Our headline specification therefore aligns the two, measuring the change in non-applicants' outcomes and the growth in market employment over the same three years (Appendix B.3.3).

The coefficient π measures the response of firms that never applied per unit of market employment growth. We estimate π separately for wages (π_w) and employment (π_ℓ). The signs expected under three distinct labor-market environments are:

1. **Neoclassical market clearing with imperfect mobility:** Local labor supply is upward-sloping (Moretti, 2010; Kline and Moretti, 2014a). An outward shift in labor demand bids up the equilibrium market wage ($\pi_w > 0$). As non-subsidized employers

[22] This definition is motivated by evidence that job search is predominantly local (Manning and Petrongolo, 2017) and by the empirical pattern that voluntary job changers overwhelmingly remain within the same province (Appendix D.4).

are poached by expanding firms, their employment contracts ($\pi_\ell < 0$).

2. **Spatial equilibrium with frictionless mobility:** If workers migrate freely across space (Roback, 1982), local labor supply is perfectly elastic. A demand expansion leaves local wages unchanged ($\pi_w = 0$), but employment growth is accommodated entirely by population inflows and in-migration from other regions.
3. **Labor rationing under binding wage floors:** The contractual floor holds the wage above its market-clearing level, so employment is set by demand and a queue of job seekers would accept work at the floor (Section 2). Subsidized firms hire from this queue without bidding up wages ($\pi_w = 0$). With the wage channel shut, non-subsidized employers face no cost-push squeeze and may expand ($\pi_\ell \geq 0$) through input-output linkages or local demand externalities (Proposition 3).

The identification problem. In potential-outcomes terms, the unit is a market m, the treatment is the growth in market employment caused by funded projects, and the outcomes are the employment and wage of the firms in m that never applied. Comparing markets that received more funded jobs with markets that received fewer does not recover the effect of the treatment, because exposure is not assigned at random: a market with a large plant generates more and larger applications, and it may already have been growing, or declining, for reasons that also move its other firms. The raw exposure of a market therefore mixes the jobs it was given with the reasons it applied, and ordinary least squares estimates of equation (14) are biased.

To overcome this endogeneity, we construct an instrument based on the quasi-random ranking of marginal projects near pool cutoffs, in the spirit of Borusyak and Hull (2023) and Borusyak and Kolerman-Shemer (2024). This design provides two key advantages. First, it isolates exogenous variation in market-level labor demand that is orthogonal to local economic fundamentals. Second, the first stage directly tests for full within-market displacement. Under full crowd-out, job gains at subsidized firms would be offset one-for-one by job losses at non-subsidized firms, leaving total market employment unchanged and yielding a first-stage coefficient of zero. A positive first stage rejects full displacement within the measured market, conditional on the validity of the instrument. Subsequent two-stage least squares estimates then determine whether non-applicant firms expand or contract, and whether that adjustment is accompanied by wage pressure.

5.2 Instrument Construction and Identification

Let $g_f \in \{0,1\}$ indicate grant receipt for applicant f, and let L_f denote promised jobs. The observed policy-induced employment shock to market m is:

$$Z_m^{\text{obs}} \;=\; \log\left(E_{m,1995} + \textstyle\sum_{f\in m} L_f\, g_f\right) \;-\; \log E_{m,1995}. \tag{15}$$

The raw shock Z_m^{obs} is systematically correlated with market application rates and regional budgets.[23] Following the local-randomization approach to regression discontinuity designs (Cattaneo et al., 2015, 2017; Abdulkadiroğlu et al., 2017, 2022), we exploit the fact that within a narrow bandwidth around each pool's funding cutoff, the precise ranking of projects is quasi-random.[24]

Assignment Mechanics and the Recentered Shock. Within each pool, we hold fixed applicant characteristics, requested grants, promised jobs, and inframarginal awards. The only variation we exploit is the ordering of projects within a narrow band around the cutoff, which we treat as random. Recentering removes the predictable exposure a market could expect simply from having more applications near the cutoff, isolating the idiosyncratic "surprise jobs" allocated to that market (Appendix E.3 provides a stylized numerical walkthrough). Formally, in funding pool p, let S_f denote the standardized score of applicant f relative to the pool cutoff, r_f its requested grant, L_f its promised jobs, and $g_f \in \{0,1\}$ its funding status. Define the marginal band as $\mathcal{C}_p = \{f \in p : |S_f| \le \bar{S}\}$, containing $n_p = |\mathcal{C}_p|$ projects, and let b_p denote the budget available to this band—the total pool budget minus the requests of inframarginal applicants scoring above the band, who win in every ordering. Pre-determined characteristics are collected in:

$$\mathbf{w} = \big(\{\mathcal{C}_p, b_p\}_p,\ \{r_f, L_f\}_f,\ \{g_f\}_{f\notin\cup_p\mathcal{C}_p},\ \{E_{m,1995}\}_m\big).$$

Let σ_p denote a permutation of the projects in $\mathcal{C}_p$. The statutory allocation rule G_p funds applicants sequentially down σ_p until b_p is exhausted:

$$\sigma_p \sim \text{Unif}(\mathcal{R}_p) \text{ independently across pools}, \qquad \mathbf{g}_p \equiv (g_f)_{f\in\mathcal{C}_p} = G_p(\sigma_p; \mathbf{w}),$$

[23] As a robustness check, Appendix B.3.2 replaces baseline employment with a Poisson pseudo-maximum-likelihood (PPML) prediction of endpoint employment; the estimates are virtually identical.

[24] Appendix E.1 details what the design treats as random, its connection to the asymptotic framework of Abdulkadiroğlu et al. (2022) and to the approximate propensity score of Narita and Yata (2023), and bandwidth selection.

where $\mathcal{R}_p$ is the set of $n_p!$ permutations. The propensity score of an applicant is $e_f \equiv \Pr(g_f = 1 \mid \mathbf{w})$.[25] Because Z_m^{obs} is non-linear in log employment, expected market exposure depends on the joint probability of multi-project funding allocations. We compute the expected counterfactual shock $\mu_m \equiv \mathbb{E}[Z_m^{\text{obs}} \mid \mathbf{w}]$ via $S = 250$ Monte Carlo simulations over the exchangeable ranking distributions:

$$\widehat{\mu}_m \;=\; \frac{1}{S}\sum_{s=1}^{S} Z_m^{(s)},$$

which approximates μ_m up to simulation error. Following Borusyak and Hull (2023), the instrument is the recentered shock:

$$\tilde{Z}_m \;=\; Z_m^{\text{obs}} - \widehat{\mu}_m, \tag{16}$$

which isolates the unexpected shock allocated to market m net of its mathematical expectation.[26]

Identifying Assumptions. Causal identification rests on three formal conditions:

Assumption 1 (Local Randomization). Conditional on pool observables $\mathbf{w}$, grant allocation within the marginal band is orthogonal to counterfactual market potential outcomes $y_m(\mathbf{g}')$, the outcome of market m under band assignment $\mathbf{g}'$:

$$\mathbf{g} \perp\!\!\!\perp \{y_m(\mathbf{g}')\}_{\mathbf{g}'} \mid \mathbf{w},$$

where $\mathbf{g}$ stacks the band assignments across all funding pools.

Assumption 2 (Exchangeable Order). Conditional on $\mathbf{w}$, every ordering $\sigma_p \in \mathcal{R}_p$ of near-cutoff applicants is equally likely. Because the continuous score contains idiosyn-

[25] The propensity score is derived analytically in Appendix E.2, with worked knapsack examples in Appendix E.3; simulation error affects only first-stage power (Remark 2).

[26] This construction relates to the regression discontinuity aggregation of Borusyak and Kolerman-Shemer (2024), who build market-level treatments from multiple threshold events under continuity assumptions. Our setting differs in two key ways. First, the treatment shock is non-linear: it measures the percentage expansion in log market employment, so an additional funded project contributes less when other projects in the same market are also funded. A linear specification would yield $\mathbb{E}[\sum_{f\in m}(L_f/E_{m,1995})g_f] = \sum_{f\in m}(L_f/E_{m,1995})e_f$ regardless of joint assignment; in contrast, the log specification requires modeling joint allocations, which our permutations capture by reapplying the budget rule. Second, we treat scores within the marginal band via local randomization rather than continuity, without controlling for aggregated polynomials of the running variable (Appendix E.1).

cratic evaluation noise, project ranks within the narrow bandwidth $\bar{S}$ approximate a uniform draw (Appendix E.1).

Assumption 3 (Many Strata and Weak Dependence). As the number of markets M grows, the cross-market covariance of recentered shocks vanishes:

$$\frac{1}{M^2} \sum_{m \neq m'} \left| \mathrm{Cov}(\tilde{Z}_m, \tilde{Z}_{m'} \mid \mathbf{w}) \right| \longrightarrow 0.$$

This weak-dependence condition ensures the consistency of the recentered instrument (Borusyak and Hull, 2023).

The design conditions on the realized band $\mathcal{C}_p$, holding the candidate set fixed while the effective cutoff moves endogenously with project requests and budget exhaustion. Conditioning on the available budget b_p is valid because it is determined prior to the marginal draw by subtracting the fixed requests of inframarginal winners. In contrast, the amount actually disbursed within the band varies across permutations due to discrete project request sizes, exactly as dictated by the statutory allocation rule (Appendix E.2).

Statistical Power and Balance. Identifying variation within a market is driven by the granularity of project requests: promised jobs among near-cutoff applicants account for a median of 15% (and 45% at the 75th percentile) of baseline applicant employment in their market. Within a market, identification is driven by a median of three near-cutoff firms, with the single largest project carrying a median weight of 0.59. Conversely, each funding pool spans an average of 43 distinct markets (median 23), ensuring that one market's idiosyncratic draw remains approximately independent of another's (Appendices E.4 and E.5).

The recentered shock $\tilde{Z}_m$ shows no correlation with pre-existing market trajectories. At the project level, neither the density of the score (McCrary $p = 0.54$) nor individual evaluation criteria exhibit discontinuities at the threshold (Appendix B.1). At the market level, regressing nine pre-policy outcomes (1991–1995 changes in employment, firm counts, wages, skill composition, and turnover) on $\tilde{Z}_m$ yields an insignificant joint balance test ($p = 0.891$ across 1,520 markets; Appendix Table A.7). These tests are supporting diagnostics: they are consistent with an exchangeable order within the band, but they do not establish it.

First-Stage Estimates. Figure 11 plots market employment growth from 1995 to 2001 against $\tilde{Z}_m$, net of fixed effects. The first-stage coefficient is 1.06 (standard error 0.16; Kleibergen–Paap $F = 40.1$; randomization-inference $p = 0.01$). Appendix Figure E.3 checks

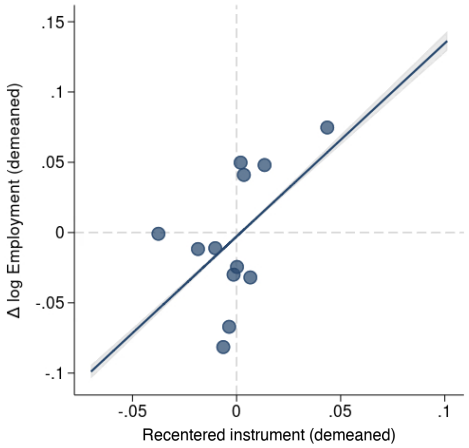
the assignment model against the data, and Appendix Figure E.4 reports randomization-inference confidence intervals for π. Both variables are changes in log employment: a market whose funded promised jobs exceed counterfactual expectations by one percent of baseline employment grows by approximately 1.06 percent more. Because total market employment includes non-applicants, a positive coefficient directly rejects full within-market displacement, under which the estimate would equal zero. It does not, on its own, bound partial displacement.[27] For this reason, we examine the response of non-applicant firms directly.

5.3 Market-Level Evidence on Non-Applicants

Employment and Wage Spillovers. Figure 12 presents two-stage least squares estimates of equation (14). Employment at firms that never applied expands steadily over the disbursement period without any accompanying wage pressure. In our baseline specification, for every 10% of market employment growth caused by the policy, employment at non-applicant firms grows by 2.4% ($\pi_\ell = 0.241$, s.e. 0.095), while their wage rate does not detectably move ($\pi_w = 0.014$, s.e. 0.031). Employment expanding at an unchanged wage matches the prediction of a binding floor in Section 2, under which firms recruit from a queue and competitors face no upward wage bidding. It is inconsistent with a clearing market with upward-sloping local supply, which predicts rising wages and contracting competitor employment; on its own, it does not exclude a perfectly elastic supply, which we discuss in Section 5.5. On the calendar clock of equation (14), which measures cumulative growth from 1995 to $\bar{t}$, the employment estimate is smaller ($\pi_\ell = 0.149$, s.e. 0.067) because late-cycle disbursements have not yet fully materialized (Appendix B.3.3). The wage coefficient is not statistically distinguishable from zero at either horizon ($\pi_w = 0.021$ at 2001; 0.067 at 2007; Appendix B.3.3). In terms of job counts—given that non-applicants hold approximately 90% of baseline employment—every 100 jobs created at applicant firms generate an additional 15 to 28 jobs at non-applicant firms in the same market. Non-applicants expand in headcount at the prevailing contractual floor: weeks worked and pay per worker remain unchanged, and hiring draws on the same local pools as before.

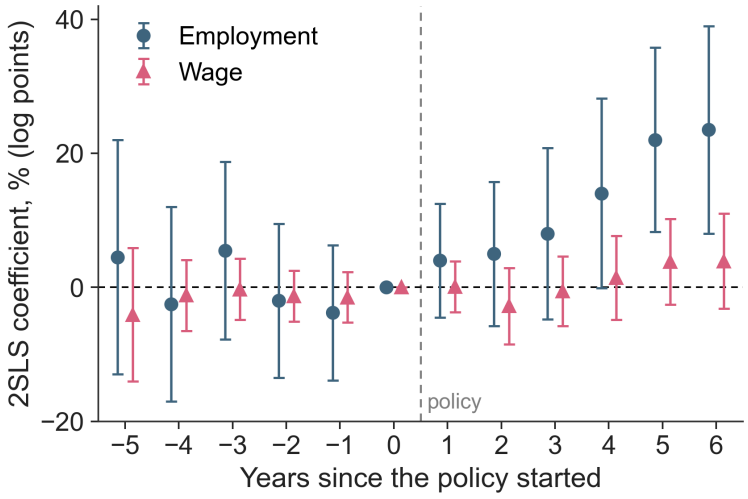
[27] Because the shock scales funded promised jobs relative to baseline employment, the coefficient approximates the actual jobs per promised job, but only roughly: the conversion depends on shock size, and two-stage least squares weights markets by the variance of their recentered shock.

Figure 11: Market-Level First Stage



Notes: This figure plots the first stage of the market-level design: market log employment growth from 1995 to 2001, $\Delta \log E_m$, against the recentered instrument $\tilde{Z}_m$, both residualized on province and sector fixed effects, as a binned scatter plot with the linear fit. The first-stage coefficient is 1.06 (s.e. 0.16; Kleibergen–Paap $F = 40.1$; randomization-inference $p = 0.01$).

Figure 12: Non-Participating Firms: Employment and Wages



Notes: This figure plots year-by-year two-stage least squares estimates of the effect of instrumented market employment growth on the employment and the wage rate of firms that never applied to the program, in log points, at the 2001 endpoint, with 95% confidence intervals; the year before the program is the reference. Employment rises with no pre-trend; the wage is flat throughout. The wage-bill and survival panels, and the 2007-endpoint versions, are in Appendix B.3.

Local Employment Multipliers. Denoting by $s_N \approx 0.9$ the employment share of non-applicant firms, the aggregate local employment multiplier is:

$$M \equiv 1 + \frac{\Delta E_N}{\Delta E_A} = \frac{1}{1 - \pi_\ell \, s_N}. \tag{17}$$

Our empirical estimates of $\pi_\ell \in [0.149, 0.241]$ imply an aggregate multiplier of $M \in [1.15, 1.28]$: each subsidized job creates 0.15 to 0.28 additional jobs at local non-applicant firms (Appendix E.6).

Balance sheet records from Cerved confirm that this non-applicant expansion occurred without shifts in profitability or labor productivity: two-stage least squares estimates for the change in log wage bill (0.066, s.e. 0.113) and log value added (−0.041, s.e. 0.094) are statistically insignificant (Appendix Table A.8). Furthermore, flat wages are not a result of workforce composition, which is unaffected.

5.4 Worker-Level Evidence: Benefits, Wages, and Mobility

The firm-level estimates document employer responses, but do not trace the workers themselves. We therefore implement the same recentered-instrument design at the worker level using the 13% longitudinal INPS panel, which captures comprehensive income sources (Section 4.2.3). We restrict the sample to workers employed in market m in 1995 who were never employed by an IDA applicant, ensuring that policy exposure is predetermined and free from direct treatment.[28]

Benefits, Employment, and Pay. Unemployment benefit receipt falls among workers in markets the policy expanded, with flat pre-trends (Appendix Figure A.40). The two-stage least squares coefficients are scaled to a 100 log point increase in market employment: between 2000 and 2004 they average about −4.5 percentage points, and the largest yearly coefficient is close to −5.5. In the units of Section 5.3, for every 10% of market employment growth caused by the policy, the share of workers drawing benefits falls by about 0.45 percentage points. For the shocks the instrument actually generates, the effect is modest. In the median market, weighting markets by their exposure to near-cutoff projects, the recentered shock is 13 log points (interquartile range 1 to 35); with a first stage of 1.06 it raises market employment

[28]The market-level worker sample is restricted to men. Each worker's outcome is regressed on the employment growth of their 1995 market, instrumented with $\tilde{Z}_m$ as in equation (14), including industry, region, and calendar-year fixed effects interacted with full-time status, with standard errors clustered by market.

by about 14 log points and lowers benefit receipt by roughly 0.6 percentage points, and by 1.7 at the 75th percentile.[29] The decline goes in the same direction as within funded firms (-1.6 to -4.0 percentage points; Section 4.2.3), but the two are not on the same scale: the firm estimate is the effect of a grant on the firm's own incumbents. Dependent employment earnings rise imprecisely, while individual weekly wages are unchanged. The probability of private-sector employment shifts by less than one percentage point, indicating that the shock primarily shortens non-employment durations for attached workers rather than moving individuals into work on an annual basis.

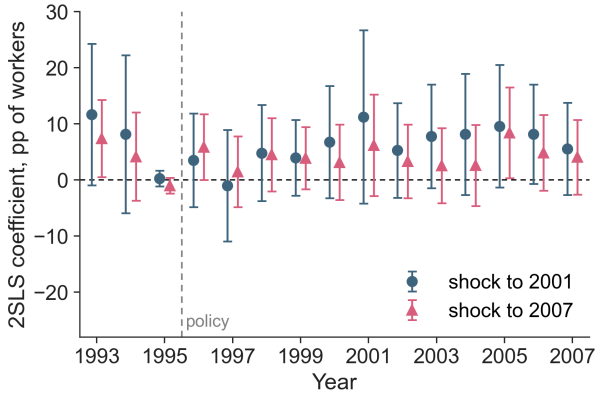
Mobility and Migration. We find no detectable increase in geographic mobility among baseline residents. Exposed workers are, if anything, slightly less likely to reside outside their baseline province (point estimates range from 0 to -14 percentage points, statistically insignificant) and more likely to continue working within it (Appendix Figure A.41), with no residential divergence over a 24-year tracking window. At funded firms, approximately 80% of new hires had no formal private employment in the prior year, and the share hired from other local employers is unchanged (Figure 13); at non-applicant firms, the share of locally born hires remains constant. While tracking the 1995 cohort provides direct evidence on out-migration and retention rather than gross inward migration, the baseline immobility of southern labor (Caponi, 2008; Ciani et al., 2019) indicates that the expansion was overwhelmingly absorbed by resident labor.

5.5 Discussion

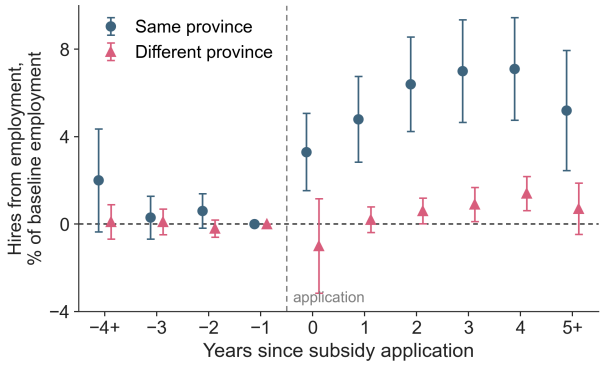
Evaluating market-level spillovers addresses the displacement question: the policy expanded employment in the market as a whole without generating upward wage pressure. The positive first-stage estimate (1.06, s.e. 0.16) rejects full within-market displacement, while non-applicant firms expand alongside the market ($\pi_\ell = 0.241$, s.e. 0.095) with zero detectable wage growth ($\pi_w = 0.014$, s.e. 0.031). Set against the theoretical benchmarks of Section 5.1, these results are inconsistent with the competitive prediction that competitors contract ($\pi_\ell < 0$), and the wage response is too small to be rationalized by market clearing with plausible local supply elasticities (see below). Frictionless spatial mobility (Rosen, 1979;

[29] Medians across markets with at least 50 employees in 1995. Each market is weighted by the sum over its near-cutoff projects of the squared ratio of promised jobs to 1995 employment, a proxy for the variance of its recentered shock, which governs its weight in two-stage least squares. Unweighted, the median market receives almost no surprise (0.1 log points), because its near-cutoff projects are funded or not with high probability.

Figure 13: Exposed Workers: Location and Mobility



(i) Workers still employed in their 1995 province



(ii) Funded firms: hires from employment, by province

Notes: This figure plots evidence on where exposed workers live and where funded firms hire. Panel (i) plots two-stage least squares estimates of the effect of the recentered market shock on the probability that a worker is employed in their 1995 province, by calendar year, in percentage points, for both endpoints of the shock (2001 and 2007), with 95% confidence intervals; 1993–95 are placebo years. Panel (ii) plots a firm-level event study at the funding cutoff for cumulative hires from employment whose previous job was in the same province or in a different province, as a share of baseline employment, winners minus near-cutoff losers, with 95% confidence intervals.

Roback, 1982) could generate a flat wage, but requires employment to be filled by inward migrants-a prediction inconsistent with the low mobility of southern workers (Appendix D), zero residential sorting among exposed cohorts, and the fact that 80% of hires are recruited from local non-employment. The evidence is consistent with the third benchmark: job rationing at a binding statutory floor ($\pi_w = 0$, $\pi_\ell \geq 0$).

Bounding labor supply. Because π_w is precisely estimated near zero, it bounds the labor supply elasticity that a market that clears would require. If the market clears along a stable supply curve, π_w equals the inverse elasticity of local labor supply, $1/\varepsilon_s$. The upper end of its 95% confidence interval, 0.075, means that a 10% employment expansion raised wages by at most 0.75%, which requires $\varepsilon_s \geq 13$ ($\varepsilon_s \geq 15$ at the 90% level). The bound holds only under the null that the market clears: it says how elastic local supply would have to be for a clearing market to produce our estimate, not that supply is that elastic.

This is large relative to estimates of the supply elasticity facing individual firms (Card et al., 2018; Sokolova and Sorensen, 2021; Lamadon et al., 2022), and the supply curve facing a whole local market should, if anything, be less elastic. As in Breza et al. (2021), the flat wage is therefore hard to reconcile with a market that clears; under rationing, π_w reflects the contractual floor rather than a supply elasticity.

Interpreting results. Unobserved compositional changes are ruled out by the stability of observable hiring characteristics, and confounding local trends are addressed by the orthogonality of $\tilde{Z}_m$ to pre-policy trajectories. A remaining alternative is an economy with elastic labor force participation, in which many non-employed residents are indifferent between working and not at the going wage. Wages and employment alone cannot separate this case from rationing, and the distinction matters for welfare: under elastic participation the marginal worker's reservation wage equals the wage, and the new jobs carry no worker surplus. Two pieces of evidence separate them: most job seekers in the South report reservation wages below the entry wage (Section 6.1), and weekly wages respond only where workers are paid above their contractual floor (Section 4.3), while incumbents paid near it work more weeks at an unchanged wage (Section 4.2.2). Together, they indicate that employment is rationed.

Finally, we do not separately identify the specific product-market channels (intermediate input purchases, local consumption spillovers, or agglomeration externalities) through which demand propagates to non-applicants; our welfare accounting in Section 6 does not depend on this decomposition, since it hinges on total employment changes. Together, the evidence is consistent with the local labor market absorbing the policy expansion by drawing underutilized workers into employment at an unchanged statutory wage.

6 Worker Surplus and Labor Market Efficiency

The previous sections show that the program created jobs at an unchanged wage, that net market employment rose and non-applicant firms expanded on average, and that most of the workers it hired came from local non-employment. This section asks what those jobs are worth to the workers who fill them. When wages clear the market, the marginal worker is indifferent between working and not, so evaluations of place-based subsidies usually do not consider a surplus from the jobs themselves, and focus instead on wages and on the cost of the program. When a wage floor holds pay above what job seekers would accept, jobs are rationed, and a job is worth more to the worker who obtains it than the time it takes. We measure this surplus, taking the contractual floor as given; together with the additional weeks worked by incumbents, it is the labor-market part of the welfare effect in equation (6). That equation also includes the distortion of capital and the excess burden of public funds, which arise under any wage-setting regime. We compare the labor-market gains with the cost of financing the program and do not estimate the distortion of capital. Appendix F.1 details the derivations. Table 4 lists the inputs of the calculation and whether each is observed in the data, estimated in Sections 4–5, or calibrated.

Table 4: Inputs of the Welfare Calculation

	Object	Value	Source	
Observed				
w^e	Net pay of a new hire	€974 a month	INPS records	Sec. 6.1
ω^R	Reservation wages	p5 €516; mean below w^e €727	Labour Force Survey, 1996	Tab. F.1
G	Grant paid per direct job	€176,000	IDA records	Sec. 6.2
Estimated				
J	Worker-years per direct job	3.46 over six years	Firm-level RD	Sec. 4
	Jobs at non-applicant firms	15–28% of direct jobs	Market instrument	Sec. 5.3
dh	Incumbents' added weeks	About two a year	Application cohorts	Sec. 4.2.2
dK	Investment per grant euro	About €1	Balance sheets	Sec. 6.2
Calibrated				
λ	Excess burden of public funds	0.25		App. F.4
τ	Income tax on new earnings	15% of gross pay (provisional)		Sec. 6.2
b	Benefits no longer paid	27% of pay for six months	Upper bound	Sec. 6.2
m	Cost of a week / floor	0.5–0.6	Lachowska et al. (2026)	
C_K	Capital distortion per job	€37,700	Subsidy rate about one third	Sec. 6.2

Notes: Observed inputs come directly from administrative or survey data, estimated inputs from Sections 4–5, calibrated inputs from outside sources or assumption. Current euros, not discounted; per-job amounts refer to direct jobs over six years.

6.1 The Surplus of the New Jobs

Under labor rationing, the net value of a job equals the prevailing wage minus the hired worker's reservation wage, making worker surplus dependent on which job seekers obtain employment (Bartik, 2012). Existing applications typically rely on strong assumptions to quantify this surplus: early cost-benefit studies set the opportunity cost of time to zero-effectively valuing new jobs by their full earnings (Haveman and Krutilla, 1967; Lefeber, 1968)-while later work relies on calibrated labor supply elasticities (Bartik, 2012, 2020). We instead incorporate direct survey evidence on reservation wages to evaluate an identified employment expansion where wage floors bind.

Formally, assuming every marginal position pays a common benchmark w^e-the average net earnings of new hires-and integrating this surplus over all created jobs across their duration T, total private worker surplus is:

$$WS \equiv \Delta \text{Worker surplus} = \int_0^T \int_{Q_t} \left(w^e - \omega^R(\ell)\right) d\ell \, dt, \tag{18}$$

where Q_t denotes the set of workers holding the additional positions in year t, measured in full-year equivalents, and $\omega^R(\ell)$ is the reservation wage of worker ℓ. Both pay and reservation

wages are annualized (twelve times the monthly amounts reported below). Because firms hire until a worker's net marginal product covers the gross labor cost, which exceeds net pay, WS is a lower bound on the full gain from employing these workers (Appendix F.3). We report all figures per direct job: we first value the jobs created at funded firms, and Section 6.2 adds the jobs created at non-applicant firms and the fiscal gain.

We measure reservation wages in the Italian Labour Force Survey (Appendix Table F.1), in which job seekers report the minimum net monthly wage they would accept for a job that meets their requirements (Sestito and Viviano, 2011), and compare them with the average net pay of new hires in our sample, €974 per month ($w^e =$ €11,700 a year), about 1.17 times the net contractual minimum. In the targeted areas, 79% of job seekers report a reservation wage below w^e.[30] Because we do not observe which job seekers fill the new positions (Luttmer, 2007), we evaluate WS under a linear benchmark and two allocation rules (Lee and Saez, 2012; Bartik, 2012).

A linear benchmark. Our most conservative calculation approximates supply by a straight line from the fifth percentile of reservation wages, $\underline{\omega}^R =$ €516, to the pay of the jobs, as if the jobs created exhausted the queue so that the last worker hired values the job at its pay (Figure 14). The average surplus is then half the gap,

$$\overline{WS}_{\text{linear}} \;=\; \tfrac{1}{2}\big(w^e - \underline{\omega}^R\big) \;\approx\; \text{€}229 \text{ per month}, \tag{19}$$

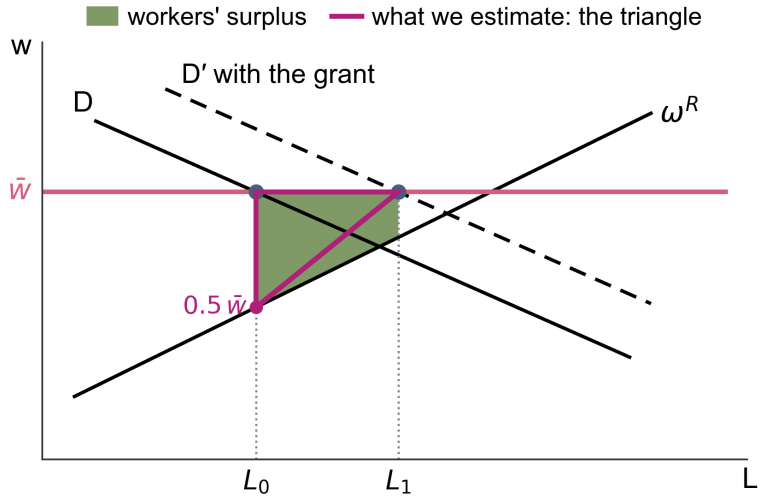
or 23.5% of net pay: about €9,500 per direct job over the six years of our analysis horizon (Section 6.2). This is the average surplus at stake when employment at w^e expands until the queue is exhausted. However, the queue is in fact far longer than the jobs created, so the surplus the program realizes depends on the share q of the queue it reaches and on who is hired first, which the two allocation rules below address.

Efficient rationing. If the additional jobs go first to the job seekers with the lowest reservation wages, then for any share q of workers hired at w^e the average surplus can be written as

$$\overline{WS}_{\text{efficient}}(q) \;=\; w^e - \frac{1}{q}\int_0^q F_R^{-1}(u)\,du, \tag{20}$$

[30] Survey evidence from the United States compares stated reservation wages with the unemployed's previous wages and finds them close (Feldstein and Poterba, 1984; Jones, 1989; Krueger and Mueller, 2011). Our comparison is with the pay of the jobs created, so the two are not directly comparable.

Figure 14: Graphical Analysis: The Value of Job Creation under Rationing



Notes: This figure illustrates, in stylized form, the value of the jobs a grant creates when the floor binds. Supply is the schedule of job seekers' reservation wages ω^R, ranked from lowest to highest, and $\bar{w}$ is the pay of the jobs. The grant shifts labor demand from D to D', and employment rises from L_0 to L_1 at that pay. The shaded area is the workers' surplus from the new jobs when they go to the job seekers with the lowest reservation wages: for each worker hired, the gap between pay and the worker's reservation wage. The outlined triangle is the linear benchmark of Section 6.1, the most conservative of the calculations; that section also computes the surplus from the observed distribution under two allocation rules.

where F_R is the distribution of the job seekers' reservation wages. This is the linear benchmark with the survey distribution in place of the straight line. Evaluating it requires the share q, and the program is small relative to the queue. Over 1999-2005 the unemployment rate in the targeted areas averaged 18.3% (Appendix Table D.1), and 79% of job seekers there report a reservation wage at or below w^e. Job seekers willing to accept w^e therefore make up about 14.5% of the labor, or 18 for every 100 employed.[31] This is a queue of willing workers, not a measure of the unemployment caused by the floor. The survey covers only job seekers, so the non-employed who have stopped searching, whose reservation wages and availability are unobserved, are not counted. Scaled to the whole program, the direct jobs created would employ 6 to 11 percent of them. Under efficient rationing, the jobs therefore go to the lowest 5 to 9 percent of all job seekers, every one of whom reports a reservation wage of at most €516. Evaluating equation (20) on this group gives a trapezoid rather than a triangle: at least €458 per month (47% of net pay), or at least €19,000 per direct job over six years. Including the jobs created at non-applicant firms, they reach about the lowest 11 percent. The bound of €458 is established for the direct jobs; applying it also to the indirect jobs, which reach slightly beyond the tenth percentile, is an additional assumption.

[31]The first number is $0.79 \times 18.3\%$, the second 14.5/81.7.

Uniform rationing. If the jobs go to willing job seekers at random, the average surplus is net pay minus their mean reservation wage, €727 in the survey:

$$\overline{WS}_{\text{uniform}} \;=\; w^e - E\left[\,\omega_i^R \,\middle|\, \omega_i^R \le w^e\,\right] \;\approx\; €247 \text{ per month}, \tag{21}$$

or 25% of net pay: about €10,000 per direct job over six years.

Average worker surplus therefore ranges from 23.5% of net pay under the linear benchmark to 25% under uniform rationing and at least 47% under efficient rationing.

Our baseline calculation may omit two additional sources of surplus. First, reported reservation wages reflect both the disutility of work and the option value of continued search (Luttmer, 2007, fn. 2); hence, the static opportunity cost of hours worked may be lower, though this depends on expected job duration and on-the-job search. Second, hiring a worker from the queue frees up job opportunities for remaining job seekers (Bartik, 2012). To the extent that these search spillovers are not already reflected in employment at non-applicant firms in Q_t, they represent an additional welfare gain.

6.2 Labor-Market Gains and the Cost of the Program

Evaluations of place-based policies usually report the cost per job, the budgetary outlay divided by the number of jobs created. However, the outlay is not equivalent to the policy's social cost: in this case, most of the grant is a transfer from taxpayers to firm owners. To compare the gains in the labor market with what the program costs society, we write the welfare change from one additional direct job as the sum of four terms: the surplus WS gained by workers, measured on net pay and including market spillovers; the fiscal externality F, the taxes on new earnings and the benefits no longer paid; the net gain to firm owners, who receive the grant G less the distortion of the investment it induced, C_K; and the net outlay $G - F$, funded by taxpayers at a cost of $1 + \lambda$ per euro.

Summing these terms, the grant itself cancels:

$$\Delta W \;=\; WS + (G - C_K) - (1+\lambda)(G - F) \;=\; \underbrace{WS + (1+\lambda)F}_{\text{labor-market gains}} \;-\; \lambda G \;-\; C_K. \tag{22}$$

Because WS is measured on net pay, taxes on new earnings enter only once, through F; and because reservation wages already include the benefits a worker gives up, forgone benefits also enter only through F (Appendix F.2). The social cost of the program is therefore not G but the excess burden of raising the funds, λG, and the capital distortion C_K. We first

measure the cost of financing and the labor-market gains, then compare them, and finally add the capital distortion and summarize all terms as a marginal value of public funds.

The cost of financing. The grants awarded to the winners in our estimation sample amount to about €231,000 per direct job created. Because 76 percent of winners receive funds (Figure 3), the outlay actually paid is about €176,000 per job, within the range reported for US cash incentives (Bartik, 2020). At $\lambda = 0.25$, raising these funds costs society about €44,000 per direct job.

Labor-market gains. The first term of equation (22) combines the worker surplus of Section 6.1 with the fiscal externality. Each direct job corresponds to 3.46 worker-years over the six years after the application, the horizon of our analysis, because funded firms hire gradually.

The fiscal externality F consists of the taxes paid on new earnings and the transfers no longer paid (Lee et al., 2019). Benefit receipt falls by 1.6 to 4.0 percentage points among treated incumbents (Section 4.2.3) and by approximately 0.45 percentage points per 10% of policy-induced market employment growth (0.6 percentage points in the median exposed market; Section 5.4). We count personal income tax, about 15 percent of gross pay or €2,300 per worker-year, and the unemployment benefits no longer paid, 27 percent of pay for six months or about €1,580 per hire, an upper bound for that component since not every hire was eligible.[32] Over six years, the fiscal externality is about €9,500 per direct job. It counts with weight $1 + \lambda$, since each euro recovered also reduces the funds that must be raised. For the direct jobs alone, the labor-market gains are therefore about €21,400 per direct job under the linear benchmark and at least €30,900 under efficient rationing.

Market spillovers. The program also created jobs at local non-applicant firms. Counting them raises worker surplus by 15 to 28 percent (Section 5.3), assuming indirect positions generate comparable average surplus, and raises the fiscal externality to €11,000–12,200 per direct job. Including them, the labor-market gains over six years are €24,600–27,400 per direct job under the linear benchmark, and at least €35,600–39,600 under efficient rationing (Table 5, Panel A).

[32] We leave out social contributions: employer contributions for new hires in these regions were waived in 1999–2001, and pension contributions are largely deferred wages.

Table 5: Labor-Market Gains, the Cost of Financing, and the MVPF per Direct Job

	Share of net pay	Per direct job, six years	Share of the excess burden	Break-even λ
Panel A. Labor-market gains and the cost of financing ($\lambda = 0.25$), no capital distortion				
Worker-years per direct job		3.46		
Grant paid, G		€176,000		
Worker surplus, direct jobs				
Linear benchmark	23.5%	€9,500		
Uniform rationing	25%	€10,000		
Efficient rationing	≥47%	≥€19,000		
Fiscal gain with indirect jobs, weighted by $1 + \lambda$				
Income tax and unemployment benefits		€13,700–15,300		
Measured labor-market gains: worker surplus with indirect jobs, plus weighted fiscal gain				
Linear benchmark		€24,600–27,400	56–62%	13–15%
Uniform rationing		€25,500–28,400	58–65%	14–16%
Efficient rationing		≥€35,600–39,600	≥81–90%	≥20–22%
Cost of financing				
Excess burden of the outlay, λG		€44,000		
Calibrated capital distortion, C_K		€37,700		

	No capital distortion	Calibrated capital distortion
Panel B. Marginal value of public funds, six years		
Workers' willingness to pay, linear	€10,900–12,200	
Workers' willingness to pay, uniform	€11,800–13,100	
Workers' willingness to pay, efficient	≥€21,900–24,300	
Owners' willingness to pay, $G - C_K$	€176,000	€138,300
Net cost to the government, $G - F$	€163,800–165,000	
MVPF		
Linear benchmark	1.13–1.15	0.90–0.92
Uniform rationing	1.14–1.15	0.91–0.92
Efficient rationing	≥1.20–1.22	≥0.97–0.99
Workers' share of measured willingness to pay		
Linear benchmark	6–7%	7–8%
Efficient rationing	≥11–12%	≥14–15%

Notes: Per direct job at funded firms, over the six years after the application; amounts are not discounted. Ranges reflect the jobs at non-applicant firms (15 to 28 percent of direct jobs, with the same surplus and fiscal gain each). Worker surplus is net pay of new hires (€974 a month) minus reservation wages from the 1996 Labour Force Survey, over 3.46 worker-years per direct job (Section 6.1); efficient-rationing figures are lower bounds, established for direct jobs and applied to indirect jobs by assumption. The fiscal gain counts income tax and unemployment benefits no longer paid. Panel A abstracts from the capital distortion; its shares hold at $\lambda = 0.25$ only, and the break-even λ solves $WS + (1 + \lambda)F = \lambda G$. Panel B reports the MVPF of equation (23) with F unweighted; the capital distortion is calibrated (Section 6.2). All figures leave out incumbents' added weeks (€4,500–5,500 per job). Appendix Table F.2 extends the calculation to twelve years.

Additional margins. Two further margins are not included in the baseline figures, but we include them in subsequent calculations.

1. **Additional weeks:** Incumbents at funded firms also work more weeks per year (Section 4.2.2). Because employment at funded firms near the cutoff rises by about 13 percent, each additional job comes with about eight incumbents. At €165–205 per incumbent-year, their added weeks are worth about €1,300–1,600 per additional job for each year the expansion lasts, or €4,500–5,500 over six years: about half the surplus from the job itself under uniform rationing.[33]
2. **Job preservation:** Four to five years after the application, about one third of the employment expansion at funded firms consists of incumbents who would otherwise have separated (Appendix Table A.5). Valuing these positions at the full gap between pay and reservation wage is right for workers who would have become non-employed, but overstates the surplus for those who would have moved directly to another employer, with no change in market employment. We therefore report a scenario that scales the surplus of retained positions by the share of separations that end in non-employment.

Labor-market gains against the cost of financing. Leaving aside the capital distortion, our exercise compares the two remaining terms of equation (22): the labor-market gains, net worker surplus plus the fiscal externality valued at $1 + \lambda$, and the cost of financing the grant, λG. Both are measured per direct job over six years. Under the linear benchmark, the labor-market gains of €24,600–27,400 cover 56 to 62 percent of the €44,000 cost of financing; under efficient rationing, they cover at least 81 to 90 percent. Equivalently, each euro of grant generates 14 to 16 cents of labor-market gains under the linear benchmark, and at least 20 to 23 cents under efficient rationing, against an excess burden of 25 cents.[34] The efficiency gain from relaxing rationing is first order, but within six years it does not cover the cost of financing the grant. Whether the program improves labor-market efficiency on net therefore

[33] Valuing added weeks requires incumbents' opportunity cost of a week of work, $\bar{\zeta}_{\text{week}}$, which we calibrate as a share $m \equiv \bar{\zeta}_{\text{week}}/\bar{w} \in [0.5, 0.6]$ of the net contractual floor $\bar{w}$ of €190 per week (Lachowska et al., 2026). A gain of two weeks per year on a baseline of 39 weeks (Section 4.2.2) then yields an annual surplus of €165–205 per incumbent. Since several incumbents gain weeks for each additional job, we express this margin per additional job, the same unit as the surplus from the jobs themselves.

[34] Under the linear benchmark, the program would break even only if the excess burden were below 13 to 15 percent. Under uniform rationing, the gains are €25,500–28,400 per direct job and cover 58 to 65 percent of the cost of financing (break-even excess burden of 14 to 16 percent); the MVPF is 1.14 to 1.15 without the capital distortion and 0.91 to 0.92 with it, and workers receive 6 to 9 percent of the measured value. Incumbents' added weeks, not included in these totals, would raise all coverage shares by about ten percentage points.

depends on how it is financed.[35]

The capital distortion. These comparisons have so far left out the distortion of capital, C_K. In our model, firms choose capital optimally at the contractual wage, so a subsidy to capital induces investment whose private return falls short of its cost. Applying the procedure we use for employment to firms' balance sheets, funded firms invest about one additional euro for each euro of grant, or about €236,000 per direct job. With an effective subsidy rate of about one third (firms requested on average 64 percent of the maximum subsidy allowed) and a linear response, the loss is about half the subsidy on that investment, €37,700 per direct job. Adding it to the cost lowers the coverage to about a third under the linear benchmark. This calibration assumes no other distortion in capital markets; financing constraints or externalities from investment would change it.

6.3 The marginal value of public funds

These components can be synthesized into a marginal value of public funds (MVPF; Hendren and Sprung-Keyser, 2020), defined as the total willingness to pay of the program's beneficiaries divided by its net cost to the government:

$$\text{MVPF} \;=\; \frac{WTP_{\text{workers}} + WTP_{\text{owners}}}{G - F}, \qquad WTP_{\text{workers}} = WS, \qquad WTP_{\text{owners}} = G - C_K, \tag{23}$$

where F is the fiscal externality. Inframarginal owners-those who would have invested regardless of the subsidy-value the grant at face value (G), whereas marginal owners value it less by the capital distortion (C_K). Absent capital distortions, the six-year MVPF ranges between 1.13 and 1.15 under the linear benchmark; incorporating the calibrated distortion reduces it to approximately 0.9 (Table 5, Panel B).

The MVPF measures the social benefit generated per euro of net public outlay rather than a net welfare change, eliminating the need to take an *ex ante* stand on the marginal cost of public funds (λ). The policy improves social welfare whenever its MVPF exceeds the marginal cost of raising revenue, which equals $1 + \lambda$ for a generic tax (Appendix F.4). Whether an MVPF of 0.9 represents a desirable use of public resources thus depends on two factors: the efficiency of the financing tax and the social welfare weight assigned to firm owners relative to rationed workers.

[35] A uniform λ treats every euro as raised by a generic tax. If the grants are funded by distortionary payroll taxes, part of the excess burden falls on the labor market itself, and it may differ from 25 cents when wage floors bind.

Who gains. Following Hendren and Sprung-Keyser (2020), we decompose willingness to pay across beneficiary groups so that readers can apply their own distributional weights (Table 5, Panel B). Over a six-year horizon, workers' willingness to pay is €10,900–12,200 per direct job under the linear benchmark, rising to at least €21,900–24,300 under efficient rationing. Owners' aggregate willingness to pay is €176,000 without capital distortions and €138,300 with them. Consequently, workers capture only 6 to 8 percent of total measured benefits under the linear benchmark (and 11 to 15 percent under efficient rationing), with the remainder accruing to capital owners. Expressed per euro of net public expenditure, workers receive roughly 7 cents in the linear case and 13 to 15 cents under efficient rationing.

The decomposition shows a tension: while worker surplus and fiscal externalities raise the MVPF, the vast majority of the financial transfer flows to firm owners. Because capital owners capture 85 to 94 percent of measured willingness to pay, any distributional welfare function that downweights owners' rents relative to the surplus of rationed workers will yield an effective program value well below 0.9.

6.4 Discussion and Comparison with Other Policies

Empirical evaluations of business incentives routinely assess policy effectiveness using fiscal cost per job. Existing estimates span a wide range. In the United Kingdom, Criscuolo et al. (2019) estimate costs for the Regional Selective Assistance program from $3,500 per job in area-level instrumental-variable specifications to $26,600 among small firms. In Germany, GRW investment subsidies generate employment at roughly $19,000 per job (Siegloch et al., 2024). In the United States, estimates average $8,500 for the California Competes Tax Credit (Hyman et al., 2022) and $51,000 for the New Markets Tax Credit (Freedman, 2012). Japanese accelerated depreciation schemes entail a fiscal cost of approximately $16,000 per job (LaPoint and Sakabe, 2022), while discretionary state incentive packages in the US average $120,000 per promised job (Slattery and Zidar, 2020). Yet while cost per job provides a practical metric of fiscal cost-effectiveness, it is silent on economic efficiency: it measures what the government spent to induce employment, not what that employment is worth to society.

The social value of a job depends on workers' opportunity cost of time. In an unconstrained, market-clearing equilibrium, the marginal worker is indifferent between employment and non-employment; under the envelope theorem, additional hiring yields zero first-order surplus to labor. When centralized wage floors bind, however, access to employment is rationed. Expanding labor demand in a slack market draws workers directly from an involuntary queue,

generating a first-order welfare gain equal to the wedge between the contractual wage and their reservation wage. Consequently, two policies with identical fiscal costs per job can generate vastly different welfare consequences depending on whether newly created jobs absorb rationed workers.

While traditional cost-benefit analyses acknowledge the opportunity cost of labor, they typically rely on calibrated or assumed parameters (Bartik, 2012). We give empirical content to this margin by combining institutional wage floors with direct survey evidence on local reservation wages. Our findings demonstrate that in institutional labor markets, flat wage responses do not signal an absence of worker gains. Because compensation is anchored to national agreements, the benefits of local demand stimulus manifest not through prices, but through quantities: expanded job access, greater employment continuity, and more weeks worked. This quantity-driven surplus mechanism is not unique to Italy; recent evidence from Germany similarly shows that regional investment subsidies expand employment out of unemployment without bidding up wages (Siegloch et al., 2024).

Accounting for worker surplus thus uncovers a primary source of economic value that conventional cost-per-job evaluations leave unmeasured. Our recruitment and mobility findings indicate that these gains accrue predominantly to local residents, with little evidence of offsetting spatial displacement. Ultimately, whether these worker gains translate into net social welfare improvements depends on the economic costs required to deliver them: the excess burden of taxation raised to finance the subsidies, and the factor-market misallocation induced by subsidizing equipment. The welfare case for place-based capital subsidies hinges on whether the surplus generated by relaxing employment rationing outweighs these fiscal and investment distortions.

7 Conclusion

Place-based policies rest on the premise that labor in distressed regions is underutilized. Whether capital subsidies can mobilize that labor-and what doing so is worth-depends on how wages are set.

In a large European place-based program, quantities adjust and wages do not. Subsidized firms expanded employment by 13 percent with virtually no change in average pay. Incumbent workers captured 7 percent higher annual earnings, driven almost entirely by additional weeks of work and job continuity rather than higher wage rates. Non-applicant firms in the same local labor markets expanded alongside them without wage growth or a shift toward

hiring workers born elsewhere, and funded firms did not increase hiring from other provinces. Weekly wages rose only among workers paid further above national contractual floors, and by about two log points.

These findings are hard to reconcile with competitive clearing, where expanding labor demand bids up wages and draws workers from neighboring firms. Instead, compensation is anchored to centralized collective agreements. While these agreements are designed to protect workers' earnings and guard against downward wage competition, their spatial uniformity means nominal floors do not vary with local economic conditions. Taking these protections as given, the jobs the program created deliver a first-order surplus to the workers who fill them: most job seekers in the targeted areas would accept less than the jobs pay, a gain that an evaluation assuming an undistorted, clearing market sets to zero. Measured against survey reservation wages, this private surplus is 23.5 percent of the net pay of new hires under the most conservative, linear calculation, or about €9,500 per direct job over six years; it is about a quarter if jobs go to job seekers at random, and at least 47 percent if they go first to those who value them most (Section 6). Adding the jobs created at other firms and the fiscal gain, these gains cover 56 to 62 percent of the cost of financing the grants over six years under the linear benchmark, and at least 81 to 90 percent under efficient rationing, before accounting for the capital the grants distorted. Counted as a marginal value of public funds, which includes owners' valuation of the grant, most of the value the program delivers accrues to the owners of subsidized firms: workers receive 6 to 9 percent of it under the linear and random benchmarks.

Beyond this specific context, our findings point to an important second-best complementarity between national labor regulations and spatial policy, offering two broad lessons.

First, where institutional wage floors bind, local wage growth is a flawed criterion for evaluating place-based policies. Substantial gains can accrue to workers through expanded employment and job continuity even when wage rates remain completely flat; an evaluation focusing solely on wage growth will systematically miss these gains.

Second, national labor regulations and spatial policies cannot be analyzed in isolation. Centralized labor institutions-such as statutory minimum wages and sector-wide agreements-are designed to protect worker earnings. Yet a spatially homogeneous rule generates spatially heterogeneous distortions: uniform wage floors can hold pay above what lower-productivity regions would otherwise sustain, and our evidence is consistent with this leaving pools of underutilized workers. Where such floors bind, targeted demand subsidies function as a second-best policy response: national agreements protect earnings, while place-based subsidies mobilize employment at that floor. Investigating how centralized labor market institutions

shape local economic vulnerabilities-and how spatial instruments should be targeted to address the distortions they create-is a natural agenda for future research.

References

Abdulkadiroğlu, Atila, Joshua D. Angrist, Yusuke Narita, and Parag A. Pathak, "Research Design Meets Market Design: Using Centralized Assignment for Impact Evaluation," *Econometrica*, 2017, *85* (5), 1373–1432.

_ , _ , _ , and _ , "Breaking Ties: Regression Discontinuity Design Meets Market Design," *Econometrica*, 2022, *90* (1), 117–151.

Addario, Sabrina Di, Patrick Kline, Raffaele Saggio, and Mikkel Sølvsten, "It Ain't Where You're From, It's Where You're At: Hiring Origins, Firm Heterogeneity, and Wages," *Journal of Econometrics*, 2023, *233* (2), 340–374.

Ahluwalia, Montek S., "Taxes, Subsidies, and Employment," *Quarterly Journal of Economics*, 1973, *87* (3), 393–409.

Anderson, T. W. and Herman Rubin, "Estimation of the Parameters of a Single Equation in a Complete System of Stochastic Equations," *Annals of Mathematical Statistics*, 1949, *20* (1), 46–63.

Arnold, David, "Mergers and Acquisitions, Local Labor Market Concentration, and Worker Outcomes," *SSRN Electronic Journal*, 2019.

Atalay, Enghin, Ali Hortaçsu, Mustafa Runyun, Chad Syverson, and Mehmet Fatih Ulu, "Micro- and Macroeconomic Impacts of a Place-Based Industrial Policy," June 2023.

Austin, Benjamin, Edward Glaeser, and Lawrence Summers, "Jobs for the Heartland: Place-Based Policies in 21st-Century America," *Brookings Papers on Economic Activity*, 2018, pp. 151–232.

Autor, David, David Dorn, Gordon H. Hanson, Maggie R. Jones, and Bradley Setzler, "Places versus People: The Ins and Outs of Labor Market Adjustment to Globalization," in "Handbook of Labor Economics," Vol. 6, Elsevier, 2025, pp. 549–653.

Autor, David H, David Dorn, and Gordon H Hanson, "The China Syndrome: Local Labor Market Effects of Import Competition in the United States," *American Economic Review*, October 2013, *103* (6), 2121–2168.

Bartik, Timothy J., "Including Jobs in Benefit-Cost Analysis," *Annual Review of Resource Economics*, 2012, *4*, 55–73.

_ , "Using Place-Based Jobs Policies to Help Distressed Communities," *Journal of Economic Perspectives*, August 2020, *34* (3), 99–127.

Berger, David, Kyle Herkenhoff, and Simon Mongey, "Labor Market Power," *American Economic Review*, April 2022, *112* (4), 1147–1193.

_ , _ , and _ , "Minimum Wages, Efficiency, and Welfare," *Econometrica*, 2025, *93*, 265–301.

Berkowitz, Peter R., Michael Storper, and Max Herbertson, "Place-Based Policies of the European Union: Contrasts and Similarities to the US Experience," February 2025.

Bertrand, Marianne and Sendhil Mullainathan, "Is there Discretion in Wage Setting? A Test Using Takeover Legislation," *The RAND Journal of Economics*, 1999, *30.*

Bhagwati, Jagdish and V. K. Ramaswami, "Domestic Distortions, Tariffs and the Theory of Optimum Subsidy," *Journal of Political Economy*, 1963, *71* (1), 44–50.

Bhuller, Manudeep, Lucas Delgado-Prieto, Santiago Hermo, and Linnea Lorentzen, "Wage-Setting Constraints and Firm Responses to Demand Shocks," 2025. arXiv:2512.19622.

Bilal, Adrien, "The Geography of Unemployment," *The Quarterly Journal of Economics*, August 2023, *138* (3), 1507–1576.

Blanchard, Olivier Jean and Lawrence F. Katz, "Regional Evolutions," *Brookings Papers on Economic Activity*, 1992, *1992* (1), 1.

Boeri, Tito, Andrea Ichino, Enrico Moretti, and Johanna Posch, "Wage Equalization and Regional Misallocation: Evidence from Italian and German Provinces," *Journal of the European Economic Association*, December 2021, *19* (6), 3249–3292.

Borusyak, Kirill and Matan Kolerman-Shemer, "Regression Discontinuity Aggregation, with an Application to the Union Effects on Inequality," arXiv:2501.00428 2024. Working paper.

_ **and Peter Hull**, "Nonrandom Exposure to Exogenous Shocks," *Econometrica*, 2023, *91* (6), 2155–2185. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.3982/ECTA19367.

Bovini, Giulia, Raffaella Nizzi, and Gaetano Basso, "La contrattazione collettiva in Italia negli ultimi venticinque anni: evidenze dall'archivio della Banca d'Italia sui CCNL maggiormente rappresentativi," Questioni di Economia e Finanza (Occasional Papers) 1070, Banca d'Italia 2026.

Breza, Emily, Supreet Kaur, and Yogita Shamdasani, "Labor Rationing," *American Economic Review*, October 2021, *111* (10), 3184–3224.

Bronzini, Raffaello and Guido De Blasio, "Evaluating the impact of investment incentives: The case of Italy's Law 488/1992," *Journal of Urban Economics*, September 2006, *60* (2), 327–349.

Burdett, Ken and Melvyn Coles, "Equilibrium Wage-Tenure Contracts," *Econometrica*, 2003, *71* (5), 1377–1404.

Busso, Matias, Jesse Gregory, and Patrick Kline, "Assessing the Incidence and Efficiency of a Prominent Place Based Policy," *American Economic Review*, April 2013, *103* (2), 897–947.

Calonico, Sebastian, Matias D. Cattaneo, and Rocio Titiunik, "Robust Nonparametric Confidence Intervals for Regression-Discontinuity Designs," *Econometrica*, 2014, *82* (6), 2295–2326.

Caponi, Vincenzo, "Centralized Wage Determination and Regional Unemployment Differences: The Case of Italy," Discussion Paper 3592, IZA 2008.

Card, D., F. Devicienti, and A. Maida, "Rent-sharing, Holdup, and Wages: Evidence from Matched Panel Data," *The Review of Economic Studies*, January 2014, *81* (1), 84–111.

Card, David, Ana Rute Cardoso, Joerg Heining, and Patrick Kline, "Firms and Labor Market Inequality: Evidence and Some Theory," *Journal of Labor Economics*, January 2018, *36* (S1), S13–S70.

Cattaneo, Matias D., Brigham R. Frandsen, and Rocío Titiunik, "Randomization Inference in the Regression Discontinuity Design: An Application to Party Advantages in the U.S. Senate," *Journal of Causal Inference*, 2015, *3* (1), 1–24.

_ , Michael Jansson, and Xinwei Ma, "Simple Local Polynomial Density Estimators," *Journal of the American Statistical Association*, 2020, *115* (531), 1449–1455.

_ , Rocío Titiunik, and Gonzalo Vazquez-Bare, "Comparing Inference Approaches for RD Designs: A Reexamination of the Effect of Head Start on Child Mortality," *Journal of Policy Analysis and Management*, 2017, *36* (3), 643–681.

Cerqua, Augusto and Guido Pellegrini, "Do subsidies to private capital boost firms' growth? A multiple regression discontinuity design approach," *Journal of Public Economics*, January 2014, *109*, 114–126.

Ciani, Emanuele, Francesco David, and Guido de Blasio, "Local Responses to Labor Demand Shocks: A Re-Assessment of the Case of Italy," *Regional Science and Urban Economics*, 2019, *75*, 1–21.

Cingano, Federico, Filippo Palomba, Paolo Pinotti, and Enrico Rettore, "Making Subsidies Work: Rules versus Discretion," *Econometrica*, 2025, *93* (3), 747–778.

Criscuolo, Chiara, Ralf Martin, Henry G. Overman, and John Van Reenen, "Some Causal Effects of an Industrial Policy," *American Economic Review*, January 2019, *109* (1), 48–85.

Daruich, Diego, Martino Kuntze, Pascuel Plotkin, and Raffaele Saggio, "The Consequences of Domestic Outsourcing on Workers: New Evidence from Italian Administrative Data," Working Paper 2024. August 2024.

Dustmann, Christian, Chiara Giannetto, Lorenzo Incoronato, Chiara Lacava, Vincenzo Pezone, Raffaele Saggio, and Benjamin Schoefer, "Opting Out of Centralized Collective Bargaining: Evidence from Italy," July 2025.

Fanfani, Bernardo, "The Employment Effects of Collective Wage Bargaining," *Journal of Public Economics*, 2023, *227*, 105006.

Feldstein, Martin and James Poterba, "Unemployment Insurance and Reservation Wages," *Journal of Public Economics*, 1984, *23*, 141–167.

Feyrer, James, Erin T. Mansur, and Bruce Sacerdote, "Geographic Dispersion of Economic Shocks: Evidence from the Fracking Revolution," *The American Economic Review*, 2017, *107* (4), 1313–1334.

Fort, Margherita, Andrea Ichino, Enrico Rettore, and Giulio Zanella, "Multi-Cutoff Rd Designs with Observations Located at Each Cutoff: Problems and Solutions," January 2022.

Freedman, Matthew, "Teaching New Markets Old Tricks: The Effects of Subsidized Investment on Low-Income Neighborhoods," *Journal of Public Economics*, 2012, *96* (11–12), 1000–1014.

_ **, Shantanu Khanna, and David Neumark**, "Combining rules and discretion in economic development policy: Evidence on the impacts of the California Competes Tax Credit," *Journal of Public Economics*, January 2023, *217*, 104777.

Friedrich, Benjamin, Lisa Laun, Costas Meghir, and Luigi Pistaferri, "Earnings Dynamics and Firm-Level Shocks," NBER Working Paper 25786, National Bureau of Economic Research 2024. Revised February 2024.

Garin, Andrew and Filipe Silvério, "How Responsive Are Wages to Firm-Specific Changes in Labour Demand? Evidence from Idiosyncratic Export Demand Shocks," *The Review of Economic Studies*, May 2024, *91* (3), 1671–1710.

Gaubert, Cecile, "Firm Sorting and Agglomeration," *American Economic Review*, 2018, *108* (11), 3117–3153.

_ , Patrick Kline, Damian Vergara, and Danny Yagan, "Place-Based Redistribution," *American Economic Review*, 2025, *115* (10), 3415–3450.

Gelman, Andrew and Guido Imbens, "Why High-Order Polynomials Should Not Be Used in Regression Discontinuity Designs," *Journal of Business & Economic Statistics*, July 2019, *37* (3), 447–456.

Gibbons, Robert and Kevin J. Murphy, "Optimal Incentive Contracts in the Presence of Career Concerns: Theory and Evidence," *Journal of Political Economy*, 1992, *100* (3), 468–505.

Glaeser, Edward L. and Joshua D. Gottlieb, "The Economics of Place-Making Policies," *Brookings Papers on Economic Activity*, 2008, *2008*, 155–239.

Greenstone, Michael, Richard Hornbeck, and Enrico Moretti, "Identifying Agglomeration Spillovers: Evidence from Winners and Losers of Large Plant Openings," *Journal of Political Economy*, June 2010, *118* (3), 536–598.

Grunau, Philipp, Florian Hoffmann, Thomas Lemieux, and Mirko Titze, "Who Benefits from Place-Based Policies? Evidence from Matched Employer-Employee Data," NBER Working Paper 33785, National Bureau of Economic Research 2025.

Guiso, Luigi, Luigi Pistaferri, and Fabiano Schivardi, "Insurance within the Firm," *Journal of Political Economy*, 2005, *113* (5), 1054–1087.

Hahn, Jinyong, Petra Todd, and Wilbert Van der Klaauw, "Identification and Estimation of Treatment Effects with a Regression-Discontinuity Design," *Econometrica*, 2001, *69* (1), 201–209.

Hanson, Gordon, Dani Rodrik, and Rohan Sandhu, "The US Place-Based Policy Supply Chain," in "Economics of Place-Based Policies," University of Chicago Press, November 2024.

Haveman, Robert H. and John V. Krutilla, "Unemployment, Excess Capacity, and Benefit-Cost Investment Criteria," *Review of Economics and Statistics*, 1967, *49* (3), 382–392.

Hendren, Nathaniel and Ben Sprung-Keyser, "A Unified Welfare Analysis of Government Policies," *Quarterly Journal of Economics*, 2020, *135* (3), 1209–1318.

Howell, Sabrina T., "Financing Innovation: Evidence from R&D Grants," *American Economic Review*, April 2017, *107* (4), 1136–1164.

Howell, Sabrina T and J David Brown, "Do Cash Windfalls Affect Wages? Evidence from R&D Grants to Small Firms," *The Review of Financial Studies*, May 2023, *36* (5), 1889–1929.

Hyman, Benjamin, Matthew Freedman, Shantanu Khanna, and David Neumark, "Firm Responses to State Hiring Subsidies: Regression Discontinuity Evidence from a Tax Credit Formula," NBER Working Paper 30664, National Bureau of Economic Research 2022.

Imbens, Guido W. and Paul R. Rosenbaum, "Robust, Accurate Confidence Intervals with a Weak Instrument: Quarter of Birth and Education," *Journal of the Royal Statistical Society: Series A*, 2005, *168* (1), 109–126.

Jones, Stephen R. G., "Reservation Wages and the Cost of Unemployment," *Economica*, 1989, *56*, 225–246.

Jovanovic, Boyan, "Job Matching and the Theory of Turnover," *Journal of Political Economy*, 1979, *87* (5), 972–990.

Jäger, Simon, Jörg Heining, and Nathan Lazarus, "How Substitutable Are Workers? Evidence from Worker Deaths," June 2024.

Kline, Patrick and Enrico Moretti, "Place Based Policies with Unemployment," *NBER Working Paper 18758*, 2013.

_ **and** _ , "Local Economic Development, Agglomeration Economies, and the Big Push: 100 Years of Evidence from the Tennessee Valley Authority," *The Quarterly Journal of Economics*, February 2014, *129* (1), 275–331.

_ **and** _ , "People, Places, and Public Policy: Some Simple Welfare Economics of Local Economic Development Programs," *Annual Review of Economics*, 2014, *6* (1), 629–662. _eprint: https://doi.org/10.1146/annurev-economics-080213-041024.

_ **, Neviana Petkova, Heidi Williams, and Owen Zidar**, "Who Profits from Patents? Rent-Sharing at Innovative Firms*," *The Quarterly Journal of Economics*, August 2019, *134* (3), 1343–1404.

Krueger, Alan B. and Andreas Mueller, "Job Search and Job Finding in a Period of Mass Unemployment: Evidence from High-Frequency Longitudinal Data," Working Paper 562, Industrial Relations Section, Princeton University 2011.

Lachowska, Marta, Alexandre Mas, Raffaele Saggio, and Stephen A. Woodbury, "Work Hours Mismatch," *Econometrica*, 2026, *94* (3), 991–1025.

Lamadon, Thibaut, Magne Mogstad, and Bradley Setzler, "Imperfect Competition, Compensating Differentials, and Rent Sharing in the US Labor Market," *American Economic Review*, January 2022, *112* (1), 169–212.

LaPoint, Cameron and Shogo Sakabe, "Place-Based Policies and the Geography of Corporate Investment," 2022. Working paper.

Lee, David and Emmanuel Saez, "Optimal Minimum Wage Policy in Competitive Labor Markets," *Journal of Public Economics*, 2012, *96* (9-10), 739–749.

Lee, David S., Pauline Leung, Christopher J. O'Leary, Zhuan Pei, and Simon Quach, "Are Sufficient Statistics Necessary? Nonparametric Measurement of Deadweight Loss from Unemployment Insurance," NBER Working Paper 25574, National Bureau of Economic Research 2019.

Lefeber, Louis, "Planning in a Surplus Labor Economy," *American Economic Review*, 1968, *58* (3), 343–373.

Lewis, W. Arthur, "Economic Development with Unlimited Supplies of Labour," *The Manchester School*, 1954, *22* (2), 139–191.

Lindenlaub, Ilse, Ryungha Oh, and Michael Peters, "Spatial Firm Sorting and Local Monopsony Power," *American Economic Journal: Macroeconomics*, 2025. Forthcoming.

Lipsey, Richard G and Kelvin Lancaster, "The general theory of second best," *The review of economic studies*, 1956, *24* (1), 11–32.

Loertscher, Simon and Ellen V. Muir, "Optimal Labor Procurement under Minimum Wages and Monopsony Power," 2025. Working paper.

Lombardi, Stefano, Antoine Bertheau, Raffaele Saggio, Oona Tuominen, Edoardo Maria Acabbi, Andreas Gulyas, Marco Guido Palladino, Tobias Renkin, and Salvatore Lattanzio, "The Consequences of Labor Demand Shocks Across Countries," Working Paper 2026. March 2026.

Luttmer, Erzo F. P., "Does the Minimum Wage Cause Inefficient Rationing?," *The B.E. Journal of Economic Analysis & Policy*, 2007, *7* (1), Article 49.

Maibom, Jonas and Rune Vejlin, "Passthrough of Firm Performance to Income and Employment Status," *Journal of Political Economy Microeconomics*, 2026.

_ **and** _ , "Passthrough of Firm Performance to Income and Employment Status," *Journal of Political Economy Microeconomics*, 2026.

Manning, Alan, "Monopsony in Labor Markets: A Review," *ILR Review*, 2021, *74* (1), 3–26.

_ **and Barbara Petrongolo**, "How Local Are Labor Markets? Evidence from a Spatial Job Search Model," *American Economic Review*, 2017, *107* (10), 2877–2907.

McCrary, Justin, "Manipulation of the Running Variable in the Regression Discontinuity Design: A Density Test," *Journal of Econometrics*, 2008, *142* (2), 698–714.

Michelacci, Claudio and Vincenzo Quadrini, "Financial Markets and Wages," *The Review of Economic Studies*, April 2009, *76* (2), 795–827.

Moretti, Enrico, "Local Multipliers," *American Economic Review*, May 2010, *100* (2), 373–377.

Nakamura, Emi and Jón Steinsson, "Fiscal Stimulus in a Monetary Union: Evidence from US Regions," *American Economic Review*, March 2014, *104* (3), 753–792.

Narita, Yusuke and Kohei Yata, "Algorithm as Experiment: Machine Learning, Market Design, and Policy Eligibility Rules," 2023. arXiv:2104.12909.

Neumark, David and Helen Simpson, "Place-Based Policies," in "Handbook of Regional and Urban Economics," Vol. 5, Elsevier, 2015, pp. 1197–1287.

Pennings, Steven, "Cross-Region Transfer Multipliers in a Monetary Union: Evidence from Social Security and Stimulus Payments," *American Economic Review*, May 2021, *111* (5), 1689–1719.

Postel-Vinay, Fabien and Jean-Marc Robin, "The Distribution of Earnings in an Equilibrium Search Model with State-Dependent Offers and Counteroffers," *International Economic Review*, November 2002, *43* (4), 989–1016.

Reenen, John Van, "The Creation and Capture of Rents: Wages and Innovation in a Panel of U. K. Companies*," *The Quarterly Journal of Economics*, February 1996, *111* (1), 195–226.

Roback, Jennifer, "Wages, Rents, and the Quality of Life," *Journal of Political Economy*, 1982, *90* (6), 1257–1278.

Rosen, Sherwin, "Wage-Based Indexes of Urban Quality of Life," in Peter Mieszkowski and Mahlon Straszheim, eds., *Current Issues in Urban Economics*, Baltimore: Johns Hopkins University Press, 1979, pp. 74–104.

Schoefer, Benjamin, "Eurosclerosis at 40: Labor Market Institutions, Dynamism, and European Competitiveness," in "ECB Forum on Central Banking, Sintra," European Central Bank, 2025.

Sen, Amartya K., *Choice of Techniques: An Aspect of the Theory of Planned Economic Development*, Oxford: Basil Blackwell, 1960.

Serrato, Juan Carlos Suárez and Philippe Wingender, "Estimating Local Fiscal Multipliers," July 2016.

Sestito, Paolo and Eliana Viviano, "Reservation Wages: Explaining Some Puzzling Regional Patterns," *LABOUR*, 2011, *25* (1), 63–88.

Shapiro, Carl and Joseph E. Stiglitz, "Equilibrium Unemployment as a Worker Discipline Device," *The American Economic Review*, 1984, *74* (3), 433–444.

Shoag, Daniel, "Using State Pension Shocks to Estimate Fiscal Multipliers since the Great Recession," *American Economic Review*, May 2013, *103* (3), 121–124.

Siegloch, Sebastian, Nils Wehrhöfer, and Tobias Etzel, "Spillover, Efficiency and Equity Effects of Regional Firm Subsidies," *American Economic Journal: Economic Policy*, 2024.

Slattery, Cailin and Owen Zidar, "Evaluating State and Local Business Incentives," *Journal of Economic Perspectives*, May 2020, *34* (2), 90–118.

Sokolova, Anna and Todd Sorensen, "Monopsony in Labor Markets: A Meta-Analysis," *ILR Review*, 2021, *74* (1), 27–55.

Sun, Liyang and Sarah Abraham, "Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects," *Journal of Econometrics*, 2021, *225* (2), 175–199.

Wilson, Daniel J., "Fiscal Spending Jobs Multipliers: Evidence from the 2009 American Recovery and Reinvestment Act," *American Economic Journal: Economic Policy*, April 2012, *4* (3), 251–282.

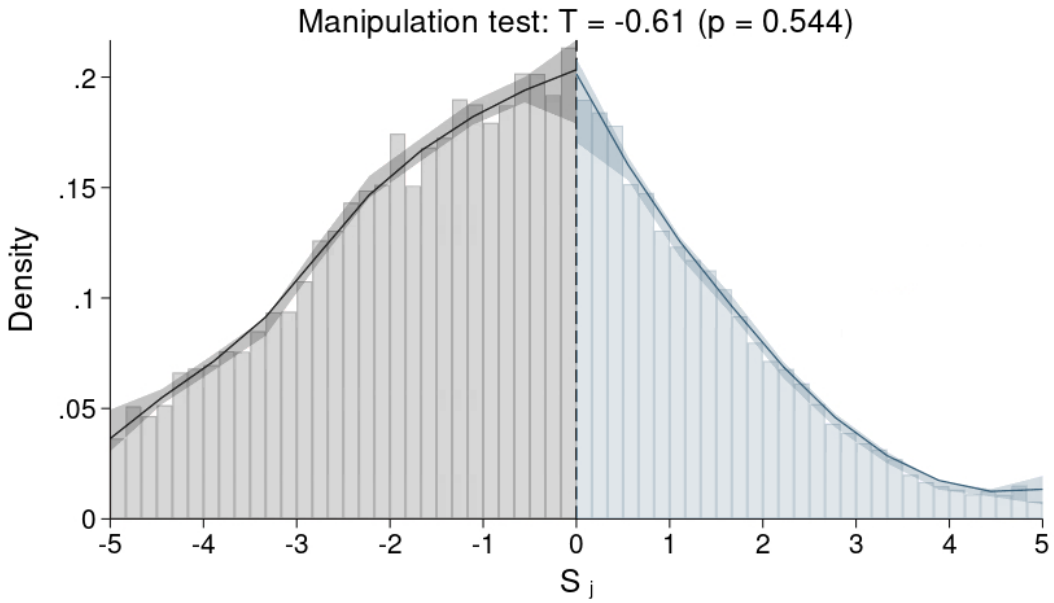
Appendix

A Additional Tables and Figures

This appendix collects the additional figures (Appendix A.1) and tables (Appendix A.2) referred to in the text and in Appendix B, in the order in which they are first discussed.

A.1 Figures

Figure A.1: Regression Discontinuity Validity: Density of Scores at the Cutoff



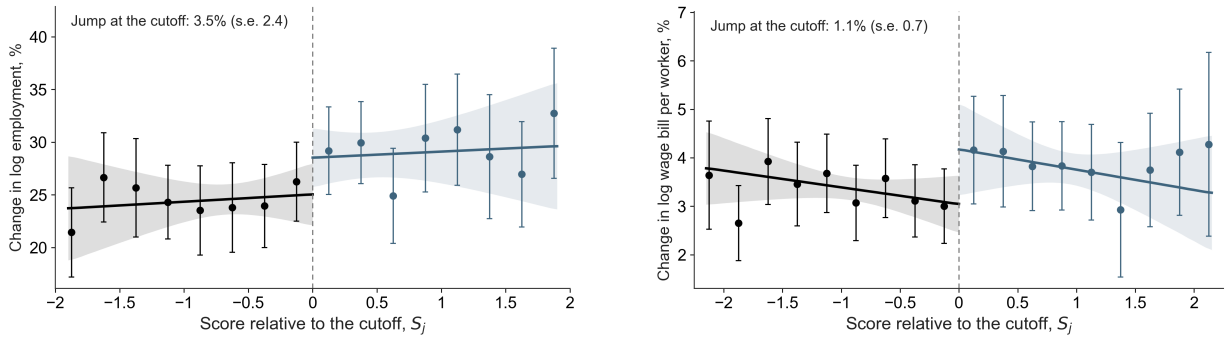
Notes: This figure plots the distribution of the project score relative to the funding cutoff of its pool, S_j, for all applicants. Bars are the histogram; lines are local-polynomial density estimates on each side of the cutoff, with 95% confidence bands. The test of a discontinuity in the density at the cutoff (McCrary, 2008; Cattaneo et al., 2020) gives $T = -0.61$ ($p = 0.544$): firms do not bunch just above the cutoff, consistent with their being unable to manipulate their scores precisely to cross it. [Back to Section 4.1]

Table A.1: Balance at the cutoff: score components and the firm before the application

	(1) Linear $\lvert S_j\rvert < 5$	(2) Quadratic $\lvert S_j\rvert < 5$	(3) Local linear MSE-optimal h
Score components (standardized)			
i_1: own capital / investment	−0.000	−0.027	−0.012
	(0.028)	(0.025)	(0.064)
i_2: promised jobs / investment	−0.074**	−0.002	−0.024
	(0.034)	(0.033)	(0.042)
i_3: aid requested, share of maximum	−0.032	−0.037	−0.006
	(0.026)	(0.030)	(0.051)
Regional priority	0.081	0.011	0.034
	(0.050)	(0.033)	(0.060)
Environmental impact	0.029	0.046*	0.049
	(0.027)	(0.026)	(0.035)
Before the application			
Log employment, t^*-1	0.021	0.063	0.070
	(0.048)	(0.053)	(0.056)
Log aid requested per promised job	0.045*	0.004	0.011
	(0.027)	(0.030)	(0.043)

Notes: Regression-discontinuity estimates of the jump at the funding cutoff in each component of the project score, in baseline log employment, and in the aid requested per promised job, for Southern applicants, under the three implementations of Table 1. Columns (1) and (2): a linear or quadratic polynomial in the score on each side of the cutoff over $|S_j| < 5$, with funding-pool and market (province × two-digit industry) fixed effects. Column (3): local linear estimate with a triangular kernel and the MSE-optimal bandwidth of Calonico et al. (2014) (between 1.55 and 1.97), on outcomes net of the same fixed effects, with the robust bias-corrected standard error. Standard errors clustered by funding pool. Score components are standardized within the funding pool, so their jumps are in standard deviations. Of the 21 estimates, three are significant at the 10% level and one at the 5% level, about what chance alone would produce, and no variable is significant in more than one column. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

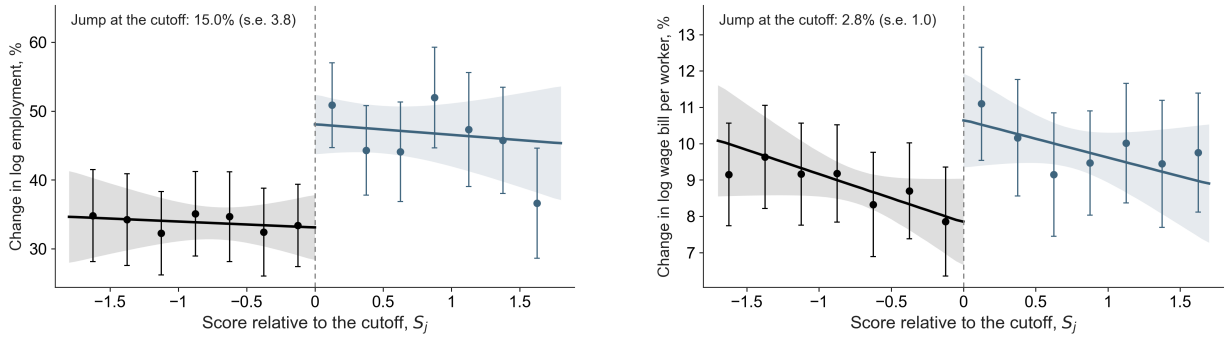
Figure A.2: Employment and wage effects emerge already at t^*+1.



(i) Employment (t^*+1) (ii) Wages (t^*+1)

Notes: RD plots of the change in log employment (left) and log wage bill per worker (right) one year after the application, constructed as in Figure 4. The robust discontinuity estimates are 0.035 (s.e. 0.024, bandwidth 1.89) for employment and 0.011 (s.e. 0.007, bandwidth 2.15) for the wage bill per worker. Effects are visible but smaller than at t^*+2 (Figure 4 in the main text), consistent with the phased nature of capital investment in new plant construction.

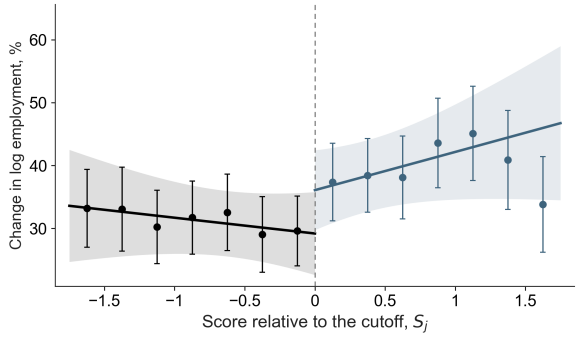
Figure A.3: Effects continue to grow through t^*+3 as investment matures.



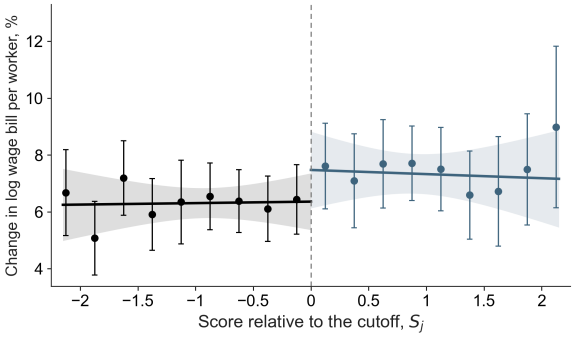
(i) Employment (t^*+3) (ii) Wages (t^*+3)

Notes: RD plots of the change in log employment (left) and log wage bill per worker (right) three years after the application, constructed as in Figure 4. The employment discontinuity has grown to 0.150 log points (s.e. 0.038, bandwidth 1.80), roughly double the t^*+2 estimate, consistent with effects that build over time as capital-intensive projects reach full capacity; the wage bill per worker rises by 0.028 (s.e. 0.010, bandwidth 1.71).

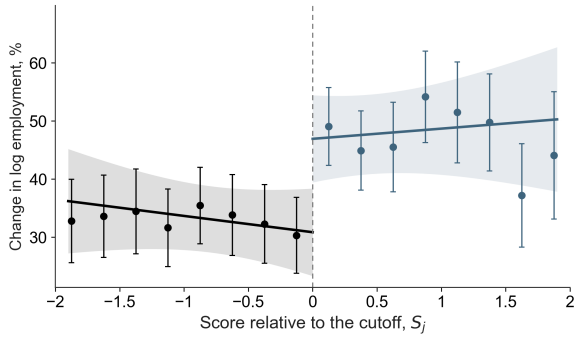
Figure A.4: Employment and pay per worker at the cutoff: raw data



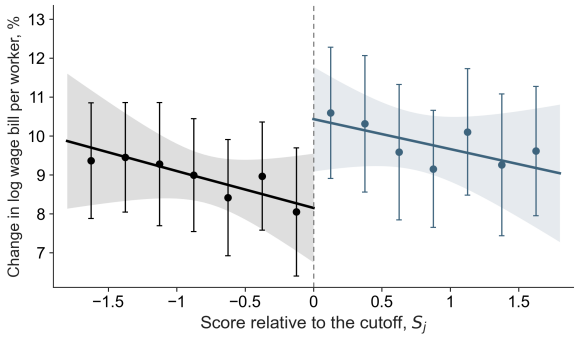
(i) Employment (t^*+2)



(ii) Wage bill per worker (t^*+2)



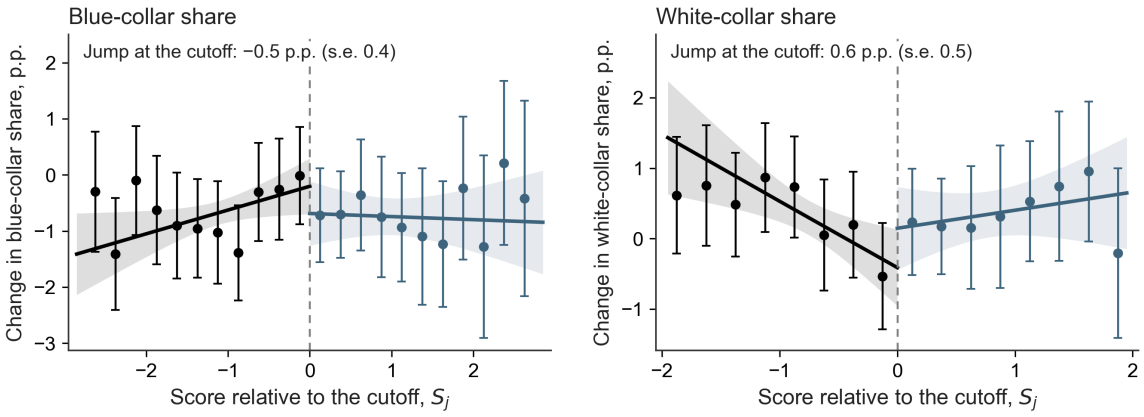
(iii) Employment (t^*+3)



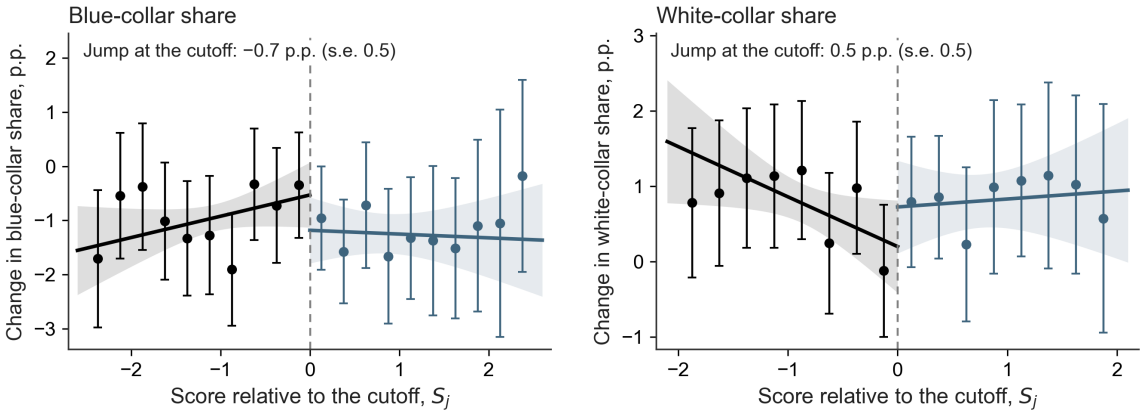
(iv) Wage bill per worker (t^*+3)

Notes: RD plots of the change in log employment and in log wage bill per worker between t^*-1 and two (top) and three (bottom) years after the application, in percent, against the score relative to the funding cutoff, S_j, for Southern applicants. Unlike Figure 4, the outcomes are not net of funding-pool or market fixed effects: dots are raw averages within bins of 0.25 inside the bandwidth on each side of the cutoff, with 95% confidence intervals, in black for applicants below the cutoff and in blue for those above it. Lines are local linear fits on each side, with triangular weights within the MSE-optimal bandwidth of Calonico et al. (2014), and shaded areas are their 95% confidence bands, clustered by funding pool; the dashed line is the cutoff.

Figure A.5: Subsidized firms do not upgrade their occupational mix.



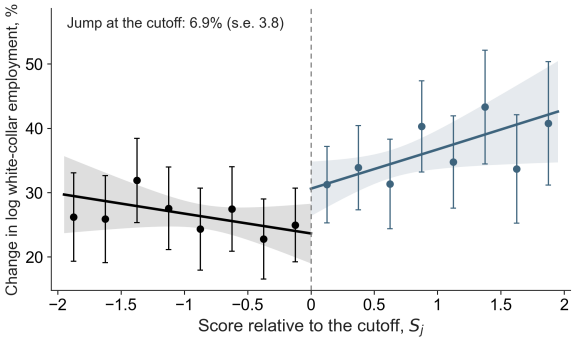
(i) Skill composition (t^*+2)



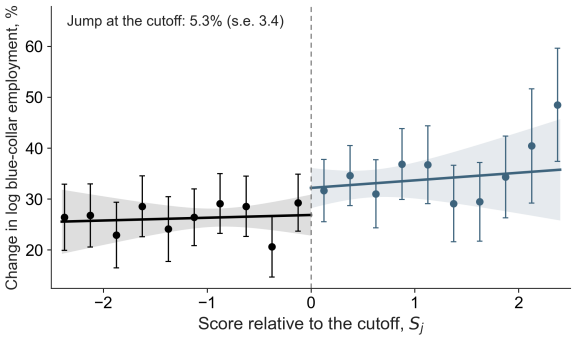
(ii) Skill composition (t^*+3)

Notes: RD plots of the change in the blue-collar and white-collar shares of employment between t^*-1 and t^*+2 (panel i) and t^*+3 (panel ii), in percentage points, constructed as in Figure 4: outcomes net of funding-pool and market fixed effects, bins of 0.25 inside the bandwidth with 95% confidence intervals, local linear fits with triangular weights, and robust discontinuity estimates (Calonico et al., 2014), clustered by funding pool. The jumps are −0.5 (s.e. 0.4) and 0.6 (0.5) percentage points at t^*+2, and −0.7 (0.5) and 0.5 (0.5) at t^*+3, for the blue-collar and white-collar shares respectively; none is significant. The occupational mix does not shift, consistent with an expansion at scale rather than selective recruitment of higher-skilled workers. Because the white-collar share does not move, a shift across occupations is unlikely to account for the wage gains of stayers documented in the main text, although this does not rule out selection within occupations.

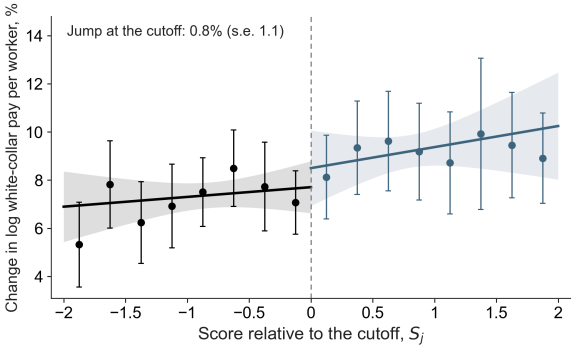
Figure A.6: Employment rises for both skill groups; pay per worker moves for neither.



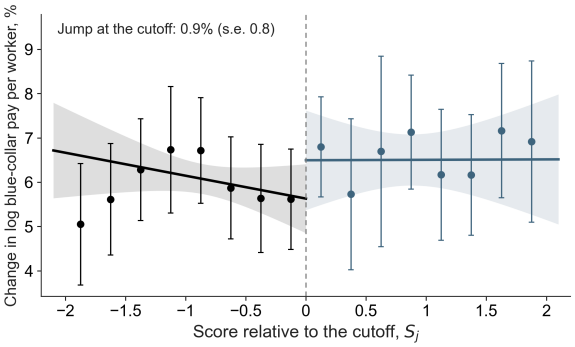
(i) White-collar employment



(ii) Blue-collar employment



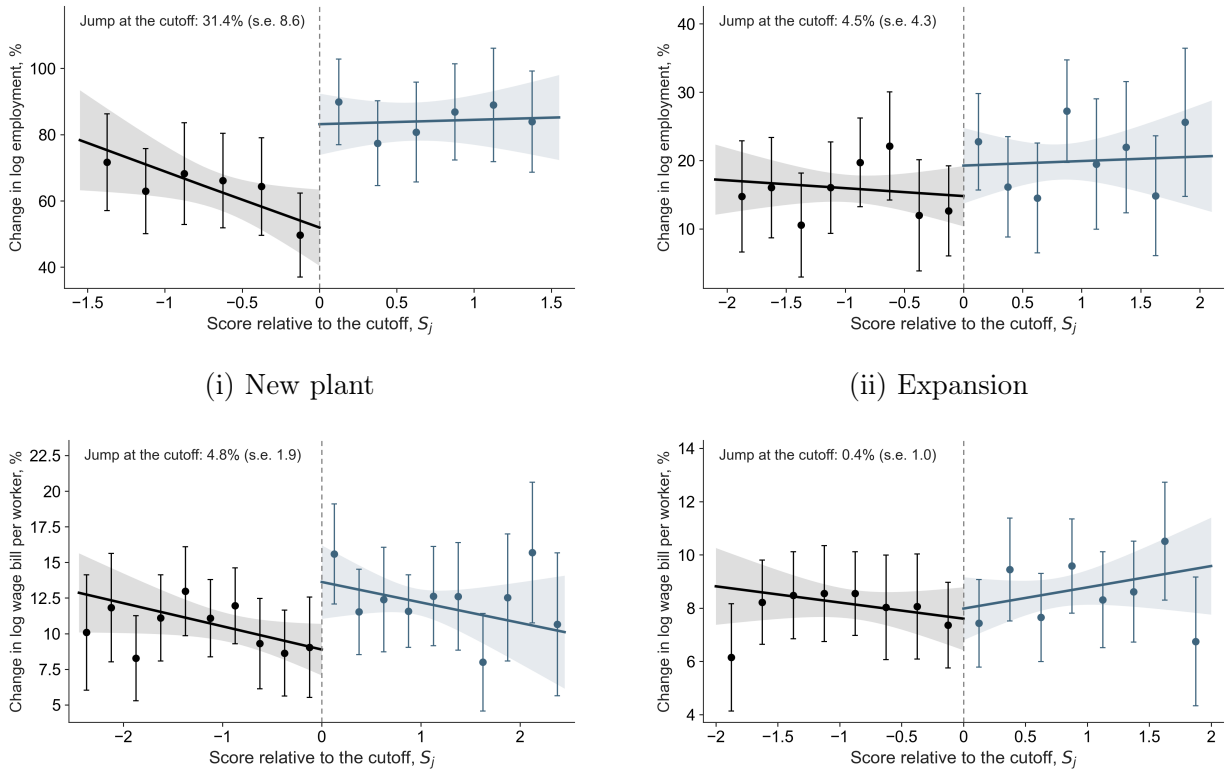
(iii) White-collar wages



(iv) Blue-collar wages

Notes: RD plots of the change between t^*-1 and t^*+2 by skill group: log employment of white-collar (a) and blue-collar (b) workers, and log pay per worker of white-collar (c) and blue-collar (d) workers, constructed as in Figure 4: outcomes net of funding-pool and market fixed effects, bins of 0.25 inside the bandwidth with 95% confidence intervals, local linear fits with triangular weights, and robust discontinuity estimates (Calonico et al., 2014), clustered by funding pool. Employment jumps by 6.9% (s.e. 3.8) for white-collar and 5.3% (3.4) for blue-collar workers; pay per worker by 0.8% (1.1) and 0.9% (0.8), neither significant.

Figure A.7: Employment effects are driven by new-plant creation.

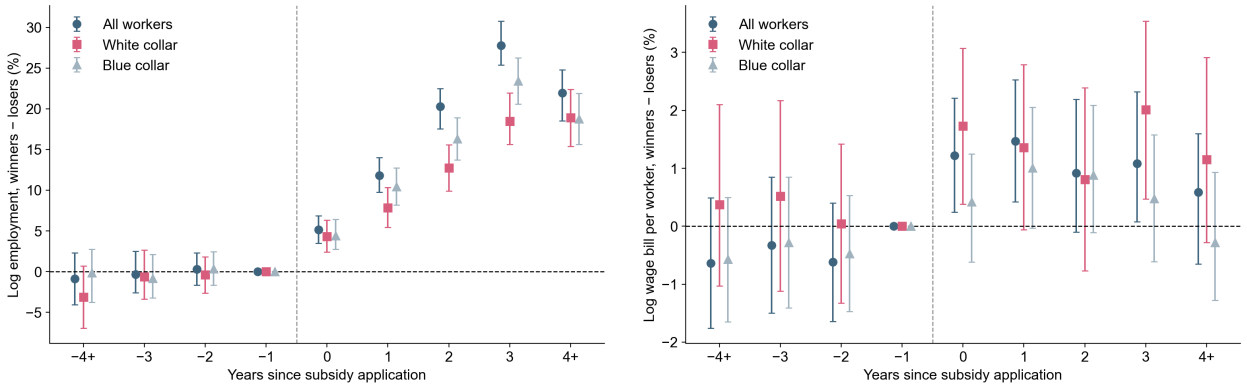


(i) New plant (ii) Expansion

(iii) New plant (iv) Expansion

Notes: RD plots of the change between t^*-1 and t^*+3 in log employment (top) and log wage bill per worker (bottom), estimated separately for new-plant projects (left) and plant-expansion projects (right), constructed as in Figure 4: outcomes net of funding-pool and market fixed effects, bins of 0.25 inside the bandwidth with 95% confidence intervals, local linear fits with triangular weights, and robust discontinuity estimates (Calonico et al., 2014), clustered by funding pool. Employment jumps by 31.4% (s.e. 8.6) at new plants and 4.5% (4.3) at expansions; the wage bill per worker by 4.8% (1.9) and 0.4% (1.0). The wage bill per worker of a plant built from scratch reflects whom it hires, so its jump measures the composition of the new workforce rather than raises for the same workers; the weekly wage of incumbents does not move (Section 4.2.2).

Figure A.8: RDD versus event study: winners against losers near the cutoff, without score controls

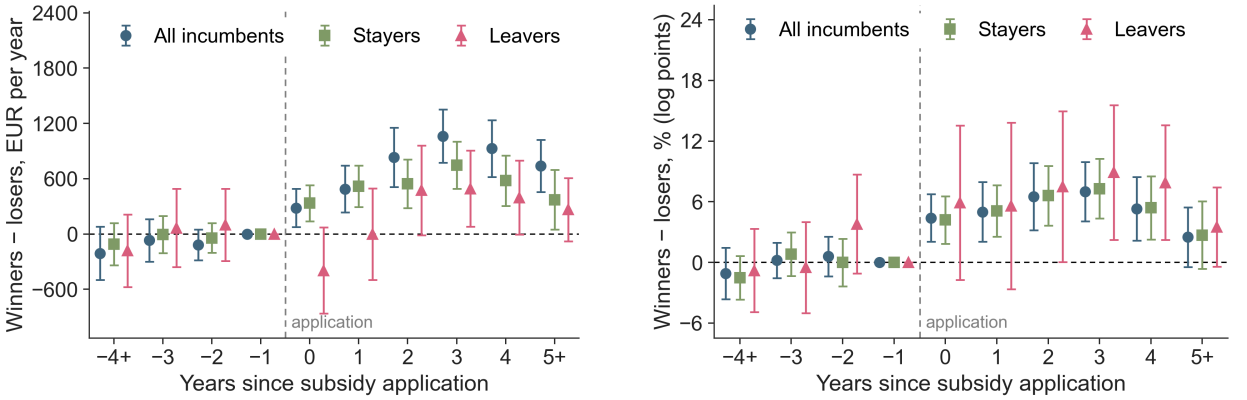


(i) Log employment

(ii) Log wage bill per worker

Notes: Event-study coefficients, winners minus losers, in log points ×100, with 95% confidence intervals; $t^* - 1$ is the reference year and the end points pool earlier and later years. Firms that applied more than once are assigned to their first winning application. Firm and calendar-year fixed effects and a market trend; no control for the score.

Figure A.9: Earnings by group: the gain is concentrated on workers who stay

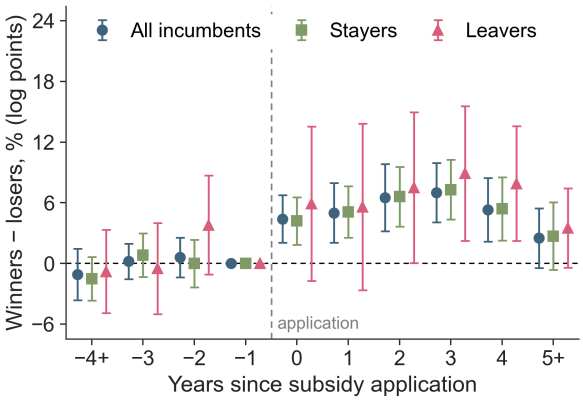


(i) Annual earnings

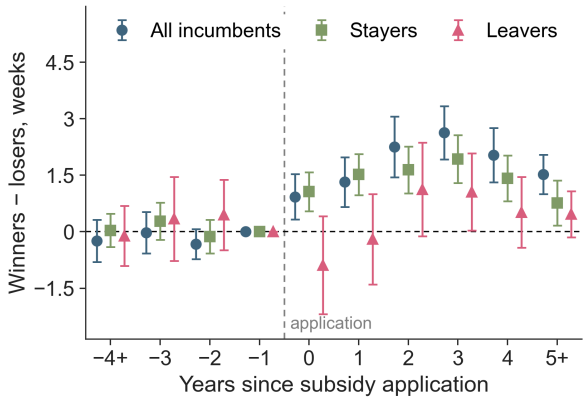
(ii) Log annual earnings

Notes: Event-study estimates of annual earnings for the application cohort, split by whether the worker stays with the applicant firm, leaves it, or re-enters after an interruption.

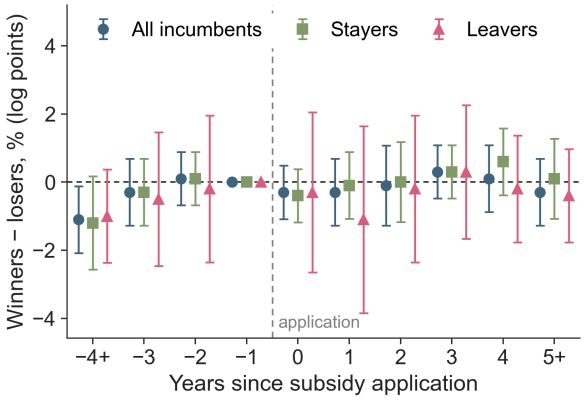
Figure A.10: The same decomposition by group, worker by worker



(i) Log earnings



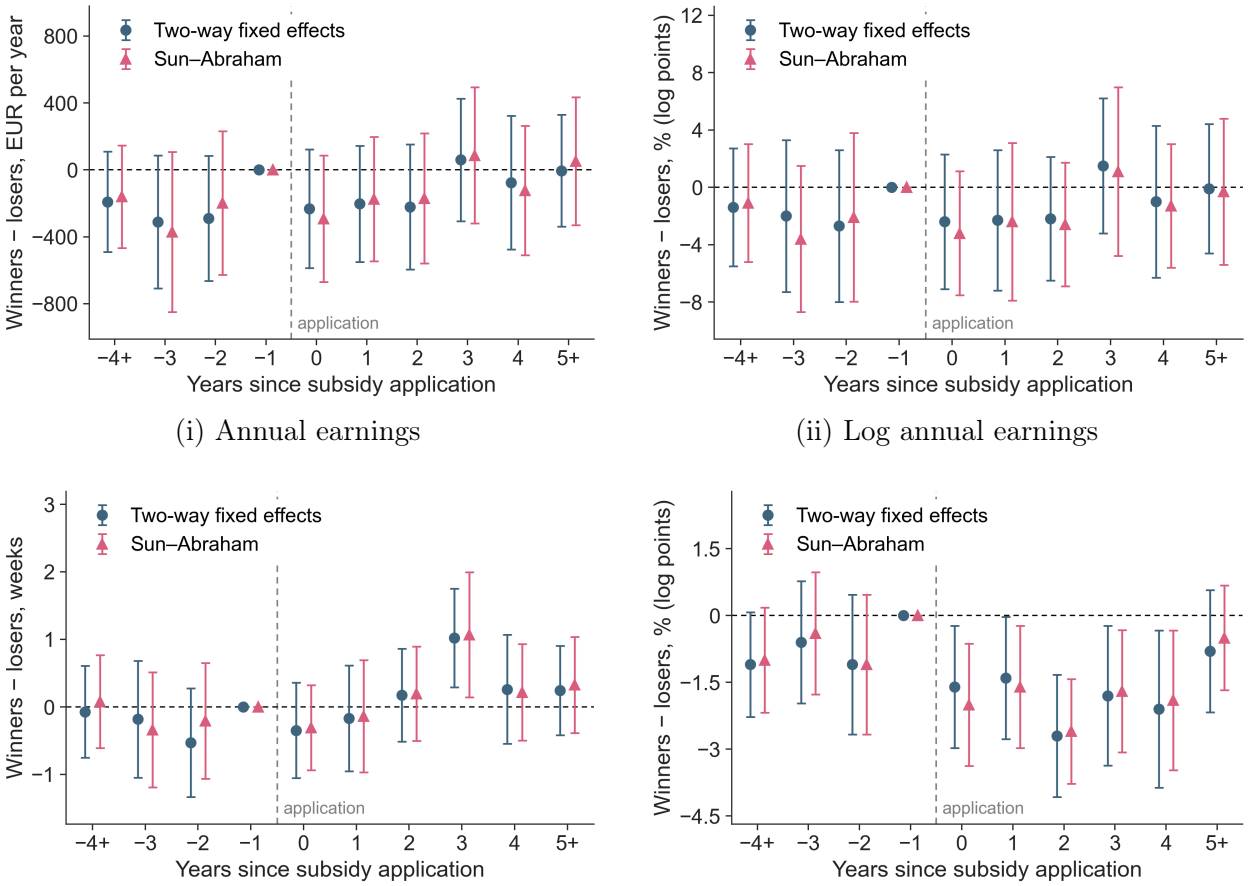
(ii) Weeks worked



(iii) Log weekly wage

Notes: Worker-level long differences from the baseline year, by group: earnings, weeks, and the weekly wage rate. The wage rate is flat in every group.

Figure A.11: New hires: earnings, weeks worked, and the entry wage



(i) Annual earnings

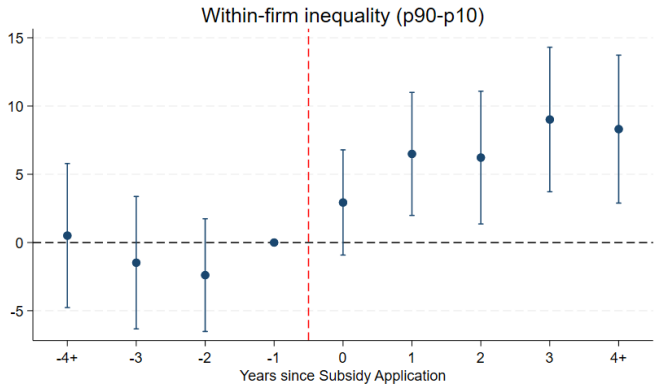
(ii) Log annual earnings

(iii) Weeks worked

(iv) Log entry weekly wage

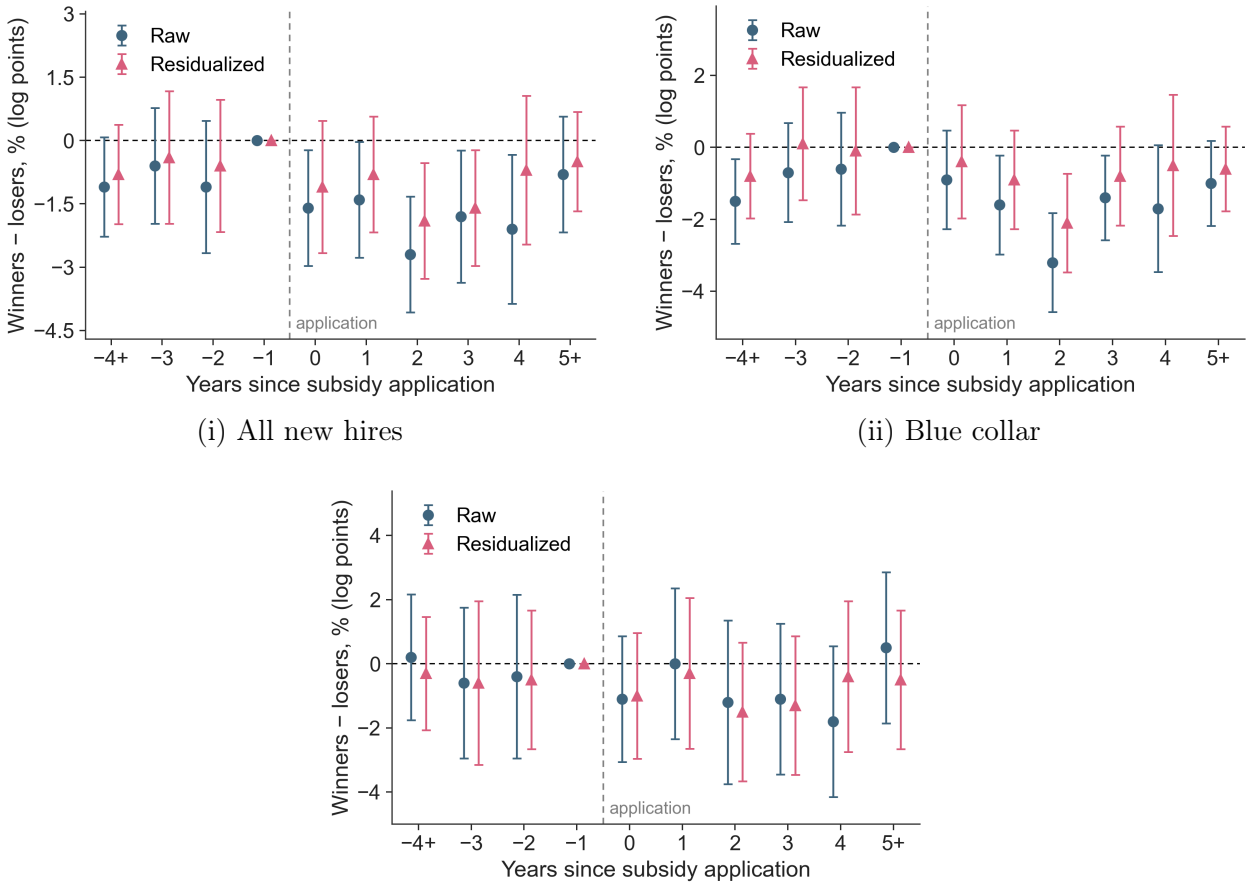
Notes: Firm-level event-study estimates of equation (11) for the workers each firm hires in year t: annual earnings in euros, log annual earnings, weeks worked, and the log entry weekly wage, with 95% confidence intervals. The year before the application is the reference. Earnings and weeks of new hires do not rise; their entry weekly wage is 1.4 to 2.7% lower at winning firms (Table A.10). Each panel overlays the two-way fixed-effects estimates (circles) and those of the interaction-weighted estimator of Sun and Abraham (2021), with near-cutoff losers as controls (triangles).

Figure A.12: Within-firm wage inequality rises at funded firms



Notes: Firm-level event-study estimates of equation (11) for the within-firm 90–10 ratio of log wages, with 95% confidence intervals. The year before the application is the reference. The ratio is flat before the application and rises from the first year after it.

Figure A.13: Entry wage: composition or price? Raw versus residualized



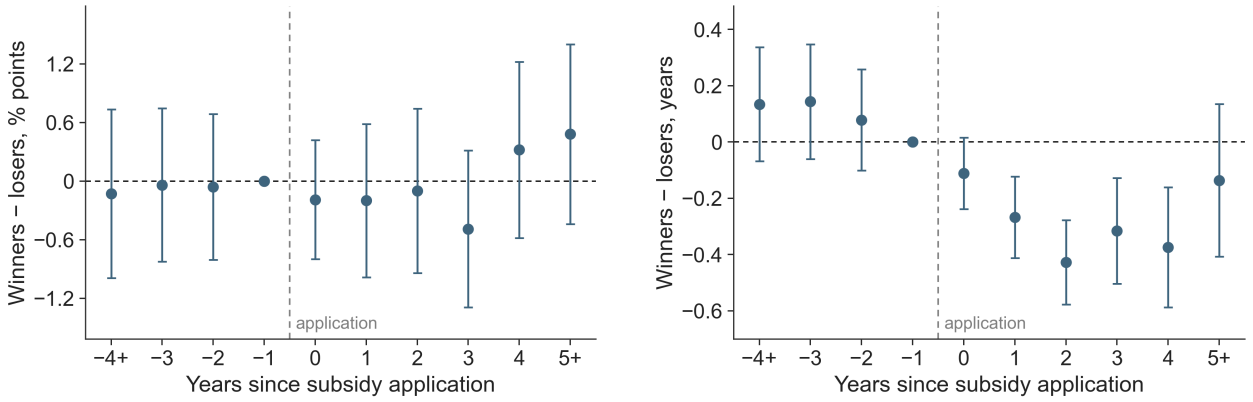
(i) All new hires

(ii) Blue collar

(iii) White collar

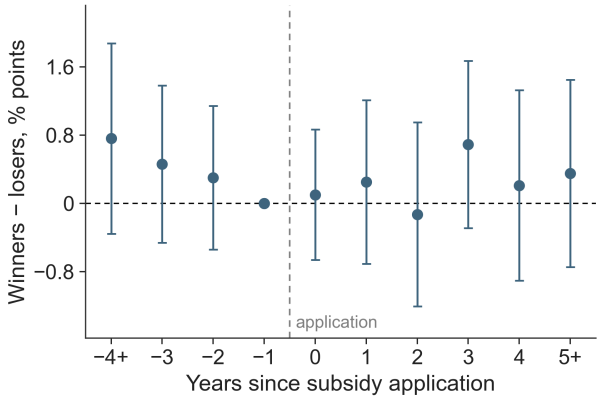
Notes: Entry weekly wage of new hires, raw and residualized on worker observables, overall and by collar. Residualizing halves the entry-wage decline and removes its significance except at the expansion peak: the raw drop is largely composition, except when the queue is drawn down, where the price of new hires falls too.

Figure A.14: Workforce Composition at Funded Firms

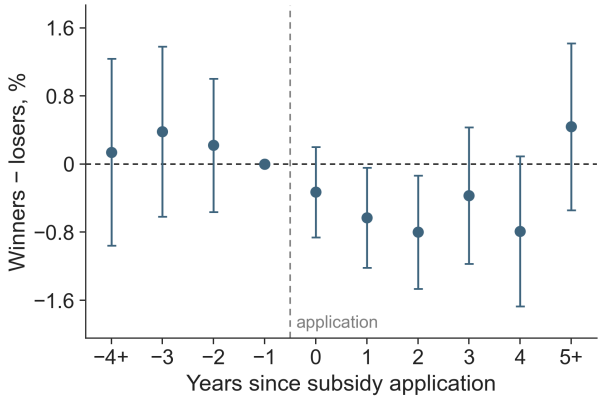


(i) Female share

(ii) Average age



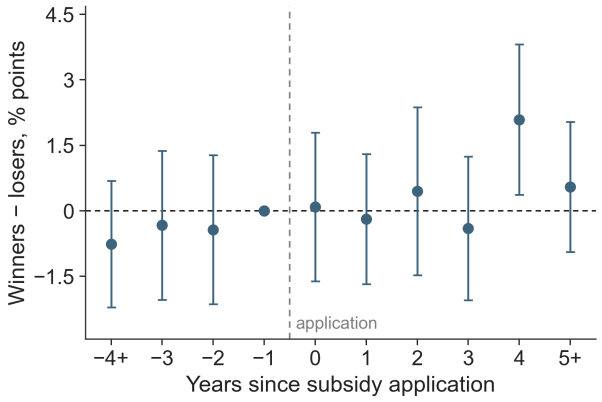
(iii) Blue-collar share



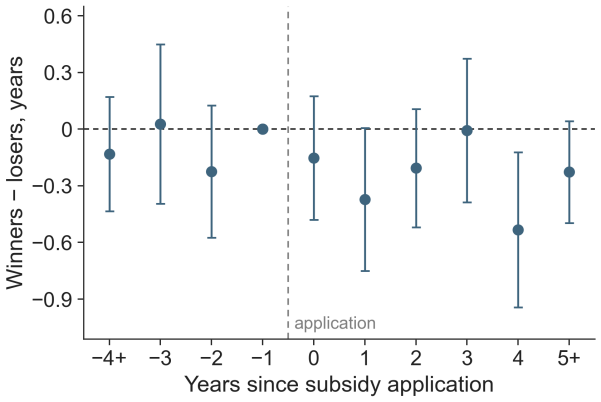
(iv) Quality index (predicted earnings)

Notes: This figure plots event-study estimates of equation (11), with a firm fixed effect, for four measures of the composition of the workforce at funded firms: the female share (panel i), average age in years (panel ii), the blue-collar share (panel iii), and an index of average worker quality, measured by predicted earnings in log points (panel iv). Winners minus near-cutoff losers, with 95% confidence intervals; t^*-1, the year before application, is the reference year. [Back to Section 4.2.1]

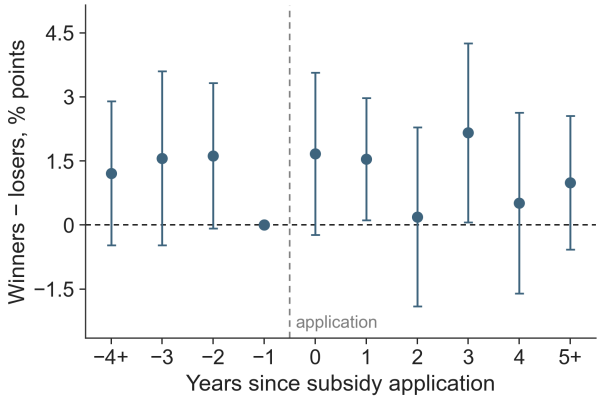
Figure A.15: Composition of the new hires



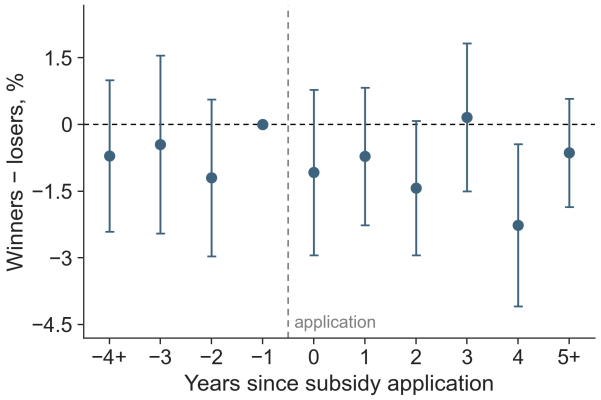
(i) Female share



(ii) Average age



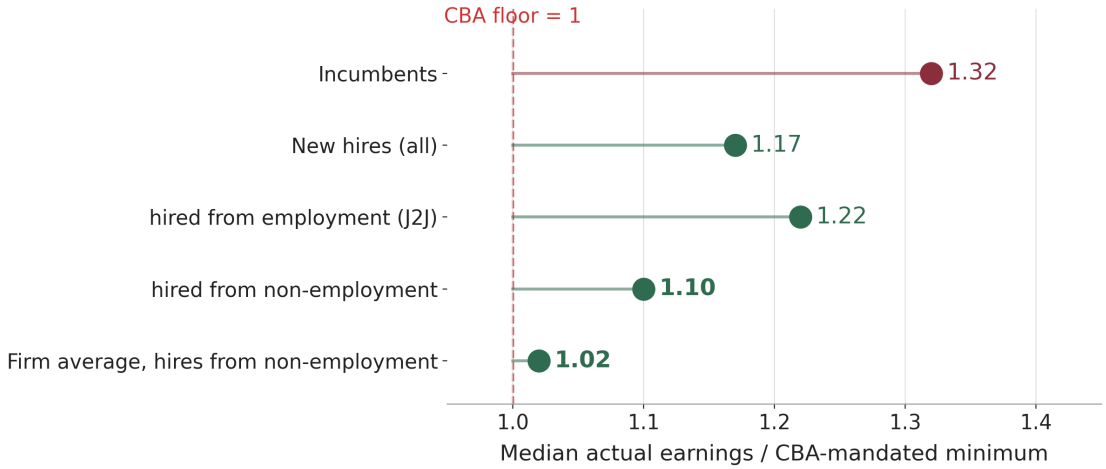
(iii) Blue-collar share



(iv) Quality index (predicted earnings)

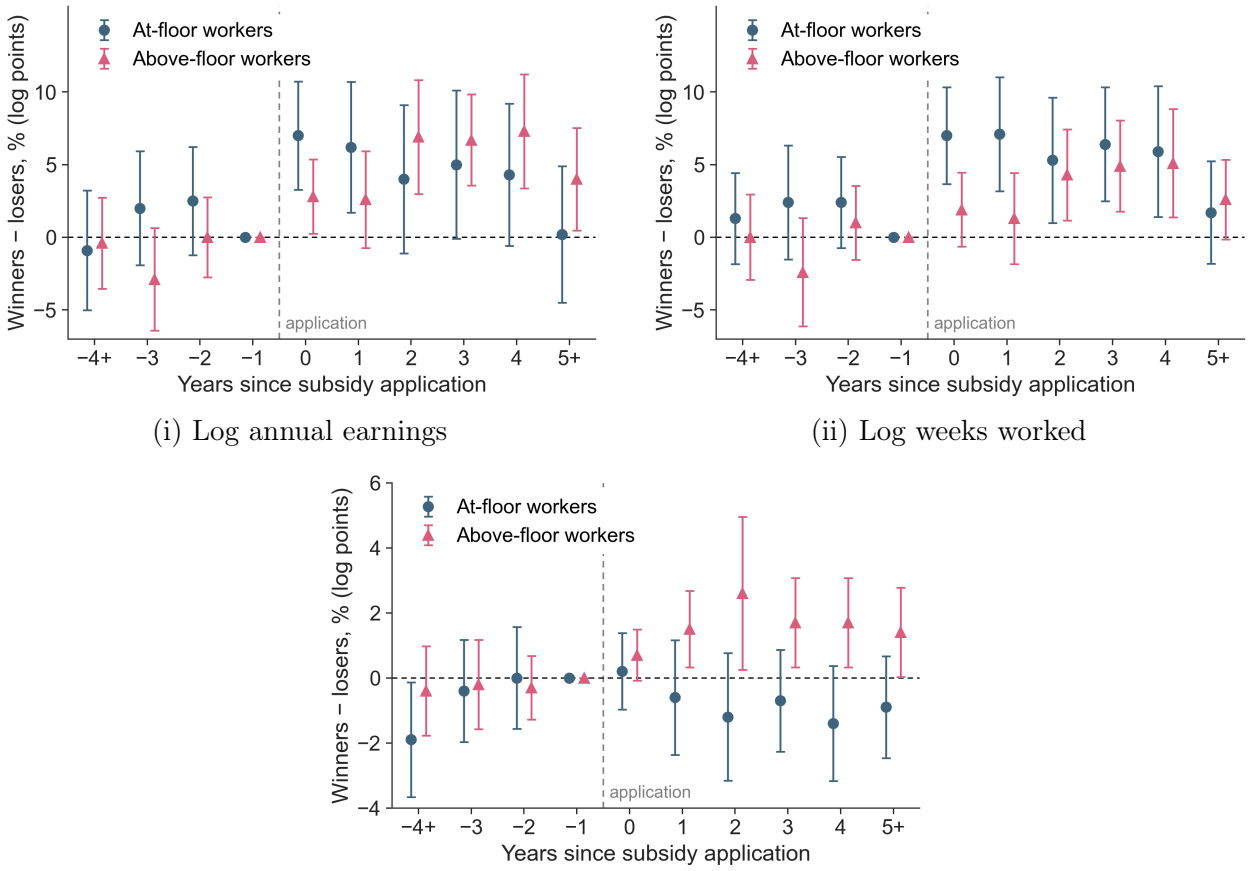
Notes: Event-study estimates for the observable composition of new hires: female share, age, blue-collar share, and predicted earnings. New hires are marginally younger and more blue-collar, and predicted quality dips modestly-the observable-composition counterpart of the entry-wage decline.

Figure A.16: The cushion ladder: pay relative to the worker's own contractual minimum



Notes: Median actual earnings divided by the worker's own contractually mandated minimum, by group; the values are those of Appendix Table A.13. A ladder down to the floor: hires from non-employment are paid close to the minimum, and the average firm hiring from non-employment pays 1.02 times it, while incumbents sit on a bargained top-up tier.

Figure A.17: Responses by Distance to the Contractual Floor: Event-Study Estimates



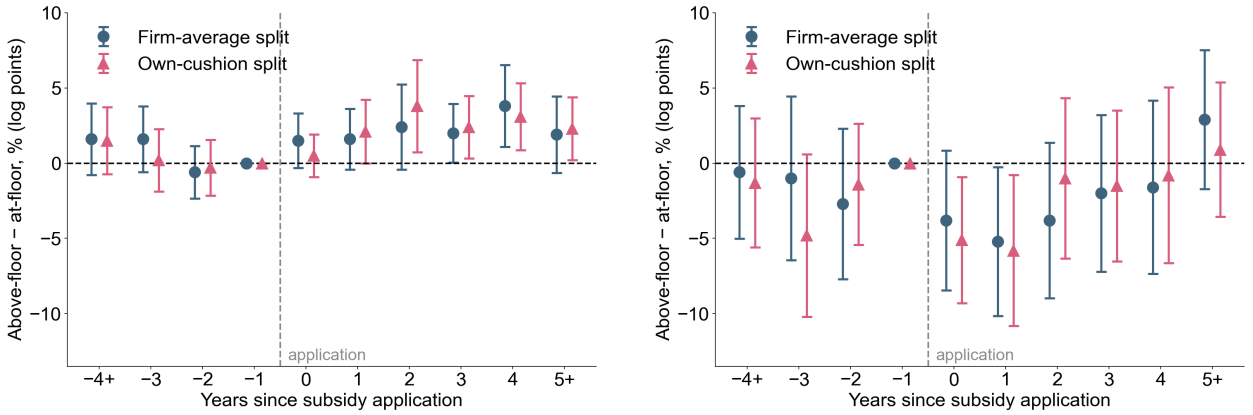
(i) Log annual earnings

(ii) Log weeks worked

(iii) Log weekly wage

Notes: This figure plots worker-level event-study estimates, in log points, separately for incumbents paid close to and above their contractual minimum. Incumbents are split at the median distance between their baseline pay and their own contractual minimum into near-floor and cushioned workers. Winners minus near-cutoff losers, with 95% confidence intervals; t^*-1, the year before application, is the reference year. Figure 10 in the main text reports the average of the year 0–4 coefficients.

Figure A.18: The Gap Between Cushioned and Near-Floor Workers

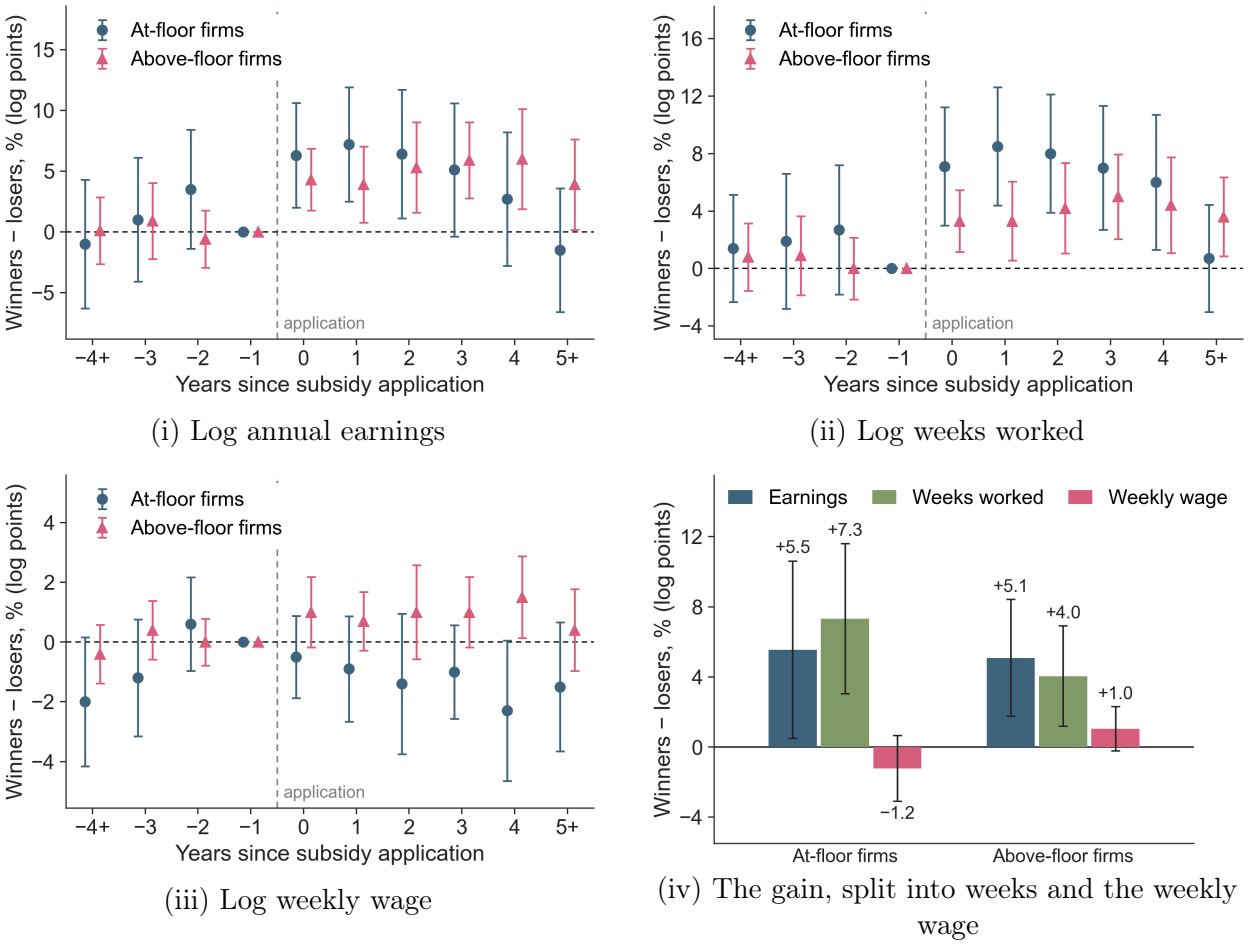


(i) Log weekly wage

(ii) Log weeks worked

Notes: This figure plots the difference between the event-study coefficients of cushioned and near-floor workers for the weekly wage (panel i) and weeks worked (panel ii). The coefficients come from separate regressions in which every fixed effect is estimated by group, so that each point equals the interaction coefficient of a pooled regression. In the firm-average split, firms are ranked by the average cushion of their full-time incumbents at t^*-1 and every incumbent takes the firm's label; in the own-cushion split, workers are ranked by their own cushion. Tables A.14–A.16 report the estimates.

Figure A.19: The floor selects the margin: firm-level split



(i) Log annual earnings

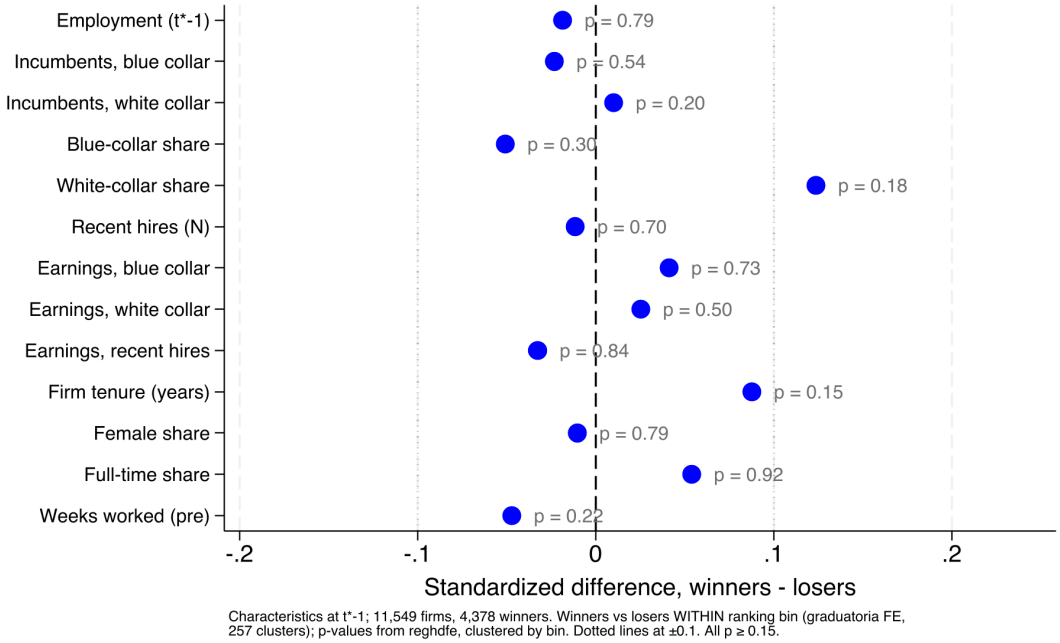
(ii) Log weeks worked

(iii) Log weekly wage

(iv) The gain, split into weeks and the weekly wage

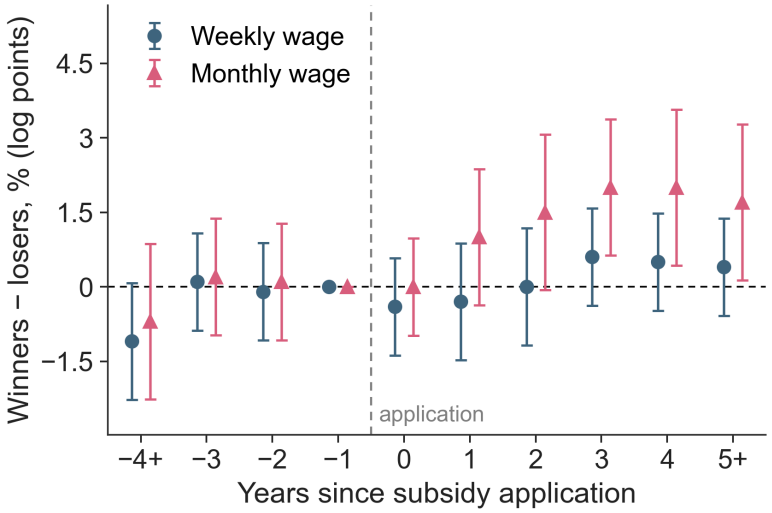
Notes: Firms split at the median of the average distance between their incumbents' baseline pay and the relevant contractual floor; every incumbent takes the firm's label. The split is by the firm's average cushion, not each worker's own (for that split, see Figure 10 and Appendix Figure A.17). Panels (i)–(iii) are event-study estimates in log points; winners minus near-cutoff losers, with 95% confidence intervals; t^*-1 is the reference year. Panel (iv) reports the average of the year 0–4 coefficients for each outcome; its intervals use the average of the yearly standard errors, which is conservative because the yearly estimates are positively correlated. At firms whose pay is close to the floor, weeks respond and the weekly wage does not; at firms with larger cushions, the weekly wage responds.

Figure A.20: The application cohort is balanced at baseline



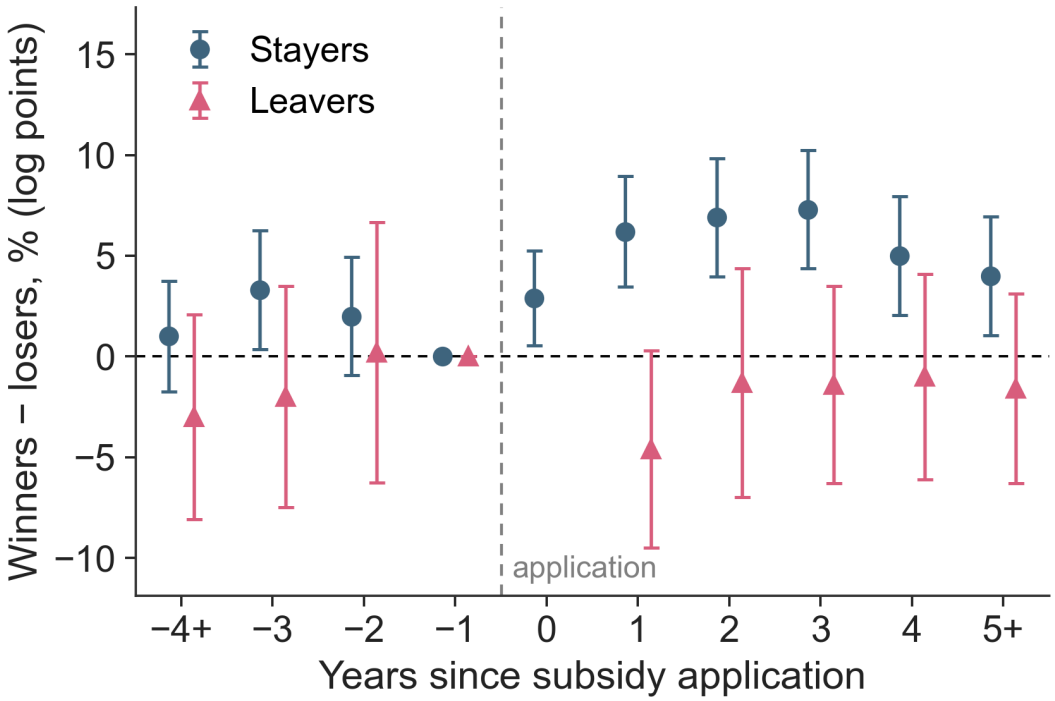
Notes: Pre-application means for the cohort at marginally funded and marginally rejected applicants.

Figure A.21: Incumbents' weekly and monthly wage



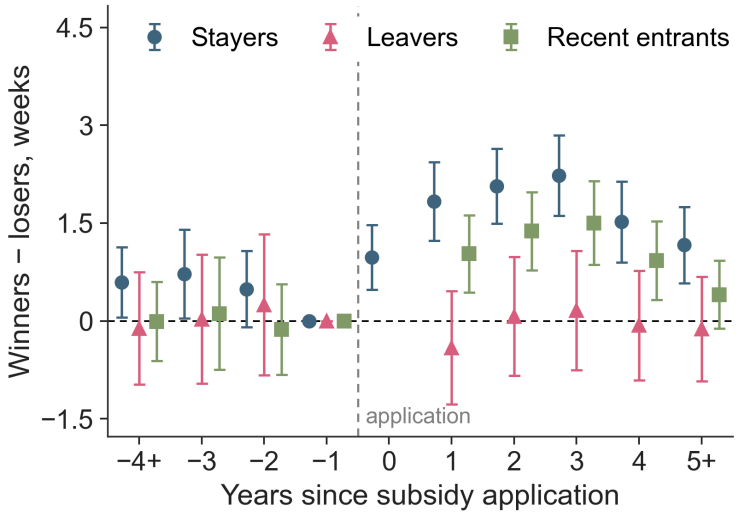
Notes: Event-study estimates for the incumbent cohort's log weekly wage and log monthly wage. Compare the earnings path in Figure 7: the difference between them is weeks worked.

Figure A.22: Incumbent Workers: Stayers and Leavers



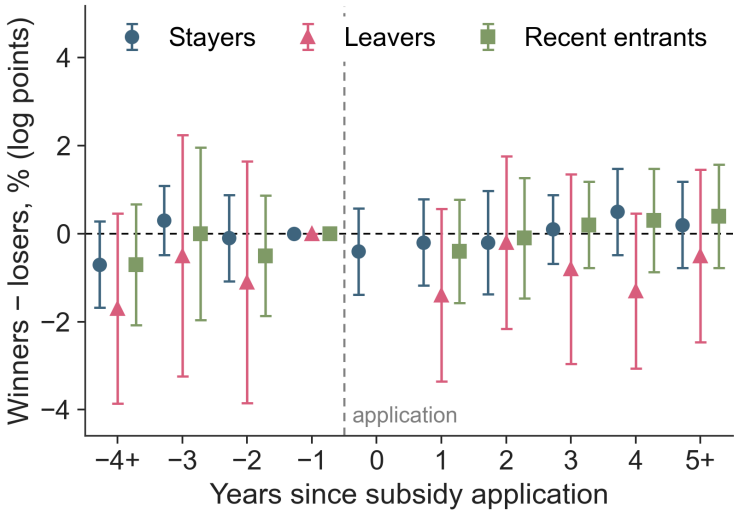
Notes: This figure plots event-study estimates of log annual earnings for the application cohort, split into workers who remain with the applicant firm (stayers) and workers who leave it (leavers). Winners minus near-cutoff losers, with 95% confidence intervals; t^*-1, the year before application, is the reference year. Because staying is itself affected by treatment, the split describes where the cohort's gain sits rather than a causal comparison between comparable groups. [Back to Section 4.2.2]

Figure A.23: Weeks worked, by group



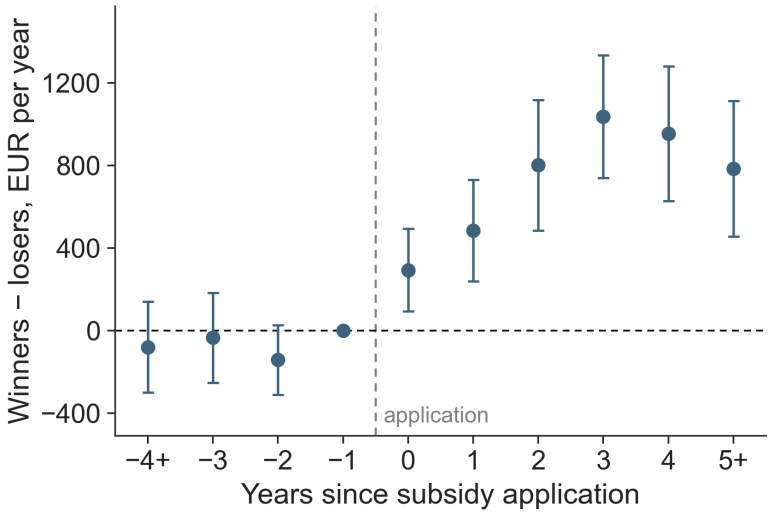
Notes: Weeks worked for stayers, leavers and recent hires. The quantity margin is where the groups differ.

Figure A.24: The weekly wage rate, by group



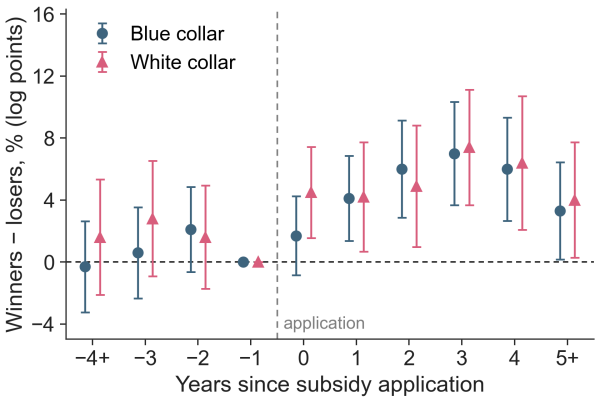
Notes: The weekly wage rate by group: flat in all of them.

Figure A.25: Earnings in levels (euros per year)

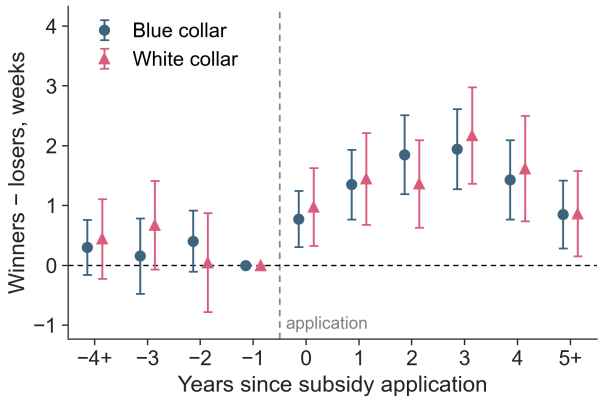


Notes: The same cohort event study in euros rather than logs, so that zeros enter.

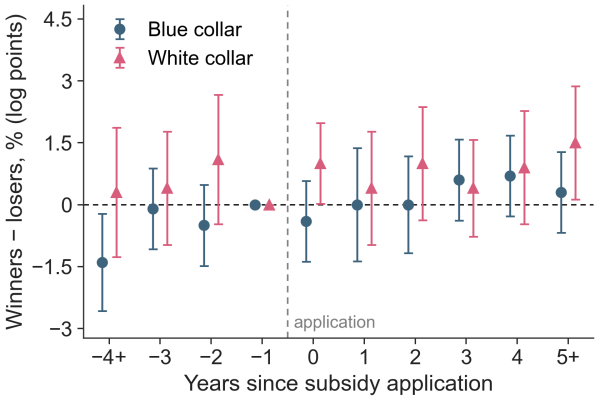
Figure A.26: Blue- and white-collar incumbents



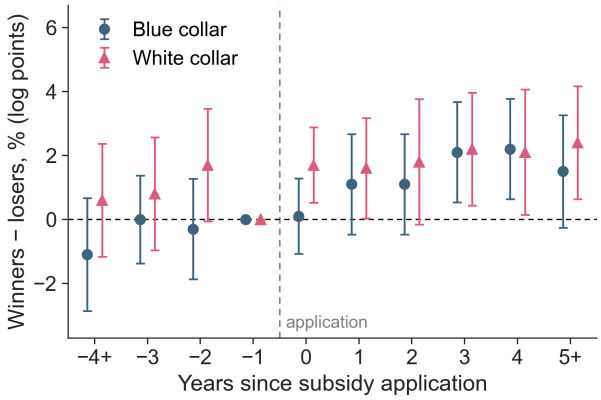
(i) Log earnings



(ii) Weeks worked (FTE)



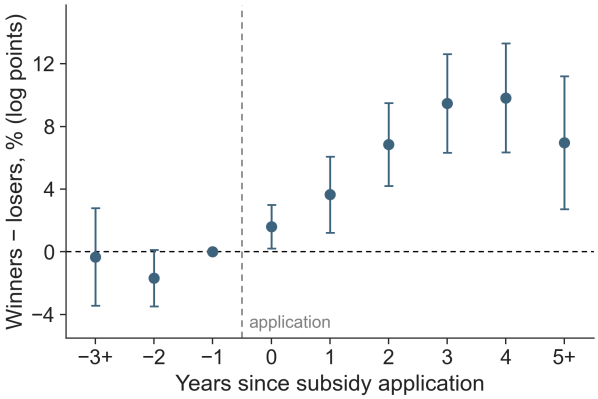
(iii) Log weekly wage



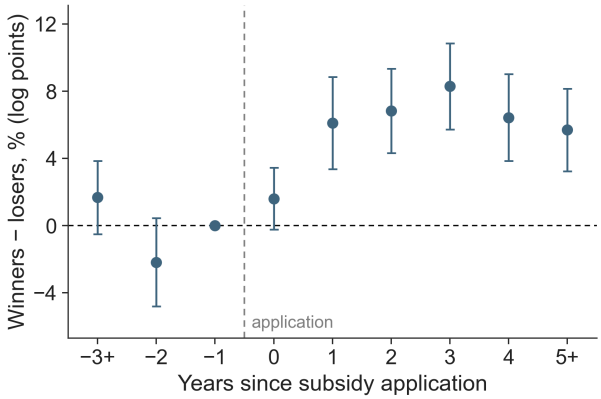
(iv) Log monthly wage

Notes: Earnings, weeks and the weekly rate, separately for blue- and white-collar incumbents.

Figure A.27: Retention and stayers' earnings



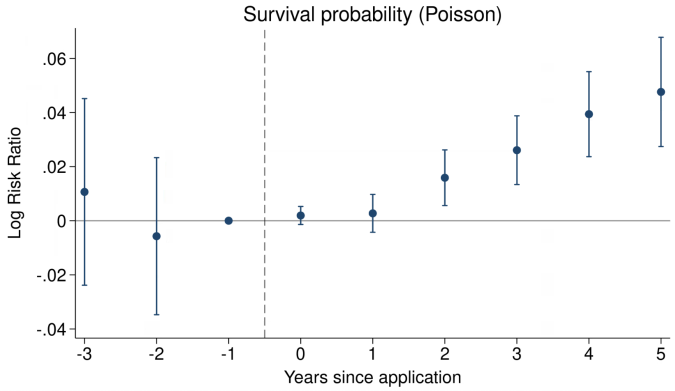
(i) Log retention rate



(ii) Log earnings of stayers

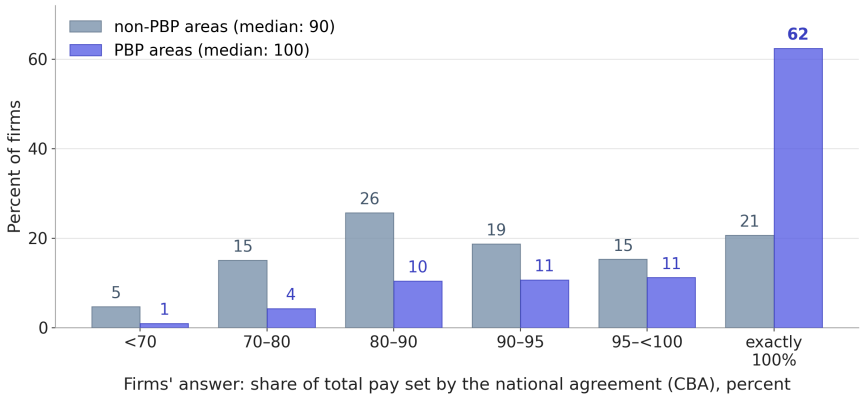
Notes: Retention at the applicant firm alongside the earnings of workers who stay.

Figure A.28: Subsidized firms are more likely to survive.



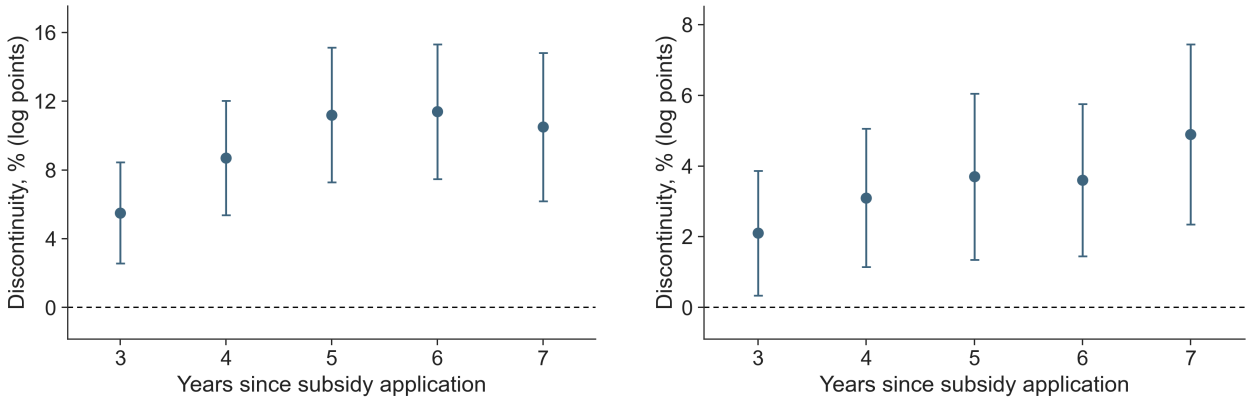
Notes: Event-study estimates of the effect of subsidy receipt on the probability that the firm remains active, from a PPML survival model, with 95% confidence intervals.

Figure A.29: How firms say they set wages: PBP-eligible vs non-eligible areas



Notes: Distribution of the share of the salary set by the national contract as reported by firms, Bank of Italy Invind survey, Centre-North versus South. The median Southern firm reports 100%.

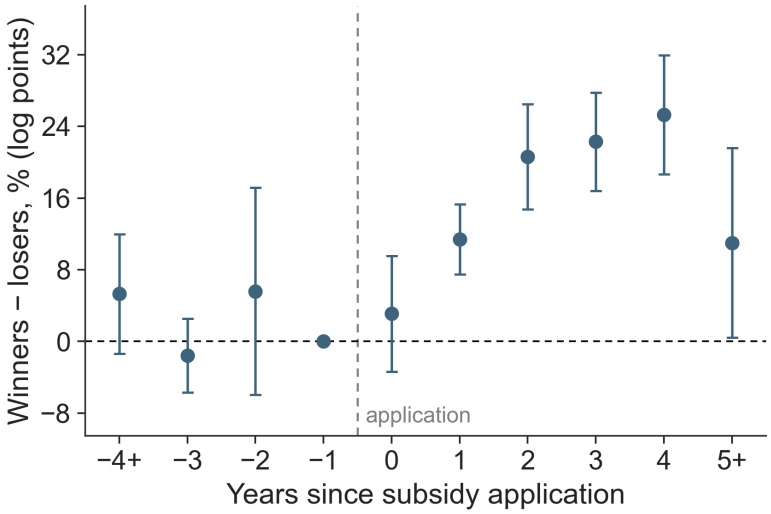
Figure A.30: The firm-level employment effect at every horizon.



(i) Log employment (ii) Log wage bill per worker

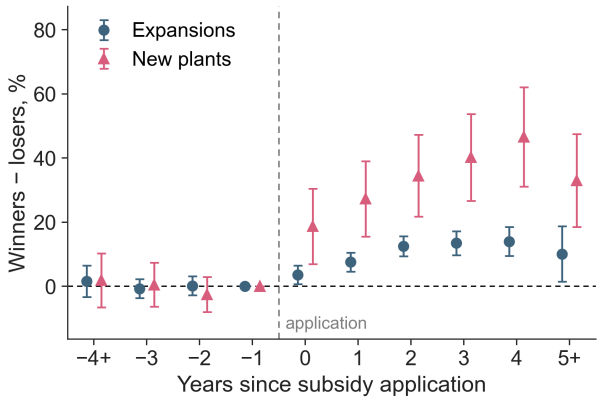
Notes: Regression-discontinuity estimates of the effect of crossing the funding cutoff on log employment (panel a) and log wage bill per worker (panel b), at horizons of three to seven years after the application, with 95% confidence intervals.

Figure A.31: Firm employment, interaction-weighted event study.

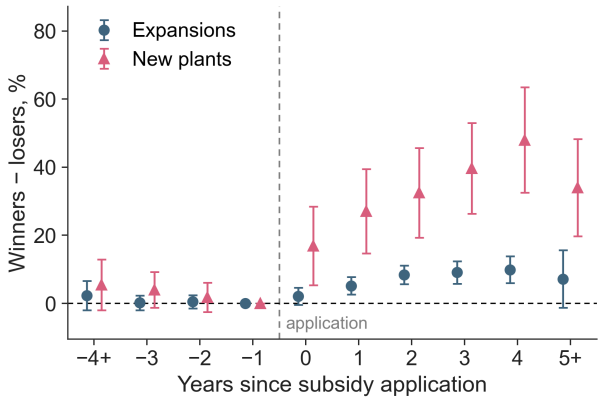


Notes: Log firm employment, interaction-weighted estimator of Sun and Abraham (2021), treatment defined as winning at the first application.

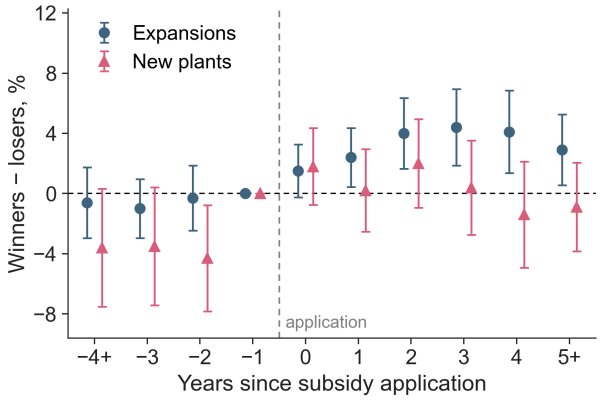
Figure A.32: Expansions versus new plants: retention is a margin of existing establishments.



(i) Employment



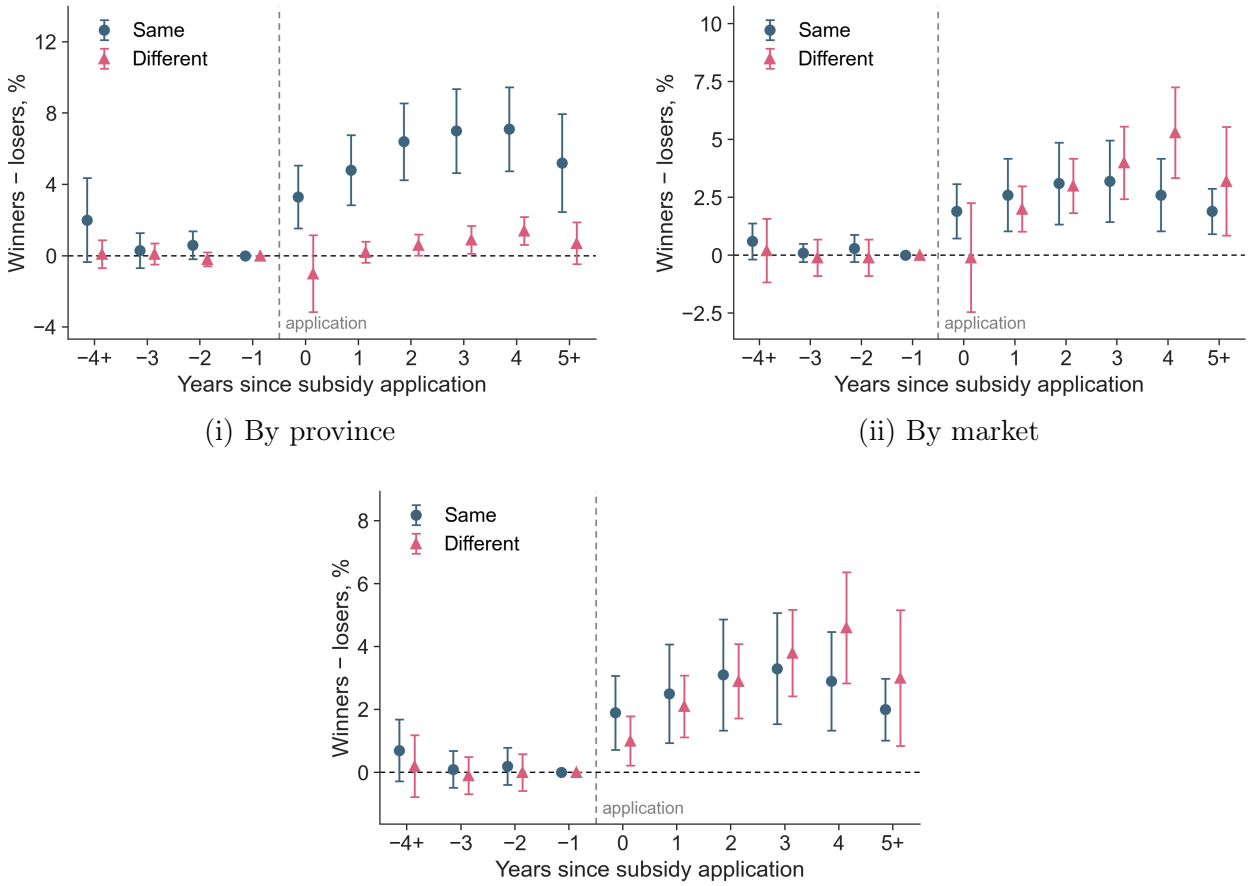
(ii) Due to hiring



(iii) Due to retention

Notes: The employment decomposition split by project type. New plants grow entirely through hiring, with retention a precise zero-where there are no incumbents to keep, no retention appears. Expansions draw about a third of their growth from retention.

Figure A.33: The geography of hiring.



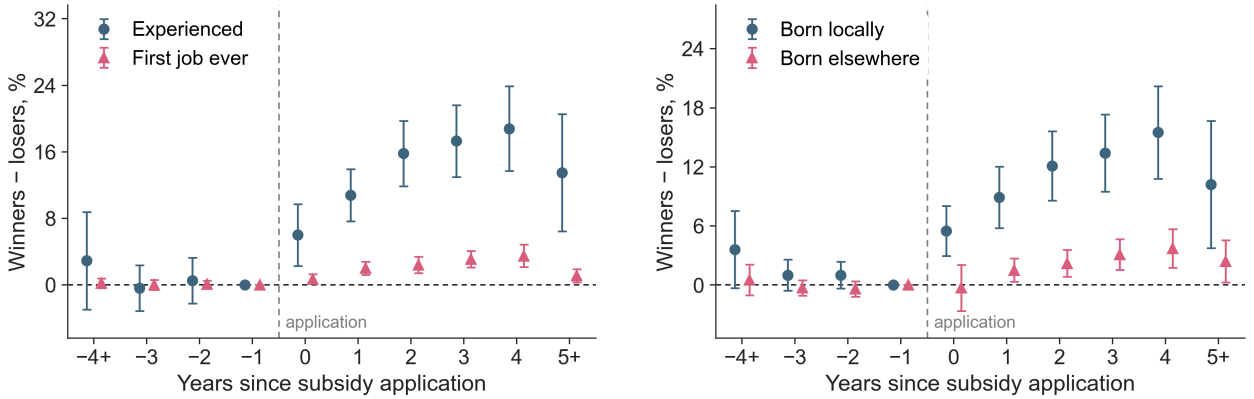
(i) By province

(ii) By market

(iii) By sector

Notes: Share of hires from the same province and from other provinces. Space binds: the hiring response is local.

Figure A.34: Who the new hires are.

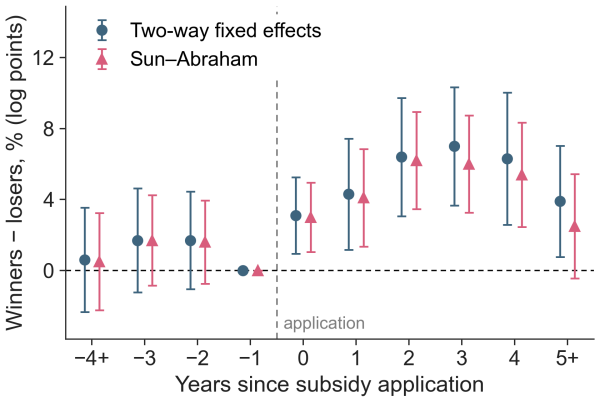


(i) First job ever versus experienced

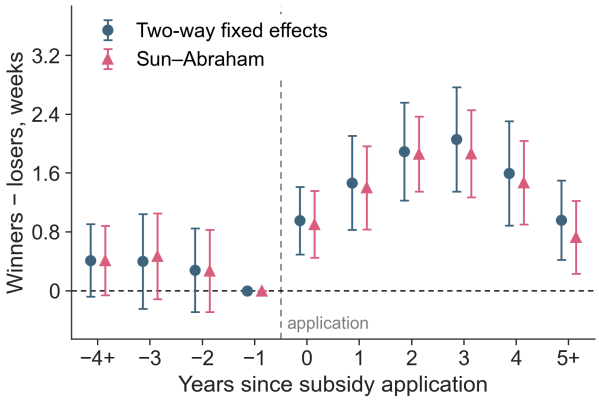
(ii) Born in the plant's province versus elsewhere

Notes: Characteristics of the additional hires at treated firms.

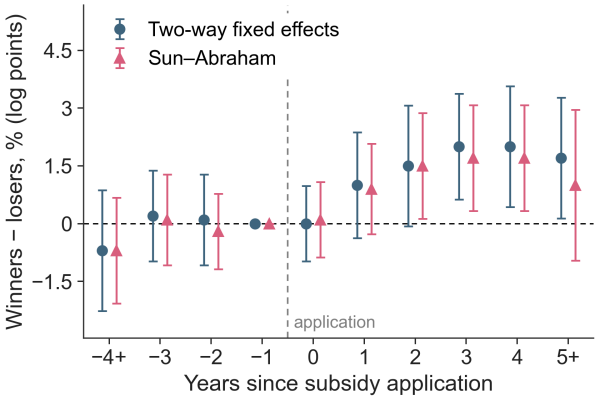
Figure A.35: The application cohort, interaction-weighted estimator.



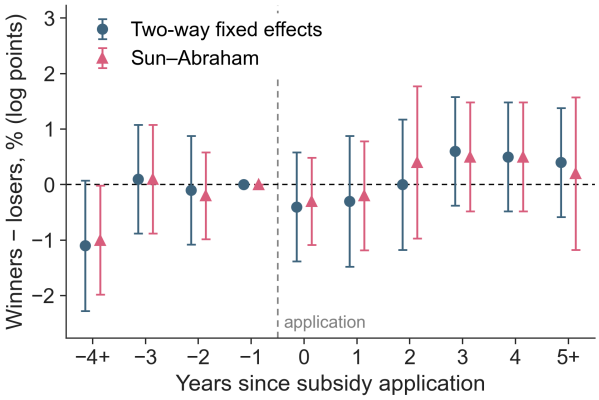
(i) Log earnings



(ii) Weeks worked (FTE)



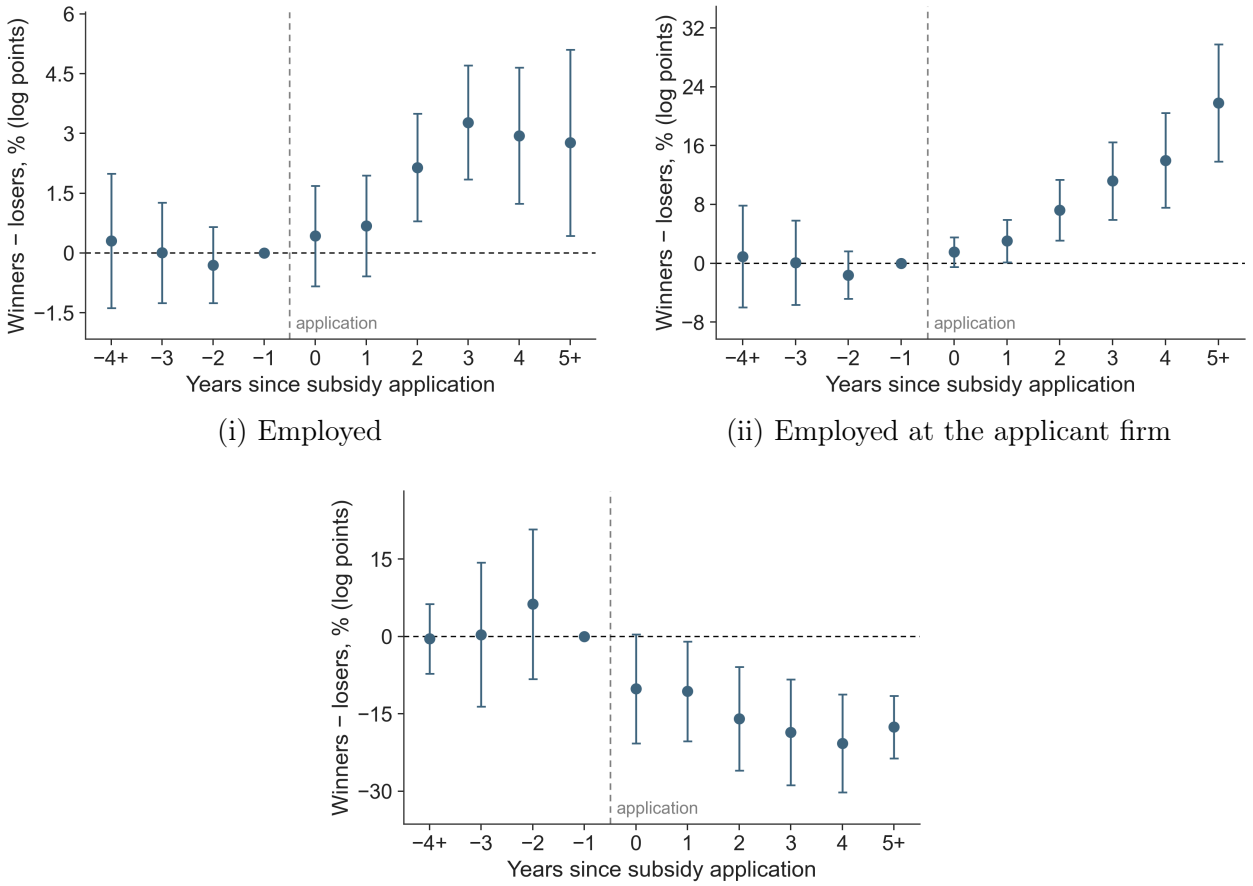
(iii) Log monthly wage



(iv) Log weekly wage

Notes: The two-way fixed-effects estimates of Figure 7 and Appendix Figure A.21 (circles), overlaid with those of the interaction-weighted estimator of Sun and Abraham (2021) (triangles).

Figure A.36: Employment margins for incumbent workers.



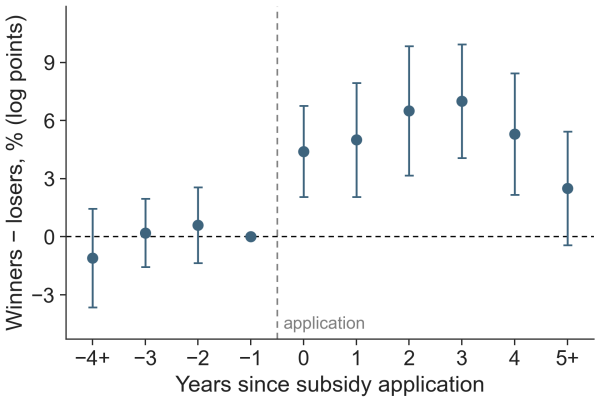
(i) Employed

(ii) Employed at the applicant firm

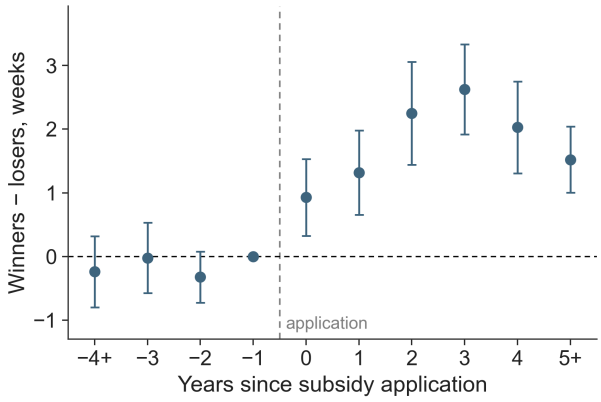
(iii) Employed at a different firm

Notes: Worker-level event-study estimates in differences from t^*-1: employed anywhere in the private sector, employed at the applicant firm, and employed at a different firm. Pre-trends shown.

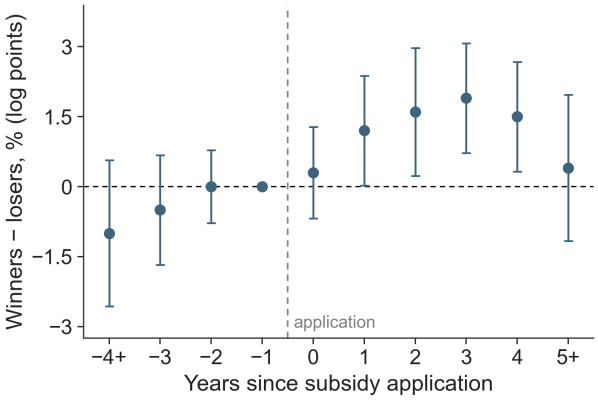
Figure A.37: The same decomposition, worker by worker.



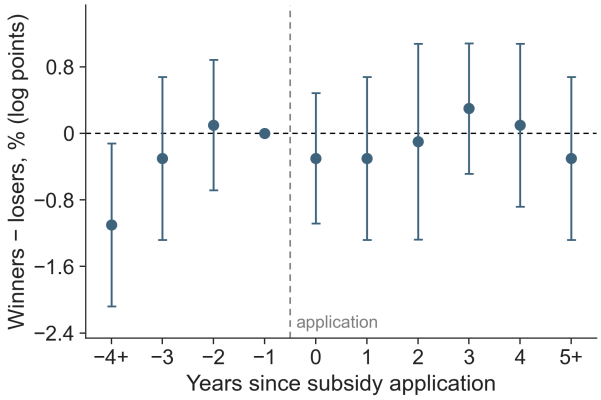
(i) Log earnings



(ii) Weeks worked



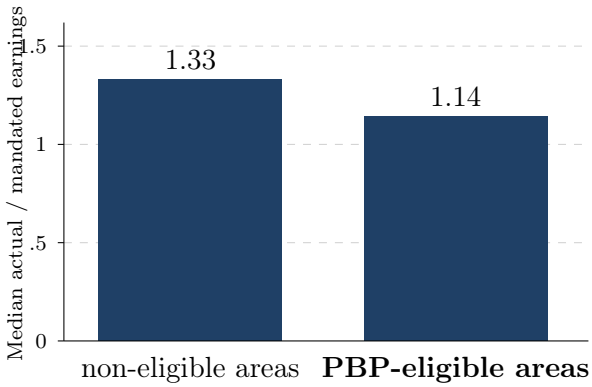
(iii) Log monthly wage



(iv) Log weekly wage

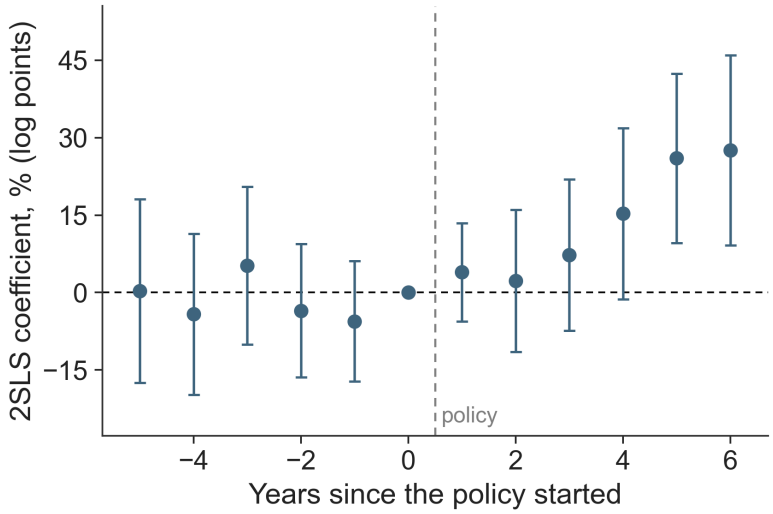
Notes: Worker-level estimates in differences from the baseline year for earnings, weeks worked and the weekly wage rate. The wage rate is flat; the earnings response is the weeks response.

Figure A.38: The contract binds harder where the policy is targeted



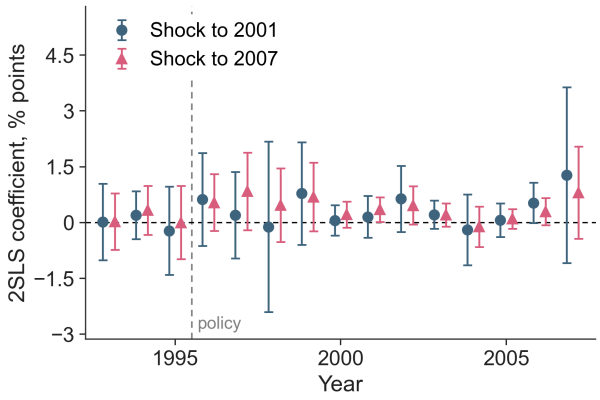
Notes: Median of actual earnings divided by contractually mandated earnings, in areas eligible for the policy and outside them. The same national schedule leaves a third of pay above the minimum outside the targeted areas and roughly a seventh inside them. These medians are computed over all workers in each area, and so differ from the medians for new hires and incumbents at applicant firms in Appendix Table A.13.

Figure A.39: Non-participating firms: the wage bill

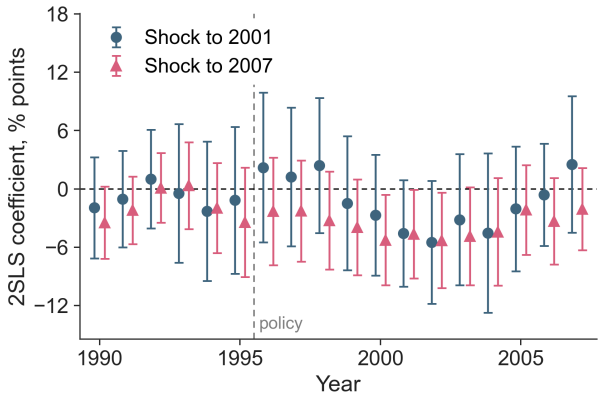


Notes: The market-level event study for the wage bill, which tracks the employment response.

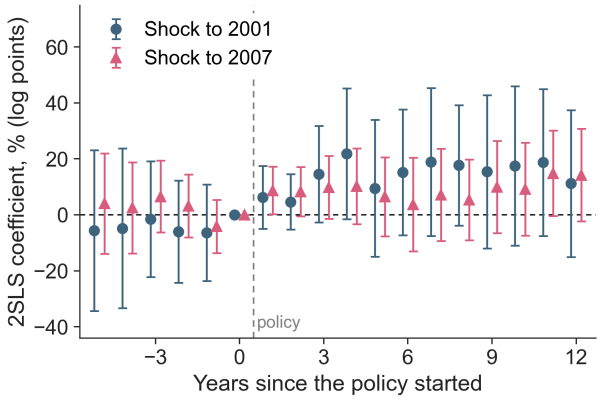
Figure A.40: The market shock, worker by worker



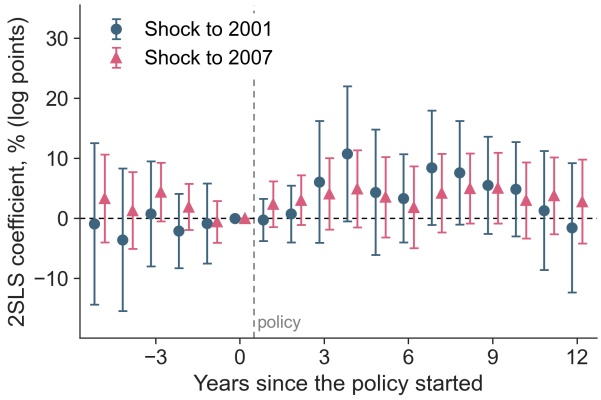
(i) Employed in the private sector



(ii) Receiving unemployment benefits



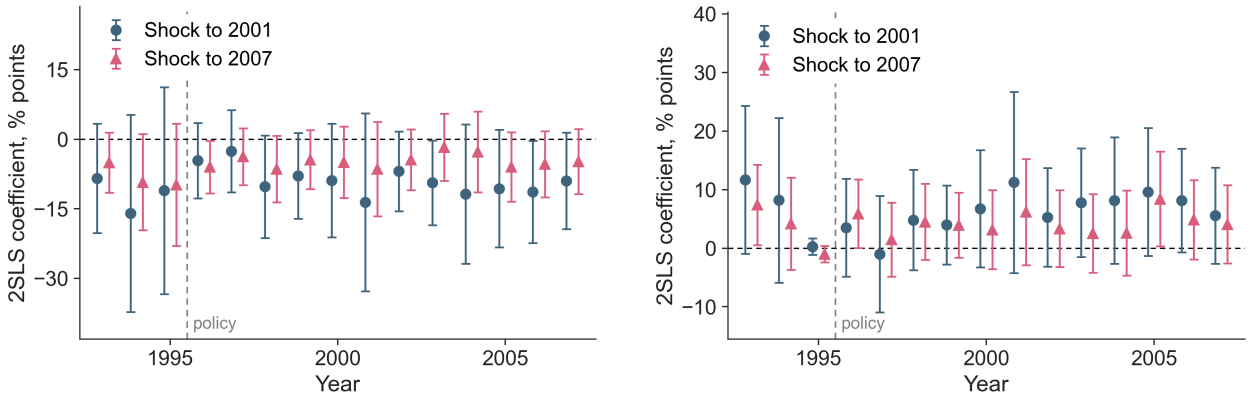
(iii) Log income from dependent employment



(iv) Log weekly wage

Notes: Worker-level effects of the market shock in the 13% subsample: wages, weeks, income, and benefit receipt.

Figure A.41: Out-migration does not respond

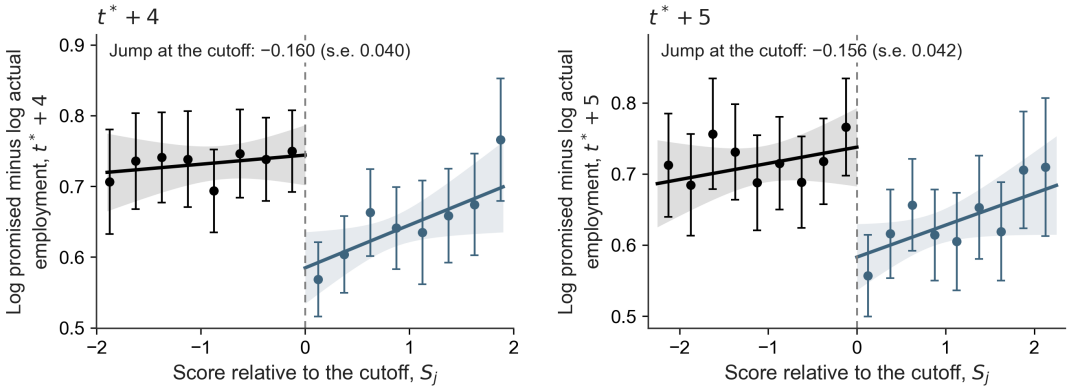


(i) Lives outside the 1995 province

(ii) Works in the 1995 province

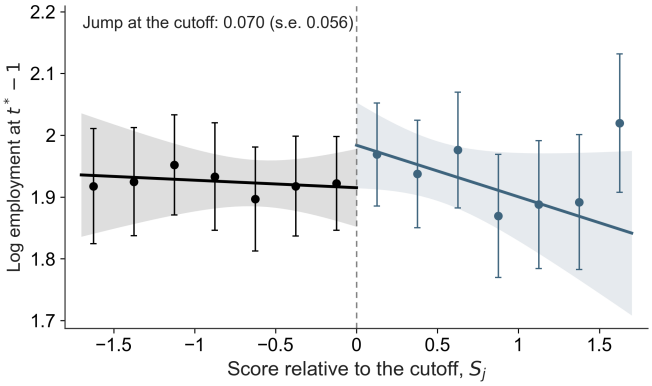
Notes: Probability of being observed working outside the 1995 province, by horizon. Every post-period point is negative.

Figure A.42: RD validation: promised vs. actual employment



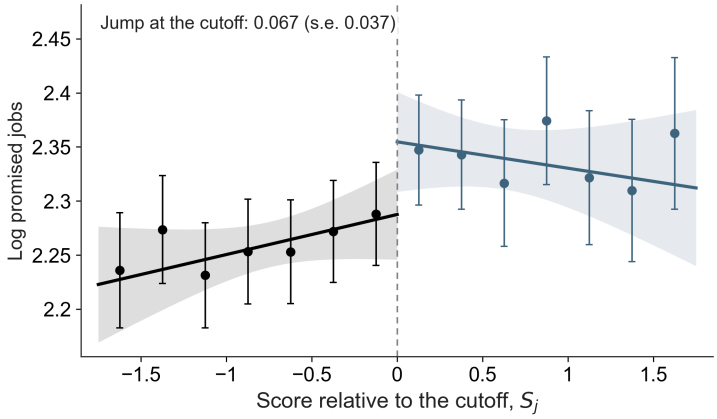
Notes: RD plots of the gap between promised and actual employment, the log of baseline employment plus promised jobs minus log actual employment, four years (left) and five years (right) after the application, constructed as in Figure 4: outcomes net of funding-pool and market fixed effects, bins of 0.25 inside the bandwidth with 95% confidence intervals, local linear fits with triangular weights, and robust discontinuity estimates (Calonico et al., 2014), clustered by funding pool. The gap is 16.0 log points smaller at winning firms at t^*+4 (s.e. 4.0) and 15.6 at t^*+5 (4.2).

Figure A.43: Balance check: baseline log employment at the funding cutoff. No significant discontinuity.



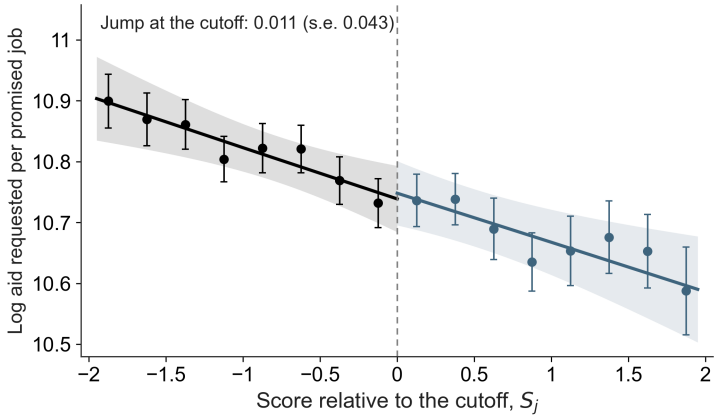
Notes: RD plot of log employment at t^*-1, constructed as in Figure 4: outcomes net of funding-pool and market fixed effects, bins of 0.25 inside the bandwidth with 95% confidence intervals, local linear fits with triangular weights, and robust discontinuity estimates (Calonico et al., 2014), clustered by funding pool. The jump is 0.070 (s.e. 0.056).

Figure A.44: Log promised jobs at the funding cutoff



Notes: RD plot of the log number of jobs promised in the application, constructed as in Figure 4: outcomes net of funding-pool and market fixed effects, bins of 0.25 inside the bandwidth with 95% confidence intervals, local linear fits with triangular weights, and robust discontinuity estimates (Calonico et al., 2014), clustered by funding pool. The local estimate is 0.067 (s.e. 0.037). Over $|S_j| < 5$ with the same fixed effects, as in Table 1, it is 0.031 (0.022) with a linear and 0.062 (0.029) with a quadratic polynomial on each side.

Figure A.45: Aid requested per promised job at the funding cutoff



Notes: RD plot of the log of the subsidy requested per promised job, constructed as in Figure 4: outcomes net of funding-pool and market fixed effects, bins of 0.25 inside the bandwidth with 95% confidence intervals, local linear fits with triangular weights, and robust discontinuity estimates (Calonico et al., 2014), clustered by funding pool. The jump is 0.011 (s.e. 0.043).

A.2 Tables

Table A.2: Summary Statistics

Panel A. Applications

	All applications in the South	Matched to social security	Near the cutoff
Eligible grant (thousand euros)	954.3 [426.5]	887.0 [437.2]	888.6 [442.9]
Grant awarded (thousand euros)	314.7	289.7	279.0
Grant disbursed (thousand euros)	217.0	211.5	200.7
Grant awarded, % of eligible grant	36.9	37.3	36.0
Promised jobs	15.9 [9]	14.5 [8]	14.8 [8]
New-plant projects (%)	46.6	31.0	31.2
Manufacturing (%)	61.0	63.2	63.6
Funded (%)	36.5	37.5	36.0
Applications	53,174	26,239	23,372

Panel B. Applicant firms

	Mean	10th pct.	Median	90th pct.	Firms
Funds requested (thousand euros)	871.2	103.5	437.2	1,837.2	25,932
Score above the cutoff (%)	37.5				25,932
Above the cutoff and funded (%)	29.3				25,932
Employees, year before application	27.8	0.8	8.0	54.1	25,932
Manufacturing (%)	63.4				25,932
Employment growth to six years after (log change)	0.31	−0.74	0.26	1.52	21,387
Wage growth to six years after (log change)	0.19	−0.09	0.20	0.45	21,386

Panel C. Workers at applicant firms, year before application

	Winners	Losers	Normalized difference
Annual earnings, private sector (euros)	15,472	14,632	−0.019
Log weekly wage	5.861	5.841	−0.007
Weeks worked	38.9	37.5	−0.022
Age (years)	33.0	33.5	−0.027
Tenure (years)	2.69	2.84	−0.043
Firm size (employees)	24.0	18.6	−0.003
Blue-collar full-time (share)	0.661	0.642	−0.017
Female (share)	0.261	0.272	0.004
Recent hire (share)	0.582	0.605	0.034
Workers		230,392	

Notes: This table reports summary statistics. Panel A: applications to IDA calls for projects in the South; applications by firms matched to social security records; and matched applications near the funding cutoff. Means, with medians in brackets for amounts and promised jobs. Panel B: applicant firms matched to social security records, all located in the South; growth is the log change in employment and in the average wage from the year before application to six years after. Panel C: workers employed at applicant firms in the year before application (the application cohort), by whether the firm won or lost. The normalized difference is the difference between winners and losers within ranking bins, divided by the pooled standard deviation; the joint test of no difference across all rows has $p = 0.465$.

Table A.3: McCrary test at the funding cutoff

McCrary Density Test (jackknife)		
T statistic	-0.607	
p-value	0.544	
	Left	*Right*
Density at cutoff	0.198	0.189
Standard error	0.010	0.010
Bandwidth	1.208	1.071
Observations	14,770	8,310
Effective N within bw	5,423	4,003

Notes: This table reports the test of McCrary (2008) for a discontinuity in the density of the running variable at the funding cutoff, estimated with the local-polynomial density estimator of Cattaneo et al. (2020) and jackknife standard errors. The running variable is the project score relative to the cutoff of its funding pool, standardized within the pool, S_j; all applicants are pooled across funding pools. The top panel reports the test statistic for the difference between the estimated densities just to the left and just to the right of the cutoff and its p-value; the bottom panel reports, on each side, the estimated density at the cutoff, its standard error, the bandwidth chosen by the estimator, the number of applications, and the number within the bandwidth. Continuity is not rejected ($p = 0.544$): applicants do not bunch just above the cutoff, consistent with their being unable to manipulate their score precisely enough to cross it. Figure A.1 plots the density. [Back to Section B.1]

Table A.4: Firm-level RD estimates by horizon and occupational group

	t^*+1	t^*+2	t^*+3
Panel (a): Δ log employment			
All workers	0.029	0.067***	0.126***
	(0.020)	(0.024)	(0.030)
White-collar	0.044**	0.083***	0.126***
	(0.022)	(0.025)	(0.031)
Blue-collar	0.026	0.049*	0.103***
	(0.021)	(0.025)	(0.031)
Panel (b): Δ log average wage			
All workers	0.006	0.016***	0.015***
	(0.004)	(0.005)	(0.005)
White-collar	0.014***	0.019**	0.022**
	(0.005)	(0.007)	(0.009)
Blue-collar	0.003	0.006	0.007
	(0.005)	(0.006)	(0.006)
Observations: all workers	22,473	22,140	21,611
Observations: white-collar	17,428	16,963	16,427
Observations: blue-collar	20,425	19,998	19,465

Notes: This table reports regression discontinuity estimates of β_1 in equation (9) one, two and three years after the application. Each outcome is the change between the year before the application, $t_j^* - 1$, and year $t_j^* + k$: in Panel (a), the change in the log number of employees, for all workers and separately for white-collar and blue-collar employees; in Panel (b), the change in the log average wage per worker for the same groups. The specification is linear in the score, with slopes allowed to differ on either side of the cutoff, estimated within a fixed bandwidth around each pool's cutoff, with funding-pool and market (province × industry) fixed effects. The sample is applicant firms in the South still active at each horizon, so the number of firms declines with the horizon; observations by group are reported at the bottom and are the same in the two panels. Standard errors, clustered by funding pool, in parentheses. Employment expands in both occupational groups and grows with the horizon; wage gains are concentrated among white-collar workers. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. [Back to Section 4.1]

Table A.5: Jobs created and jobs maintained, by horizon

Share of baseline employment	t^*	+1	+2	+3	+4	+5
Hires present (creation)	0.052	0.104	0.143	0.166	0.192	0.126
Baseline workers kept (maintenance)	0.016	0.031	0.055	0.079	0.089	0.068
Total jobs	0.068	0.135	0.198	0.245	0.281	0.194
Maintenance share	24%	23%	28%	**32%**	**32%**	35%

Notes: Treated-minus-control stocks at each horizon as shares of baseline employment, from the decomposition in Figure 5. [Back to Section 4.2.1]

Table A.6: The ratio of the wage to the employment response at funded firms

	(1)	(2)
Horizon	2 years	3 years
$\Delta \log$ Employment ($\hat{\beta}$)	0.245	0.119
	(0.107)	(0.048)
Wald 95% CI	[0.036, 0.454]	[0.025, 0.213]
AR 95% CI	[0.047, ∞)	[0.000, ∞)
Kleibergen–Paap F	3.27	6.63
Hansen J (p-value)	0.932	0.703
Observations	22,140	21,611
Clusters	340	337

Notes: This table reports two-stage least squares estimates of $1/\eta$ in equation (24), the ratio of the wage response to the employment response at funded firms that a monopsony reading of the firm-level evidence would interpret as the inverse labor supply elasticity. The dependent variable is the change in the log wage bill per worker between $t_j^* - 1$ and $t_j^* + k$; the endogenous regressor is the change in log employment over the same horizon, instrumented with the funding indicator and its interaction with the running variable, in the regression discontinuity sample and specification of equation (9), with funding-pool and market fixed effects. Column (1) is two years and column (2) three years after the application. The table reports the coefficient with its standard error, clustered by funding pool, in parentheses; the Wald 95% confidence interval; the Anderson and Rubin (1949) (AR) 95% confidence set, obtained by inverting the joint test that the two instruments are excluded from the regression of $\Delta \log W_{jk} - b_0 \Delta \log E_{jk}$ over a grid of b_0, which remains valid however weak the first stage is; the Kleibergen–Paap F statistic for the excluded instruments, which is low in both columns; the p-value of the Hansen test of overidentifying restrictions; and the number of funding pools (clusters). Read literally, the estimates imply a firm-level labor supply elasticity between about 4 and 8; Appendix B.1 explains why we do not adopt that reading where the floor binds. [Back to Section B.1]

Table A.7: Balance of the Recentered Instrument

	Recentered $\tilde{Z}_m$
Δ log total employment	0.000
	(0.017)
Δ log number of firms	0.023
	(0.025)
Δ log average wage	0.010
	(0.016)
Δ white-collar share	0.075
	(0.062)
Δ entry share	0.001
	(0.018)
Δ exit share	0.017
	(0.054)
Δ share of firms < 5 employees	-0.095^{*}
	(0.058)
Δ log white-collar premium	-0.000
	(0.004)
Δ log 90/10 wage ratio	0.009
	(0.010)
Joint F p-value	**0.891**
Markets	1,520

Notes: This table reports balance tests for the recentered instrument. Each row regresses a pre-policy market characteristic, measured in changes over the years before the program, on the recentered instrument, with province and sector fixed effects. Standard errors in parentheses. The instrument is uncorrelated with every characteristic bar one marginally significant coefficient, and the joint test does not reject. [Back to Section 5.1]

Table A.8: Non-Participating Firms: Balance-Sheet Outcomes

	$\Delta \log$ WB	$\Delta \log$ VA
Panel A: 2SLS (β on $\Delta \log E$)		
$\Delta \log E$	0.066	-0.041
	(0.113)	(0.094)
KP F	23.6	20.1
Panel B: Reduced form (on Z_m^*)		
Z_m^*	0.106	-0.064
	(0.179)	(0.144)
Panel C: OLS ($\Delta \log E$)		
$\Delta \log E$	0.054**	0.036*
	(0.025)	(0.022)
Observations	21,366	20,064

Notes: This table reports two-stage least squares of firm-level balance-sheet outcomes on instrumented market employment growth, at the 2001 endpoint, with region fixed effects; Panel B reports the reduced form on the recentered instrument. Outcomes are the change in the log wage bill and the change in log value added. The sample is firms that never applied to the program. [Back to Section 5.3]

Table A.9: The application cohort: who they are, and balance at baseline

	Winners	Losers	Diff. (within bin)	p
Size and workforce (at $t^* - 1$)				
Employment	22.40 (36.19)	23.12 (40.47)	−0.71	0.79
Incumbents, blue collar	16.01 (26.65)	16.68 (30.92)	−0.67	0.54
Incumbents, white collar	5.30 (10.05)	5.20 (9.04)	0.10	0.20
Recent hires	8.67 (10.72)	8.83 (17.03)	−0.16	0.71
Composition				
Blue-collar share	0.66 (0.11)	0.67 (0.09)	−0.01	0.30
White-collar share	0.27 (0.08)	0.26 (0.07)	0.01	0.18
Female share	0.26 (0.10)	0.26 (0.10)	−0.00	0.79
Full-time share	0.94 (0.05)	0.94 (0.06)	0.00	0.92
Earnings (euros) and attachment				
Earnings, blue collar	15,636 (3,291)	15,505 (3,054)	131	0.73
Earnings, white collar	17,643 (4,276)	17,538 (4,005)	105	0.50
Earnings, recent hires	12,440 (2,934)	12,539 (3,145)	−99	0.84
Firm tenure (years)	11.34 (3.27)	11.08 (2.76)	0.27	0.15
Weeks worked (pre)	38.78 (4.47)	38.99 (4.53)	−0.21	0.22
Firms	4,378	7,171	11,549 total	

Notes: This table reports pre-application characteristics of the application cohort-all workers employed at the applicant firm in the year before the application, $t_j^* - 1$-at near-cutoff winning and losing firms. The first two columns report means, with standard deviations in parentheses: firm size and workforce, the composition of the workforce, earnings in euros per year, firm tenure and weeks worked in the year before the application. The third column is the difference between winners and losers within bins of the ranking, which compares firms that applied with similar scores, and the fourth the p-value of the test that it is zero. Differences are small relative to the standard deviations and jointly insignificant. The last row reports the number of firms in each group. [Back to Section B.2]

Table A.10: New hires: the estimates

	Δ Earnings (euros)	Δ log earnings	Δ weeks	Δ log **entry weekly wage**
t^*	−232	−0.024	−0.35	**−0.016****
t^*+1	−203	−0.023	−0.17	**−0.014****
t^*+2	−221	−0.022	0.18	**−0.027*****
t^*+3	60	0.015	1.02***	**−0.018****
t^*+4	−77	−0.010	0.26	**−0.021****
t^*+5	−5	−0.001	0.25	−0.008

Notes: This table reports event-study estimates for the workers newly hired at applicant firms: the change in their annual earnings in euros and in logs, in the weeks they work, and in their log entry weekly wage. Firm-level event-study estimates of equation (11), near-cutoff winners against near-cutoff losers, with funding pool × year fixed effects, standard errors clustered by funding pool; $k = -1$, the year before the application, is the reference. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Outcomes are averages over the workers each firm hires in each year. New hires at winning firms earn no more than new hires at near-cutoff losing firms and work about the same number of weeks, while their entry weekly wage is 1.4 to 2.7% lower at most horizons, a difference that largely reflects the composition of the hires (Appendix Figure A.15). [Back to Section 4.2.2]

Table A.11: Retention and stayers' earnings

	Log retention	Log stayers' earnings	Ratio
Panel (a): event-study coefficients			
$k = -3$	−0.003 (0.016)	0.017 (0.011)	
$k = -2$	−0.017 (0.009)	−0.022 (0.013)	
$k = 0$	0.016 (0.007)	0.016 (0.009)	0.99
$k = 1$	0.037 (0.012)	0.061 (0.014)	0.60
$k = 2$	0.069 (0.014)	0.068 (0.013)	1.00
$k = 3$	0.095 (0.016)	0.083 (0.013)	1.14
$k = 4$	0.098 (0.018)	0.064 (0.013)	1.53
$k = 5$	0.070 (0.022)	0.057 (0.013)	1.22
Short run, $k = 1$	0.037	0.061	0.60
Long run, mean of $k \in \{1, \ldots, 5\}$	0.074	0.067	1.10
Panel (b): two-stage least squares			
Log retention on log stayers' earnings		1.444 (0.270)	
Kleibergen–Paap F		11.68	
Hansen J (p-value)		4.69 (0.196)	
Observations (firm-years) / firms		140,772 / 8,338	

Notes: Firm-level estimates for applicant firms with more than two employees at baseline. Log retention is the log of the number of incumbents still at the firm in year t over employment the year before the application; log stayers' earnings is the log of their mean annual earnings. Panel (a): event-study coefficients from the interaction-weighted estimator of Sun and Abraham (2021), with near-cutoff losers as controls and firm, year, funding pool × year and market × year fixed effects; k is years since the application, and $k = -1$ is the reference. The ratio column divides the retention coefficient by the earnings coefficient; the short- and long-run windows follow Jäger et al. (2024), whose retention elasticity is 0.73 in the short run and 0.48 in the long run. Panel (b): log retention regressed on log stayers' earnings, instrumented with treatment-by-event-time indicators, with firm, event-time, year and funding pool × year fixed effects. Standard errors, clustered by firm, in parentheses.

Table A.12: Contractual minima and actual pay

	Incumbents		New hires	
	Mean	Median	Mean	Median
All workers	0.784	0.752	1.054	0.887
White-collar	0.744	0.687	1.003	0.888
Blue-collar	0.797	0.767	1.094	0.886

Notes: Entries report the share of total earnings accounted for by the contractual pay floor implied by the relevant collective bargaining agreement. A share above one means that measured pay lies below the imputed floor. Mean shares above one for new hires reflect a minority of such observations; the median new hire's share is below one (0.89). Full distributions and complementary measures of pay above the contractual floor are reported in Appendix Tables B.2.1 and B.2.1. [Back to Section 4.3]

Table A.13: The cushion ladder: the underlying statistics

	Median earnings/minimum	Median cushion
Incumbents	1.32	0.275
New hires (all)	1.17	0.156
hired from employment (J2J)	1.22	0.197
hired from **non-employment**	**1.10**	**0.099**
Firm average, hires from non-employment	**1.02**	**0.012**

Notes: This table reports how far pay sits above the contractual floor for different groups of workers at applicant firms in the South. For each worker and year, mandated earnings are the monthly minimum of the worker's own national collective agreement and grade, summed over the months worked; the first column is the median of actual annual earnings divided by mandated earnings, and the second the median cushion, log(earnings) − log(mandated earnings). Both actual and mandated earnings enter this ratio in nominal euros, as the agreements set the minima. The sample is full-time blue- and white-collar workers who work at least three months in the year; both ratios are trimmed at the 1st and 99th percentiles. Incumbents are workers employed at the firm in the year before the application; new hires are workers who join the firm, split by whether they were employed (job-to-job) or not employed in the previous year. The last row averages, for each firm, the cushion of the workers it hires from non-employment in the application year, and reports the median across firms. The ladder runs down to the floor: workers hired from non-employment are paid close to the minimum, while incumbents are paid on a bargained top-up tier. Figure A.16 plots the medians with their interquartile ranges in the application-year window.[Back to Section 4.3]

Table A.14: The floor selects the margin: worker-level estimates

	Δ log earnings	Δ log weeks	Δ log **weekly wage**
Near-floor workers (own cushion below median; 76k workers)			
t^*	0.070***	**0.070*****	0.002
t^*+1	0.062***	**0.071*****	−0.006
t^*+2	0.040	**0.053****	−0.012
t^*+3	0.050*	**0.064*****	−0.007
Cushioned workers (own cushion above median; 74k workers)			
t^*	0.028**	0.019	0.007*
t^*+1	0.026	0.013	**0.015****
t^*+2	0.069***	0.043***	**0.026****
t^*+3	0.067***	0.049***	**0.017****
Cushioned − near-floor			
t^*	−0.042*	−0.051**	0.005
	(0.023)	(0.021)	(0.007)
t^*+1	−0.036	−0.058**	0.021*
	(0.029)	(0.026)	(0.011)
t^*+2	0.029	−0.010	**0.038****
	(0.033)	(0.027)	(0.016)
t^*+3	0.017	−0.015	**0.024****
	(0.031)	(0.026)	(0.011)

Notes: This table reports worker-level event-study estimates separately for incumbents paid close to and above their contractual minimum. Incumbents are split at the median of their own baseline cushion, the distance between their pay in the year before the application and the minimum of their own agreement and grade: the near-floor group (below the median) and the cushioned group (above it), with the number of workers reported in each panel heading. Worker-level event-study estimates of equation (25): each outcome is the change from the worker's own baseline year $t_j^* - 1$, regressed on event-time indicators interacted with winning, and funding pool × year fixed effects. Every incumbent employed at the applicant firm at $t_j^* - 1$ is kept whether or not they stay, with outcomes set to zero in years without work where the outcome is defined in levels, and each worker is weighted by the inverse of the number of incumbents at their firm, so that every firm counts equally. $k = -1$ is the reference year; standard errors are clustered by firm. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Every fixed effect is estimated separately by group. For near-floor workers, the earnings gain arrives as weeks and the weekly wage does not move; for cushioned workers the weekly wage rises. The last panel is the difference between the two groups' coefficients, which equals the coefficient on the interaction with the cushioned indicator in a pooled regression. [Back to Section 4.3]

Table A.15: The floor selects the margin: firm-level estimates

	Δ log earnings	Δ log weeks	Δ log weekly wage	% of gain from weeks
Above-median firms (2.19M worker-years, 224 clusters)				
t^*	0.043***	0.033***	0.010*	77%
t^*+1	0.039**	0.033**	0.007	85%
t^*+2	0.053***	0.042***	0.010	79%
t^*+3	0.059***	0.050***	0.010*	85%
At-floor firms (713k worker-years, 157 clusters)				
t^*	0.063***	**0.071****	−0.005	113%
t^*+1	0.072***	**0.085****	−0.009	118%
t^*+2	0.064**	**0.080****	−0.014	125%
t^*+3	0.051*	**0.070****	−0.010	137%
Above-median − at-floor				
t^*	−0.020	−0.038	0.015	
	(0.026)	(0.024)	(0.009)	
t^*+1	−0.033	−0.052**	0.016	
	(0.029)	(0.025)	(0.010)	
t^*+2	−0.011	−0.038	**0.024***	
	(0.033)	(0.026)	(0.014)	
t^*+3	0.008	−0.020	**0.020****	
	(0.032)	(0.027)	(0.010)	
t^*+4	0.033	−0.016	**0.038*****	
	(0.035)	(0.029)	(0.014)	

Notes: Firms split at the median of the average cushion of their full-time incumbents at t^*-1; every incumbent takes the firm's label. The split is by the firm's average cushion, not by each worker's own: an incumbent of an above-median firm may itself be paid at the floor, and the reverse. The estimates therefore describe firms whose pay structures leave more or less room above the floor; Appendix Table A.16 repeats the split with each worker's own cushion. The last column is the change in log weeks as a share of the change in log earnings; above 100% the weekly wage falls slightly. The last panel is the difference between the two separate regressions. [Back to Section 4.3]

Table A.16: The wage gap and the weeks gap across split definitions and tenure groups

	Δ log weekly wage: above − at floor			Δ log weeks: above − at floor		
	t^*+2	t^*+3	t^*+4	t^*+2	t^*+3	t^*+4
All incumbents						
Firm-average cushion	0.024*	0.020**	0.038***	−0.038	−0.020	−0.016
	(0.014)	(0.010)	(0.014)	(0.026)	(0.027)	(0.029)
Own cushion	0.038**	0.024**	0.031***	−0.010	−0.015	−0.008
	(0.016)	(0.011)	(0.011)	(0.027)	(0.026)	(0.030)
Own cushion, median by year	0.032**	0.019*	0.030**	−0.017	−0.020	−0.023
	(0.016)	(0.011)	(0.013)	(0.026)	(0.026)	(0.029)
Incumbents with tenure below four years						
Firm-average cushion	0.024	0.008	0.047***	−0.046	−0.032	−0.032
	(0.017)	(0.014)	(0.016)	(0.035)	(0.034)	(0.037)
Own cushion	0.051**	0.033***	0.033**	0.027	0.003	−0.008
	(0.022)	(0.013)	(0.015)	(0.034)	(0.032)	(0.039)
Incumbents with tenure of four years or more						
Firm-average cushion	0.003	0.000	0.004	−0.022	−0.008	0.010
	(0.018)	(0.015)	(0.015)	(0.027)	(0.024)	(0.027)
Own cushion	0.033**	0.010	0.018	−0.011	0.022	0.027
	(0.016)	(0.012)	(0.013)	(0.019)	(0.022)	(0.023)

Notes: This table reports the difference between the event-study coefficients of the cushioned and the near-floor group, for the log weekly wage and log weeks worked, two to four years after the application, under alternative definitions of the split. Each entry comes from separate regressions for the two groups with the specification of Table A.14, in which every fixed effect is estimated by group, so it equals the interaction coefficient of a pooled regression. "Firm-average cushion" ranks firms by the average cushion of their full-time incumbents at $t_j^* - 1$ and assigns every incumbent the firm's label; "own cushion" ranks workers by their own cushion; "median by year" splits workers at the median cushion of their application year. The tenure panels repeat the comparison for shorter- and longer-tenured incumbents, with the tenure-specific weights of the main worker analysis. Standard errors, clustered by firm, in parentheses. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. [Back to Section 4.3]

Table A.17: Earnings in levels (euros per year), by group

	t^*	t^*+1	t^*+2	t^*+3	t^*+4	t^*+5
Application cohort (all)	+294***	+484***	+801***	**+1,037***	+954***	+784***
Stayers	+244***	+564***	+549***	+741***	+580***	+457***
Recent entrants	-	+230*	+646***	**+891***	+732***	+450***
Leavers	-	−52	+183	+172	+88	+155

Notes: This table reports event-study estimates of the effect of winning on annual earnings in levels, in 2010 euros, for the application cohort-every worker employed at the applicant firm in the year before the application, $t_j^* - 1$-and for groups within it, at each year since the application. Estimates come from a firm-level event study on the application cohort: for each applicant firm and year, the outcome is averaged over the cohort members of the group, and the firm-year means are regressed on event-time indicators interacted with winning, with firm and funding pool × year fixed effects; near-cutoff losers-applicants just below the cutoff that were never funded-are the controls. The sample is near-cutoff applicant firms in the South with at most 1,000 employees at $t_j^* - 1$. Earnings are annual private-sector earnings in 2010 euros (deflated by the CPI), summed over all employers, winsorized at the 1st and 99th percentiles, and set to zero in years without private-sector work. $k = -1$ is the reference year. Standard errors, clustered by funding pool (about 220 clusters), in parentheses or not reported. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Stayers are cohort members still employed at the applicant firm in year t; recent entrants are cohort members hired less than three years before the application year; leavers are incumbents not observed at the applicant firm in year t. These groups are defined by outcomes the treatment affects and describe where the gain sits; the application cohort is the causal comparison. Because earnings are in levels and years without work enter as zeros, the estimates combine changes in pay with changes in whether, and for how many weeks, the worker is employed. Three years after the application, the cohort's earnings are €1,037 higher, about 6.4% of baseline pay.[Back to Section 4.2.2]

Table A.18: New hires' entry wages fall

Δ log weekly wage, new hires	All	Blue collar	White collar
t^*	−0.016**	−0.009	−0.011
$t^* + 1$	−0.014**	−0.016**	−0.000
$t^* + 2$	**−0.027*****	−0.032***	−0.012
$t^* + 3$	−0.018**	−0.014**	−0.011
$t^* + 4$	−0.021**	−0.017*	−0.018
$t^* + 5$	−0.008	−0.010	0.005

Notes: This table reports event-study estimates for the entry weekly wage of workers newly hired at applicant firms, for all new hires and by occupational group. Firm-level event-study estimates of equation (11), near-cutoff winners against near-cutoff losers; $k = -1$, the year before the application, is the reference. The outcome is the mean log weekly wage, in 2010 euros, of the workers each firm hires in each year, measured in the year of hiring. A worker's entry wage is observed only once, so worker fixed effects are not available; the residualized version, in Appendix Figure A.13, removes a quartic in age, a female indicator and its interaction with the age quartic, with year fixed effects. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Pre-application coefficients are flat. The composition of the new hires is in Appendix Figure A.15.[Back to Section 4.2.2]

Table A.19: Broader income and the safety net

	Δ log earnings (dep.)	UI receipt (pp)	CIG receipt (pp)
t^*	0.039**	−1.6**	−1.0***
t^*+1	0.043**	−2.7***	−0.9***
t^*+2	0.042**	−2.7***	−0.7**
t^*+3	0.061***	−3.3***	−0.6*
t^*+4	0.037*	−4.0***	−0.8**
t^*+5	0.047**	−1.3**	0.2
Control mean	-	9.5%	3.1%

Notes: This table reports worker-level estimates on the 13% random subsample of the INPS archives in which every source of income is observed, for the 31,393 incumbents of applicant firms in that subsample. Worker-level event-study estimates of equation (13): each outcome is the change from the worker's own baseline year $t_j^* - 1$, regressed on event-time indicators interacted with winning, and funding pool × year fixed effects. Every incumbent employed at the applicant firm at $t_j^* - 1$ is kept whether or not they stay, with outcomes set to zero in years without work where the outcome is defined in levels, and each worker is weighted by the inverse of the number of incumbents at their firm, so that every firm counts equally. $k = -1$ is the reference year; standard errors are clustered by firm. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The first column is the change in log earnings from private-sector dependent employment; the second and third are the probability of receiving unemployment benefits (UI) and short-time work benefits (CIG) in the year, in percentage points, against the control-group means in the last row. Appendix A.1 reports total income including transfers and the components separately. [Back to Section 4.2.3]

Table A.20: Cross-market variation in employment growth over the policy period

	Mean	St. Dev	Median	P25	P75	P90
Scale						
Log Emp. (1995)	5.623	1.753	5.764	4.550	6.831	7.752
Log Emp. (2001)	5.868	1.619	5.953	4.794	7.031	7.941
Δ Scale (2001-1995)	0.270	0.805	0.254	-0.025	0.578	1.035
Instrument (recentered shock)						
Scale IV (2001)	0.158	0.359	0.048	0.000	0.172	0.437

Notes: This table describes the markets-province × 2-digit industry cells-used in the market-level design of Section 5. The first rows report the distribution across markets of log total employment, applicants and firms that never applied together, in 1995, the last year before the program, and in 2001, and its change between the two years, $\Delta \log E_m$, the endogenous regressor of equation (14). The last row reports the distribution of the recentered instrument $\tilde{Z}_m$ for the 2001 endpoint, equation (16): the observed policy-induced change in log market employment from promised jobs, net of its expectation under the assignment rule. The instrument is zero or close to zero in most markets and concentrated in a few, which is where the identifying variation comes from. [Back to Section 5.3]

B Additional Results

This appendix reports the additional results referred to in the text. The figures and tables it discusses are in Appendix A.

B.1 Additional RDD Figures and Tables

Design validity. Firms could not fine-tune their position at the cutoff. The composite score standardizes each indicator against the *realized* distribution of competitors' bids within the competition, so a firm would have had to know the bids of every other applicant; pools and their budgets were not observable at bidding time; and enforcement rules-clawback of grants on unmet commitments, site inspections-discouraged inflated promises.

Employment and wages at other horizons. The figures below show RD plots at t^*+1 and t^*+3 (the main text reports t^*+2). The employment effect grows from approximately 0.03 at t^*+1 to 0.13-0.14 at t^*+3; the wage response is consistent across horizons. Figure A.4 shows the comparisons at t^*+2 and t^*+3 on the raw data, without fixed effects.

The firm-level RD estimates under three implementations are in Table 1. Table A.4 reports the estimates at horizons t^*+1, t^*+2 and t^*+3, for all workers and separately for white-collar and blue-collar workers.

Skill composition. Figure A.5 shows the change in the blue-collar and white-collar shares of employment at t^*+2 and t^*+3. Neither share moves by more than one percentage point at the cutoff, and none of the four jumps is significant.

Firm balance sheets: scale without margins. Matching the applicant firms to the Cerved accounts lets us ask whether the subsidy changed the firm's scale or its profitability. The answer is scale rather than margins: capital, revenue and value added all rise with employment (revenue by 9.0% at $t^* + 3$, s.e. 3.4, and 13.4% at $t^* + 5$; value added by 7.8% and 8.5%), while value added *per worker* is close to zero and precisely estimated at every horizon, the labor share is unchanged, profit per worker does not move, and the implied return on capital falls. The firm gets bigger, not more profitable per worker, so there is no widening margin to share.

Heterogeneous RD effects. We report the firm-level RDD estimates separately by skill group and by project type, referenced from Section 4.

Heterogeneity by skill group. Figure A.6 decomposes both employment and wages by skill group (white-collar and blue-collar) at t^*+2. Employment rises by similar amounts for both groups, 6.9% for white-collar and 5.3% for blue-collar workers, each estimated imprecisely on its own. Pay per worker does not move for either group: 0.8% (s.e. 1.1) for white-collar and 0.9% (0.8) for blue-collar workers. Funded firms grow without changing whom they hire or what they pay.

Heterogeneity by project type. The IDA program financed two main types of investment: the creation of new plants and the expansion of existing establishments. Opening a new plant typically requires creating a new set of employer-worker matches from scratch and may therefore involve stronger hiring needs and greater upward pressure on wages than scaling up an existing plant. Figure A.7 shows that the employment effect is driven by new-plant projects, 31.4% (s.e. 8.6) against 4.5% (4.3) for expansions. The wage bill per worker also rises at new plants, by 4.8% (1.9); because a new plant's workforce is hired from scratch, this reflects whom it hires rather than raises for the same workers.

RDD versus event study. As a check on the regression discontinuity, we estimate a firm-level event study that compares winners with losers close to the funding cutoff over event time, without controlling for the score. A firm that applied more than once is assigned to its first winning application. The outcomes are log employment and the log wage bill per worker at the applicant firm, in levels; the regressions include firm and calendar-year fixed effects and a market trend, and the year before the application is the reference year. Figure A.8 reports the estimates for all workers and separately for white- and blue-collar workers.

Winners and losers follow the same path before the application, for employment and for pay. Afterwards, employment at winning firms grows steadily: by 5% in the year of the application, 12% after one year, 20% after two and 28% after three, before settling at 22%. The increase is similar for blue-collar (23% at $t^* + 3$) and white-collar workers (18%). The wage bill per worker rises by 1 to 1.5%, with the gain concentrated among white-collar workers (up to 2%); for blue-collar workers it is never significantly different from zero.

The wage response is the same as in the regression discontinuity, which finds 1.5 to 2.9% after three years (Table 1). The employment response is about twice as large as the discontinuity estimate of 12.6 to 15.2%. Because this comparison does not condition on the score, it averages over projects at different distances from the cutoff and includes any growth difference

correlated with the score, while the discontinuity isolates the effect at the cutoff. The two designs therefore agree on the price and on the shape of the quantity response, and differ only in its size.

Labor supply elasticity If funded firms moved along upward-sloping labor supply curves, the ratio of their wage response to their employment response would be the inverse elasticity of labor supply to the firm. We compute that ratio by two-stage least squares,

$$\Delta \log W_{jk} \;=\; \tfrac{1}{\eta}\, \Delta \log E_{jk} \;+\; \varepsilon_{jk}, \tag{24}$$

where W_{jk} is the wage bill per worker and E_{jk} employment, and the change in log employment is instrumented with the funding discontinuity of Section 4.1 (Win_j and its interaction with the score). Table A.6 reports the estimates: 0.245 (s.e. 0.107) at t^*+2 and 0.119 (s.e. 0.048) at t^*+3. Read literally, they imply a firm-level labor supply elasticity between about 4 and 8, in the range of the estimates collected by Sokolova and Sorensen (2021) and of Berger et al. (2022) and Lamadon et al. (2022). The first stage is weak, with Kleibergen–Paap F-statistics of 3.27 and 6.63, and the Hansen J-test does not reject the overidentifying restriction in either column. With a first stage this weak, the Wald intervals understate the uncertainty, so the table also reports Anderson and Rubin (1949) confidence sets, which are valid whatever the strength of the instrument. They are $[0.047, \infty)$ at t^*+2 and $[0.000, \infty)$ at t^*+3: the data rule out a negative ratio, but place no upper bound on it. In terms of the elasticity, the confidence set at two years caps η at about 21 and admits any value below it, and the set at three years does not bound η at all. The elasticities of 4 to 8 implied by the point estimates are therefore not informative on their own.

We report this ratio because it is what a standard monopsony reading of the firm-level evidence would deliver, not because we adopt it. The reading assumes that the firm sets employment on its labor supply curve. Where the contractual floor binds, the firm instead hires at the floor from a queue of workers willing to work at that wage, and the small wage response measures how little of pay is free to move above the floor, not the slope of a supply curve (Section 4.3).

B.2 Additional Worker-Level Tables and Figures

How these magnitudes compare. Three years after the application, revenue at funded firms is 9.0% higher (Appendix B.1). Expressed as a pass-through-the change in the outcome divided by the change in sales over the same horizon-the weekly wage moves by 0.07 per

unit of sales, not significantly different from zero. That is the same order as the daily-wage pass-through of positive sales shocks in Italy, about 0.08 (Lombardi et al., 2026), and as the 0.05–0.09 estimated for permanent shocks in Denmark (Maibom and Vejlin, 2026b) and the 0.07 for Italy (Guiso et al., 2005), and below the 0.10–0.15 that Garin and Silvério (2024) estimate for the wages of incumbents after idiosyncratic export shocks in Portugal. Annual earnings are different. With zeros kept for years without work, as in Lombardi et al. (2026), incumbents' earnings rise by €1,037 at $t^* + 3$ (Appendix Table A.17), about 6.4% of their baseline pay, for a pass-through of about 0.7: four to five times the 0.16 they report for positive shocks in Italy, and ten times the Danish estimates. The difference is weeks worked. Their sample is larger firms, mostly in manufacturing, whose workers are already employed most of the year; ours are small Southern firms whose incumbents work about 39 weeks a year at baseline, so additional demand has room to be paid out as additional work. Nor can we compute a rent-sharing elasticity in the sense of Kline et al. (2019), who find that earnings rise by 29 cents per dollar of patent-induced surplus, an elasticity of 0.35, and of 0.56 for stayers: value added per worker at funded firms does not move (Appendix B.1), so there is no first stage.

Earnings–retention elasticity. A growing literature infers employer market power from the response of retention or separations to pay (Kline et al., 2019; Sokolova and Sorensen, 2021; Jäger et al., 2024). Following Kline et al. (2019), we compute for each applicant firm the log retention rate-the incumbents still at the firm in year t as a share of those employed there the year before the application-and the log mean annual earnings of those stayers, and estimate the ratio of the two responses by two-stage least squares, instrumenting stayers' earnings with treatment-by-event-time indicators. The estimate is 1.44 (s.e. 0.27; Kleibergen–Paap $F = 11.7$; Appendix Table A.11). Following Jäger et al. (2024), we also separate the short run, the first year after the application, from the long run, the average over the first five years: the ratio is 0.60 in the short run and 1.10 in the long run. The ratio rises with the horizon, because retention keeps building after the earnings gain has levelled off (Appendix Figure A.27).

Reading this ratio as a labor supply elasticity relies on an implicit assumption: that labor input per retained worker is fixed, so that a change in earnings is a change in the wage. Our setting is the case in which that assumption fails, since the floor fixes the weekly wage and the shock moves weeks: the weekly wage of incumbents does not move (Section 4.2.2), so the same ratio with the wage in the denominator has no first stage. The well-identified ratio can therefore reflect the value of additional weeks and of job security to rationed workers

rather than wage-setting power: at a frozen wage, a retention response is a response to the package of weeks and job security, not to a wage. Because most administrative earnings data lack hours or weeks, the same conflation can arise wherever labor demand shocks move hours or weeks: an earnings-based elasticity combines a price margin with a quantity margin, and only the first speaks to market power.

This appendix reports the worker-level material that supports Section 4, drawn from the current INPS extraction. Every figure and table below is produced on the same matched sample, the same event-study specification (13), and the same weighting as the main text.

B.2.1 Individual worker evidence

The cohort results above are firm-level averages over workers. We now estimate the same objects worker by worker, in differences from the fixed baseline year, which lets us put the extensive margin-whether the worker is employed at all, and where-on the same footing as earnings, and lets us retain zeros rather than conditioning on employment.

The worker-level specification. We estimate the effect in *differences* from the worker's own baseline year. For worker i attached to applicant firm j at $t_j^* - 1$, we estimate the event study

$$\underbrace{y_{ijt} - y_{ij,t_j^*-1}}_{\Delta y_{ijt}} = \sum_{k \neq -1} \tilde{\beta}_k \, \mathbf{1}\{t - t_j^* = k\} \cdot \text{Win}_j + \tilde{\delta}_{c(j),t} + \eta_{ijt}, \qquad \text{weights } 1/N_j, \tag{25}$$

on the sample of workers present at j in $t_j^* - 1$, with funding pool $\times$ year fixed effects $\tilde{\delta}_{c(j),t}$, which also absorb year and event-time effects, as in equation (11). Two features of (25) matter, and both differ from the standard rent-sharing template.

First, there is no worker fixed effect, and there does not need to be one: differencing from $t_j^* - 1$ removes any time-invariant worker component by construction, so $\tilde{\beta}_k$ is identified from *within-worker* changes, and with an unbalanced panel a within transformation would demean each worker over a different window. What keeps workers who stop working in the estimation is not the absence of fixed effects but the outcomes and the sample: the cohort is fixed at $t_j^* - 1$, every member stays in it whether or not they work, and outcomes defined in levels or as indicators-earnings, weeks, employed at all, employed at the applicant firm, employed elsewhere-take the value zero in years without work. That is why the extensive margin is visible here. Outcomes in logs, and wage rates, are defined only in years with work, so they are conditional on employment whatever the fixed effects; we report them as such.

Second, we weight each worker by $1/N_j$, the inverse of the number of workers at their application firm, so that every *firm* contributes equally and the estimates are not driven by the largest applicants. The cost of the first choice is that we cannot separate worker from match effects; the benefit is that the quantity margins-employment, weeks, continuity-are in the estimand rather than conditioned away.

Figure A.36 shows the employment margins. Treated incumbents are more likely to be employed anywhere in the private sector, substantially more likely to be still at the applicant firm, and *less* likely to be working at a different firm-the last falling by about six percentage points on a base of 0.38. The three together say that the retention response is not a reshuffling of who works where: it is a reduction in transitions out, both to other employers and to non-employment.

Figure A.37 then reproduces the decomposition at the worker level: earnings rise, weeks rise, the weekly wage does not move, and the pattern holds separately for stayers and leavers in the expected way. One caveat belongs here rather than in an appendix. The event-study sample is unbalanced in event time: cohorts enter at different calendar dates and are observed for different numbers of post-application years, so a coefficient at a long horizon is estimated on earlier cohorts only, and any drift in composition across cohorts is absorbed into the path rather than into the level. We handle this by holding the estimation sample fixed across the three decomposition regressions and re-estimating on a balanced panel of cohorts observed at every horizon; the weeks-versus-rate split is unchanged, and the level of the earnings path moves by less than a percentage point. Estimates by occupational group, by tenure, and the matched-sample version with exactly balanced cells are reported in Appendix B.2, together with the retention estimates.

The pension-regime split. Workers with eighteen years of contributions by 1995 remained in the more generous defined-benefit regime, which creates a within-firm discontinuity in the return to *reported* pay. Splitting the incumbent sample at that threshold, the quantity responses are indistinguishable across regimes, and the only pay responses that appear are in the defined-benefit group-consistent with reporting incentives rather than wage setting, and a caution for any Italian design that reads pensionable pay as a price.

B.3 Market-level analysis

B.3.1 Validation of Promised Jobs Using RDD

We begin by validating the use of firms' promised job creation as an input for the counterfactual employment measures used in the permutation exercises. The key concern is whether the number of jobs firms commit to create at the application stage is informative about their realized post-subsidy employment trajectory once funding is awarded. To assess this, we compare the counterfactual employment level implied by baseline employment plus promised jobs to actual observed employment at winning and non-winning firms. We do so four and five years after the application, using the log of baseline employment plus promised jobs minus log actual employment at t^*+4 and t^*+5. If promised jobs are a meaningful proxy for realized employment expansion, then among subsidized firms the discrepancy between promised and actual employment should be smaller than among firms that applied but did not receive funding.

We test this prediction using the same regression discontinuity design employed in the main analysis. Figure A.42 shows a clear downward discontinuity at the funding cutoff at both horizons: the gap is 16.0 log points smaller at winning firms at t^*+4 (s.e. 4.0) and 15.6 at t^*+5 (4.2). This implies that winning firms exhibit a significantly smaller gap between promised and realized employment than losing firms, as one would expect if promised jobs contain information about the employment response induced by the subsidy. Put differently, the firms that receive the subsidy appear substantially closer to delivering the employment expansion embedded in their application, which supports our use of promised jobs to construct counterfactual post-treatment employment in the simulation exercise.

As an additional check on the validity of the design, Figure A.43 shows that baseline log employment does not display any significant discontinuity at the funding threshold, consistent with smooth sorting around the cutoff. Table A.1 reports the jump at the cutoff in each component of the project score under the three implementations of Table 1; no component is significant in more than one.

We complement this validation with two further pieces of evidence. First, Figure A.44 shows the number of jobs firms pledged. Winners' projects promise somewhat more jobs: 6.2 log points (s.e. 2.9) with a quadratic and 6.7 (3.7) with a local linear fit, but 3.1 (2.2), not significant, with a linear fit over the whole range. Second, Figure A.45 shows no discontinuity in the subsidy requested per promised job, 0.011 (s.e. 0.043), and the jobs promised per euro invested, one of the scored components, is balanced as well (Table A.1). Any difference in

the size of the pledges is too small to account for the employment effect: measured against each firm's own promise, winners close a gap of 16 log points, more than twice the largest estimate of the difference in what they promised.

Overall, these validation exercises provide support for the simulation strategy used in the paper. Promised jobs are not simply cheap talk: conditional on being near the cutoff, they are informative about subsequent employment performance, and they become more tightly aligned with realized outcomes precisely for firms that obtain the subsidy. This makes them a reasonable and empirically grounded ingredient for constructing counterfactual employment paths in the permutation-based analysis.

B.3.2 An alternative baseline: predicted endpoint employment

The shock in (15) adds the promised jobs to employment in 1995. An alternative adds them to the employment each market would have reached at the endpoint without the program. We predict endpoint employment of incumbent firms with a Poisson pseudo-maximum-likelihood (PPML) regression, with an offset for log employment in 1995, covariates dated 1995 or earlier, a polynomial in firm size, and additive province and sector fixed effects; the ratio of predicted to baseline employment is winsorized. The predicted baseline B_m replaces $E_{m,1995}$ in the shock, and recentering proceeds as in the main text, $\tilde{Z}_m = \log(B_m + \sum_{f \in m} L_f g_f) - \frac{1}{S}\sum_s \log(B_m + \sum_{f \in m} L_f g_f^{(s)})$.

The prediction must not contain market fixed effects. With the 1995 level fixed by the offset, a province×sector fixed effect absorbs each market's realized growth from 1995 to the endpoint: the PPML first-order conditions set the residuals to zero within each fixed-effect group, so the fitted value reproduces the market's realized endpoint employment, $\hat{E}_{m,T} \approx E_{m,1995}\, e^{\hat{\alpha}_m} \approx E_{m,T}$. The baseline then contains the endogenous regressor, including the employment of the firms that never applied, and the instrument inherits it. In that specification the wage rate of firms that never applied appeared to rise by 0.09 log points ($p < 0.05$). The coefficient reflects the construction of the baseline, not a wage response, and it disappears once the market fixed effects are replaced by additive province and sector fixed effects.

With the predicted baseline, at the 2001 endpoint and the six-year horizon, a 10 percent increase in market employment raises employment at firms that never applied by 2.4 percent ($\pi_\ell = 0.236$, $p < 0.01$) and their wage bill by 2.8 percent (0.276, $p < 0.01$), while the wage rate does not move (0.040, not significant). The first stage is strong ($F = 31.6$). The wage-bill effect is the employment effect, $0.276 \approx 0.236 + 0.040$, and the market-level instrument built

on the predicted baseline is correlated at 0.99 with the one built on 1995 employment. Both baselines give the same answer: market employment rises and the wage does not.

B.3.3 The timing of the shock, and the wage at every horizon

The size of π_ℓ depends on aligning the window over which the shock is measured with the window over which the response is measured. On a single calendar window, the shock a market has received by 2001 includes disbursements made late in the window, whose employment response has not yet arrived, and this dilutes the estimate: on the calendar clock, $\pi_\ell = 0.149$ (0.067), against 0.241 (0.095) in the headline specification. With the endpoint at 2007, the year-by-year path has the same shape, with $\pi_\ell = 0.198$ ($p < 0.05$) at the six-year horizon. The wage row is zero on every clock and at every horizon: estimated on firms that never applied, π_w is 0.021, 0.039 and 0.067 at the 2001, 2003 and 2007 endpoints, none of them significant. The wage bill of these firms rises with their employment (Figure A.39).

We do not separate the channels through which the program's demand reached other local firms-purchases from the funded firms, spending by newly employed workers, agglomeration- and the welfare accounting does not need them, since under rationing each one raises employment at other firms without raising their wage.

C A model of place-based subsidies under a uniform wage floor

This appendix states the model behind Section 2 and derives its results. The core is the environment of Section 2: a firm that pays a contractual wage, has more willing applicants than it employs, and chooses capital, headcount and weeks (Sections C.1–C.3). Everything else is labelled as an extension or a benchmark and claims less: a separate posting benchmark in which firms set wages and weeks, which describes the regimes and the flexible-wage benchmark but does not derive the queue (Section C.4); and spillovers onto other firms, which rest on that benchmark for the flexible-wage case (Section C.5). It proves Propositions 1, 2 and 3 of Section 2.2. The model does not need to explain every empirical margin; it needs welfare terms that correspond to the quantities we estimate.

C.1 The core environment

A labor market contains a mass of workers and a finite set of firms j. Time is one year.

Assumption 1 (Preferences)**.** Worker i has a time endowment of $\bar{h}$ weeks and quasi-linear utility. Working h weeks costs her $\psi_i(h)$, the value of the non-market time she gives up, with $\psi_i(0) = 0$, $\psi_i' > 0$, $\psi_i'' \geq 0$ and $\psi_i'(h) \to \infty$ as $h \to \bar{h}$, so that desired weeks are interior. If not employed she receives the transfer b_i. Her private reservation weekly wage for a job offering h weeks is $\zeta_i^R(h) \equiv (\psi_i(h) + b_i)/h$, which survey reservation wages measure; the resource cost of the same weeks is $\psi_i(h)$, and the cost of one more week is the marginal cost $\psi_i'(h)$.

Assumption 2 (Technology)**.** Firm j produces $F\big(K_j,\, n_j\, g(h_j)\big)$ with capital K_j and effective labor $L_j = n_j\, g(h_j)$, where n_j is headcount and h_j weeks per worker. F is increasing and strictly concave, with $F_{KL} > 0$; $g' > 0$, $g'' < 0$. Expanding headcount from its predetermined level n_{0j} costs $\kappa(n_j - n_{0j})$, with $\kappa' > 0$, $\kappa'' \geq 0$. Output sells at a fixed price, the numeraire. The subsidy pays a share σ of the user cost of capital ρ. At the optimum, capital is interior and the Hessian of F in (K, L) is negative definite.

The concavity of g has a CES interpretation: with $g(h) = h^\eta$, $\eta \in (0, 1)$, the composite $n\, g(h)$ is a monotone transform of the CES aggregator $\big(\int_0^n h_i^\eta\, di\big)^{1/\eta}$ over workers' weeks, so that weeks are imperfect substitutes across workers and the firm values spreading them over more people. Strict concavity of F requires a departure from constant returns in (K, L) jointly: a

fixed factor (managerial span, land, organizational capacity). Concentrating out capital,

$$R(L;\sigma) \;\equiv\; \max_{K}\big\{F(K,L)-(1-\sigma)\rho K\big\}, \qquad R_L = F_L, \qquad R_{L\sigma} = \frac{\rho\, F_{KL}}{-F_{KK}} > 0, \qquad R_{LL} = \frac{F_{KK}F_{LL} - F_{KL}^2}{F_{KK}}$$

where the sign of R_{LL} uses the negative-definite Hessian, so the subsidy raises the marginal product of labor at a given labor input. R is output net of *privately paid* capital costs, not output; the welfare accounting below therefore keeps capital explicit.

Assumption 3 (The institution)**.** The national sectoral contract requires $w_j \geq \bar{w}$ per week, uniformly across space, and $d\bar{w}/d\sigma = 0$. Firms post a wage and a number of weeks and commit to them within the year, but cannot commit to multi-year wage paths.

Assumption 4 (Excess applicants)**.** Firm j pays $\bar{w}$ and its pool of willing applicants at the posted package, $\mathcal{A}_j(\bar{w}, h_j)$, exceeds its employment, $n_j < \mathcal{A}_j$, over the policy change considered. We treat $\mathcal{A}_j$ as a reduced-form pool of workers willing to take the job; we do not model how applicants are allocated across rationed jobs, their probability of being hired, or what they do after a rejection.

Assumption 5 (Underemployment)**.** At the firm's chosen weeks, employed workers would work more weeks at the contractual wage on average: $m_j \equiv \bar{\psi}'_j(h_j)/\bar{w} < 1$, where $\bar{\psi}'_j$ is the average of $\psi'_i(h_j)$ over the firm's workers.

Assumption 5 is not implied by the others. A worker can strictly prefer a job to non-employment, which is the participation condition $\bar{w} \geq \zeta_i^R(h)$, and still not want another week at its wage, which requires $\psi'_i(h) < \bar{w}$. Excess applicants therefore do not by themselves imply that those hired are underemployed. We impose the condition and calibrate m rather than derive it: Lachowska et al. (2026) estimate $m \approx 0.5$–0.6 for workers whose hours are set by employers.

C.2 The firm in the demand-constrained regime

Under Assumption 4 the applicant constraint is slack and the firm solves

$$\max_{K,n,h}\; F\big(K,\, n\, g(h)\big) - (1-\sigma)\rho K - \bar{w}\, nh - \kappa(n - n_0),$$

with first-order conditions

$$\text{(capital)}\;\; F_K = (1-\sigma)\rho, \qquad \text{(headcount)}\;\; F_L\, g(h) = \bar{w}h + \kappa'(n-n_0), \qquad \text{(weeks)}\;\; F_L\, g'(h) = \bar{w}. \tag{26}$$

Proof of Proposition 1. Dividing the headcount and weeks conditions in (26),

$$H(h) \;\equiv\; \frac{g(h)}{g'(h)} - h \;=\; \frac{\kappa'(n-n_0)}{\bar{w}}, \qquad H'(h) = -\frac{g(h)\,g''(h)}{g'(h)^2} > 0. \tag{27}$$

Consider an interior optimum at which Assumption 4 holds and $h < \bar{h}$. (i) By Assumption 4 the firm pays $\bar{w}$ and has applicants to spare, so the wage does not respond, $\delta_w = 0$. Differentiating (27), along the optimum $dh/dn = \kappa''/\big(\bar{w}H'(h)\big) \geq 0$, so headcount and weeks move together. (ii) Write $h = h(n)$ for the locus (27) and substitute into the weeks condition, with capital concentrated out: $\Phi(n,\sigma) \equiv R_L\big(n\,g(h(n));\sigma\big)\,g'(h(n)) - \bar{w} = 0$. Then $\Phi_\sigma = R_{L\sigma}\,g' > 0$ and $\Phi_n = R_{LL}\,\big(g + n\,g'\,h'\big)\,g' + R_L\,g''\,h' < 0$, since $R_{LL} < 0$, $g'' < 0$ and $h' \geq 0$. Hence $dn/d\sigma = -\Phi_\sigma/\Phi_n > 0$. (iii) $dh/d\sigma = h'(n)\,dn/d\sigma \geq 0$, strictly if $\kappa'' > 0$. With $\kappa'' = 0$ the right side of (27) is constant, so h is fixed and the adjustment falls entirely on headcount. ■

Proposition 1 is static. Convex hiring costs explain why weeks can respond alongside headcount, and part (iv) means that, within this specification, a weeks response requires increasing marginal hiring costs; other technologies or constraints could also produce one. The model does not say which margin moves first over calendar time, and it does not model vacancies, quits, layoffs, firm exit, or which incumbents gain the most weeks; reading the timing of the estimates through it is an interpretation, not a prediction.

C.3 Welfare and incidence

Index workers by the order in which they obtain employment as the firm expands, so that at employment n the employed are $i \in [0,n]$, and assume that this allocation is locally nested. Aggregate opportunity cost is $C(n,h) = \int_0^n \psi_i(h)\,di$, with $C_n \equiv \partial C/\partial n = \psi_{i=n}(h)$, the cost to the n-th worker, and $C_h = \int_0^n \psi_i'(h)\,di = n\bar{\psi}'(h)$. With quasi-linear utility and equal welfare weights, wages, the grant and benefits are transfers. Let $G = \sigma\rho K$ be the gross outlay and λ the excess burden per euro raised, the marginal cost of public funds minus one; the euro itself is a transfer. The financing cost here depends on gross expenditure only; benefits no longer paid and taxes on new earnings reduce the net financing requirement, and the empirical fiscal accounting nets them in the same way. Social surplus is

$$W \;=\; F\big(K,\, n\,g(h)\big) - \rho K - \kappa(n-n_0) - C(n,h) - \lambda G. \tag{28}$$

Capital enters at its full resource cost ρ; only the firm's private cost is reduced by the subsidy.

Proof of Proposition 2. Social surplus (28) is equation (5). Differentiating it,

$$\frac{dW}{d\sigma} = (F_K - \rho)\frac{dK}{d\sigma} + \big(F_L g - \kappa' - C_n\big)\frac{dn}{d\sigma} + \big(F_L\, n\, g' - C_h\big)\frac{dh}{d\sigma} - \lambda\frac{dG}{d\sigma}.$$

By (26), $F_K - \rho = -\sigma\rho$, $F_L g - \kappa' = \bar{w}h$ and $F_L\, n\, g' = n\bar{w}$. Substituting, with $\bar{\zeta}_{\text{entrant}} = C_n/h = \psi_{i=n}(h)/h$ and $\bar{\zeta}_{\text{week}} = C_h/n = \bar{\psi}'(h)$, gives equation (6). There is no separate term in $d\sigma$: the subsidy changes surplus only through the allocation, because it lowers the private, not the social, cost of capital. Under Assumption 5 the weeks term equals $n(1-m)\bar{w}\, dh/d\sigma > 0$; the employment term is positive when the worker hired strictly prefers the job. ■

The proof uses no indifference condition and no taste shocks: the labor terms are positive because the floor holds the wage above the opportunity cost of the workers hired and of the weeks added. In the undistorted competitive benchmark, with no benefits forgone and no labor taxes, the headcount and weeks conditions equate the value of output with $\psi_{i=n}(h)$ and $\bar{\psi}'(h)$, and both labor terms vanish; only the two cost terms remain. With benefits or labor taxes, private reservation pay differs from the resource cost of work even where wages clear the market, and the zero-wedge result requires switching those fiscal wedges off. The decomposition is local; a finite policy change requires integrating it along the policy path, and the capital term is first order in the subsidy rate, which for IDA is not small.

C.4 The posting benchmark and the regimes

This section sets out a separate posting benchmark, which describes the regimes conditional on the applicant pool $\mathcal{A}(w,h)$, the mass of workers willing to take a package. The pool describes workers' preferences over offered packages, not their probability of being hired, so the benchmark does not derive the equilibrium queue.

Lemma 1 (Regimes)**.** *Let the firm choose (w,h,n,K) subject to $w \geq \bar{w}$ (multiplier $\nu \geq 0$) and $n \leq \mathcal{A}(w,h)$ (multiplier $\mu \geq 0$). The headcount and weeks conditions are*

$$F_L g - wh - \kappa' - \mu = 0, \qquad n\big(F_L g' - w\big) + \mu\,\mathcal{A}_h = 0.$$

We define the regimes by the constraints, not the multipliers: (R1) $w > \bar{w}$ and $n = \mathcal{A}$: the floor is slack and the firm employs its whole pool; (R2) $w = \bar{w}$ and $n = \mathcal{A}$: the firm pays $\bar{w}$ and employs everyone willing to work at its package; (R3) $w = \bar{w}$ and $n < \mathcal{A}$: the firm pays $\bar{w}$ and has excess applicants, $q_j = \mathcal{A}_j - n_j > 0$. Away from the boundaries between regimes, R1 has $\nu = 0 < \mu$, R2 has $\nu, \mu > 0$, and R3 has $\mu = 0 < \nu$, so that in R3 the conditions reduce to (26); at a boundary a multiplier can be zero while its constraint binds. In R2 a floor

binds without rationing; in R2 the firm may also recruit by raising weeks beyond the point where $F_L g' = \bar{w}$, since $\mu \mathcal{A}_h > 0$.

Proof. Kuhn–Tucker conditions of the firm's problem. In R3 the applicant constraint holds with strict inequality, so complementary slackness gives $\mu = 0$, which removes the applicant term from both conditions; they then coincide with (26). ■

Lower productivity makes R3 more likely at a given national floor, which is why the same contract can leave a queue in lagging regions and be slack in productive ones. It does not assign regions to regimes mechanically: amenities, outside options, market structure and hiring costs also shift $\mathcal{A}_j$ and the firm's demand. The empirical counterparts are the share of firms in targeted areas reporting that the national contract sets their whole wage bill (the median firm reports 100%; Section 4.3) and an entry cushion close to zero; the median new hire is paid 1.17 times the floor (Appendix Table A.13) because the median mixes occupational grades, and at the blue-collar hiring margin the 25th percentile is 0.955 times the floor.

C.5 Extensions

The results in this section are conditional statements or illustrations. Each adds assumptions to the core, which we state, and they are not derived predictions of the core.

Corollary 1 (Illustration: magnitudes under constant-elasticity supply)**.** Suppose that supply to the firm has a constant elasticity ε, that weeks are fixed, that hiring costs are zero at the margin, and that, with capital concentrated out, output net of capital costs is $A(\sigma)\,\ell^{\beta}$ with $\beta < 1$ and $d\ln A/d\sigma = a > 0$. Then

$$\text{(R3):}\quad \frac{d\ln w}{d\sigma} = 0, \quad \frac{d\ln \ell}{d\sigma} = \frac{a}{1-\beta}; \qquad \text{(R1):}\quad \frac{d\ln w}{d\sigma} = \frac{a}{1+\varepsilon(1-\beta)}, \quad \frac{d\ln \ell}{d\sigma} = \frac{\varepsilon a}{1+\varepsilon(1-\beta)},$$

so the employment response is larger, and the wage response smaller, where the floor binds. These expressions illustrate the contrast under a deliberately simple supply curve.

Proof. R3: differentiate $\beta A \ell^{\beta-1} = \bar{w}$. R1: substitute $\ell \propto w^{\varepsilon}$ into the markdown condition $\beta A \ell^{\beta-1} = w(1 + 1/\varepsilon)$ and differentiate. ■

Proof of Proposition 3. Let k be an untreated firm in the treated firm's market. If the treated firm expands, raising its weeks or the market's inclusive value, $\mathcal{A}_k$ can fall. (i) If

k is in R3 and remains strictly demand-constrained after its pool changes-the fall in $\mathcal{A}_k$ is smaller than its own queue $\mathcal{A}_k - n_k$-complementary slackness keeps its applicant multiplier at zero (Lemma 1), so its choices solve (26), which do not involve $\mathcal{A}_k$: its employment and wage do not respond through the labor market. The condition concerns k's own queue, not only the aggregate queue. (ii) If k is in R1, the same expansion lowers its pool; in the constant-elasticity benchmark of Corollary 1, and absent other channels, k then raises its wage and lowers its employment. These signs are predictions of that benchmark, not general implications of the posting model. ■

Beyond Proposition 3, product-market and agglomeration linkages shift k's demand and enter additively in both regimes. They are not implied by the floor, and they can be positive (demand for k's output, spending by newly employed workers) or negative (business stealing).

The pair $(\pi_w = 0, \pi_\ell > 0)$ is therefore informative against a clearing market with finite supply elasticity, in which $\pi_\ell > 0$ would require k to bid for workers and so $\pi_w > 0$; it is not a universal test of the regime, and given the regime the positive π_ℓ says that net linkages are positive.

C.6 Remarks

Remark 1 (Conduct)**.** In R3 the core results do not depend on the posting protocol: employment is read off (26), which bargaining with a binding floor, oligopsony with a binding floor and perfect competition with a minimum wage all deliver once the firm pays $\bar{w}$ and has excess applicants. Conduct matters for the R1 predictions and for the counterfactual against which the floor itself is judged, which this paper does not evaluate.

C.7 Predictions and estimates

Table C.1 collects four coefficients, what the posting benchmark (R1) and the demand-constrained regime (R3) predict for each, and what we estimate. The wage responses are informative about the regime relative to these alternatives; given the regime, the positive response of employment at competitors additionally says that net demand and agglomeration linkages are positive.

Table C.1: The four coefficients: what each regime predicts, and what we estimate

Coefficient	Object	Flexible wage	Binding floor	Estimate
δ_ℓ	Employment at funded firms	+, dampened	**+, full**	0.126^{***} (0.028)
δ_w	Wage rate at funded firms	> 0	**0**	0.000 (n.s., every horizon)
π_ℓ	Employment at competitors	− (poaching)	≥ 0	0.241^{**} (0.095)
π_w	Wage rate at competitors	> 0	**0**	0.014 (0.031)

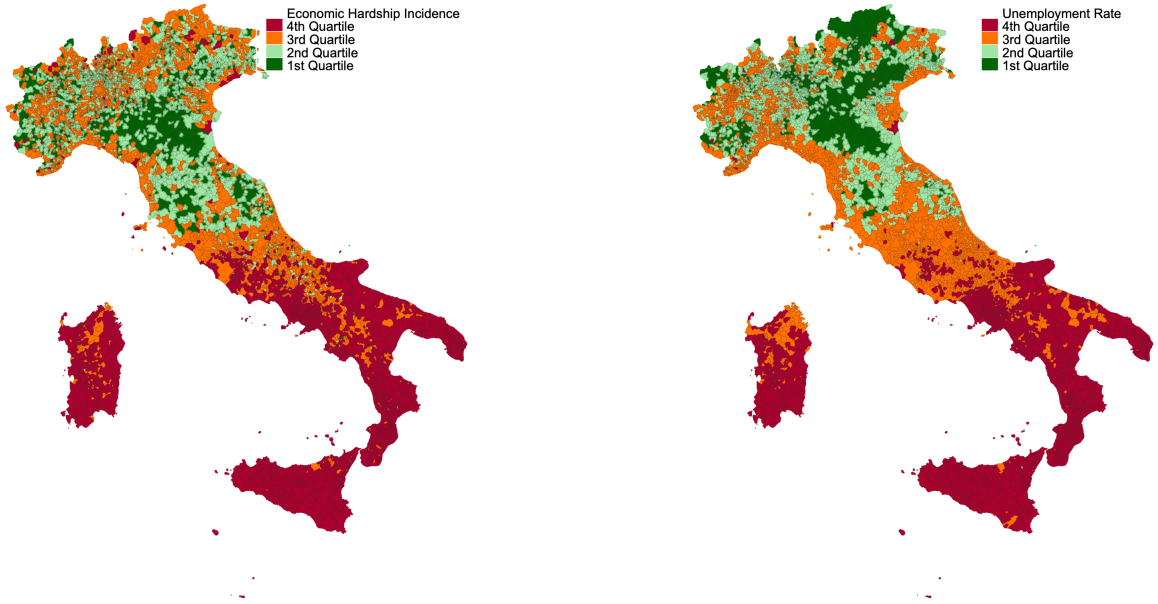
Notes: Predictions from Corollary 1, Proposition 1 and Proposition 3; estimates from the pooled two-stage least squares design of Section 5 (Section 5.3), on each firm's own three-year window. δ_w is the incumbent *weekly wage rate* from the worker-level decomposition of Section 4.2-the object the model speaks to-rather than the firm-average wage, which mixes in composition; the firm-level average-pay response is 0.011^{*} (0.006). The six-year direct effect for the earliest cohort is 0.396^{***} (0.092) with $\pi = 0.149^{**}$ (0.067) on the calendar clock. Standard errors cluster on the 1995 province × industry market. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

D Context

D.1 The Italian Regional Disparities

Labor market dualism. Italy's regional policy debates have historically centered on the lagging southern regions (*Mezzogiorno*), comprising Abruzzo, Molise, Campania, Apulia, Basilicata, Calabria, Sicily, and Sardinia. In contrast, the Center-North includes Valle d'Aosta, Piedmont, Lombardy, Liguria, Veneto, Trentino-Alto Adige, Friuli-Venezia Giulia, Emilia-Romagna, Tuscany, Umbria, Marche, and Lazio. Italian regions are administratively comparable to US states: in 1995, Italy had a population of approximately 56.84 million, with an average regional population of 2.84 million inhabitants. Figure D.1 illustrates these structural imbalances using municipal and provincial census records from 1991. The spatial distribution of economic hardship (left panel) and unemployment rates (right panel) shows a sharp territorial divide: Northern regions generally exhibit low hardship and low unemployment rates (first and second quartiles, in green), whereas the South exhibits deep, widespread economic hardship and double-digit unemployment (fourth quartile, in red).

Figure D.1: Spatial Distribution of Hardship and Unemployment at Baseline (1991)



(i) Economic Hardship Index

(ii) Unemployment Rate

A defining feature of southern labor markets is the extreme persistence of joblessness. Table D.1 uses the European Labour Force Survey to compare unemployment dynamics across Italian

macro-regions and peer European economies. Between 1999 and 2005, the unemployment rate in the South averaged 18.3%, compared to 5.3% in the Center-North. Furthermore, 64.0% of all unemployed southern workers had been actively seeking work for a year or more. Long-term unemployment alone accounted for 11.8% of the southern labor force-five times the rate of the Center-North (2.4%), more than double the rates in Germany (4.7%) and Spain (4.3%), and ten times the rate in Denmark (1.1%). In Italy as a whole, 40% of unemployed workers with reported search durations had been unemployed for at least two years, and 23% for four years or more (against 21% and 6% in France). This structural slack persisted long after our sample window: in 2023, the southern long-term unemployment rate remained at 9.1% (Schoefer, 2025). In an economy with rigid wage floors, these protracted spells reflect labor market rationing: workers queue for jobs at statutory contractual wages rather than bidding wages down.

Table D.1: Long-Term Unemployment: Southern Italy vs. European Benchmarks

	1999–2005 average			2023		
	Unemp.	Long-term		Unemp.	Long-term	
	rate	share	rate	rate	share	rate
Italy, South (covered areas)	18.3	64.0	11.8	14.0	64.7	9.1
Italy, Centre-North	5.3	45.2	2.4	5.1	43.8	2.2
Italy (Total)	9.4	56.7	5.4	7.7	54.8	4.2
Germany	9.3	50.1	4.7	3.1	31.1	1.0
France	9.7	39.3	3.8	7.3	24.5	1.8
Spain	11.8	35.6	4.3	12.2	35.0	4.3
Denmark	4.8	22.1	1.1	5.1	10.7	0.5
EU-27	9.7	46.9	4.5	6.1	35.0	2.1

Notes: Percentages, population aged 15–74. Long-term unemployment is defined as continuous job search of 12 months or more. Long-term share is the share of the unemployed; long-term rate is the share of the active labor force. South encompasses NUTS-1 regions Sud and Isole; Centre-North encompasses Nord-Ovest, Nord-Est, and Centro. Source: Eurostat (EU Labour Force Survey).

Internal migration, housing, and spatial equilibrium. Despite substantial earnings and unemployment differentials, labor mobility between the South and the Center-North remained remarkably limited during our sample period. In 1997, net inter-regional migration from the South to the Center-North was approximately 95,000 individuals, representing less than 0.5% of the southern population (Caponi, 2008; Ciani et al., 2019). Even when workers change employers, most voluntary job-to-job moves stay within the same province (64%; Appendix D.4).

Other papers show that, in this period, Italian labor markets absorbed employment shocks with minimal population adjustment. Between 1971 and 2001, local employment creation generated an estimated population elasticity of 0.46, substantially below estimates for the United States (0.85) and France (0.50–0.80). Consequently, local labor demand shifts are predominantly absorbed by resident workers rather than inward migrants. Two pricing margins explain why spatial arbitrage fails to equalize economic conditions. First, local nominal wages exhibit near-zero responsiveness to local labor demand shocks; in the 1990s, the elasticity of southern wages to local shocks was 0.14 and statistically indistinguishable from zero (Ciani et al., 2019). Second, housing markets generate a powerful wedge: local demand shocks capitalize into housing costs at higher rates in the Center-North (elasticity of 1.8) than in the South (0.6) (Ciani et al., 2019). Accounting for local consumer price and housing indices, real purchasing-power wages are estimated to be 9.2% *higher* in the South than in the North (Boeri et al., 2021). A southern worker contemplating relocation faces identical nominal contractual wage floors combined with substantially higher northern rental costs, dampening the incentives for inter-regional migration.

D.2 Institutional Collective Bargaining Details

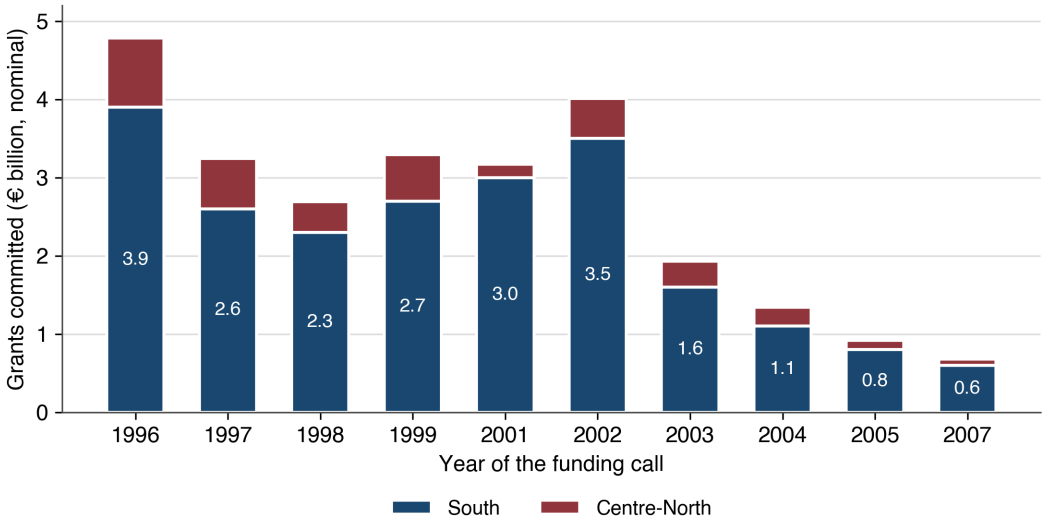
National collective bargaining agreements (*contratti collettivi nazionali di lavoro*, CCNL) cover 97.7% of dependent employment and 99.3% of non-agricultural firms in Italy (Boeri et al., 2021). Among enterprises with ten or more employees, coverage reaches 98%, contrasting with 45% in Germany (Bovini et al., 2026). As of 2021, 346 distinct national agreements were active across sectors. National contracts define salary grids based on monthly earnings and standard weekly hours. For full-time employees, the contract establishes both the base monthly wage and the implied hourly rate. The territorial uniformity of these minima is an institutional policy choice: territorial wage zones (*gabbie salariali*), which indexed wages to local living costs in the postwar period, were phased out following the 1968–1969 national agreements and formally eliminated by 1972. National contract negotiations take place on staggered, multi-year cycles uncoupled from local business cycles. Contract expirations are routinely followed by extended renegotiation intervals, during which nominal wage floors remain legally frozen at expired rates. Between 2000 and 2004, the contractual minima of the fifty largest national agreements were updated on average once every eleven months, and 61% of renewals took place at least three months after agreement expiration (Bovini et al., 2026). Because national employer confederations and labor unions are headquartered in northern manufacturing districts—two-thirds of agreements are executed in Lombardy, Emilia-Romagna, Veneto, and Piedmont, while less than 10% are executed in the South—statutory

minima track northern labor productivity. This produces marked geographic decoupling. For example, in the national construction contract, nominal floors do not vary across provinces, and hourly labor costs exhibit a spatial coefficient of variation of only 2.5%. Across Italian provinces, the elasticity of wages to local value added is 0.20 (versus 0.74 in Germany), while the elasticity of the non-employment rate to local value added is −1.43 (versus −0.25 in Germany) (Boeri et al., 2021). Under rigid wage floors and strict dismissal protections, cost adjustments occur along the employment margin. When Italian firms outsource production units, transferred workers incur a 22% drop in annual earnings, four-fifths of which is driven by lost weeks of work rather than daily wage reductions, with many workers shifting into agreements with lower contractual minima (Daruich et al., 2024).

D.3 The IDA Program

Funding over time. Appendix Figure D.2 shows the grants committed in each funding call. Of the €26 billion committed, about €22 billion (85 percent) went to the South, most of it in the calls up to 2002, about €2.6 billion a year; commitments then fell below €1 billion a year. About a quarter of commitments was later withheld for non-completion, so payments to the South amounted to roughly €16–17 billion.

Figure D.2: IDA Grants Committed by Funding Call



Notes: Grants committed in each funding call, in billions of nominal euros, by the region of the funded project. Totals: €26.0 billion, of which €22.0 billion in the South and €4.0 billion in the Centre-North. There was no call in 2000.

D.3.1 Policy Framework and Eligibility

European Cohesion Policy and US place-based policy. The IDA program served as Italy's primary instrument for executing European Cohesion Policy, which aims to reduce regional disparities across member states under the Single European Act (1986). Cohesion funding is disbursed through structural funds, primarily the European Regional Development Fund (ERDF) and the European Social Fund (ESF). Objective 1 status targeted regions with GDP per capita below 75% of the EU average, while Objective 2 targeted areas undergoing industrial restructuring characterized by high unemployment and sector-specific employment contraction. Together with transitional support zones, eligible regions encompassed 48% of the Italian population. The architecture of IDA mirrors prominent place-based interventions in the United States. Similar to the Opportunity Zones initiative established under the 2017 Tax Cuts and Jobs Act and the manufacturing incentives under the 2022 CHIPS and Science Act, IDA was structured around incentivizing private physical capital formation rather than underwriting recurring labor expenses. In its competitive scoring auction design, IDA closely resembles the Small Business Innovation Research (SBIR) grant competitions evaluated by Howell (2017); Howell and Brown (2023). Across both institutional settings, public subsidies are predicated on the premise that capital market frictions depress local investment in lagging regions, raising the empirical question of whether investment subsidies generate durable employment gains or pass through into local rents (Slattery and Zidar, 2020; Kline and Moretti, 2014a; Busso et al., 2013).

Eligible investment categories. Firms submitted investment projects under distinct administrative categories:

1. **New Establishments:** Construction of new, autonomous manufacturing plants or operating facilities.
2. **Capacity Expansions:** Expansion of production capacity for existing product lines (horizontal expansion) or integration of upstream and downstream production phases within an existing facility (vertical expansion).
3. **Modernization:** Installation of innovative technologies to enhance process efficiency or environmental compliance.
4. **Restructuring and Rationalization:** Organizational and technological restructuring of active production processes.
5. **Product Conversion:** Reconfiguring production lines to shift output toward different industrial sub-sectors.
6. **Reactivations and Transfers:** Resuming operations at idle production facilities

or relocating facilities mandated by environmental, zoning, or public administration directives.

Maximum subsidy schedules set by the European Commission. State-aid regulations enforced by the European Commission established statutory caps on subsidy intensities based on regional economic disadvantage and establishment size. Tables D.2, D.3, and D.4 detail the maximum subsidy schedules across regions, time periods, and firm size classes.

Table D.2: Maximum Allowable Subsidy Rates: Objective 1 Areas (1995–1999)

Territorial Areas	**Small-Medium**	**Large**
Provinces of Benevento, Potenza, Catanzaro, Cosenza, Crotone, Reggio Calabria, Vibo Valentia, Agrigento, Caltanissetta, Enna, Messina, Trapani, Nuoro, Oristano	65%	50%
Provinces of Avellino, Caserta, Napoli, Salerno, Matera, Bari, Brindisi, Foggia, Lecce, Taranto, Catania, Palermo, Ragusa, Siracusa, Cagliari, Sassari	55%	40%
Abruzzo	30%	25%
Molise (prior to 30/06/1995)	55%	40%
Molise (01/07/1995–31/12/1996)	45%	35%
Molise (01/01/1997–31/12/1998)	40%	30%
Molise (01/01/1999 onward)	30%	25%

Table D.3: Maximum Allowable Subsidy Rates: Non-Objective 1 Areas (1995–1999)

Designation Category	**Small Enterprises**	**Medium-Large Enterprises**
Objective 2 and 5b (derogation Art. 92.3.c)	20%	15% (medium), 10% (large)
Other Objective 2 and 5b Areas	15%	7.5%
Non-Objective Areas (derogation Art. 92.3.c)	20%	15% (medium), 10% (large)

Table D.4: Maximum Allowable Subsidy Rates: 2000–2006 Regulatory Cycle

Macro-Region	Statutory Classification	Small	Medium	Large
Southern Regions	Art. 87.3.a (Calabria)	60%	60%	50%
	Art. 87.3.a (Other southern areas)	45%	45%	35%
Molise	Art. 87.3.c	27%	27%	20%
	Non-derogation areas	10%	5%	–
Abruzzo	Art. 87.3.c	27%	27%	20%
	Non-derogation areas	10%	5%	–
Centre-North	Art. 87.3.c	15%	12%	8%
	Non-derogation areas	10%	5%	–

Example. Statutory ring-fencing required at least 50% of regional budgets to be allocated to small and medium enterprises, capped service-sector allocations at 5%, and prioritized EU co-financed projects. Within each regional competition, projects competed within separate funding pools f defined by firm size, sector, EU co-financing, and objective designation. Table D.5 presents an illustrative comparison of two projects applying in the 1996 Sicily competition within Pool 1 (small manufacturing firms in Objective 1 areas with EU co-financing). While competing for the same pool budget, the two firms operated in distinct labor markets: an automotive component firm in Palermo and a wood products manufacturer in Catania. Project 1000 was a marginal winner ($S_j = +0.018$) with an own-capital share of 0.925 and zero promised additional employment, whereas Project 22245 was a marginal loser ($S_j = -0.003$) with an own-capital share of 0.728. Across all calls, the average pool includes applicants from 27 distinct labor markets. Because promised jobs are heavily right-skewed across projects, marginal winners near the threshold generate plausibly exogenous variation in labor demand shocks across labor markets.

Table D.5: Two Applications from the Same Competition

Competition n. 2: Sicily x 1996								
Project	**Pool**	**Province**	**Sector**	**Specialty**	$i_1 = \frac{OwnK}{Inv}$	$i_2 = \frac{Jobs}{Inv}$	$i_3 = \frac{100}{Subsidy}$	**ProjectScore**
1000	1	Palermo	Manuf.	Auto	.925	0	1.33	2.26
22245	1	Catania	Manuf.	Wood	.728	.006	1.50	2.234

D.3.2 Descriptive evidence

Funding-pool descriptives As the funding pools generate every piece of variation used below, Table D.6, Panel A, reports their anatomy. Funding pools are wide and winners are sparse: the average funding pool draws applicants from 27 distinct labor markets (median 10, up to 261), while each market hosts only a handful of winners. The allocation of funded jobs *across* markets is therefore decided inside the funding pools, by who happens to clear a pool-specific bar. Panel A of Table D.6 also describes the job bids themselves: promised jobs are right-skewed across projects, so which near-cutoff project wins moves market-level promised employment and not merely the identity of the winner. This is the raw material of the market-level design in Section 5; Appendix B.1 reports the McCrary density test and the balance of each score component at the cutoff.

Table D.6: The IDA Competitions and the Funded Projects: Descriptive Statistics

Panel A. Competitions, markets and promised jobs

	N	Mean	Median	p10	p90	Min	Max
Markets per funding pool	495	26.5	10	1	70	1	261
Applicants per funding pool	495	73.6	16	1	198	1	1,248
Winners per funding pool	495	29.1	6	0	75	0	469
Applicants per market	2,253	16.2	6	1	41	1	540
Winners per market	2,253	6.4	2	0	16	0	304
Promised jobs per market	2,253	245.1	81	6	652	0	6,286
Promised jobs per market × year	7,877	70.1	28	2	176	0	2,344
Promised jobs per market, before 2001	1,515	158.3	53	4	408	0	3,353

Panel B. Funded projects by investment type

	Subsidies		Projects	Spending	Share pledging (%)		
	Total (€M)	Mean (€)		share (%)	Jobs	Own capital	Both
New plants	6,145	616,807	9,963	51	99.3	99.1	98.5
Expansions and other	5,997	530,837	11,298	49	82.9	97.0	81.2

Panel C. Projects by firm size

	All projects		Funded projects		Funded new plants		
	N	New plants (%)	N	New plants (%)	N	Pledging jobs (%)	Manufacturing (%)
Small	57,820	54.4	23,482	54.1	9,237	99	77
Medium	7,090	21.1	3,849	20.3	508	98	79
Large	4,467	19.5	2,255	18.4	222	92	56

Notes: This table describes the IDA competitions and the projects they funded. Panel A: the distribution, across funding pools and across labor markets (1995 province × 2-digit industry), of the number of markets represented in each funding pool, of applicants and winners, and of jobs promised in project bids; promised jobs total 552,128 across markets. Promised jobs had to be new hires at the project site: workers transferred from the firm's other plants did not count. Winners promised roughly 290,000 jobs in total and delivered about half, which is why the analysis measures realized employment and uses promised jobs only to build market-level exposure. Each funding pool draws on many markets, while each market hosts few winners. Panel B: approved projects by investment type; the shares pledging jobs (the employment criterion), own capital (the capital criterion), or both are computed on the winning projects in the application records. Panel C: applications and funded projects by firm size, and funded new-plant projects with the share committing to create employment and the share in manufacturing. Of the 9,967 funded new-plant projects, small firms account for 9,237.

Descriptives on funded markets, projects Moreover, we use our dataset on subsidy applications to document three key facts: (i) the majority of subsidies are allocated to firms opening new plants, which typically promise more jobs per unit of investment; (ii) new plants are predominantly built by small firms, with notable variation across labor markets; (iii) firms prioritize specific labor market characteristics when applying for new-plant subsidies, often in ways inversely correlated with what would be socially desirable.

When linking the data to the administrative archives, we retrieve 40,683 firms, of which 17,487 (36%) received financing, covering 19,900 projects. Among these, 8,607 firms were financed specifically to open new plants (8,824 projects); more than half operate in manufacturing (4,818 firms, 4,911 projects). We retrieve 2,610 new plants opened in a new province where the firm was not active before 488 funding-half in manufacturing-appearing on average 2.6 years after the funding date. Defining a market as a province, the average number of financed projects per market is 200 (median 135); restricting to manufacturing, the mean is 129 (median 92). At baseline (1995), manufacturing markets receiving funded projects have 156 firms on average (median 57).

Table D.6, Panel B, summarizes spending patterns on funded projects. Projects for constructing new plants receive half of the spending even though they are relatively fewer. The mean grant for opening a new plant is approximately €617,000.

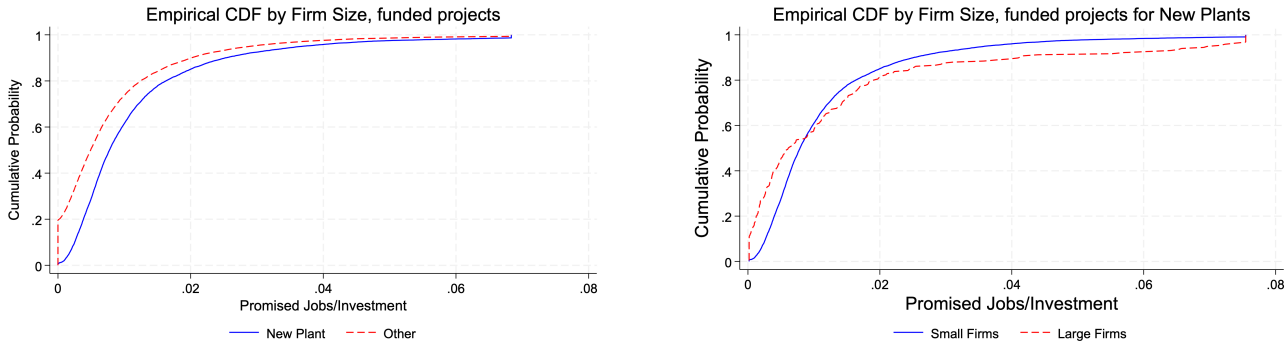
We then turn to examine the commitments in terms of employment and capital that firms make in the application process. In Table D.6, Panel B, we show that for new plant creation, nearly all firms commit to investing their own capital and creating jobs. However, this pattern does not hold for projects classified as other types, where the commitment to employment and capital investment is weaker.

In Figure D.3, panel (i), we show that new plants consistently generate more jobs per unit of investment compared to other projects. However, job creation per unit of capital varies significantly with firm size: larger firms appear less likely to create jobs relative to their investment levels (panel ii).

In Figure D.4, the left panel displays the distribution of promised jobs by investment type: new plants average 20.64 promised jobs versus 12.66 for expansions/other investments. The right panel shows total investment distributions, which are more similar across project types.

Table D.6, Panel C, shows variation in the type of submitted projects by firm size: small firms submit more new-plant projects, while medium and large firms undertake other project types. The manufacturing sector receives the largest support, and funds are concentrated in the South of Italy, with variation across municipalities.

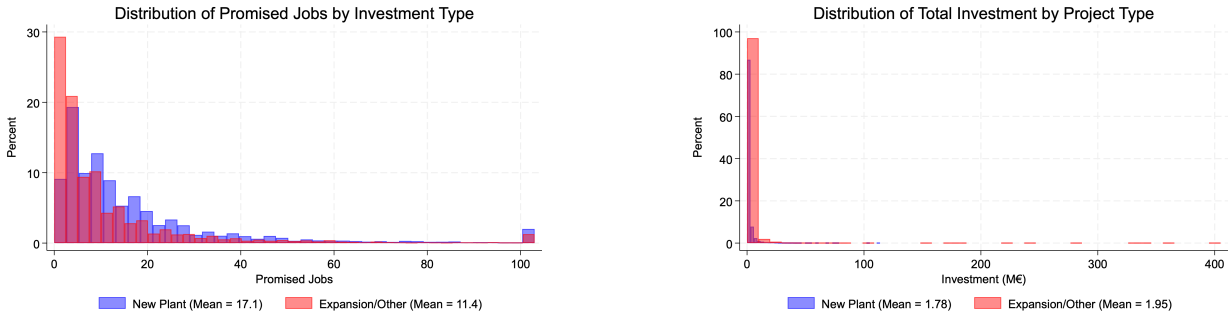
Figure D.3: New plants promise more jobs per euro invested, especially among small firms.



(i) (ii)

Notes: Empirical CDFs of promised jobs per unit of investment (left panel: by project type; right panel: by firm size for new plants only). New-plant projects consistently dominate expansions in employment intensity across the distribution. Among new plants, small firms promise more jobs per euro than medium or large firms, consistent with the program's intent to channel employment-intensive investment to lagging regions.

Figure D.4: New-plant projects target more jobs but similar investment amounts as expansions.

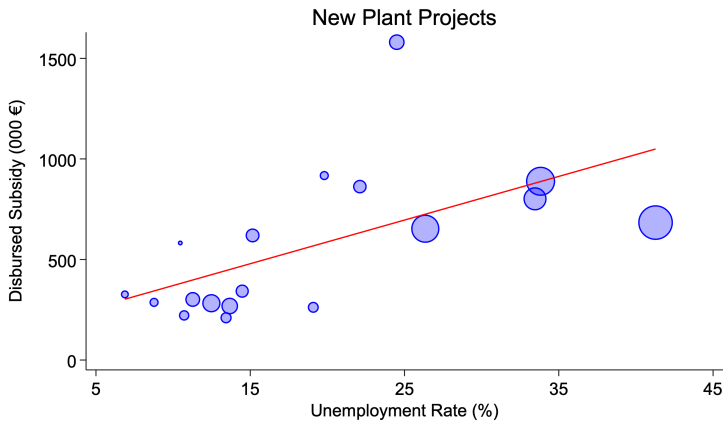


(i) (ii)

Notes: Distribution of promised job counts (left) and total investment amounts (right) for winning projects, by project type. New plants average 20.6 promised jobs versus 12.7 for expansions-a 63% gap-while the investment distributions are more similar, suggesting that the employment intensity advantage of new plants is not simply an artifact of project size.

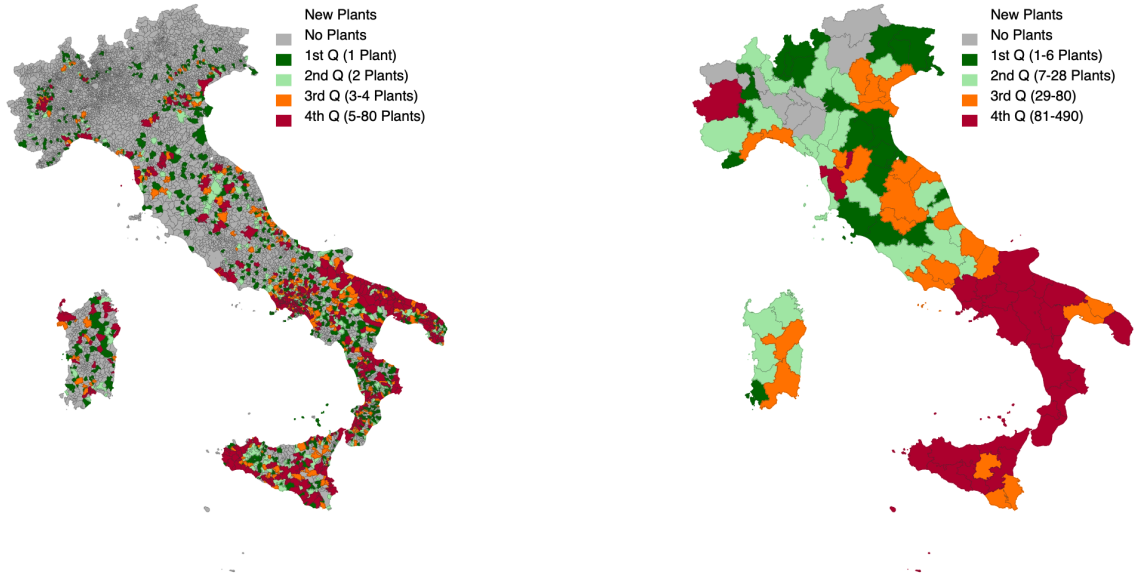
A total of 47,200 firms submitted applications (average 1.33 projects per firm). The vast majority (45,651 firms) applied in only one municipality. Figure D.5 shows that regions with higher pre-policy unemployment received larger per-plant grants and more plants overall. Figure D.6 maps the geographic distribution of funded new manufacturing plants, showing a clear concentration in the South.

Figure D.5: Higher-unemployment regions attract more new-plant projects and receive larger grants.



Notes: Binned scatter plots of mean subsidy amount (left) and number of funded new-plant projects (right) against the regional pre-policy unemployment rate. Both series slope upward, indicating that the program's spatial allocation broadly tracks the geographic distribution of labor market distress-though this is partly mechanical given that eligible areas were defined by EC Objective Area classifications based on economic indicators.

Figure D.6: IDA-funded new plants are heavily concentrated in Southern Italy.



(i) Municipality level (ii) Province level

Notes: Geographic distribution of approved new manufacturing plant projects at the municipality level (left) and province level (right). The Southern regions (Campania, Calabria, Sicily, Sardinia, Puglia, Basilicata) account for the overwhelming majority of funded plants, reflecting both the program's explicit targeting of EC Objective 1 areas and the higher pre-policy unemployment rates in the Mezzogiorno.

D.4 Data Construction and Administrative Record Linkage

INPS Matched Employer-Employee Records. We access the administrative universe of employer-employee social security records through the VisitINPS program. Reporting is mandatory for all non-agricultural private employers via UNIEMENS filings. The data cover approximately 93% of private dependent employees (74% of all private employment including the self-employed). Annual records span 1990–2013, with monthly records available beginning in 2005. A firm is defined by a unique Tax Identification Number (TIN). Within multi-establishment firms, INPS assigns distinct establishment identifiers whenever operating units conduct activities classified under different sectoral codes (for instance, an enterprise operating a gasoline service station and an adjacent food service establishment receives two separate establishment identifiers).

The worker panel and the dominant employer. We link the INPS employer–employee records to the IDA applicant firms and construct a person-year panel in which each worker i is assigned a unique *dominant employer* in each year, denoted $J_{it} \in \mathcal{J} \cup \{\emptyset\}$: the establishment associated with the worker's longest employment spell that year, so that $J_{it} = j$ means that worker i's dominant job in year t is at firm j. The panel covers 1.2 million workers who passed through an applicant establishment, and 1.3 million who passed through an applicant firm. Incumbents, stayers, leavers and new hires (Section 4.2.1) are all defined on the dominant employer.

VisitINPS Linkage Protocol and RD Design. Because of administrative privacy regulations, the record linkage between IDA application registries and INPS employer-employee files was executed using establishment identifiers, application year, and official subsidy eligibility status, without directly exporting individual continuous application scores into the confidential worker-level environment. For this reason, our worker-level analyses evaluate winners and losers adjacent to the funding cutoff using an event-study framework centered on the application year—analogous to a local-constant regression discontinuity design—rather than conditioning parametrically on the continuous running variable.

Digitization of Contractual Minima and Supplementary Sources. To measure the contractual wage floor $\underline{w}_{\ell s,t}$ applicable to each worker-year, we digitized wage schedules from national collective agreements registered with the National Council for Economics and Labor (CNEL). Each contract records the nominal monthly base wage, cost-of-living allowances (*indennità di contingenza*), and mandatory periodic increments by occupational grade and

calendar date. Matching these tables to job spells enables us to observe both the statutory minimum and the discretionary wage cushion, defined as $\text{Wage}_{it} - \underline{w}_{\ell(i)s(i),t}$. We supplement these records with Cerved balance sheets (capital stock, value added, EBITDA, sales), the Bank of Italy's Invind survey, and Istat labor force surveys.

Market definition. We follow Manning and Petrongolo (2017) in treating labor markets as local. They show that the appeal of a vacancy to a job seeker decays sharply with distance, so that each worker's effective market is small, but neighboring markets overlap and a local shock ripples beyond the area where it occurs. Two implications guide our choice. First, the market should be drawn at the scale at which workers search, which we take to be the province, the level at which Italy's public employment services were organized, and should be restricted to employers that use similar skills, which the 2-digit sector approximates. Second, because markets overlap, any bounded definition leaves some spillovers outside it; when neighboring markets also receive part of a shock, comparisons across markets with different shocks tend to understate the spillovers rather than inflate them. To assess whether province $\times$ 2-digit sector is a meaningful definition of the labor market in our data, we use matched employer-employee records to study voluntary job-to-job transitions. Specifically, we focus on male workers in permanent and temporary contracts whose jobs start between 2005 and 2008, and we construct transition matrices across provinces and 2-digit NACE sectors, optionally also considering 1-digit occupations. The idea is that a plausible market definition should capture the set of firms that workers realistically view as outside options. Reassuringly, we find that 27% of voluntary transitions occur within the same province $\times$ 2-digit sector cell, with an even higher share for workers on permanent contracts. This suggests that labor mobility is substantially concentrated within these cells, consistent with the view that province $\times$ 2-digit sector captures an economically relevant market boundary. This pattern is also in line with the evidence in Appendix Table A1 of Arnold (2019), which documents substantial within-industry and within-occupation mobility in the CPS. Taken together, these findings support our use of province $\times$ 2-digit sector pairs as the baseline market definition.

Table D.7 reports the full set of shares. Even conditional on choosing to change jobs, workers rarely leave their province: 64% of voluntary moves end in the same province, 66% for permanent contracts. In the South, about a third of voluntary moves stay within the same province $\times$ 2-digit sector cell (31% for high-skill and 33% for low-skill workers). Movers change sector far more often than they change province: only 41% stay in the same 2-digit sector. Geographic mobility is therefore limited even among the workers most likely to move, which is consistent with the absence of a detectable mobility response among workers attached

to the market in Section 5.4; that evidence follows baseline workers and does not measure arrivals from elsewhere.

Voluntary moves are also rarer in the South. Among blue-collar workers, 43% of separations are voluntary in the South against 78% in the Centre-North, and the gap between two jobs lasts 220 days against 146; among white-collar workers the shares are 54% and 82%, and the gaps 186 and 80 days. Workers in the South quit less and wait longer for the next job, as in the Labour Force Survey evidence of Section 4.3. Together with the worker-level evidence of Section 5.4, this supplies the object Gaubert et al. (2025) identify as the sufficient statistic for the efficiency cost of place-based redistribution, estimated here rather than assumed.

Table D.7: Where voluntary job-to-job movers go

	All contracts		Permanent contracts	
Share of moves to a job in the same…	Share	N	Share	N
Province	0.638	2,036,000	0.659	1,145,000
2-digit sector	0.405	2,036,000	0.467	1,145,000
3-digit sector	0.359	2,036,000	0.421	1,145,000
Province × 2-digit sector	0.270	2,038,000	0.319	1,146,000
Province × 2-digit sector, by skill				
High-skill	0.262	430,710	0.295	331,003
High-skill, South	0.308	63,455		
Low-skill	0.274	1,586,000	0.333	796,690
Low-skill, South	0.330	360,387		

Notes: Voluntary job-to-job transitions of male workers whose new job starts between 2005 and 2008, from the INPS employer-employee records as organized by Di Addario et al. (2023). Each entry is the share of moves in which the new job is in the same unit as the old one. Sample sizes rounded to the nearest thousand where above one million.

The social safety net and benefit measurement. We set out the institutional features of the Italian social safety net during our sample period, which inform our welfare accounting and fiscal externality calculations. Ordinary unemployment benefits (*indennità di disoccupazione ordinaria*) were available to workers involuntarily separated from employment who met statutory social security contribution thresholds. By international standards, ordinary unemployment benefits offered low replacement rates (approximately 30% of reference wages in the early 1990s) and brief entitlement durations. For workers experiencing temporary reductions in operating hours, the *Cassa Integrazione Guadagni* (CIG) subsidized wages without formal employment termination. Workers displaced in collective redundancy procedures qualified for mobility allowances (*indennità di mobilità*; Law 223/1991), which provided longer

entitlement durations for older workers and those in the South. Crucially, the most protective income support instruments-extraordinary wage supplementation (CIG Straordinaria) and mobility allowances-were legally restricted to firms employing more than 15 workers. In our sample of IDA applicants, the median firm employs 8 workers. As a consequence, most workers in our sample qualified solely for modest ordinary unemployment benefits, and younger labor market entrants in the southern queuing pool possessed no entitlement history. In our empirical analysis using the 13% longitudinal administrative subsample, we observe contribution-type codes that capture ordinary unemployment benefits, CIG allocations, and public transfers. We evaluate three distinct margins: benefit receipt (an indicator for positive transfers), total transfer amounts, and comprehensive worker income (earnings net of forgone transfers). Because statutory maximum durations vary by age, region, and instrument, where we analyze duration we standardize receipt spells against the applicable legal ceiling.

Informality. In southern Italian labor markets, a potential concern is whether the measured employment response to investment grants represents the formalization of previously informal workers (*emersione*) rather than genuine job creation. Three empirical patterns make this interpretation unlikely to account for the response. First, subsidy receipt generates significant increases in both headcount employment and total weeks worked among incumbent workers who were already formally registered at baseline. The regularized reporting of informal workers cannot account for intensive-margin expansions among pre-existing formal personnel. Second, tracking worker histories in the INPS database shows that approximately 80% of newly hired workers at winning establishments had no formal employment record in the preceding year. Generating the observed employment response solely through formalization would require transition rates from informality to formality orders of magnitude greater than any empirical benchmark observed during this period. Third, from a public finance and welfare accounting perspective, formalization generates positive fiscal externalities identical to new employment creation: formalization induces social security contribution payments, broadens the tax base, and regularizes safety net coverage. This symmetry is fiscal only. It does not make the worker-surplus calculation robust to formalization: a move from informal to formal work may have a very different opportunity cost from a move out of non-employment, and the calibration of Section 6.1 applies only to the latter.

E Market-Level Design

This appendix collects the econometric details of the market-level design of Section 5. Appendix E.1 states what the design treats as random and the choices this requires. Appendix E.2 writes the assignment down as an algorithm and derives the propensity scores, the dispersion of the instrument and the cost of simulation, and Appendix E.3 works through an example. Appendix E.4 documents the dependence between markets and the randomization inference, Appendix E.5 the anatomy of the assignment in the data, and Appendix E.6 derives the local employment multiplier.

E.1 What is random, and the choices the design requires

Why the observed shock is not enough. Even if funding near the cutoff were a coin flip, Z_m^{obs} in (15) would not be random across markets. A market that put more projects near the cutoff, or projects promising more jobs, receives more jobs in expectation whoever wins, and the probability of winning itself differs across markets with the requests and budgets of the pools their applicants entered. Comparing markets by Z_m^{obs} would therefore compare markets that applied more with markets that applied less-the comparison ordinary least squares already makes. What is random is not how exposed a market was, but whether its near-cutoff projects did better or worse than their odds.

What is random. Applications were ranked within funding pools and funded in score order until each pool's budget ran out (Section 3.1). Far from the cutoff, the score settled the outcome: very strong projects were funded under any plausible ordering and very weak ones under none. Near it, scores were so close that small differences decided which projects the budget reached. We treat the *order* of the projects inside a band around each pool's cutoff as random. This requires taking a view on the score: not as a fixed characteristic of the project, but as one realization of an evaluation that could have come out slightly differently, as a test score could on another day. Neither the scores nor the cutoff are drawn. Because the budget is fixed, a different order moves the cutoff: a project asking for a large grant may no longer fit, and two smaller ones take its place. The number of projects funded, as well as their identity, is therefore an outcome of the draw.

The protocol. Each funding pool is a stratum. Within a stratum, the program treats as many projects as its budget allows, without replacement: once a project is funded, its request

is subtracted from what is left for the others. It is a stratified randomized trial that treats a fixed amount in each stratum, the structure of the centralized school-assignment lotteries in Abdulkadiroğlu et al. (2017), whose approach we follow. Because the assignment rule is known, the probability that each applicant is funded, and the exposure each market should expect, can be computed by re-running the rule on re-drawn orders rather than modelled.

Relation to Abdulkadiroğlu et al. (2022). The closest formal result is theirs: when assignment is decided by a running variable with a random component, the propensity score of an applicant at the cutoff converges to one half as the bandwidth shrinks. Their result holds the cutoff fixed and lets a single applicant approach it. In our setting several applicants remain near the cutoff, and the cutoff moves with the order in which the budget meets them. The natural extension is that, when k grants of equal size are at stake among n applicants near the cutoff, each applicant's propensity converges to k/n, and that with unequal requests it converges to the probabilities the award rule implies (Proposition 4). We do not prove this extension. We take it as the intuition for the design and implement a discrete approximation to it: we fix a band, treat the order within it as uniformly random, and re-apply the rule.

Relation to Narita and Yata (2023). They study algorithms that recommend treatment to each unit as a known function of that unit's own observed inputs. They define the approximate propensity score as the average probability of a treatment recommendation over a shrinking neighborhood of the unit's inputs, and show that conditional on it the recommendation is as good as random. Our simulated probability of funding plays the same role, with two differences. The neighborhood is a fixed band rather than one that shrinks. And an applicant's funding depends on the other applicants in its pool, because the budget ties their outcomes together, a case their framework does not cover. We therefore compute the probability pool by pool, re-applying the award rule to the whole pool in each draw.

A propensity-score model, not a continuity design. Once the order inside the band is uniform, everything else follows mechanically: each ordering maps, through the rule, into an allocation of the pool's budget, and each allocation maps into the jobs each market receives. The randomization is at the level of the pool and the treatment at the level of the market, so the potential outcomes that matter are those of markets, $y_m(\mathbf{g})$. The firm-level estimates of Section 4 take the continuity approach to the discontinuity, under which the score may carry information right up to the cutoff. A continuity version of the market design would estimate, inside the band, how outcomes vary with the score, and would deliver propensities that vary

within the band. The local-randomization version is simpler, and its one assumption is easier to state and to examine.

Exchangeable order is an approximation. Assumption 2 is an approximation, and we state it as one. The score is a deterministic function of the application, and even if it measured quality with error, a project at the bottom of the band would remain less likely to come out on top than one at the top, so no model of measurement error delivers exactly uniform ranks. The approximation improves as the band narrows. The model could be tilted, so that the applicant at the bottom of the band is less likely than the one at the top to come out first, though not impossible; drawing the order from a noisy score, $S_f + \varepsilon_f$ with $\varepsilon_f \sim N(0, \sigma^2)$, does exactly that. We keep the uniform order as the model: it is the transparent benchmark, and a tilt would have to be chosen rather than estimated.

The band. The width of the band is the design's bias–variance choice. A wider band takes in applicants whose odds of winning rise with their score, which strains Assumption 2; a narrower one leaves fewer projects whose funding could have gone either way, and so less variation across markets. There is no simple data-generating process from which an optimal band could be derived for an assignment of this kind, and we use the band of the firm-level design, $\bar{S} = 5$ (Section 4). The band is fixed around the realized cutoff, but within it the cutoff is allowed to move: under an order in which a large request comes last, every smaller project above it is funded and the cutoff falls just above the large one. Had the band been drawn around that cutoff instead, it would have contained a different set of applicants. The design therefore conditions on the realized band, as local-randomization designs condition on the observed window. This is also why the design is not a permutation of the cutoff, as in designs that draw the threshold itself: the cutoff moves only as a consequence of the re-drawn order and the fixed budget.

E.2 The award rule and the win probabilities

The design rests on a *known* assignment process. This subsection writes that process down as an algorithm and derives what it implies for the win probabilities, for the dispersion of the instrument, and for simulation error.

The rule. Fix a funding pool p. Let $\mathcal{C}_p$ be the set of near-cutoff applicants in that pool, $n \equiv |\mathcal{C}_p|$, and for each $f \in \mathcal{C}_p$ let $r_f > 0$ be the requested grant and $L_f \geq 0$ the promised

jobs. Let b_p be the budget available to the near-cutoff band, the amount actually awarded to near-cutoff winners. Let $\sigma \in \mathcal{R}_p$ be a ranking of $\mathcal{C}_p$, one of the $n!$ orderings. The award map $G_p : \mathcal{R}_p \to \{0,1\}^n$ is the program's own procedure: start with residual budget $B = b_p$ and walk down σ; on reaching f, if $r_f \le B$ set $g_f = 1$ and $B \leftarrow B - r_f$; at the first f with $r_f > B$, stop, leaving f and every applicant ranked after it unfunded.[36] Assumption 2 of Section 5 is the statement that, conditional on $\mathbf{w}$, $\sigma_p \sim \text{Unif}(\mathcal{R}_p)$ independently across pools, and $\mathbf{g}_{\mathcal{C}_p} = G_p(\sigma_p; \mathbf{w})$. The assumption is on rankings, not on allocations: a uniform draw from all budget-feasible allocations would put mass on allocations the program never makes, such as funding no one, whereas the administration funded down the list until the budget ran out. Simulating the mechanism rather than the set of allocations is the approach Abdulkadiroğlu et al. (2017) take for centralized assignment.

Proposition 4 (The award rule is an exact sampler)**.** Fix $\mathbf{w}$ and let $\sigma \sim \text{Unif}(\mathcal{R}_p)$. Then

(i) $G_p(\sigma)$ funds the longest initial segment of σ whose requests fit in b_p.

(ii) The win probability of $f \in \mathcal{C}_p$ is

$$e_f = \Pr(g_f = 1 \mid \mathbf{w}) = \sum_{\mathcal{P} \subseteq \mathcal{C}_p \setminus \{f\}} \omega_{|\mathcal{P}|}\, \mathbf{1}\Big\{ r_f + R(\mathcal{P}) \le b_p \Big\}, \qquad \omega_k = \frac{k!\,(n-1-k)!}{n!}, \quad R(\mathcal{P}) = \textstyle\sum_{j \in \mathcal{P}} r_j .$$

Proof. (i) Requests are positive, so cumulative requests rise along σ: the rule funds $\sigma(1), \ldots, \sigma(K)$ and stops at $\sigma(K+1)$, the first request that does not fit. (ii) f is funded if and only if the set $\mathcal{P}$ ranked ahead of it fits together with f, since every earlier initial segment then fits as well; exactly $|\mathcal{P}|!\,(n-1-|\mathcal{P}|)!$ orderings put the set $\mathcal{P}$ ahead of f, a share $\omega_{|\mathcal{P}|}$ of all $n!$. For example, with $\mathcal{C}_p = \{1,2,3\}$, $(r_1, r_2, r_3) = (2,1,1)$ and $b_p = 2$, the rule funds $\{1\}$ or $\{2,3\}$ with probability $1/3$ each and $\{2\}$ or $\{3\}$ with probability $1/6$ each, so the win probabilities are $1/3$, $1/2$ and $1/2$. ∎

Win probabilities and the dispersion of the instrument. The sum in Proposition 4(ii) runs over the 2^{n-1} sets of other applicants in the band; with a median band of 41 applicants that is about 10^{12} terms, which is why μ_m is computed by simulation rather than evaluated. An applicant that asks for more fits behind fewer sets, and so is funded less often. With equal requests the sum reduces to k/n:

[36] This is what the *decreto* did: applications were funded in score order until the pool's budget ran out. The alternative convention-passing over a project that does not fit and continuing down the list-would be equally easy to simulate; our simulations implement the stopping rule.

Corollary 2 (Equal requests)**.** Suppose $r_f \equiv r$ and let $k = \lfloor b_p/r \rfloor$. Then $G_p(\sigma)$ funds the top k applicants in σ, so $\mathbf{g}_{\mathcal{C}_p}$ is a uniformly drawn k-subset of $\mathcal{C}_p$, $e_f = k/n$, and for the near-cutoff firms $A = \mathcal{C}_p \cap m$ of market m,

$$\operatorname{Var}\Big(\sum_{f\in A} L_f g_f\Big) \;=\; \frac{k(n-k)}{n(n-1)}\left[\sum_{f\in A} L_f^2 \;-\; \frac{1}{n}\Big(\sum_{f\in A} L_f\Big)^2\right]. \tag{29}$$

Equation (29) says where the identifying variation comes from. Taking $A = \mathcal{C}_p$, it collapses to $\frac{k(n-k)}{n(n-1)}\sum_f (L_f - \bar{L})^2$: if every near-cutoff applicant in a pool promised the same number of jobs, the shock would have no variance, however many firms were in the band. The first stage exists because promised jobs are right-skewed (Table D.6, Panel A); the design combines a fair lottery with an unequal prize. Because each funding pool reaches across many markets (a median of 23; Figure E.2), the negative correlation that the budget induces between wins in the same market is small: in the median market the variance is about 2.5% below what independent assignment would give. The shock is largest relative to baseline employment where few, large near-cutoff bids sit in a small market, so the estimates weight small markets, and sectors that are thin in their province, more heavily than their share of employment.

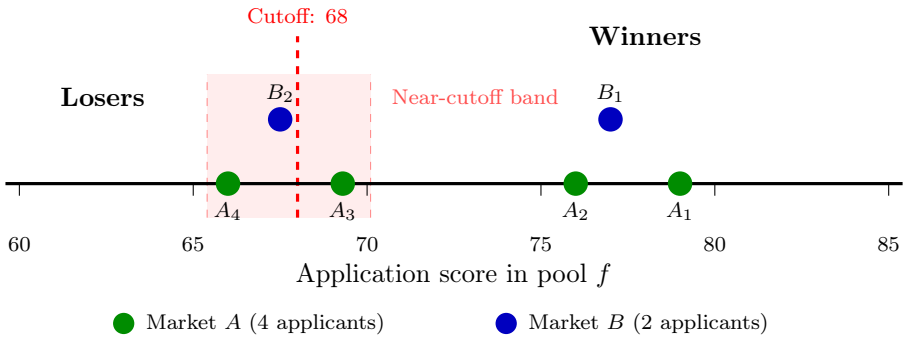
Simulation error. With S draws, $\widehat{\mu}_m = S^{-1}\sum_s Z_m^{(s)}$ and the instrument is $\widetilde{Z}_m = Z_m^{\text{obs}} - \widehat{\mu}_m = \widetilde{Z}_m^{\star} - \nu_m$, where $\widetilde{Z}_m^{\star} = Z_m^{\text{obs}} - \mu_m$ is the infeasible ideal and $\nu_m = \widehat{\mu}_m - \mu_m$ is Monte Carlo error.

Remark 2 (What finite S costs)**.** Conditional on $\mathbf{w}$, ν_m is mean zero and independent of $\widetilde{Z}_m^{\star}$ and of the outcome, since it comes from the researcher's random number generator, with $\operatorname{Var}(\nu_m \mid \mathbf{w}) = \operatorname{Var}(Z_m \mid \mathbf{w})/S$. Validity is therefore unaffected for any $S \geq 1$, and so is the probability limit of the 2SLS estimate. Only first-stage power is lost: the first-stage coefficient attenuates by approximately $S/(S+1)$, which at our $S = 250$ is 0.996.

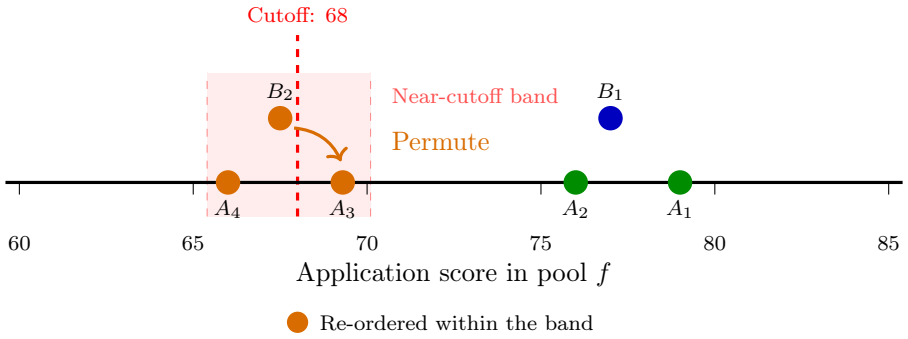
E.3 An example

Figure E.1 works the design through on a single funding pool with six applicants from two markets. Panel (a) shows the observed allocation: market A has two applicants in the near-cutoff band and market B one, so market A is more exposed to the program whoever wins, and the observed shock Z_m^{obs} mixes that exposure with luck. Panel (b) shows the fix: only the applicants inside the band are re-ordered, the funding rule is run on each ordering, and the average over orderings, μ_m, is subtracted. Example 1 puts numbers on it.

Figure E.1: The observed shock, and what the recentering removes



(i) The observed allocation. Market A is systematically more exposed: it has more applicants in the band.



(ii) Only the three applicants inside the band are re-ordered; the rule $G_f(\sigma; \mathbf{w})$ is then run on each ordering, and μ_m is the average of the resulting $Z_m(\sigma)$.

Notes: A single funding pool with six applicants from two markets. Vertical position carries no meaning and is used only to separate the two markets. The near-cutoff band is the caliper $|S_j| \leq \bar{S}$ of Section 5; the budget b_f and every applicant's requested amount and promised jobs are held fixed across draws, so the only thing that varies is which near-cutoff applicants the money reaches.

Example 1. Put numbers on the pool of Figure E.1. The budget is 300, and every project requests a grant of 75 and promises one job. The three inframarginal projects, A_1, A_2 and B_1, score far above the cutoff, consume 225 and are funded under every permutation.

The marginal band $\mathcal{C}_p$ contains three nearly tied projects—A_3, B_2, and A_4—competing for the remaining budget of 75 ($300 - 225 = 75$). Because only one project can be funded from the band, each applicant has an identical win probability of $e_f = 1/3$. Before the ranking draw, market A has two marginal candidates and expects to receive $2 \times (1/3) = 2/3$ of a job, while market B has one candidate and expects $1/3$ of a job.

Suppose applicant A_3 ranks first in the realized draw and receives the grant. Realized project jobs from the marginal band are 1 for market A and 0 for market B. The naive shock Z^{obs}

would credit market A with a larger shock simply because it submitted more applications. The recentered shock $\tilde{Z}_m$ removes this baseline exposure, isolating the unexpected policy surprise:

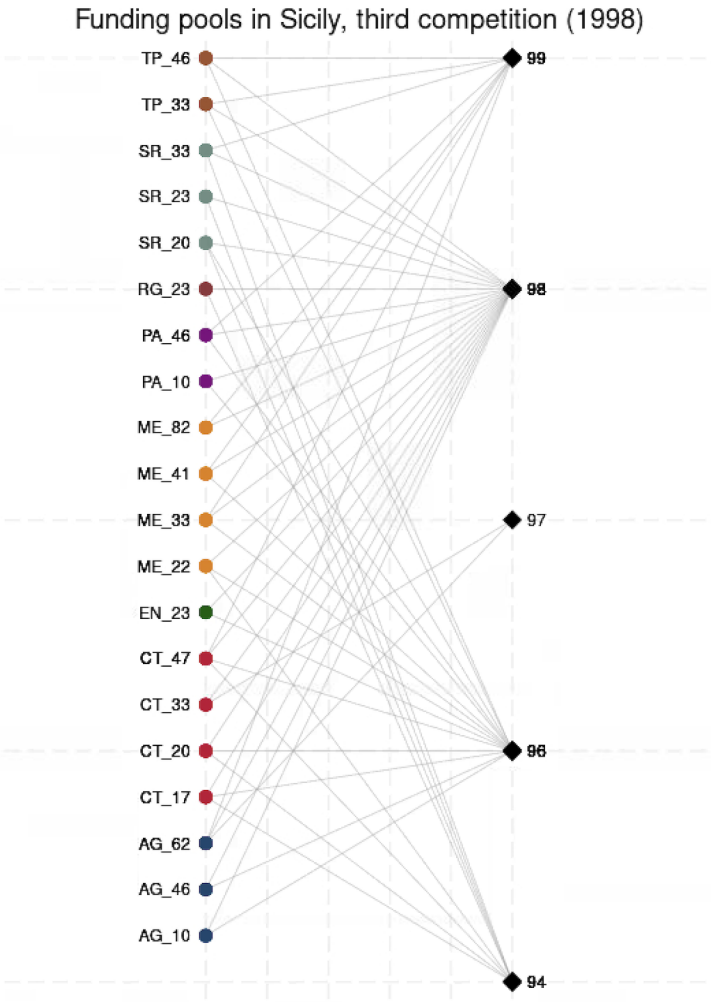
$$\tilde{Z}_A \;=\; 1 - \frac{2}{3} \;=\; +\frac{1}{3}, \qquad \tilde{Z}_B \;=\; 0 - \frac{1}{3} \;=\; -\frac{1}{3}.$$

E.4 Cross-market dependence and randomization inference

Assumption 3 (Many Strata and Weak Dependence) requires that idiosyncratic policy shocks average out across labor markets. Because multiple markets compete within the same regional funding pool, shocks across markets sharing a pool are negatively correlated in finite samples: one market winning a grant consumes budget that could have funded a project in another market.

Figure E.2 visualizes the bipartite structure linking markets (left nodes) to funding pools (right nodes) for one call, the third (1998), in Sicily, the largest recipient region; each edge connects a market to a funding pool in which firms from that market applied. The graph shows substantial cross-market mixing: most funding pools include near-cutoff firms from multiple distinct markets, and most markets participate in multiple funding pools. This imperfect matching between markets and funding pools is what generates useful identifying variation: if each funding pool were effectively market-specific, there would be no cross-market variation to exploit.

Figure E.2: Markets and funding pools are cross-linked, not siloed (Sicily, 1998 call).



Notes: Bipartite graph for the funding pools of the third call (1998) in Sicily. Left nodes (circles) are labor markets, labeled by province and 2-digit sector (e.g. CT_20 is sector 20 in the province of Catania) and colored by province. Right nodes (diamonds) are the call's funding pools. An edge connects market m to funding pool f whenever at least one firm from market m applied in funding pool f. The graph shows that most funding pools include near-cutoff firms from several distinct markets, and most markets are exposed through multiple funding pools, showing imperfect market–funding-pool matching-the key requirement for cross-market identifying variation.

We verify that the recentered instrument satisfies the Weak Mutual Dependence condition of Borusyak and Hull (2023). Because two markets sharing a funding pool are correlated through the same permutation draw, the instruments $\{\tilde{Z}_m\}$ are not independent across markets. WMD requires only that the *average* pairwise covariance vanishes, which holds through two channels: (i) each individual market–funding-pool concentration share $\theta_{m,f} = J_{m,f}/n_f$ is small, making any single pairwise covariance small in magnitude; and (ii) each market shares pools with only a limited number of other markets, restricting how many non-zero covariances exist.

These two channels combined imply that the average pairwise covariance term vanishes, well below conventional thresholds, so that a law of large numbers operates across markets despite the within-pool dependence.

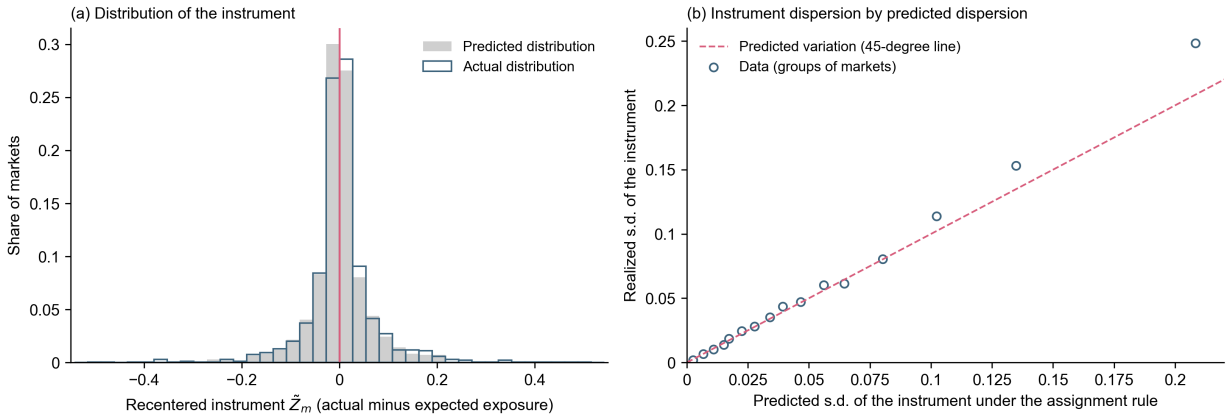
We measure this dependence directly on the simulated allocations. Across pairs of markets that share a funding pool, the average absolute covariance between the recentered shocks $\tilde{Z}_m$ and $\tilde{Z}_{m'}$ is 8.0×10^{-5} against an average variance of 1.8×10^{-3}, so the ratio the weak-dependence condition restricts is about 0.045. Across all pairs of markets nationally, the covariance is less than 0.1% of the variance. The largest neighbourhood-the set of markets linked to a given market through shared pools-covers 26% of markets. The maximum pairwise covariance is larger, as it must be for markets that share a small pool, but the design's asymptotics depend on the average rather than the maximum.

To guard against finite-sample clustering or reliance on asymptotic normality with a modest number of regional strata, we compute exact p-values via *randomization inference*. Specifically, for each simulation draw $s \in \{1, \ldots, 250\}$, we recompute the market shock vector $\tilde{Z}^{(s)}$, re-estimate the first-stage regression on the pseudo-shocks, and record the resulting test statistics. The empirical p-value is the share of counterfactual simulations that produce a first-stage F-statistic exceeding the realized value of 40.1, which yields $p = 0.01$.

Randomization inference inherits the same property in a stronger form. Because the S draws are exact draws from the assignment law (Proposition 4), the p-value obtained by comparing the realized statistic to its value on the simulated allocations is finite-sample exact for any S, up to the $1/(S+1)$ granularity of the reference distribution, with no appeal to asymptotics in the number of markets or pools. This matters here for a specific reason: the identifying weight within a market rests on a median of three near-cutoff firms (Figure E.6), so market-level asymptotic approximations are the weakest link in the inference and are worth avoiding where an exact alternative exists.

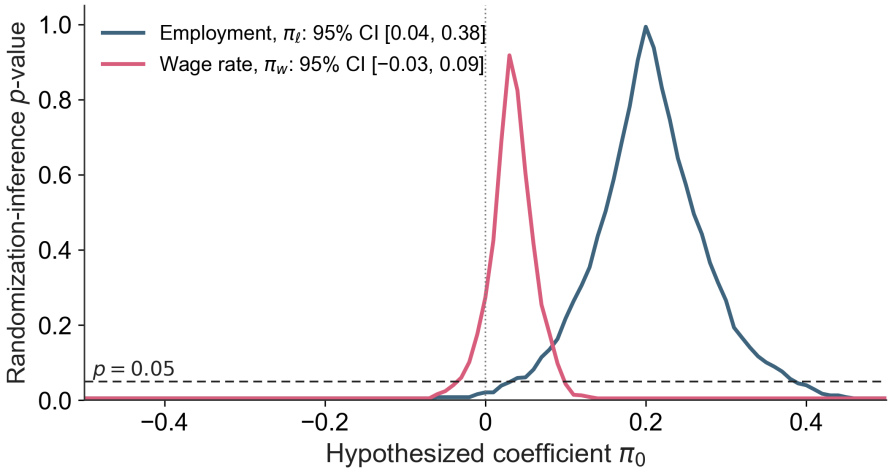
Figures E.3 and E.4 report the checks of the assignment model and the randomization inference. Figure E.3 compares the recentered instrument with the deviations the assignment rule generates in simulation, and checks that the instrument's dispersion matches the dispersion the rule predicts. Figure E.4 inverts the randomization test to obtain confidence intervals for π.

Figure E.3: The assignment model against the data



Notes: The allocation itself is illustrated in Figure E.1. Panel (a) plots the share of markets in each bin of the recentered instrument $\tilde{Z}_m$, the actual minus the expected exposure (outline), against the same share for the deviations that the assignment rule generates in simulated allocations (shaded). By construction the instrument is mean zero; most markets receive a small surprise, because their near-cutoff projects are funded or not with high probability, and a few receive a large one. Panel (b) groups markets into 20 bins by the standard deviation of the instrument predicted by the assignment rule and plots, for each bin, the realized standard deviation of $\tilde{Z}_m$, with the 45-degree line as the prediction. The three bins with the smallest predicted dispersion lie on top of one another at the origin, and the bin with the largest (predicted 0.76, realized 0.70) lies outside the plotted range.

Figure E.4: Randomization-inference confidence intervals for π



Notes: For each hypothesized π_0, the figure plots the randomization-inference p-value of the test that the adjusted outcome $y - \pi_0 x$ is unrelated to the recentered instrument (Imbens and Rosenbaum, 2005), where x is the change in log market employment and y is the change in log employment (blue) or in the log wage rate (rose) of firms that never applied, 1995 to 2001. The reference distribution is the coefficient on the placebo instrument $Z_m^{(s)} - \mu_m$ over 250 draws from the assignment rule, with region and industry fixed effects, and the p-value is $(\#\{|b_s| \geq |b_{\text{obs}}|\} + 1)/(S + 1)$. The 95% confidence interval is the set of π_0 with $p > 0.05$: $[0.04, 0.38]$ for employment and $[-0.03, 0.09]$ for the wage rate. The dashed line marks $p = 0.05$ and the dotted line $\pi_0 = 0$.

E.5 The anatomy of the assignment

Four features of this assignment determine what the instrument can identify, and the tables below document each. *Scale*: the promised jobs of near-cutoff firms are large relative to the markets they land in-a median of 15% of baseline applicant employment in the market and 45% at the 75th percentile (Table E.1)-so a reshuffle of who wins moves market-level employment materially rather than marginally. *Tightness*: among the pools that generate any variation at all, the budget covers on average 44% of the requested amounts, which is the range in which which-firm-wins is genuinely open (Table E.4). *Mixing*: the average funding pool spans 43 distinct labor markets (median 23), so the same pool's budget constraint links firms in many different markets (Table E.6). *Concentration*: within a market, the identifying weight is concentrated, with a median of three near-cutoff firms and a median top-firm weight of about 0.59 (Table E.7).

Table E.1: The scale of the near-cutoff variation

	N	Mean	P25	P50	P75	P90	Max
Baseline employment m,1995	1,803	1,936	105	384	1,213	3,826	106,575
Promised jobs (near-cutoff firms)	1,803	128.7	14	48	127	328	2,925
Scale factor $\sum_{f \in \mathcal{C}_m} L_f \,/\, E_{m,1995}$	1,803	0.500	0.037	0.153	0.445	1.167	5.482

Notes: Scale factor $= \sum_{f \in \mathcal{C}_m} L_f \,/\, E_{m,1995}$, where is L_f firm's promised new-hire count and $E_{m,1995}$ is total 1995 employment at IDA-applicant firms in market m (province × sector cell). Scale factor winsorised at the 2nd and 98th percentiles. Near-cutoff bandwidth $\delta = 5$ SD of the standardised score.

Table E.2: Funding-pool descriptives

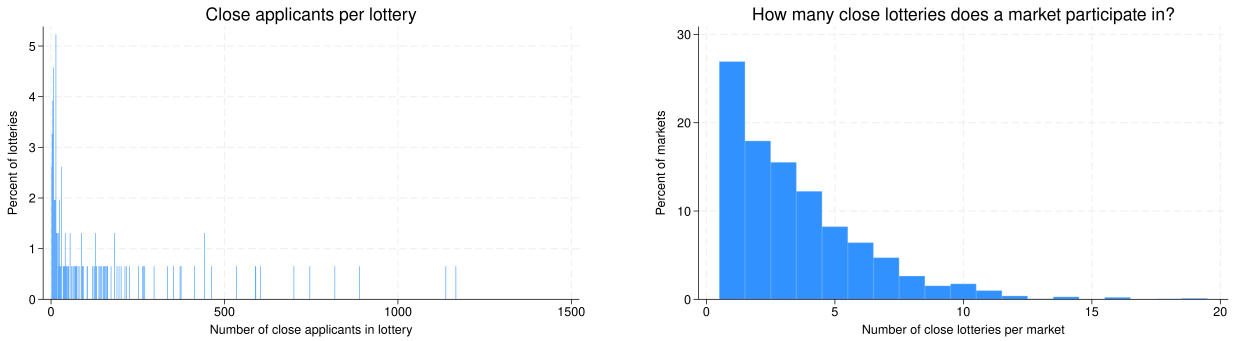
Variable	N	Mean	P25	P50	P75	P90	P95	Max	SD
n_apps	153	134.830	12.000	44.000	155.000	381.000	616.000	1248.000	218.669
n_close	153	127.399	11.000	41.000	150.000	372.000	590.000	1167.000	207.732
n_close_win	153	54.137	4.000	15.000	64.000	137.000	282.000	455.000	90.733
jobs_close	153	1519.007	140.000	466.000	1623.000	4492.000	7721.000	14086.000	2611.960

Notes: One observation per funding pool. `n_apps`: applicants; `n_close`: applicants inside the near-cutoff band; `n_close_win`: near-cutoff winners; `jobs_close`: jobs promised by near-cutoff applicants.

Table E.3: Market-level descriptives

Variable	N	Mean	P25	P50	P75	P90	P95	Max	SD
close_lotteries	1,823	3.500	1.000	3.000	5.000	7.000	9.000	19.000	2.640
close_firms	1,823	10.692	2.000	4.000	11.000	24.000	43.000	206.000	19.103
close_jobs	1,823	127.487	14.000	47.000	126.000	324.000	522.000	2925.000	250.371

Figure E.5: How many near-cutoff firms per funding pool, and how many funding pools per market



(i) Near-cutoff firms per funding pool

(ii) Funding pools per market

Table E.4: Budget tightness, and the share of the pool's requests the budget covers

Variable	N	Mean	P25	P50	P75	P90	P95	Max	SD
budget_tightness	152	0.440	0.285	0.405	0.549	0.833	0.891	0.986	0.228

Notes: Pools that generate variation among near-cutoff firms. Budget tightness is the share of near-cutoff requested amounts the remaining budget can finance; values near zero or one leave little room for which-firm-wins to matter.

Table E.5: Exposure of a market to a given funding pool

Variable	N	Mean	P25	P50	P75	P90	P95	Max	SD
theta_mr	6,379	0.024	0.004	0.008	0.022	0.054	0.095	0.667	0.050

Notes: θ_{mr} is the share of market m's baseline employment that near-cutoff firms in funding pool r could add.

Table E.6: Cross-market mixing: markets spanned by a funding pool

Variable	N	Mean	P25	P50	P75	P90	P95	Max	SD
n_markets_in_lot	143	42.559	8.000	23.000	53.000	129.000	155.000	221.000	49.254
n_markets_variation	143	42.559	8.000	23.000	53.000	129.000	155.000	221.000	49.254

Notes: Number of distinct labor markets represented among the near-cutoff applicants of a funding pool. High mixing is what makes the budget constraint link firms across markets, and is what the weak-dependence condition needs.

Figure E.6: Identifying weights across markets

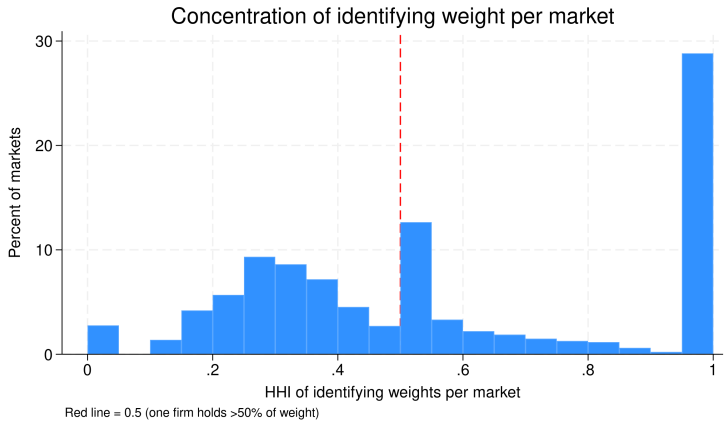


Table E.7: Concentration of the identifying weight within a market

Variable	N	Mean	P25	P50	P75	P90	P95	Max	SD
hhi_weight	1,811	0.571	0.312	0.500	1.000	1.000	1.000	1.000	0.314
top1_share	1,761	0.654	0.424	0.593	1.000	1.000	1.000	1.000	0.268
n_close_firms_mkt	1,811	3.361	1.000	3.000	5.000	7.000	8.000	18.000	2.447

Notes: `hhi_weight` is a Herfindahl index of near-cutoff firms' contributions to a market's shock; `top1_share` the share contributed by the largest one; `n_close_firms_mkt` the number of near-cutoff firms in the market.

E.6 Derivation of the Local Employment Multiplier

This subsection derives the aggregate local employment multiplier M presented in Section 5.3.

Let total baseline employment in labor market m be decomposed into employment at IDA applicant firms, E_A, and employment at firms that never applied, E_N:

$$E_m \;=\; E_A + E_N.$$

Let $s_N \equiv E_N/E_m$ denote the employment share of non-applicant firms, so that $1 - s_N = E_A/E_m$. Log total employment growth in the market can be approximated as the share-weighted average of employment growth in the two sectors:

$$\Delta \log E_m \;\approx\; (1 - s_N)\,\gamma_A \;+\; s_N\,\gamma_N, \tag{30}$$

where $\gamma_A \equiv \Delta \log E_A$ is the growth rate of applicant firms and $\gamma_N \equiv \Delta \log E_N$ is the growth rate of non-applicant firms.

Equation (14) defines non-applicant employment growth as an empirical elasticity π_ℓ with respect to total market growth:

$$\gamma_N \;=\; \pi_\ell\,\Delta \log E_m. \tag{31}$$

Substituting (30) into (31) yields:

$$\gamma_N \;=\; \pi_\ell\big[(1 - s_N)\,\gamma_A \;+\; s_N\,\gamma_N\big].$$

Rearranging terms to isolate γ_N as a function of applicant growth γ_A:

$$\gamma_N\big(1 - \pi_\ell s_N\big) \;=\; \pi_\ell(1 - s_N)\,\gamma_A \;\Longrightarrow\; \frac{\gamma_N}{\gamma_A} \;=\; \frac{\pi_\ell(1 - s_N)}{1 - \pi_\ell s_N}.$$

Converting these percentage growth rates into net job changes ($\Delta E_N \approx E_N \gamma_N$ and $\Delta E_A \approx$

$E_A\gamma_A$):

$$\frac{\Delta E_N}{\Delta E_A} = \frac{E_N}{E_A} \cdot \frac{\gamma_N}{\gamma_A} = \frac{s_N}{1-s_N}\left[\frac{\pi_\ell(1-s_N)}{1-\pi_\ell s_N}\right] = \frac{\pi_\ell s_N}{1-\pi_\ell s_N}. \tag{32}$$

The aggregate local employment multiplier M, defined as total net jobs created in the market per job created at applicant firms, is therefore:

$$M \equiv 1 + \frac{\Delta E_N}{\Delta E_A} = 1 + \frac{\pi_\ell s_N}{1-\pi_\ell s_N} = \frac{1}{1-\pi_\ell s_N}. \tag{33}$$

Mapping to Theoretical Benchmarks. Equation (33) defines the four economic regimes discussed in the text:

1. **Full Displacement (Pure Crowd-Out):** If subsidized firms expand entirely by poaching workers from non-subsidized competitors, $\Delta E_N = -\Delta E_A$, which implies $\Delta E_m = 0$. In this case, the first-stage regression coefficient is zero, π_ℓ is not defined, and $M = 0$ by the definition $M \equiv 1 + \Delta E_N/\Delta E_A$; equation (33) does not apply.
2. **Partial Displacement:** When market wages rise and non-applicants contract ($\pi_\ell < 0$), the ratio $\Delta E_N/\Delta E_A \in (-1, 0)$, yielding a multiplier strictly between zero and one: $M \in (0, 1)$.
3. **Queue Without Spillovers (Pure Additionality):** If subsidized firms hire exclusively from a rationed queue of non-employed workers and there are no product-market demand or agglomeration linkages to non-applicants, $\pi_\ell = 0$. In this case, non-applicant employment is unchanged ($\Delta E_N = 0$), yielding $M = 1$.
4. **Queue With Positive Demand/Agglomeration Spillovers:** When non-applicants expand alongside funded firms ($\pi_\ell > 0$), $M > 1$. In our empirical setting, non-applicants represent approximately 90% of baseline employment ($s_N \approx 0.90$). Our baseline estimates of $\pi_\ell \in [0.149, 0.241]$ imply an aggregate multiplier of $M \in [1.15, 1.28]$, indicating that each subsidized job generates roughly 0.15 to 0.28 additional jobs among non-applicant firms in the same labor market.

F The welfare accounting: the queue and the calibration

This appendix collects the calculations behind Section 6: the worker-surplus calculation, its link to equation (6), and the condition under which an investment subsidy raises employment.

F.1 Details of the worker-surplus calculation

Table F.1: Reservation Wages of Job Seekers in the Areas Targeted by the Program

Net € a month	1996	2004
1st percentile	310	300
5th percentile	516	400
10th percentile	516	
25th percentile	671	600
Median	775	800
Mean	870	871
Mean below net pay of a new hire	727	
75th percentile	930	1,000
Job seekers (observations)	40,652	8,917
Net pay of a new hire	974	
Job seekers who would accept less, 1996	79%	
Surplus per month of work, linear benchmark, 1996	€229 (23.5% of net pay)	
Surplus per month of work, uniform rationing, 1996	€247 (25% of net pay)	
Surplus per month of work, efficient rationing (lower bound), 1996	€458 (47% of net pay)	

Notes: ISTAT Labour Force Survey, job seekers resident in the regions where the areas targeted by the program are concentrated. Both waves are in current euros.

Survey reservation wages measure private reservation pay, which includes the benefits a worker gives up (equation (2)). The gap between the wage and a survey reservation wage is therefore the worker's own surplus from the job; the resource surplus exceeds it by the benefits no longer paid, which are counted on the fiscal side. Job seekers in the survey report the minimum net monthly wage they would accept for a job that meets their requirements, the standard measure of the reservation wage in these data (Sestito and Viviano, 2011). Because reservation wages are stated net of taxes, the comparison is with net pay, and the taxes collected on the new earnings are a gain to taxpayers that this measure leaves out.

The efficient-rationing bound. Let F_R be the distribution of reservation wages among job seekers willing to work at w^e, and q the fraction of them hired. Under efficient rationing the hired are the q with the lowest reservation wages, and the average surplus per job is equation (20). Because F_R^{-1} is increasing, the average of $F_R^{-1}(u)$ over $u \in [0, q]$ is at most $F_R^{-1}(q)$. With 79% of job seekers willing, a share q of the willing corresponds to the $0.79\,q$ quantile of all job seekers. Scaling the jobs created in the estimation sample to the whole program, the program's direct and indirect jobs are 7 to 14 percent of the willing job seekers, which reaches between the sixth and the 11th percentile of all job seekers. In the 1996 survey microdata (weighted), the tenth percentile is €516, equal to the fifth because answers bunch at one million lire, so when the jobs reach no further than the tenth percentile the average surplus is at least $974 - 516 \approx$ €458 a month. The bound applies to the pool from which the additional jobs recruit, and, when multiplied by the worker-years per job, to the workers who hold the jobs over time. Lefeber (1968, eq. 27) prices the same gap as the payroll subsidy that would close it, assuming the time of the unemployed has no value; we measure that value instead.

From job seekers to the workers hired. The calibration assumes that the workers hired because of the policy are drawn from the distribution of job seekers who would accept the job at w^e. The survey covers job seekers, while many of the workers hired out of non-employment may not have been searching, and the direction of the resulting bias is not signed: discouraged workers may have low reservation wages, workers with care obligations high ones. Stated reservation wages also include the value of continuing to search (Luttmer, 2007, fn. 2), so the static opportunity cost of the hours worked may be lower; how much depends on job duration and on search while employed. The 2004 wave of the survey gives a similar median and a lower fifth percentile (Table F.1).

Uniform rationing. Under uniform rationing the hired are a random draw from the willing job seekers, and the average surplus per job is equation (21): €974 minus the mean reservation wage below €974, about €247 a month, with a mean reservation wage of €727 in the weighted 1996 microdata.[37]

[37] Where the rationing rule has also determined who is in the queue, the ranking of the two rules can reverse: under efficient rationing the workers who value a job most are already employed, so the marginal job can be worth less than under uniform rationing (Lee and Saez, 2012).

F.2 Worker surplus and the welfare equation

While our survey measures private surplus WS, the overall resource gain from additional employment in equation (6) equals $WS + F$, where F is the net fiscal gain: forgone transfers plus new taxes and contributions, net of any future benefit entitlements. In turn, each euro of fiscal gain reduces the public funds the program requires, saving a further λ euros of excess burden. Accounting for grant outlays G, the integrated cost of the capital distortion C_K, and the excess burden per euro of public funds raised, λ, the extensive-margin welfare effect is:

$$\Delta W \;=\; WS + F - C_K - \lambda\,(G - F) \;=\; WS - C_K - \lambda G + (1+\lambda)\,F. \tag{34}$$

The next subsection maps this expression to the benefit-cost framework of Bartik (2012).

F.3 The labor-allocation gain and equation (6)

Worker surplus WS is part of a broader labor-allocation gain. Taking the wage floor as given and holding post-grant capital fixed, let $D_t(\ell)$ denote the annual marginal revenue product of labor net of marginal hiring costs, with additional jobs spanning L_0 to L_{1t} ($|Q_t| = L_{1t} - L_0$ full-year jobs). Valuing recruited workers' time at their reservation wages yields the labor-allocation gain:

$$\Delta A \;=\; \int_0^T \Big[\int_{L_0}^{L_{1t}} D_t(\ell)\,d\ell \;-\; \int_{Q_t} \omega^R(\ell)\,d\ell\Big]\,dt \;\geq\; \int_0^T \int_{Q_t} \big(w^e - \omega^R(\ell)\big)\,d\ell\,dt \;=\; WS. \tag{35}$$

The inequality holds year by year. Because a profit-maximizing firm sets employment where net marginal product covers the gross labor cost—which exceeds net pay w^e by taxes and social contributions—diminishing returns and convex hiring costs ensure that $D_t(\ell) \geq w^e$ for every job created. Hence, WS bounds ΔA from below without requiring estimates of labor demand.

This comparison is deliberately conditional. The gain ΔA isolates the value of expanding employment along the post-grant demand curve; it is not an add-on benefit to the full welfare framework in equation (34), where the return to capital is already netted against its cost. Furthermore, because private reservation wages include forgone social benefits, the resource gain is at least as large as ΔA. Finally, ΔA coincides with the familiar deadweight-loss triangle ($\int_0^T \int_{L_0}^{L_{1t}} (D_t - S)\,d\ell\,dt$, with S the inverse supply schedule) only in the restrictive benchmark where jobs are allocated strictly to workers with the lowest opportunity costs and private costs equal social costs.

Equation (6) is the welfare change along the subsidy path, with firms on their first-order conditions at every point. Holding weeks per job fixed, its jobs term values each additional job at the gap between its pay and the resource opportunity cost of the worker who fills it. In equation (34) this term is $WS + F$: the private surplus we measure plus the net fiscal gain. Incumbents' additional weeks are valued separately, in the weeks term.

The labor-allocation gain ΔA of equation (35) adds the excess of the net marginal product over pay on the jobs created. That excess has no counterpart among the labor terms of equation (6): with D_t the marginal product at post-grant capital, it is part of the gross return to the induced capital, which equation (6) nets against the capital's resource cost. Adding it to equation (34) would count that return twice. ΔA is therefore the value of the jobs created at post-grant demand, which we use only to read WS as a lower bound on it.

Relation to benefit-cost practice. In the framework of Bartik (2012), the efficiency benefit of a labor demand shock, relative to its direct earnings effect, is $(1-f)S_{me} + (1-h)(1-g)S_{ere}$, where S_{me} and S_{ere} are the elasticities of market wages and of the employment rate with respect to the shock, h the wage gap of the newly employed, f the share of wage gains offset by employer losses, and g the share of the new earnings that represents forgone non-work time, a resource concept. With wages that do not respond, the first term vanishes, and the benefit depends on g, which that framework sets near zero for high-unemployment areas and near one for low-unemployment ones. Our reservation wages measure private surplus, which also reflects the benefits a worker gives up. Since the resource gain is at least as large as the private surplus, our estimates imply that forgone non-work time is at most about three quarters of net pay under uniform rationing, and at most about half under efficient rationing.

F.4 The cost of financing and the MVPF

Section 6.2 compares the measured labor-market gains with the cost of financing the grant and then summarizes the same components as an MVPF. The two are linked. With equal weights on workers, owners and taxpayers, equations (34) and (23) give

$$\Delta W \;=\; (G-F)\,\big[\text{MVPF} - (1+\lambda)\big], \qquad \lambda^* \;=\; \text{MVPF} - 1 \;=\; \frac{WS + F - C_K}{G-F}, \tag{36}$$

provided $G - F > 0$. The program raises welfare if and only if its MVPF exceeds $1 + \lambda$, and the break-even excess burden reported in Table 5 is the MVPF minus one. Without a

capital distortion, λ^* is 13 to 15 percent under the linear benchmark, 14 to 16 percent under uniform rationing and at least 20 to 22 percent under efficient rationing. With the calibrated distortion, λ^* is negative under the linear and random benchmarks, so the program would not break even even if public funds were costless; under efficient rationing the lower bound is also negative, which leaves the break-even value undetermined. The shares of the excess burden in Table 5 are evaluated at $\lambda = 0.25$: because the fiscal gain enters with weight $1 + \lambda$, they change with λ and should not be read as break-even values.

F.5 Jobs that outlast the sixth year

Section 6.2 values the jobs over the six years after the application. Here we extend the calculation to the twelfth year. We apply the same procedure that yields the 3.46 worker-years per direct job: at each horizon h, the firm-level employment effect estimated near the cutoff is extended to all southern winners, converted into jobs, and summed, exactly as for the sixth-year jobs. Dividing the jobs in each year by the 26,000 jobs in the sixth year gives the profile in Panel A of Table F.2.

The employment effect does not persist in full. It peaks in the sixth year; in the seventh year, the funded firms still hold 78 percent of the sixth-year jobs, and over years eight to twelve between a quarter and a half. Summed over twelve years, each direct job yields 6.08 worker-years, rather than the 9.46 implied if every sixth-year job lasted six more years. Part of the decline could reflect firms that leave the data. Restricting the sample to the 5,254 winners observed in every year from the first to the twelfth gives a similar profile and 6.46 worker-years.

Panel B of Table F.2 repeats the accounting of Table 5 over twelve years. Under the linear benchmark and uniform rationing, the benefits now roughly equal the excess burden of €44,000: 0.95 to 1.09 times it in the full sample and 1.00 to 1.15 times in the balanced sample. Discounted at 3 percent a year, they fall below it in both samples. Under efficient rationing, the lower bound exceeds the excess burden by 38 to 63 percent, or 19 to 32 percent once discounted. The cumulative benefits first cover the excess burden in the seventh or eighth year (the eighth or ninth, discounted). Under the linear benchmark and uniform rationing, they do so only in the eleventh year with the larger multiplier, and never once discounted. Incumbents' added weeks would add a further €7,900–9,700 per job over twelve years. All these totals are before the capital distortion C_K, which can only lower them. The last rows of Panel B report the MVPF of equation (23), without the capital distortion and with the calibrated distortion of Section 6.2.

Table F.2: Worker-Years and Benefits When Jobs Outlast the Sixth Year

Panel A. Jobs in each year, as a share of the jobs in the sixth year

Year	1	2	3	4	5	6	7	8	9	10	11	12
Full sample	0.07	0.24	0.52	0.68	0.96	1.00	0.78	0.43	0.43	0.23	0.51	0.24
Cumulative worker-years per job	0.07	0.31	0.83	1.51	2.47	3.46	4.24	4.67	5.10	5.33	5.84	6.08
Balanced sample						1.00	0.82	0.46	0.49	0.27	0.65	0.32
Cumulative worker-years per job												6.46

Panel B. Benefits per direct job and share of the excess burden (€44,000)

	Six years	Twelve years		
		Full sample	Balanced sample	Full sample, discounted
Worker-years per direct job	3.46	6.08	6.46	5.19
Private worker surplus, $\widehat{WS}$				
Linear benchmark	€9,500	€16,700	€17,800	€14,300
Uniform rationing	€10,000	€18,000	€19,100	€15,400
Efficient rationing	≥€19,000	≥€33,400	≥€35,500	≥€28,500
Fiscal gain, weighted by $1+\lambda$				
Income tax and unemployment benefits	€11,900	€19,500	€20,500	€16,900
Total benefits: surplus with indirect jobs, plus fiscal gain				
Linear benchmark	€24,600–27,400	€41,600–46,300	€44,000–49,000	€35,900–39,900
Uniform rationing	€25,500–28,400	€43,100–48,000	€45,600–50,800	€37,100–41,300
Efficient rationing	≥€35,600–39,600	≥€60,800–67,700	≥€64,500–71,700	≥€52,300–58,200
Total benefits as a share of the excess burden				
Linear benchmark	0.56–0.62	0.95–1.05	1.00–1.11	0.81–0.91
Uniform rationing	0.58–0.65	0.98–1.09	1.04–1.15	0.84–0.94
Efficient rationing	≥0.81–0.90	≥1.38–1.54	≥1.46–1.63	≥1.19–1.32
MVPF, no capital distortion ($C_K = 0$)				
Linear benchmark	1.13–1.15	1.23–1.26	1.25–1.28	1.20–1.22
Uniform rationing	1.14–1.15	1.24–1.28	1.26–1.29	1.21–1.23
Efficient rationing	≥1.20–1.22	≥1.36–1.40	≥1.38–1.43	≥1.30–1.34
MVPF, calibrated capital distortion ($C_K \approx$ €37,700)				
Linear benchmark	0.90–0.92	1.00–1.02	1.01–1.04	0.96–0.99
Uniform rationing	0.91–0.92	1.01–1.03	1.02–1.05	0.97–1.00
Efficient rationing	≥0.97–0.99	≥1.12–1.16	≥1.14–1.19	≥1.07–1.10

Notes: Panel A: jobs created at funded southern winners in each year after the application, from the firm-level employment effect at that horizon, divided by the jobs in the sixth year. The full sample is the one behind the 3.46 worker-years of Table 5; the balanced sample keeps the 5,254 winners observed in every year from the first to the twelfth, with shares relative to its own sixth-year jobs. Panel B: totals and shares compare benefits with a marginal excess burden of 0.25 (Appendix F.4); the MVPF is as in Table 5, with the worker-years of Panel A. The discounted column discounts each year's worker-years and fiscal gain at 3 percent a year to the first year. Ranges reflect indirect jobs at non-applicant firms, 15 to 28 percent of direct jobs. Totals and shares are before the capital distortion C_K; all rows leave out incumbents' added weeks.

F.6 When an investment subsidy raises employment

Whether a subsidy to capital raises employment at all is the question the standard view of such subsidies turns on. Consider a firm with constant-elasticity-of-substitution technology, elasticity of substitution σ_{KL} between capital and labor and cost shares θ_K and θ_L, selling under monopolistic competition with demand elasticity η, and paying the floor. A fall in the user cost of capital c changes its inputs by

$$-\frac{\partial \ln K}{\partial \ln c} \;=\; \theta_K\,\eta + \theta_L\,\sigma_{KL}, \qquad -\frac{\partial \ln L}{\partial \ln c} \;=\; \theta_K\,\big(\eta - \sigma_{KL}\big), \qquad -\frac{\partial \ln (K/L)}{\partial \ln c} \;=\; \sigma_{KL}. \tag{37}$$

Employment rises if and only if $\eta > \sigma_{KL}$: the scale effect of cheaper capital, which raises output and with it the demand for every input, must dominate the substitution of capital for labor. The employment response at funded firms shows that it does: three years after the application, capital is 0.31 log points higher and employment 0.11 log points higher in the Cerved sample, while pay does not move (0.005). Employment rose, so the scale effect dominated.